# Selecting the Right Mode: Symmetry-Aware, Orbit-Consistent Tesseract Trees on Frozen Foundation Features for 6D Object Pose Estimation

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## Abstract

*Single-view 6D object pose estimation often fails not because the correct pose cannot be represented, but because it is not* selected*: symmetric objects have several equally correct rotations, and near-symmetric ones have look-alike rotations that are wrong. We study this selection problem with a deliberately lightweight design. Two frozen foundation backbones (DINOv3 for appearance, MoGe-2 for surface geometry) feed small autoregressive Transformers that predict a rotation as a root-to-leaf path in a hierarchical tesseract partition of $SO(3)$. The two branches are fused per tree level by a training-free product of experts, and a translation block on the same tokens completes the 6D pose. Three components address selection: (i)* symmetry-aware multi-path targets *that spread the training signal over all equivalent rotations; (ii)* orbit voting*, which scores hierarchical hypotheses for consistency across in-plane rotations of the crop through an exactly derived camera-roll transform; and (iii)* orbit distillation *of that consistency into single-view students. We evaluate on T-LESS, SYMSOL-T, YCB-V and the synthetic-to-real benchmark SPEED+, training only 2.4 M parameters per branch. Multi-path targets reduce the SYMSOL-T error from $3.82°$ to $2.07°$. On SPEED+, orbit voting and distillation reduce the real-domain error from $26.9°/36.7°$ to $11.2°/11.1°$. With synthetic training data, the validation-selected YCB-V error drops from $69°$ to $12.9°$. Absolute accuracy remains below fine-tuned state-of-the-art methods, and we report matched comparisons, pre-registered ablations and negative results throughout.*

## 1. Introduction

Estimating the 6D pose of a known object from one image underlies robotic grasping, augmented reality and autonomous rendezvous. A large part of the remaining error comes from *ambiguity*. A symmetric object has several rotations that produce the same image, and every one of them is correct. A nearly symmetric object has look-alike rotations that differ only in small details, and only one of them is correct. Under a domain shift (renders to real images, new lighting), those small details are exactly what becomes unreliable. Distributional pose methods [7, 13, 16] can represent such multi-modal posteriors, but a single prediction still has to *select* a mode.

We study this selection problem with a deliberately lightweight design built on *frozen* foundation models. DINOv3 [22] gives robust appearance features, and MoGe-2 [24] gives per-pixel surface normals. A small head predicts the rotation as a path in a hierarchical partition of $SO(3)$ derived from the recursive tesseract subdivision [15]. It is an autoregressive Transformer that first chooses one of 4 charts, then one of 8 children at each of 5 levels (131,072 cells), attending to the frozen tokens. The DINOv3 and MoGe-2 trees are fused per level by a training-free product of experts (PoE) [9], and a translation block on the same tokens completes the 6D pose. Only 2.4 M parameters per branch are trained.

Hierarchical hypotheses make selection explicit, and we attack it at three points. *During training*, the target of a symmetric object is the whole set of equivalent rotations, so the tree learns to split its probability between them instead of fitting an arbitrary label (Sec. 3.3). *At test time*, we use the fact that rotating the crop in the image plane is an exact camera roll. A wrong look-alike pose tends to win only in some rotated views, whereas the true pose, mapped back through the known transform, is consistent across all of them. *Orbit voting* collects the tree's beam candidates from four rotated views and scores each candidate by its tree likelihood as seen from every view (Sec. 3.5). Finally, *orbit distillation* trains single-view students towards the orbit system's consensus (Sec. 3.6).

An audit on the synthetic-to-real benchmark SPEED+ [19] motivates the design (Sec. 4.3). Its target spacecraft is nearly symmetric. The real-domain error of the baseline is dominated by $180°$ flips about a few body axes, and an oracle that picks the best of the tree's own candidates reaches $9.8°$, against $26.9°$ for the prediction. The correct mode is usually present; it is not selected.

Our contributions are:

- A lightweight hierarchical 6D pose predictor on frozen DINOv3 and MoGe-2 features, with training-free per-level PoE fusion. It is far more accurate than a flat classifier over a 131k-anchor SPACE-HOP Hopf grid under the same recipe, with $15\times$ fewer parameters; a same-grid control that isolates hierarchical prediction is in progress (Sec. 5).
- Symmetry-aware multi-path targets for hierarchical rotation trees. They reduce the SYMSOL-T error by 20–30% relative to single or canonical labels, and, unlike canonical labels, remain compatible with orbit voting (Sec. 5).
- Orbit voting with an exactly derived roll transform for hierarchical candidates, and orbit distillation into single-view students. On SPEED+ they repair 62–80% of severe flips while breaking 0.4% of good predictions, and reduce the real-domain error to $11.2^\circ/11.1^\circ$ (Sec. 4.3).
- An evaluation on four benchmarks (T-LESS, SYMSOL-T, YCB-V, SPEED+) with comparisons matched to published protocols where possible, three seeds, pre-registered criteria, selection on validation data only, and full reporting of negative results (Secs. 4 and 5).

Symmetry-aware losses, test-time augmentation and self-distillation are established ideas. Our claim concerns their exact geometric form for hierarchical rotation hypotheses and their measured effect on mode selection, not the ideas themselves. We do not claim a new state of the art: fine-tuned methods remain more accurate on every benchmark we use.

## 2. Related Work

**Object pose estimation.** Learning-based 6D pose estimators regress poses directly or predict 2D–3D correspondences followed by PnP [25]. The BOP benchmark [11] standardises evaluation on datasets such as T-LESS [10] and YCB-V [25], where textureless and symmetric objects remain the hardest cases. Synthetic-to-real transfer is studied in its purest form on SPEED+ [19], where models train on renders of a spacecraft and are tested on hardware-in-the-loop captures. There, keypoint-and-PnP pipelines with fine-tuned backbones lead [17, 18, 20, 23]; the most accurate recent result fine-tunes DINOv3 with LoRA and reports that a head on *frozen* DINOv3 features fails with frequent flips [3]. Direct regressors are faster but less accurate [1, 5, 26]. We keep the backbones frozen and ask how far selection-aware heads can take them.

**Discrete and distributional rotation prediction.** Uniform $SO(3)$ grids can be built from the Hopf fibration [27] or by recursive tesseract subdivision [15]; we adapt the latter to cell-centre representatives with unique paths. Grid- and implicit-density methods model multi-modal rotation posteriors, through MLP scores over grids [12, 16] or equivariant spherical features [14]; parametric alternatives include Bingham mixtures [4]. SpyroPose [7] evaluates a hierarchical $SE(3)$ pyramid coarse to fine, and LiePose [13] applies score-based diffusion on $SE(3)$ and handles symmetric objects well. Continuous rotation regression has known representation issues [28]. Riemannian [2] and discrete [6] flow matching provide generative alternatives; we tested both as refiners and report them as negative results.

**Consistency, test-time augmentation and distillation.** Averaging predictions over transformed inputs is a standard route to test-time robustness, and four-rotation test-time augmentation is also used for spacecraft keypoints [3]. Data distillation [21] turns multi-transform ensembles into training targets, a form of knowledge distillation [8]. Equivariant architectures build rotation consistency into the network [14]. Our orbit voting is a test-time augmentation over in-plane rotations, and orbit distillation is a data-distillation variant. What is specific here is the exact camera-roll transform for crops of off-axis objects, scoring with hierarchical tree likelihoods rather than averaging point estimates, and the target: a diagnosed mode-selection failure.

## 3. Method

Figure 1 gives an overview of training and inference.

### 3.1. Problem setup

Given a square crop of a known object around its bounding box (15% margin per side), we predict the rotation $q \in SO(3)$ (unit quaternion, $q \equiv -q$) of the object frame relative to the camera and the translation $t \in \mathbb{R}^3$. For objects that appear away from the optical axis (T-LESS, YCB-V), the tree predicts the *allocentric* rotation $R_{\text{ray}}(t)^\top R$, which removes the viewing-ray component so that the crop's appearance depends only on the predicted rotation; it is converted back exactly. Rotation error is $E_R = 2\arccos|\langle \hat{q}, q\rangle|$, minimised over the object's symmetry set $\mathcal{S}$ when one exists ($\min_{S\in\mathcal{S}} E_R(\hat{q}, q\,S)$), and translation error is $E_T = \|\hat{t} - t\|_2$.

### 3.2. Tesseract tree on $SO(3)$

Following the recursive tesseract subdivision [15], a quaternion is assigned to a *chart* $c = \arg\max_i |q_i|$, sign-normalised so that $q_c > 0$ (this identifies $q$ and $-q$), and projected to the cube $x = q/q_c$. Its three free coordinates in $[-1, 1]$ are halved recursively, so each level selects one of 8 children. We use *cell centres* (lifted back to the unit sphere) as representatives. Unlike corner representatives, every rotation then has a unique root-to-leaf path. With 4 charts and depth 5 there are $4 \cdot 8^5 = 131{,}072$ cells. The quantisation error of the leaf centre is $2.51^\circ$ on average ($3.97^\circ$ p95, $6.20^\circ$ worst case), and it halves per level. For objects with-

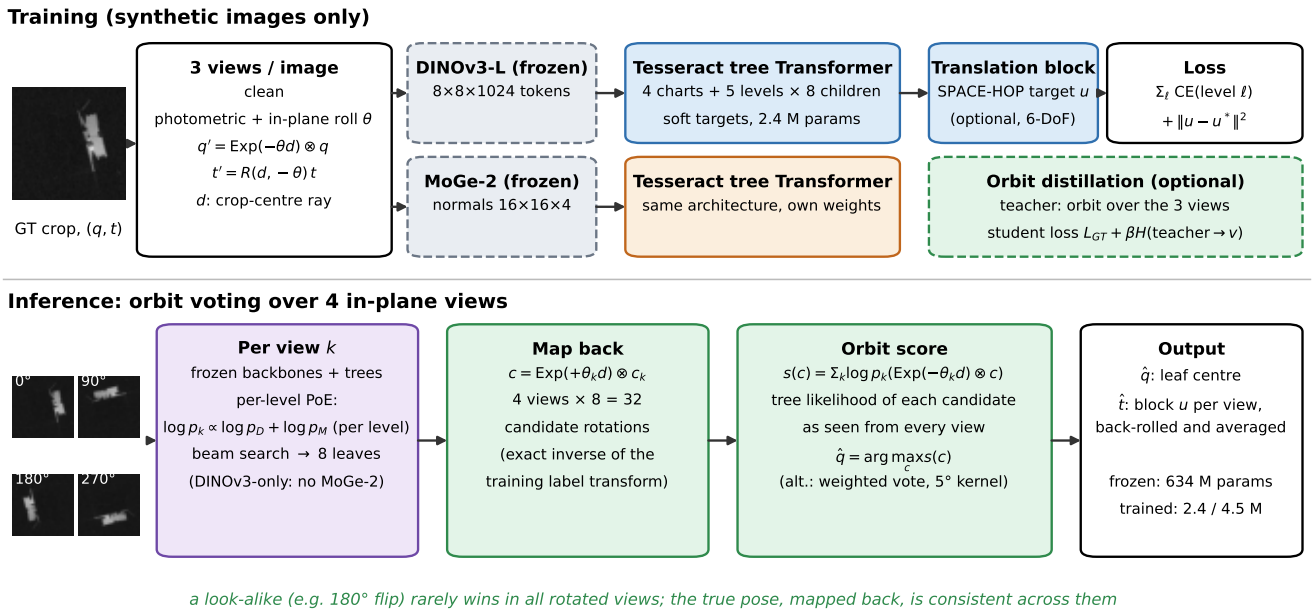



Figure 1. **Overview.** *Top:* training. Each synthetic crop is used as a clean view and as two augmented copies (photometric augmentation plus an in-plane roll $\theta$ with exactly transformed labels). Frozen DINOv3 and MoGe-2 features feed one tesseract tree Transformer each; an optional translation block reads the same tokens, and optional orbit distillation replaces the ground-truth target by an orbit teacher. *Bottom:* orbit voting at test time. Candidates from four rotated views are mapped back to a common frame and scored by their per-level PoE tree likelihood in every view.

out symmetry this floor bounds the fine accuracy of our rotation results; for symmetric objects the nearest leaf to *any* equivalent rotation is closer.

**Tree Transformer.** For each backbone, a small decoder predicts the path autoregressively. The frozen tokens are layer-normalised, linearly projected to $d = 256$, given 2-D positional encodings and passed through one encoder layer. A causal decoder with 2 layers (4 heads, FFN 512) then reads a sequence that starts with a root token followed by one token per decided level. Each such token embeds the level and a 6-frequency Fourier encoding of the current cell centre, and cross-attends to the encoded image tokens. The root position predicts the chart (4 logits) and position $\ell$ predicts the child (8 logits). The model has 2.4 M parameters. We train with geodesic soft targets: at each level, the target over the candidate cells is $\propto \exp(-(\delta/\tau_\ell)^2)$, where $\delta$ is the angle between the ground truth and each cell centre. The widths $\tau_\ell$ ($39.3^\circ$ for the root and level 1, $20.0^\circ$, $10.0^\circ$, $5.0^\circ$, $2.5^\circ$) halve with the cell size. Training uses teacher forcing on the ground-truth path and a cross-entropy at every level; inference is greedy or beam search, and the prediction is the leaf centre.

**Frozen features.** DINOv3 ViT-L/16 [22] last-layer patch tokens are average-pooled to an $8\times8$ grid ($64 \times 1024$). For geometry, MoGe-2 [24] surface normals are masked by a foreground mask on which both backbones agree (the intersection of a DINOv3 foreground estimate and the MoGe-2 mask). They are then pooled to $16\times16$ cells, each holding three masked-normal channels and the mask fraction ($256 \times 4$). This consensus mask removes most of MoGe-2's background leakage on real images (Appendix B). Features are extracted once; only the trees are trained.

**Per-level product of experts.** The two trees are fused without training. At every decoding step, the log-softmax distributions of both branches are added and renormalised: $\log p^{(\ell)} \propto \log p^{(\ell)}_{\text{DINO}} + \log p^{(\ell)}_{\text{MoGe}}$. Beam search then runs on the fused tree. We report this *PoE* variant and a faster *DINOv3-only* variant.

### 3.3. Symmetry-aware multi-path targets

For a symmetric object, the annotated rotation $q$ is one arbitrary member of the equivalence set $\{q\,s : s \in \mathcal{S}\}$, where $\mathcal{S}$ holds the object-frame symmetries (continuous symmetries discretised). Training on $q$ alone (*single label*) gives contradictory targets for identical images. The usual fix maps every label to a *canonical* member, *e.g.* the one closest to identity. This removes the contradiction but makes the target a discontinuous function of the image: two nearly identical views can receive canonical labels far apart, and the

in-plane relation of Eq. (2) no longer holds between views. Instead, we train the tree on all equivalent rotations. The loss is the soft-target cross-entropy of Sec. 3.2, averaged over the members of the set, each with teacher forcing along its own path:

$$\mathcal{L}_{\text{sym}} = \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} \mathcal{L}_{\text{GT}}(q\, s), \tag{1}$$

where $\mathcal{S}' = \mathcal{S}$ if $|\mathcal{S}| \leq 8$ and otherwise 8 members are drawn per sample and step (an unbiased estimate). The tree learns to split its probability between the equivalent paths, which is the level-marginal target of the whole set. For objects without symmetry Eq. (1) reduces exactly to the standard loss. Right-multiplication by $s$ commutes with the roll of Eq. (2), so augmented views and orbit voting are unchanged; we verified both identities numerically ($< 10^{-9}$).

### 3.4. Augmentation with exact in-plane labels

Each training image is used as a clean crop and as two pre-extracted augmented copies; one of the three is drawn at every step. The augmentations are photometric (brightness, contrast, gamma, blur, glare blobs, noise, random erasing, JPEG) plus an in-plane rotation of the crop by $\theta \sim \mathcal{U}(-180°, 180°)$ about its centre. Rotating the crop about its centre is equivalent to rotating the camera about the unit ray $d$ through the crop centre, computed with the lens distortion removed. Points therefore transform as $p' = R(d, -\theta)\, p$, and the labels as

$$q' = \text{Exp}(-\theta\, d) \otimes q, \qquad t' = R(d, -\theta)\, t. \tag{2}$$

We derived this convention from the camera geometry rather than fitting it to a predictor. Projecting model keypoints with $q'$ reproduces the warped image to $0.2°$ (p95 label error measured by PnP). For datasets whose objects lie far from the optical axis (T-LESS, YCB-V), an affine crop rotation is not exactly a camera roll (label error up to $6.7°$). There we warp with the roll homography $KR(d, -\theta)K^{-1}$ instead, which makes Eq. (2) exact (verified to $0.000°$).

### 3.5. Orbit voting

At test time we form $K{=}4$ views of the crop rotated by $\theta_k \in \{0°, 90°, 180°, 270°\}$. In view $k$ the fused tree's beam search returns 8 leaves $c_{k,1..8}$, which are mapped back to the original frame by the inverse of Eq. (2), $c = \text{Exp}(+\theta_k d) \otimes c_k$, giving 32 candidates. Each candidate $c$ is then scored by the tree likelihood of its own path when mapped into every view:

$$s(c) = \sum_{k=1}^{K} \log p_k\big(\text{path}(\text{Exp}(-\theta_k d) \otimes c)\big), \tag{3}$$

and predict $\hat{q} = \arg\max_c s(c)$. Here $\log p_k$ sums the fused root and child log-probabilities along the path, evaluated with teacher forcing. A $180°$ look-alike that wins in one view must also be likely in the other three to be selected, whereas the true pose is consistent by construction. We also evaluate a *weighted vote*, in which each candidate collects Gaussian-kernel support ($5°$) from the other candidates, weighted by their normalised beam probabilities within their view. Orbit voting has no parameters and multiplies inference cost by about $4\times$ (Tab. 5).

### 3.6. Orbit distillation

Orbit voting is more accurate than a single view, so we use it as a teacher. For every training image we run the frozen orbit system over its three available views (clean and two augmented, with known rolls). This gives 24 candidates in the clean frame with weights $w_j = \text{softmax}_j\, s(c_j)$. A new tree per branch is then trained from scratch with the same recipe and the loss

$$\mathcal{L} = \mathcal{L}_{\text{GT}}(v) - \frac{\beta}{L{+}1} \sum_j w_j \sum_{\ell=0}^{L} \log p^{(\ell)}\big(\text{pre}_\ell(\text{Exp}(-\theta_v d) \otimes c_j)\big), \tag{4}$$

with $\beta = 1$ and $\text{pre}_\ell$ the length-$\ell$ path prefix. Here $v$ is the view drawn for the image, and the teacher candidates are transported into its frame. The student is a single-view model with unchanged inference cost; at test time it can be used alone or with orbit voting. This is a data-distillation scheme [21] applied to hierarchical rotation targets.

### 3.7. Translation block (6-DoF)

For translation we keep the SPACE-HOP parameterisation [5] unchanged. With box centre $(b_x, b_y)$ and size $(b_w, b_h)$, image size $(W, H)$, intrinsics $(f_x, f_y, c_x, c_y)$ and $\alpha = 1.6$, the target is $u = (U_x, U_y, U_z)$ with

$$U_x = \tfrac{1}{b_w}\big(f_x \tfrac{X}{Z} + c_x - b_x\big), \quad U_y = \tfrac{1}{b_h}\big(f_y \tfrac{Y}{Z} + c_y - b_y\big), \tag{5}$$

$$U_z = \tfrac{1}{2}\big(\tfrac{\alpha b_w}{W} + \tfrac{b_h}{H}\big) Z, \tag{6}$$

trained with $\|u - u^*\|^2$ and inverted in closed form. A translation block (one learned query cross-attending to the tree's encoded tokens, then an MLP) predicts $u$. It is trained either jointly with the tree ($\mathcal{L}_{\text{tree}} + \mathcal{L}_{\text{trans}}$) or on a frozen tree. Rolled views use $t'$ from Eq. (2) with the unchanged box. With orbit voting, each view's $\hat{t}$ is rolled back and the four are averaged.

## 4. Experiments

### 4.1. Benchmarks and protocol

**Benchmarks.** We use four benchmarks that stress different kinds of ambiguity. **T-LESS** [10]: textureless industrial

parts with continuous, discrete or no symmetries. We use 7 objects spanning these classes (1, 4, 5, 11, 18, 21, 27), train on PBR renders and real training images, select on held-out PBR scenes, and test on all real BOP19 targets [11] of these objects. **SYMSOL-T**: five symmetric solids (tetrahedron, cube, icosahedron, cone, cylinder) with random translations, rendered with LiePose's generator and seeds [13] (so the images follow its protocol but are not guaranteed identical to its published sets). **YCB-V** [25]: household objects in cluttered real videos. We use 6 objects without declared symmetries (cracker box, mustard bottle, banana, mug, power drill, scissors), validate on held-out real sequences and test on the real keyframes. **SPEED+** [19]: a nearly symmetric spacecraft, 47,966 synthetic training and 11,994 synthetic validation images, tested on two real domains, *lightbox* (6,740) and *sunlamp* (2,791). Training uses synthetic images only.

**Protocol.** Experiments followed pre-registered protocols: question, arms, controls and pass criterion were fixed before each run, and protocol changes were recorded before the affected runs. Post-hoc analyses are labelled as such. We train 3 seeds and report mean $\pm$ seed standard deviation; a criterion passes if the difference exceeds the larger seed standard deviation. Checkpoints, hyper-parameters and variants are selected on validation data only; real test sets are never used for selection. One tree per object and branch is trained with the same recipe on every dataset. Errors are averaged per object and then over objects. Unless stated otherwise, crops come from ground-truth boxes (Sec. 6). Failed criteria and negative results are reported (Appendix C).

**Implementation.** Trees are trained for 100 epochs with batch 256, AdamW (learning rate $10^{-3}$, weight decay 0.05, one-cycle schedule) and bf16, which takes about 3–20 minutes per tree on one shared GPU (about 13 minutes for a DINOv3 tree under the locked SPEED+ recipe; Appendix K). Timings are measured end-to-end at batch 1 on a shared NVIDIA RTX 6000 Ada and are upper bounds.

### 4.2. Comparison with published results

**T-LESS (Tab. 1).** Orbit voting raises recall at all thresholds for the single-label trees (R@10 68.1 $\rightarrow$ 73.5%), and canonical labels help the single view but break orbit voting (Sec. 5). Multi-path targets (A2) are best at every threshold, with or without orbit voting: with orbit voting, R@2/5/10 rise to 5.7/48.1/79.0% and the mean error falls to 15.2°, and on validation the orbit error drops from 9.25° (A0) to 8.28°, which passes the pre-registered criterion. Unlike canonical labels, multi-path targets keep orbit voting effective. LiePose is clearly more accurate, especially at fine thresholds (R@2 34.9% vs. 5.7%). Three factors are confounded in this gap: LiePose fine-tunes its backbone on all 30 objects, it masks the crop with the ground-truth visible mask, and our depth-5 leaves quantise rotations to 2.5° on average, which caps R@2.

| Method | Inference | R@2 | R@5 | R@10 | mean (°) |
|---|---|---|---|---|---|
| LiePose [13] | 1 sample | 34.9 | 78.2 | 88.9 | – |
| Ours, A0 | 1 view | 2.3 | 27.0 | 68.1 | 17.9 |
| | orbit (4) | 4.0 | 36.0 | 73.5 | 16.6 |
| Ours, A1 | 1 view | 3.6 | 33.8 | 73.4 | 16.7 |
| | orbit (4) | 2.7 | 26.7 | 65.9 | 18.2 |
| Ours, A2 | 1 view | 4.5 | 40.6 | 76.5 | 15.7 |
| | orbit (4) | 5.7 | 48.1 | 79.0 | 15.2 |
| Ours, A2 + distill. | 1 view | | | | |
| | orbit (4) | | | | |

Table 1. **T-LESS**, real BOP19 test targets of our 7 objects. Symmetry-aware rotation recall (%) at 2/5/10° and mean error, PoE variant, 3 seeds. LiePose numbers are its per-object results (Tab. 8 of [13]) averaged over the same 7 objects; it fine-tunes its backbone jointly on all 30 objects and multiplies the crop by the ground-truth visible mask. Ours: frozen backbones, ground-truth box, no mask; continuous symmetries (objects 1 and 4) are discretised in 10° steps for the error, which can add up to 5° for those objects. A0/A1/A2: single, canonical and multi-path targets (Sec. 3.3). Blank cells: runs not yet completed.

| Method | tet | cube | icosa | cone | cyl | avg. |
|---|---|---|---|---|---|---|
| LiePose [13] | 0.59 | 0.58 | 0.64 | 0.54 | 0.41 | 0.55 |
| Ours, A0 + orbit | 3.06 | 2.90 | 3.71 | 1.79 | 1.65 | 2.62 |
| Ours, A2 + orbit | 2.46 | 2.30 | 2.52 | 1.57 | 1.48 | 2.07 |
| Ours, A2 + distill. + orbit | | | | | | |

Table 2. **SYMSOL-T** test, symmetry-aware mean rotation error (°) per shape. Ours: DINOv3 variant (selected on validation), orbit voting, 3 seeds. LiePose: SE(3) model, Tab. 3 of [13]. Blank cells: runs not yet completed.

| Training data | Feat. | Inference | val | test | test acc@10 |
|---|---|---|---|---|---|
| real videos | DINOv3 | 1 view | 64.3 ±11.6 | 71.2 ±7.0 | 8% |
| | DINOv3 | orbit (4) | 63.0 ±2.4 | 69.4 ±5.2 | 9% |
| | PoE | 1 view | 73.5 ±6.8 | 81.1 ±7.3 | 16% |
| | PoE | orbit (4) | 74.1 ±7.4 | 79.9 ±3.2 | 18% |
| + synthetic (ours) | DINOv3 | 1 view | 9.0 ±1.1 | 12.9 ±6.9 | 85% |
| | DINOv3 | orbit (4) | 11.5 ±1.8 | 13.4 ±4.7 | 83% |
| | PoE | 1 view | 10.5 ±0.9 | 10.9 ±4.1 | 87% |
| | PoE | orbit (4) | 12.7 ±1.0 | 12.2 ±5.5 | 84% |

Table 3. **YCB-V**, 6 objects, rotation error (°) on held-out real sequences (val) and real keyframes (test), and the share of test predictions within 10°. Adding synthetic renders composited on real training backgrounds turns a failure into a working model.

**SYMSOL-T (Tab. 2).** Multi-path targets lower the error on every shape, from 2.62° to 2.07° on average. For symmetric shapes this is below the 2.31° error of the ground-truth rotation's own leaf, because the tree may pick the leaf

| Method | Setting | $E_R$ (°) | | score | |
|---|---|---|---|---|---|
| | | LB | SL | LB | SL |
| TangoUnchained* [20] | detector + keypoints/PnP, real images | 3.19 | 4.30 | 0.073 | 0.090 |
| lava1302* [20] | detector + keypoints/PnP, self-training | 6.66 | 2.73 | 0.165 | 0.059 |
| SPNv2 [17] | full image, no adaptation | 6.48 | 11.07 | 0.142 | 0.230 |
| SPNv2 + ODR* [17] | + test-time adaptation | 5.58 | 9.79 | 0.122 | 0.197 |
| EagerNet [23] | detector + dense correspondences | 1.75 | 2.66 | 0.039 | 0.059 |
| SPNv3-B† [18] | GT crop, keypoints/PnP, fine-tuned | 2.03 | 3.40 | 0.047 | 0.074 |
| Church & Nikolian† [3] | GT crop, DINOv3-H+ LoRA, 12 passes | 1.17 | 1.56 | 0.027 | 0.038 |
| frozen DINOv3-L + head [3] | GT crop, frozen, 1 pass | 15.47 | 24.49 | | |
| FastPose-ViT [1] | detector, direct regression | 12.26 | 18.28 | | |
| PAID-ViT [26] | crop, direct regression | 12.29 | 11.76 | 0.237 | 0.230 |
| Ours, distilled + orbit | GT crop, frozen, 4 views | 11.21 | 11.09 | 0.227 | 0.224 |
| Ours, 6-DoF (joint PoE) + orbit | GT crop, frozen, 4 views | 12.11 | 11.73 | 0.241 | 0.238 |
| Ours, distilled + orbit | synthetic-only detector, frozen | | | | |

Table 4. **SPEED+ real domains** (lightbox LB, sunlamp SL): mean rotation error and the official SPEED+ score (rotation error in radians plus normalised translation error, averaged over images; lower is better). Published values as reported by each source. * uses real or unlabelled test-domain images during training or adaptation. † ensemble of 3 models. Settings differ (detector or ground-truth crop, fine-tuned or frozen backbone), so only rows with the same setting are matched comparisons. Blank cells: runs not yet completed.

closest to *any* equivalent rotation. LiePose remains about $4\times$ more accurate; closing the gap needs a continuous refinement on top of the selected leaf.

**YCB-V (Tab. 3).** Trained on the real training videos alone, the trees fit their training frames but memorise video context and fail on new sequences (test 69–81°; Appendix G). Adding the standard synthetic renders, pasted onto real training-sequence backgrounds, turns this into a working model. The configuration selected on validation (DINOv3, single view, 9.0° validation) reaches 12.9° on the real test keyframes, with 85% of predictions within 10°; the PoE single-view model is better on test (10.9°, 87%) but is not the validation choice. Five of the six objects reach 4–8°, and the thin, near-planar scissors remain the main failure (Tab. 17). Unlike on the other benchmarks, orbit voting does *not* help on YCB-V: it raises the validation error from 9.0° to 11.5° (DINOv3) and from 10.5° to 12.7° (PoE). We have not yet diagnosed why; one hypothesis is that, with cluttered real backgrounds and a narrow range of camera roll in the training videos, the rotated views are scored less reliably than the original view (Appendix L). Published YCB-V results use BOP or ADD(-S) metrics on all 21 objects with detected boxes, so we do not place our numbers in those tables.

**SPEED+ (Tab. 4).** With ground-truth crops, our frozen-feature system reaches $11.2°/11.1°$, close to recent direct methods (PAID-ViT $12.3°/11.8°$) and well ahead of a head on frozen DINOv3 features in the same crop setting ($15.5°/24.5°$ [3]). Fine-tuned keypoint pipelines are far more accurate (1.2°–3.4°). Our official score with ground-truth crops is $0.227/0.224$ for the distilled system (rotation from the distilled PoE students with orbit voting, translation from the jointly trained DINOv3 block) and $0.241/0.238$ for the jointly trained 6-DoF PoE model. This is similar to SPNv2 without adaptation on sunlamp and to PAID-ViT, and it is dominated by the rotation term.

### 4.3. Case study: a nearly symmetric target under domain shift

**Diagnosis: real-domain errors are flips.** Before changing the model, we audited the large real-domain errors of the clean (non-augmented) SPEED+ trees. Encoder and decoder round trips are exact, and the metric, rotation convention and data pipeline were ruled out. Chart seams were ruled out as well: flat Hopf-grid heads on the same features flip as often or more. Of the errors above 90°, 36–41% lie within 15° of 180°, against 20% for uniformly random wrong rotations. Their body-frame rotation axes cluster around a few axes of the near-symmetric spacecraft. Allowing a 180° rotation about one body axis lowers the lightbox mean of the DINOv3 tree from 38.9° to 23.5°. The correct mode is usually available: among the tree's beam candidates and their 180° flips, an oracle achieves 9.8° on lightbox, against 26.9° for the fused greedy prediction. Neither a continuous flow posterior, flow-based re-noise scoring, synthetic templates nor a discrete path flow selects it on real images (Appendix C). The problem is *mode selection* under domain shift.

**Orbit voting and distillation.** Table 5 summarises the progression. Augmentation alone (photometric plus roll,

| Training | Feat. | Inference | Synth. val | Lightbox | Sunlamp | >150° (%) | | ms |
|---|---|---|---|---|---|---|---|---|
| | | | mean $E_R$ (°), 3 seeds | | | LB | SL | |
| Clean tree (no augmentation) | PoE | 1 view | 5.54 ±0.07 | 26.91 ±0.78 | 36.66 ±0.93 | 5.9 | 7.4 | 64 |
| Augmented tree (base recipe) | DINOv3 | 1 view | 4.84 ±0.05 | 22.55 ±0.73 | 22.00 ±0.70 | 6.0 | 3.2 | 15 |
| Augmented tree (base recipe) | DINOv3 | orbit (4) | 4.21 ±0.04 | 17.40 ±0.68 | 15.27 ±0.64 | 4.8 | 2.0 | 70 |
| + orbit distillation | DINOv3 | 1 view | 4.57 ±0.05 | 21.61 ±0.82 | 20.55 ±0.94 | 5.8 | 3.1 | 15 |
| + orbit distillation | DINOv3 | orbit (4) | 3.90 ±0.03 | 16.43 ±0.75 | 13.64 ±0.57 | 4.5 | 1.6 | 70 |
| Augmented tree (base recipe) | PoE | 1 view | 4.85 ±0.03 | 17.33 ±0.31 | 20.15 ±0.57 | 4.2 | 3.3 | 64 |
| Augmented tree (base recipe) | PoE | orbit (4) | 3.87 ±0.03 | 12.30 ±0.50 | 12.25 ±0.37 | 3.0 | 1.7 | 279 |
| + orbit distillation | PoE | 1 view | 4.56 ±0.03 | 15.90 ±0.53 | 18.33 ±0.66 | 3.7 | 3.0 | 64 |
| + orbit distillation | PoE | orbit (4) | **3.69** ±0.02 | **11.21** ±0.42 | **11.09** ±0.49 | 2.6 | 1.4 | 279 |

Table 5. **SPEED+ rotation, ground-truth crops.** Mean $E_R$ over 3 seeds (± seed std), fraction of severe flips (>150°), and end-to-end latency at batch 1. The clean tree is trained without augmentation. The DINOv3 student is distilled from a DINOv3-only orbit teacher, and the PoE students from the PoE orbit teacher. Inference cost is unchanged by distillation.

single view) cuts the real-domain error of the PoE system from 26.9°/36.7° to 17.3°/20.2°. Orbit voting then brings it to 12.3°/12.3° and the synthetic error to 3.87°. It repairs 62% and 80% of the severe flips on lightbox and sunlamp while breaking only 0.4% of good predictions (Tab. 13). The median errors on real images are 5.3° and 6.3°, so the remaining means are driven by a tail of failures. Orbit distillation improves every setting. The single-view students recover 13–43% of the orbit gain at single-view cost; for example, the PoE greedy error on lightbox drops from 17.3° to 15.9°. Students with orbit voting give the best system: 3.69° synthetic and 11.21°/11.09° on lightbox/sunlamp, about twice the seed std better than the undistilled orbit system, with fewer severe flips. For distillation, synthetic validation ranks the systems in the same order. For the orbit *scoring rule* it does not: the likelihood product was fixed in the base recipe before the scoring ablation was run, and synthetic validation would have selected the weighted vote instead (Sec. 5), which gives 3.27°/11.35°/11.91° with the distilled PoE students. MoGe-2 helps on real images: PoE with orbit voting reaches 12.3° on lightbox, against 17.4° for DINOv3-only. It costs about 50 ms per view, half of which is the CPU consensus-mask step. DINOv3-only orbit voting runs at 14 FPS and a single DINOv3 view at 68 FPS (Appendix A).

**Full 6-DoF.** Table 14 (supplementary) adds translation with the SPACE-HOP target [5]. Because our crops come from ground-truth boxes, the box alone already locates the target to 0.55 m. The translation block improves on this by about 7–10× on synthetic data (both pre-registered criteria pass). Joint training does not hurt rotation and even helps slightly on real images: PoE + orbit reaches 3.86°/12.11°/11.73°, against 3.87°/12.30°/12.25°. Orbit averaging lowers translation error by 10–25%. The DINOv3 block is the most accurate on synthetic data (0.059 m) and reaches 0.185/0.183 m on the real domains; the MoGe-2 block alone is weak for translation. An evaluation with a synthetic-only detector instead of ground-truth boxes is in progress (last row of Tab. 4).

| | | T-LESS (7 obj.) | | SYMSOL-T (5) | |
|---|---|---|---|---|---|
| Training target | Inference | val | test | val | test |
| single annotated label | 1 view | 12.3 ±0.1 | 17.9 ±0.2 | 3.87 ±0.03 | 3.88 ±0.01 |
| | orbit (4) | 9.3 ±0.1 | 16.6 ±0.2 | 2.65 ±0.01 | 2.67 ±0.00 |
| canonical label | 1 view | 11.5 ±0.0 | 16.7 ±0.1 | 3.40 ±0.01 | 3.39 ±0.01 |
| | orbit (4) | 10.7 ±0.0 | 18.2 ±0.7 | 9.73 ±0.05 | 9.69 ±0.05 |
| all equivalents (ours) | 1 view | 10.8 ±0.1 | 15.7 ±0.2 | 2.72 ±0.01 | 2.71 ±0.01 |
| | orbit (4) | 8.3 ±0.0 | 15.2 ±0.3 | 2.10 ±0.01 | 2.11 ±0.01 |
| + orbit distillation | 1 view | | | | |
| | orbit (4) | | | | |

Table 6. **Training targets for symmetric objects** (symmetry-aware mean error, PoE). Single annotated label (A0), canonical label closest to identity (A1), and all equivalent rotations (A2, Eq. (1)); the last block adds orbit distillation on top of the selected target. Blank cells: runs not yet completed.

## 5. Ablations

All ablations use the PoE variant, 3 seeds and the same evaluation run within each table.

**Symmetry-aware targets.** Table 6 compares the three training targets. Canonical labels (A1) remove the contradiction between equivalent labels and improve single-view accuracy on both datasets (SYMSOL-T 3.87° → 3.40°, T-LESS validation 12.3° → 11.5°). However, they break orbit voting: a canonical label is not related across in-plane rotations by Eq. (2), so the views disagree and the SYMSOL-T orbit error rises to 9.7°. Multi-path targets (A2) keep the relation and are the best single-view and orbit models

| Head (params/branch) | Grid | Floor | Inference | Synth. | LB | SL |
|---|---|---|---|---|---|---|
| Tesseract tree (2.4 M) | 131,072 leaves | 2.51 | 1 view | 4.85 ±0.03 | 17.33 ±0.31 | 20.15 ±0.57 |
| | | | orbit | 3.87 ±0.03 | 12.30 ±0.50 | 12.25 ±0.37 |
| Flat (36.1 M) | Hopf 10,923×12 | 6.06 | 1 view | 8.95 ±0.06 | 20.56 ±0.70 | 27.16 ±1.04 |
| | | | orbit | 7.53 ±0.01 | 13.15 ±0.37 | 16.31 ±0.59 |
| Flat, same grid | 131,072 leaves | 2.51 | 1 view | | | |
| | | | orbit | | | |
| Flat, balanced Hopf | Hopf 1,771×74 | 2.29 | 1 view | | | |
| | | | orbit | | | |

Table 7. **Tree vs. flat grid** (PoE, mean error in °). Same frozen features, augmentation, training budget and orbit inference (4 views, top-8 per view). Floor: mean error of the nearest anchor (tree: its leaf) on the synthetic-validation rotations. The *same-grid* flat head classifies the tree's own 131,072 leaves, isolating hierarchical prediction; the balanced Hopf grid tests an alternative discretisation of the same size. Blank cells: runs not yet completed.

on SYMSOL-T (2.72° and 2.10°; pre-registered criterion passes by about 100× the seed std).

**Hierarchical tree vs. flat grid.** We replace the tree with a count-matched flat classifier over 131,076 Hopf-fibration anchors [27] in SPACE-HOP's configuration (10,923 directions × 12 rolls), keeping features, augmentation, training budget and orbit inference identical (Tab. 7). The tree is far more accurate (3.87° vs. 7.53° synthetic with orbit voting), better on real data in all comparisons, and uses 15× fewer parameters per branch (2.4 M vs. 36.1 M, of which 33.6 M is the 131k-way output layer). Orbit voting helps both heads, and the flat head with orbit voting even has slightly fewer severe flips on lightbox, so flip repair comes mainly from orbit voting.

This comparison does not isolate hierarchical prediction. With only 12 in-plane rolls, the SPACE-HOP grid alone has a quantisation floor of 6.06° on the synthetic-validation rotations, against 2.51° for the tree's leaves, so part of the gap is due to the grid. We therefore run a controlled comparison with two further flat heads, trained with the identical recipe and evaluated with identical orbit inference (4 views, top-8 candidates per view, orbit-product scoring). The first classifies the tree's *own* 131,072 leaf centres, and scores an orbit candidate by the probability of its Tesseract cell, exactly as the tree does; only the prediction (hierarchical vs. flat) differs. The second classifies a balanced Hopf grid of 1,771 Fibonacci directions × 74 rolls (131,054 anchors), an alternative discretisation of the same size whose floor (2.29° mean, 3.8° max) is *lower* than the tree's (2.51°, 5.8°). Both flat heads have about 33.6 M output-layer parameters per branch. Their results will fill the blank rows of Tab. 7; if the same-grid flat head matches the tree, the advantage in the first comparison is due to the grid rather than to hierarchical prediction.

**Cost of the output layer.** Measured on the prediction head alone with an earlier version of the architecture, a flat 131k-way head is about 3× faster than the tree's greedy decoding on a large GPU at batch 1 (0.40 vs. 1.23 ms), but 1.5× slower at batch 256, 2.7× slower on a small GPU and 7× slower on one CPU thread. Either head is small compared with the backbone (DINOv3 about 6 ms per view), so the tree's practical advantage is its size and its behaviour on constrained hardware, not GPU latency. The controlled comparison above also measures head and full-pipeline latency and training and inference memory for all three heads under identical conditions.

| Selection over 32 candidates | Synth. | LB | SL |
|---|---|---|---|
| 1 view (greedy) | 4.85 ±0.03 | 17.33 ±0.31 | 20.15 ±0.57 |
| max-confidence | 5.51 ±0.03 | 14.63 ±0.42 | 15.92 ±0.15 |
| top-1 vote | 4.37 ±0.03 | 13.24 ±0.42 | 13.38 ±0.57 |
| weighted vote (32) | 3.46 ±0.03 | 12.58 ±0.42 | 12.88 ±0.49 |
| orbit product | 3.87 ±0.03 | 12.30 ±0.50 | 12.25 ±0.37 |

Table 8. **Selection rule over the same 32 orbit candidates.**

| Augmentation | 1 view | | | orbit (4) | | |
|---|---|---|---|---|---|---|
| | Syn. | LB | SL | Syn. | LB | SL |
| none | 5.54 ±0.07 | 26.90 | 36.66 | 4.62 ±0.04 | 20.25 | 24.83 |
| photometric | 5.00 ±0.03 | 17.38 | 19.68 | 4.25 ±0.06 | 12.29 | 12.37 |
| in-plane roll | 4.91 ±0.00 | 24.30 | 31.52 | 3.78 ±0.03 | 18.69 | 20.70 |
| both | 4.85 ±0.03 | 17.33 | 20.15 | 3.87 ±0.03 | 12.30 | 12.25 |

Table 9. **Which augmentation matters.** *None* is the clean tree. The roll uses exact labels (Eq. (2)).

**How to select among orbit candidates.** With the same 32 candidates (Tab. 8), picking the most confident one is clearly worse than any form of agreement, and voting among the four views' top-1 predictions is better. Using all candidates helps most. The kernel-weighted vote is best on synthetic data (3.46°). The likelihood product of Eq. (3) is best on both real domains, but its advantage over the vote is within about one seed std. Its pre-registered criterion ("product beats voting on synthetic validation") *fails*, and under our selection protocol the weighted vote would be preferred. The product is reported in Tab. 5 only because it was fixed in the base recipe beforehand. We therefore claim multi-view consistency over hierarchical candidates, not a specific scoring rule.

**Photometric vs. rotation augmentation.** Table 9 separates the two augmentation families. Photometric augmentation with orbit voting reaches 12.3°/12.4° on real images, the same as the combined recipe. Rotation augmentation alone gives the best synthetic precision (3.78°) but leaves real errors at 18.7°/20.7°. Orbit voting on the clean

tree already removes a quarter to a third of the real error ($26.9° \rightarrow 20.3°$, $36.7° \rightarrow 24.8°$). The real-domain gain therefore comes from *lighting robustness plus orbit consistency*. The exact roll transform is needed for orbit voting itself; as a training augmentation it mainly improves synthetic precision. This corrects our initial hypothesis that rotation augmentation causes the flip repair.

**Can training learn orbit consistency?** On SPEED+ we compared paired views of the same image (a data control), a transported consistency loss on the leaf distribution (a standard equivariance regulariser), a *hierarchical* transported loss at every tree level, and orbit distillation (Tab. 15, supplementary). The hierarchical loss beats the base recipe and the paired-view control but not the leaf-only loss; its pre-registered criterion fails, so we make no hierarchical-transport claim. Both consistency losses mostly improve synthetic precision. Orbit distillation is the only objective that clearly improves the real domains ($11.2°/11.1°$ with orbit voting).

**Further ablations** in the supplementary material cover the decoder lineage (MLP, GRU and Transformer decoders, hard vs. soft targets, beam search), fusion (single branches, learned fusion, weighted PoE), finer tokens, continuous flow posteriors and verifiers, discrete flow matching, oracle headroom, zero-shot transfer and timing.

## 6. Limitations

- **Ground-truth crops.** The reported results use ground-truth boxes; a synthetic-only detector for SPEED+ is being evaluated (Tab. 4). For translation in particular, the box alone gives 0.55 m on SPEED+, so a deployable claim needs detected boxes.
- **Translation beyond SPEED+.** Translation blocks for T-LESS, SYMSOL-T and YCB-V are still being evaluated, so these datasets are currently compared on rotation only.
- **Matched comparisons.** Published numbers use different crops, masks, object sets and fine-tuned backbones; Tabs. 1, 2 and 4 state each setting, and only rows with the same setting are matched.
- **Hierarchy vs. grid.** The flat-baseline comparison of Sec. 5 uses a coarser grid than the tree; the same-grid control is still running, so the share of the gain due to hierarchical prediction is not yet established.
- **Quantisation floor.** The depth-5 leaf quantisation ($2.51°$ mean) bounds fine accuracy for asymmetric objects. Our best synthetic SPEED+ errors (3.3–3.7°) are close to it, and it caps fine-threshold recall on T-LESS. A continuous refinement on top of the selected mode is a natural next step; the refiners we tried helped only on synthetic data (Appendix C).
- **Per-object heads.** One tree is trained per object; there is no zero-shot transfer to new objects (Appendix H).
- **Cost.** Orbit voting costs $4\times$ inference, and MoGe-2 fusion does not help on every dataset.
- **Timing.** Timings come from a shared GPU and are upper bounds.

## 7. Conclusion

A small tesseract tree on frozen DINOv3 and MoGe-2 features is a competitive basis for 6D object pose when its hypotheses are *selected* well. Three components target selection: multi-path targets that spread the training signal over all symmetric equivalents, orbit voting that scores hierarchical hypotheses for consistency across exactly mapped in-plane rotations, and orbit distillation that moves this consistency into single-view students. Multi-path targets improve T-LESS at every recall threshold and every SYMSOL-T shape and, unlike canonical labels, remain compatible with orbit voting. On the nearly symmetric SPEED+ target, orbit voting and distillation repair most real-domain flips. Synthetic training data turns YCB-V from a failure into a working model. Fine-tuned methods remain more accurate on all four benchmarks, and our evaluation is intentionally conservative; we hope the protocol, the matched comparisons and the negative results are as useful as the method.

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# Selecting the Right Mode: Symmetry-Aware, Orbit-Consistent Tesseract Trees on Frozen Foundation Features for 6D Object Pose Estimation

Supplementary Material

## A. Timing

| System | batch 1 | | batch 64 | |
|---|---|---|---|---|
| | ms | FPS | ms/img | img/s |
| DINOv3, 1 view | 14.7 | 68.1 | 2.9 | 347 |
| DINOv3 + orbit | 69.6 | 14.4 | 12.1 | 82 |
| PoE, 1 view | 64.0 | 15.6 | 50.0 | 20 |
| PoE + orbit | 278.5 | 3.6 | 206.8 | 5 |

Table 10. **End-to-end latency** of the augmented trees (shared RTX 6000 Ada; upper bounds; image already decoded).

At batch 1 (Tab. 10), a DINOv3 forward pass takes 6.4 ms and one tree 7.4 ms. MoGe-2 adds 21 ms for the network and 20 ms for the CPU consensus-mask step per view, which is the main optimisation target. Orbit voting multiplies the cost by 4.4–4.7×. Batched evaluation reduces tree cost to 0.1–0.3 ms per image.

## B. Model lineage and fusion

| Design (clean trees, PoE greedy) | Synth. | LB | SL |
|---|---|---|---|
| MLP path decoder, hard targets | 6.24 | 31.1 | 43.7 |
| GRU path decoder | 6.15 | – | 40.6 |
| MLP decoder, geodesic soft targets | 6.07 | 28.2 | 42.1 |
| Transformer decoder, hard targets | 5.69 | 31.6 | 41.8 |
| **Transformer + soft targets** | **5.54** | **26.9** | **36.7** |
| + beam 8 instead of greedy | no gain | – | – |
| + tangent residual at the leaf | −0.13 | ±0.0 | ±0.0 |
| 16×16 instead of 8×8 DINOv3 tokens | 5.56 | 27.0 | 36.0 |
| DINOv3 tree alone | 5.35 | 37.1 | – |
| MoGe-2 tree alone | – | 43.7 | – |
| Learned concat fusion, one Transformer (8×8) | 5.34 | 40.8 | – |
| Equal per-level PoE (same grid) | 5.58 | 27.9 | – |
| Weighted PoE, $w_{\text{DINO}}$=0.8 (proxy-selected) | 5.14 | 32.0 | 36.7 |

Table 11. **Design steps** on clean (non-augmented) trees, seed means. Dashes: not evaluated in that study.

Table 11 lists the design steps that led to the base tree. The Transformer decoder with geodesic soft targets was best; beam search at full data brought no gain, and a tangent residual only refined errors within the correct cell. Fusion is strongly asymmetric across domains. A learned fusion that feeds concatenated tokens into one Transformer matches DINOv3 alone on synthetic data but is much worse on real images (40.8° vs. 27.9° for PoE on the same grid). We interpret this as the learned fusion over-fitting render statistics, while training-free PoE keeps the branches' errors partly independent. A weighted PoE selected on the rotated validation proxy ($w$=0.8) is 5° worse on lightbox than equal weights. Following our protocol, we did not use real data to choose another weight. The consensus foreground mask reduces MoGe-2's background leakage on real images from 0.52 to 0.12 (lightbox) and from 0.57 to 0.12 (sunlamp). It is what makes the MoGe-2 branch useful.

## C. Negative results

| Approach (on clean trees unless noted) | Synth. | LB | SL |
|---|---|---|---|
| Clean PoE tree (reference) | 5.54 | 26.9 | 36.7 |
| $SO(3)$ flow posterior on tree memory, KDE mode | 5.22 | 29.8 | 37.6 |
| best of 32 samples (*oracle*) | 2.60 | 9.7 | 10.6 |
| Flow re-noise verifier over beams + flips + corners | 4.78 | 26.5 | 32.7 |
| best candidate (*oracle*) | 2.25 | 7.9 | 11.4 |
| Kurz corner tree [15] | 5.20 | 28.8 | 39.1 |
| Tree distilled from flow masses ($\beta$=1, 1 seed) | 5.62 | 26.2 | 37.7 |
| Discrete flow matching on tree paths | 7.22 | 29.8 | 39.8 |
| Aug. DINOv3 tree (reference) | 4.84 | 22.6 | 22.1 |
| + $SO(3)$ flow, KDE | 4.10 | 25.2 | 24.4 |

Table 12. **Refiners and verifiers that did not fix real flips** (seed means). Oracles are not deployable and show headroom only.

Table 12 collects approaches that improved synthetic accuracy but not, or only slightly, real accuracy. A Riemannian flow-matching posterior [2] conditioned on the tree's encoder memory contains good samples: its best-of-32 oracle reaches 9.7° on lightbox. It also assigns the ground truth a much higher log-likelihood than the tree does (3.85 vs. 0.98 nats on lightbox), and its 90% regions over-cover (0.98). Yet its mode estimate is worse than the tree's. Re-scoring an expanded candidate set (beam, 180° flips, cell corners) by re-noise/return consistency helps synthetic data and sunlamp, but does not repair lightbox flips. Synthetic render-and-compare templates also failed to choose the right flip. A discrete flow on the tesseract path [6] was clearly worse. We note that interpolating path tokens is not geometry-preserving: 45–53% of intermediate paths lie more than 30° from both endpoints. Distilling flow masses into the tree did not pass its criterion. An oracle that stops decoding at the best depth would gain only 0.51° on synthetic data, so adaptive depth was not pursued. These results support the diagnosis of Sec. 4.3: the right mode is reachable, but scorers trained on clean synthetic features cannot select it on real images. Lighting-robust training and multi-

view consistency can.

## D. Flip transitions on SPEED+

| System | Lightbox | | Sunlamp | |
|---|---|---|---|---|
| | fixed | broken | fixed | broken |
| Aug. tree, 1 view | 51% of 401 | 1.0% | 59% of 206 | 1.0% |
| Aug. tree + orbit | 62% of 401 | 0.4% | 80% of 206 | 0.4% |
| Aug. DINOv3 + orbit | 52% of 401 | 1.3% | 74% of 206 | 0.5% |

Table 13. **Flip transitions relative to the clean PoE tree.** *Fixed*: share of the clean tree's $>150^\circ$ errors that become $<20^\circ$ (denominator given). *Broken*: share of its $<20^\circ$ predictions that become $>150^\circ$.

## E. Orbit distillation and 6-DoF details

| Arm | Feat. | $\hat{t}$ | $E_T$ (m) | | | $E_R$ (°), orbit | | |
|---|---|---|---|---|---|---|---|---|
| | | | Syn. | LB | SL | Syn. | LB | SL |
| Box only | – | – | 0.561 | 0.547 | 0.553 | – | – | – |
| Frozen | DINOv3 | 1 view | 0.090 | 0.233 | 0.240 | 4.21 | 17.40 | 15.27 |
| | | orbit | 0.068 | 0.205 | 0.205 | | | |
| Frozen | PoE | 1 view | 0.108 | 0.206 | 0.232 | 3.87 | 12.30 | 12.25 |
| | | orbit | 0.089 | 0.184 | 0.206 | | | |
| Joint | DINOv3 | 1 view | 0.078 | 0.209 | 0.212 | 4.19 | 17.31 | 14.87 |
| | | orbit | **0.059** | 0.185 | **0.183** | | | |
| Joint | PoE | 1 view | 0.111 | 0.196 | 0.221 | 3.86 | 12.11 | 11.73 |
| | | orbit | 0.093 | **0.175** | 0.194 | | | |

Table 14. **SPEED+ 6-DoF** with the SPACE-HOP translation target, ground-truth crops. Rotation uses PoE or DINOv3 orbit voting. Translation ($\hat{t}$) is predicted from one view or as the orbit average; seed std of $E_T \leq 0.005$ m. *Box only* uses no image: the object centre is placed at the box centre and the depth at the training mean of $U_z$.

| Training objective | 1 view, DINOv3 | | PoE + orbit | | |
|---|---|---|---|---|---|
| | Syn. | Syn.-rot. | Syn. | LB | SL |
| Base recipe (GT loss) | 4.84 ±0.05 | 9.88 ±0.22 | 3.87 | 12.30 | 12.25 |
| + paired views | 4.96 ±0.06 | 10.04 ±0.21 | 3.94 | 12.38 | 12.45 |
| + leaf-level transported consistency | 4.59 ±0.04 | 9.43 ±0.14 | 3.71 | 12.41 | 11.74 |
| + hierarchical transported consistency | 4.65 ±0.03 | 9.57 ±0.06 | 3.74 | 12.49 | 11.73 |
| + orbit distillation | 4.57 ±0.05 | 9.33 ±0.18 | **3.69** | **11.21** | **11.09** |

Table 15. **Training-time consistency.** Single-view DINOv3 trees on synthetic and rotated synthetic validation (left) and PoE + orbit (right). The consistency losses transport each view's beam distribution into another view's frame via Eq. (2).

Retention of the orbit gain by the single-view students, defined as (base greedy − student greedy)/(base greedy − base orbit), is 43%/18%/22% (DINOv3) and 30%/28%/23% (PoE) on synthetic/lightbox/sunlamp, and 13–15% on the rotated proxy. For the joint 6-DoF model, the ESA-style score (rotation error in radians plus normalised translation error, with the challenge precision floors) is 0.084/0.241/0.238 for PoE.

## F. Per-object results

| Object | PoE+orbit | DINOv3+orbit |
|---|---|---|
| *T-LESS* | | |
| 1 | 23.82 | 25.00 |
| 4 | 14.50 | 15.69 |
| 5 | 6.22 | 7.07 |
| 11 | 11.90 | 12.92 |
| 18 | 14.38 | 14.78 |
| 21 | 34.78 | 37.24 |
| 27 | 10.36 | 12.43 |
| *SYMSOL-T* | | |
| cone | 1.83 | 1.79 |
| cube | 2.72 | 2.90 |
| cyl | 1.67 | 1.65 |
| icosa | 3.81 | 3.71 |
| tet | 3.35 | 3.06 |
| *YCB-V* | | |
| 003_cracker_box | 85.27 | 74.61 |
| 006_mustard_bottle | 97.96 | 87.29 |
| 011_banana | 61.63 | 83.23 |
| 025_mug | 24.25 | 22.76 |
| 035_power_drill | 41.34 | 25.07 |
| 037_scissors | 169.07 | 123.72 |

Table 16. **Per-object test error** of the single-label (A0) trees (3-seed mean; symmetry-aware for T-LESS and SYMSOL-T; YCB-V trained on real videos only).

On T-LESS (Tab. 16), most of the test error comes from object 21 (no symmetry metadata, near-symmetric; 18% of predictions above $90^\circ$) and object 1 (continuous symmetry). Continuous symmetries are evaluated on a $10^\circ$-step discretisation of the symmetry set, which adds up to $5^\circ$ of artificial error for objects 1 and 4 and for the cone and cylinder.

## G. YCB-V diagnosis

On YCB-V we checked four things:

1. The trees fit their training frames: $3.8^\circ$ and $3.9^\circ$ on mustard bottle and mug.
2. Test poses are covered by training: the median nearest training rotation is about $5^\circ$ away for cracker box, mustard bottle and banana, whose test errors are still 75–87°.
3. Every crop contains all of the object's label pixels.
4. The ground-truth rotations agree with the segmentation: 65–98% of the projected model points fall on the object mask, against 28–83% for a transposed-rotation control.

The failure is therefore memorisation of the 16–20 cluttered training videos per object. The object covers only 5–40% of the crop and the DINOv3 tokens are not masked. This

is not a label or pipeline error. Standard YCB-V training adds the synthetic `data_syn` renders, whose backgrounds are black. We pasted each render onto a real frame drawn only from the training sequences (never validation or test), capped at 12,000 crops per object as for the real data, and retrained with the unchanged recipe. Table 17 shows the effect per object.

| | real videos | | + synthetic | |
|---|---|---|---|---|
| Object | DINOv3 | PoE | DINOv3 | PoE |
| 003_cracker_box | 77.7 | 78.7 | 5.1 | 5.7 |
| 006_mustard_bottle | 101.5 | 102.4 | 6.1 | 7.6 |
| 011_banana | 79.9 | 77.6 | 6.1 | 5.3 |
| 025_mug | 24.6 | 28.4 | 6.5 | 5.6 |
| 035_power_drill | 30.3 | 38.2 | 4.3 | 4.4 |
| 037_scissors | 113.2 | 161.5 | 49.4 | 36.9 |

Table 17. **YCB-V per-object test error** (°, single view, 3 seeds) without and with synthetic training data.

## H. Zero-shot transfer of the frozen SPEED+ trees

| Dataset | Raw $E_R$ | | Frame-free pair error | | |
|---|---|---|---|---|---|
| | frozen | chance | frozen | chance | constant |
| T-LESS | 117.5 | 103.6 | 95.4 $\pm$3.0 | 104.5 | 88.7 |
| SYMSOL-T | 49.4 | 53.7 | 47.8 $\pm$0.5 | 53.7 | 53.8 |
| YCB-V | 115.6 | 125.8 | 101.6 $\pm$2.8 | 125.8 | 57.8 |

Table 18. **Zero-shot**: frozen SPEED+ PoE trees with orbit voting, no retraining, test split. The pair error $\min_S \angle(\hat{R}_i \hat{R}_j^\top, R_i S R_j^\top)$ over random same-object pairs cancels the unknown offset between object frames. *Chance*: uniformly random predictions. *Constant*: one rotation predicted for every image.

Object frames are arbitrary, so raw errors are expected near chance. The frame-free pair error (Tab. 18) is therefore the meaningful measure. On T-LESS and YCB-V it is *worse* than predicting one constant rotation, because both datasets have strong pose priors. On SYMSOL-T, where rotations are uniform, the trees beat chance only because of one shape: the cube (27.8° vs. 41.2°), plausibly because the spacecraft body is box-like. There is no general zero-shot transfer: the trees must be retrained per object.

## I. Disclosures and reproducibility

- **Late feature-equivalence check.** The augmented and rotated views are extracted by a compact feature extractor. A pre-registered check against the full feature caches was run after the main results. It failed its strict element-wise tolerance: the maximum standardised difference was 0.51 for MoGe-2 features (p99 0.004; correlation 0.999998). Its functional follow-up showed that no PoE prediction changed on 512 crops, and that single-branch greedy leaves changed for 1/512 (DINOv3) and 3/512 (MoGe-2) crops.
- **SYMSOL-T bug.** Nine MoGe-2 SYMSOL-T trees first trained to NaN: always-empty border cells had a standard deviation near $10^{-6}$, and float16 overflowed. They were retrained with a standard-deviation floor of $10^{-2}$; all other trees were unaffected. Only the corrected evaluation is reported.
- **YCB-V paths.** A file-path error in the YCB-V manifest (colour images are JPEG) was fixed before any YCB-V result.
- **Post-hoc control.** The constant-prediction control in Tab. 18 was added after the T-LESS zero-shot result and is labelled as post hoc.

Every number in the results tables is generated by a script directly from the evaluation logs of the named runs.

# J. Extended method details

This appendix lists the implementation details that Sec. 3 leaves out. Each paragraph ends with a comment naming the source file it was taken from. Quaternions are scalar-first, $q = (w, x, y, z)$, and every stored quaternion is sign-normalised to $w \geq 0$. Products $\otimes$ are Hamilton products, and $\mathrm{Exp}(\phi\, d) = (\cos\frac{\phi}{2}, \sin\frac{\phi}{2}\, d)$ for a unit axis $d$. All angles in tables are in degrees.

## J.1. Problem setup, crops and rotation targets

**Square crop.** The crop box is computed from the projected object points. Let $(x_0, y_0)$ and $(x_1, y_1)$ be the minimum and maximum of the projected points. The crop centre is $c = \frac{1}{2}(x_0 + x_1,\ y_0 + y_1)$ and its side is

$$s = \max(x_1 - x_0,\ y_1 - y_0,\ 32\,\mathrm{px}) \cdot (1 + 2 \cdot 0.15), \quad (7)$$

which gives the 15% margin per side. The box may extend beyond the image. The crop is a single affine warp from the full image to an $N \times N$ output, with scale $N/s$, bilinear interpolation and zero padding outside the image. On SPEED+ the projected points are the 11 Tango keypoints, projected with lens distortion. The grayscale image is replicated to three channels. On T-LESS and YCB-V the same rule is applied to the projected model bounding box and to the projected model points (`points.xyz`). These images are read in colour. SYMSOL-T uses the full $224 \times 224$ frame, so the crop centre is the principal point.

**Allocentric targets.** For T-LESS and YCB-V the tree predicts the allocentric rotation. Let $t$ be the object translation in the camera frame and $\hat{t} = t/\|t\|$. Let $R_{\text{ray}}(t)$ be the minimal rotation that takes $e_z$ to $\hat{t}$. With $v = e_z \times \hat{t}$, $s = \|v\|$ and $c = \hat{t}_z$, it is

$$R_{\text{ray}}(t) = I + [v]_\times + \frac{1 - c}{s^2}[v]_\times^2, \quad (8)$$

where the factor $(1 - c)/s^2$ is replaced by $\frac{1}{2}$ when $s < 10^{-12}$. The target is $R_{\text{allo}} = R_{\text{ray}}(t)^\top R_{\text{ego}}$, and the inverse is $R_{\text{ego}} = R_{\text{ray}}(t)\, R_{\text{allo}}$. Both conversions need $t$. At test time we use the ground-truth translation, because all rotation results use ground-truth crops. SPEED+ and SYMSOL-T use egocentric targets. For SYMSOL-T we follow the LiePose renderer [13]: the camera distance is $Z_0 = 5.25/\tan 30°$, the focal length is $f = (224 - 112)\, Z_0/5.25$, the principal point is $(112, 112)$, and the annotated translation is offset by $(0, 0, Z_0)$.

**Symmetry-aware error.** Let $\mathcal{S}$ be an object's set of symmetry matrices, with the identity first. The symmetry-aware error of a prediction $\hat{R}$ is

$$E_R^{\text{sym}} = \arccos\Big(\tfrac{1}{2}\big(\max_{S \in \mathcal{S}} \mathrm{tr}(\hat{R}^\top R\, S) - 1\big)\Big). \quad (9)$$

Table 19. **Size $|\mathcal{S}|$ of the symmetry sets** used for training and evaluation. T-LESS continuous symmetries are discretised into 36 steps. Objects 18 and 21 have no symmetry in the BOP metadata.

| T-LESS object | 1 | 4 | 5 | 11 | 18 | 21 | 27 |
|---|---|---|---|---|---|---|---|
| $\lvert\mathcal{S}\rvert$ | 36 | 36 | 2 | 2 | 1 | 1 | 4 |
| SYMSOL-T shape | tet | cube | icosa | cone | cyl | | |
| $\lvert\mathcal{S}\rvert$ | 12 | 24 | 60 | 360 | 720 | | |

For T-LESS the sets come from the BOP `models_info.json`. Continuous symmetries are discretised into 36 steps about their axis, and discrete ones are used exactly. For SYMSOL-T the sets are the annotated groups. Table 19 lists the set sizes; YCB-V objects use the identity only. The same function is used as the checkpoint criterion on validation (Appendix J.4).

## J.2. Tesseract partition of $SO(3)$

**Charts and canonicalisation.** Let $q \in S^3$ be normalised. The chart is $c = \arg\max_i |q_i|$; when two coordinates tie, the first maximum is taken. The quaternion is multiplied by $\mathrm{sign}(q_c)$, so that $q_c > 0$. This makes $q$ and $-q$ land on the same chart with the same coordinates, so $\mathrm{path}(q) = \mathrm{path}(-q)$ holds exactly. The projection to the cube facet is $x = q/q_c = q/\|q\|_\infty$. Its three free coordinates are $u = (x_j)_{j \neq c}$, taken in increasing index order and clipped to $[-1, 1]$. For charts $c = 0, 1, 2, 3$ the free indices are $\{1, 2, 3\}$, $\{0, 2, 3\}$, $\{0, 1, 3\}$ and $\{0, 1, 2\}$.

**Encoding.** Every free coordinate starts on the interval $[\ell, h] = [-1, 1]$. At level $\ell' = 1, \ldots, L$ each interval is split at its midpoint $m = \frac{1}{2}(\ell + h)$. A coordinate with $u_i \geq m$ goes to the upper half and sets bit $b_i = 1$; otherwise $b_i = 0$. A value exactly on the midpoint therefore goes to the upper half. The child id at that level is

$$k = b_0 + 2b_1 + 4b_2 \in \{0, \ldots, 7\}, \quad (10)$$

and the interval is updated to $[m, h]$ or $[\ell, m]$. The leaf id at depth $L$ is

$$\mathrm{id} = c \cdot 8^L + \sum_{\ell'=1}^{L} k_{\ell'}\, 8^{L - \ell'}, \quad (11)$$

so there are $4 \cdot 8^L$ leaves, ordered by chart and then lexicographically by path. Decoding a leaf id inverts this base-8 expansion.

**Cell centres.** After $l$ decisions with bits $b_{t,i}$ ($t = 1..l$), the free coordinate $i$ of the cell centre is

$$u_i = -1 + \sum_{t=1}^{l} b_{t,i}\, 2^{1-t} + 2^{-l}. \quad (12)$$

The centre quaternion is obtained by setting $x_c = 1$ and $x_j = u$ on the free indices, then normalising $x/\|x\|$. With $l = 0$ the centre is the chart axis $e_c$. This is the representative used as the root target (Appendix J.4) and as the first decoder token (Appendix J.4). The tree uses cell centres only. The cell corners, at all $2^3$ combinations of the interval ends, are lifted the same way; they are used only to compute the exact worst-case error in Tab. 20. The GPU decoder computes the same centres in closed form and is identical to the NumPy decoder.

**Relation to Kurz *et al.*** The recursive subdivision of the cubic facets of the 4-D hypercube, the 8-way halving and the radial projection $q = x/\|x\|$ follow [15]. We differ in three ways:

- we use cube *centres*, not corners, as representatives. A corner of the subdivided surface is shared by up to 8 cubes, so it has no unique parent;
- we keep only the 4 positive facets, so $q \sim -q$ is handled by canonicalisation. This gives $4 \cdot 8^L$ cells instead of $(2^m{+}1)^4 - (2^m{-}1)^4$ corner points with antipodal duplicates;
- we add an explicit encoder, decoder, chart rule and tie rule, which a flat point set does not need.

**Invariants.** We checked the encoder on all 69,491 SPEED+ labels (47,966 synthetic train, 11,994 synthetic validation, 6,740 lightbox, 2,791 sunlamp). All of the following hold:

- $\text{path}(q) = \text{path}(-q)$ and $\text{encode}(\text{decode}(\text{path})) = \text{path}$;
- the canonical chart coordinate is positive;
- prefixes are consistent across depths 1–5;
- no label has an exact chart tie; the smallest top-2 margin is $5.0 \cdot 10^{-6}$.

Perturbing every label by $0.001°$ changes the leaf of 21 labels, one of them across charts. In every such case the jump between decoded centres is at most 0.587 times twice the cell bound, so the decoded centre is continuous up to cell size. 200,000 probes placed exactly on chart seams ($|q_i| = |q_j|$) are antipodally consistent and follow the first-maximum rule. Only 38,318 of the 131,072 L5 leaves (29%) contain a synthetic-train label, which motivates the soft targets of Appendix J.4.

**Quantisation by depth.** Table 20 lists precision and cost by depth, measured on $2 \cdot 10^6$ Haar-uniform rotations. The *own-cell* error is the angle between a rotation and the centre of its own encoded cell, which is what the tree returns. Its exact maximum is the largest centre-to-corner angle over all cells. It is exact because the angle to a fixed direction is quasiconvex on a convex cell in the projected cube, so the maximum lies at a corner. The *nearest-centre* error uses the closest centre in any cell. Its maximum is sampled, so it is a lower bound on the covering radius. Cell volumes follow from the chart-map density $(1 + \|u\|^2)^{-2}$ on $S^3$, integrated per cell with $6^3$-point Gauss–Legendre quadrature. Greedy descent scores $4 + 8L$ nodes. On SPEED+ labels the own-cell statistics match the Haar values to within $0.03°$ (L5 mean 2.51–2.53° in every domain).

The error halves with every level. The grid is not uniform: at L5 cell volumes differ by up to $14.4\times$, close to the analytic limit of $16\times$ between the chart centre (density 1) and a cube corner (density $1/16$). Small cells lie near the chart seams. As a result, a uniform prior over leaves is not a uniform prior over rotations. In about 10% of cases the decoded cell is not the one with the nearest centre, but the cost is small (L5 mean 2.51° against 2.48°). We stop at $L = 5$. L6 would halve the floor to about 1.25°, but it has 1,048,576 cells against 47,966 SPEED+ training images.

**Comparison with other grids.** Table 21 compares the Tesseract with Super-Fibonacci and Hopf [27] grids and with SPACE-HOP's anchor grid [5], all measured on $10^6$ Haar rotations. The Hopf grid uses HEALPix on $S^2$ times a uniform $S^1$, with $72 \cdot 8^r$ points. Its counts can never equal $4 \cdot 8^L$, so we list the two neighbouring levels. The SPACE-HOP grid and its balanced variants are defined in Appendix J.10. At equal count, Super-Fibonacci has a lower error than the Tesseract on every statistic, and both Hopf variants are more uniform: their spacing CV is about 0.06, against 0.21 for the Tesseract. SPACE-HOP's own grid at L5 count, $10{,}923 \times 12$, is much coarser than the Tesseract (mean 6.06° against 2.51°), because it uses only 12 in-plane rolls. A balanced Hopf grid ($1{,}771 \times 74$) is finer (2.30°). The Tesseract's advantage is search cost: greedy descent scores 44 nodes at L5, whereas a flat search scores all 131k anchors and the Hopf hierarchy needs 104 nodes at r4.

The L5 maxima differ by definition. 6.20° in Tab. 20 is the exact centre-to-corner bound. 5.90° in Tab. 21 is the largest own-cell error among the $10^6$ samples, and 5.86° is the sampled nearest-centre maximum.

## J.3. Frozen features

**Freezing.** Both backbones are put in evaluation mode and have `requires_grad` disabled. Every forward pass runs under `torch.no_grad` and is followed by an assertion that no backbone parameter requires or holds a gradient. Features are extracted once per crop and view into an immutable HDF5 cache in fp16. Each cache is keyed by a hash of its preprocessing specification and a hash of its image list. The trees read only these caches.

**DINOv3 tokens.** The DINOv3 featset is `tokens8`. We use DINOv3 ViT-L/16 [22] with Meta's LVD-1689M weights, loaded through the timm port `vit_large_patch16_dinov3.lvd1689m`. The

Table 20. **Tesseract precision and cost by depth** ($2 \cdot 10^6$ Haar rotations). Own-cell: distance to the centre of the encoded cell (the tree's output); the maximum is the exact centre-to-corner bound. Nearest: distance to the closest centre in any cell (sampled maximum). "Not own": fraction of rotations whose nearest centre is not in their own cell. Volume ratio and CV are exact. Spacing is each centre's nearest-neighbour distance.

| $L$ | cells | own mean | own median | own p95 | own max (exact) | nearest mean | nearest max | not own | vol. max/min | vol. CV | spacing mean / min / max | greedy nodes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 32 | 39.27 | 39.84 | 59.63 | 81.79 | 39.28 | 81.17 | 0.0% | 1.00 | 0.000 | 62.0 / 62.0 / 62.0 | 12 |
| 2 | 256 | 19.95 | 19.85 | 31.44 | 46.83 | 19.64 | 46.21 | 10.3% | 4.71 | 0.494 | 30.6 / 24.8 / 44.0 | 20 |
| 3 | 2,048 | 10.01 | 9.95 | 15.83 | 24.43 | 9.88 | 23.81 | 10.6% | 9.62 | 0.548 | 15.8 / 11.2 / 26.5 | 28 |
| 4 | 16,384 | 5.01 | 4.98 | 7.92 | 12.36 | 4.95 | 11.81 | 10.4% | 12.81 | 0.562 | 8.3 / 5.3 / 14.0 | 36 |
| 5 | 131,072 | 2.51 | 2.49 | 3.97 | 6.20 | 2.48 | 5.86 | 10.4% | 14.44 | 0.565 | 4.3 / 2.6 / 7.1 | 44 |

Table 21. **Grid quantisation** on $10^6$ Haar rotations (degrees). Tesseract: own-cell (decoded) error; other grids: nearest-anchor error. Maxima are sampled. The balanced Hopf rows were descriptive in EXP-101 (not pre-registered).

| grid | $N$ | mean | p95 | max |
|---|---|---|---|---|
| Tesseract L3 | 2,048 | 10.01 | 15.83 | 23.81 |
| Super-Fibonacci | 2,048 | 9.53 | 13.72 | 18.26 |
| SPACE-HOP Hopf $171 \times 12$ | 2,052 | 9.67 | 14.60 | 18.12 |
| Balanced Hopf $108 \times 19$ | 2,052 | 9.22 | 12.96 | 16.40 |
| Hopf r2 | 4,608 | 7.04 | 9.90 | 13.53 |
| Tesseract L4 | 16,384 | 5.01 | 7.92 | 11.83 |
| Super-Fibonacci | 16,384 | 4.68 | 6.72 | 8.53 |
| SPACE-HOP Hopf $1365 \times 12$ | 16,380 | 6.76 | 13.09 | 15.40 |
| Balanced Hopf $443 \times 37$ | 16,391 | 4.61 | 6.47 | 8.12 |
| Hopf r3 | 36,864 | 3.52 | 4.95 | 6.67 |
| Tesseract L5 | 131,072 | 2.51 | 3.97 | 5.90 |
| Super-Fibonacci | 131,072 | 2.47 | 3.76 | 4.52 |
| SPACE-HOP Hopf $10923 \times 12$ | 131,076 | 6.06 | 12.91 | 15.05 |
| Balanced Hopf $1771 \times 74$ | 131,054 | 2.30 | 3.22 | 3.91 |
| Hopf r4 | 294,912 | 1.76 | 2.47 | 3.24 |

crop is resized to $256 \times 256$, scaled to $[0, 1]$ and normalised with the ImageNet mean and standard deviation. The backbone runs under bf16 autocast. `forward_intermediates` returns blocks 5, 11, 17 and 23 with the final norm applied. Only the last one, block 23, is used: its $16 \times 16 \times 1024$ patch tokens (the CLS and 4 register tokens are dropped) are average-pooled with `adaptive_avg_pool2d` to $8 \times 8$. This gives a $64 \times 1024$ token grid, flattened row-major in $(y, x)$ order and stored in fp16. The unpooled $16 \times 16$ grid (`tokens16`) was tested in EXP-109 arm B. It did not help (val-aug $30.51°$ against $30.84°$, within the seed std), at $4\times$ the tokens.

**MoGe-2 normals.** We use MoGe-2 [24] with checkpoint `Ruicheng/moge-2-vitl-normal`. The same crop is resized to $518 \times 518$ and scaled to $[0, 1]$. Inference uses $37\times37$ encoder tokens, fp16, a predicted field of view (no intrinsics are given) and `apply_mask=False`, so all maps stay finite. The predicted camera-frame surface normals and the foreground mask are area-pooled to a $74 \times 74$ grid. The extractor fails loudly if any normal is non-finite.

**DINOv3 foreground and consensus mask.** The consensus mask is computed per image without training. It uses no fitted thresholds and no dataset statistics.

1. Centre the 256 DINOv3 last-layer patch tokens ($16\times16\times1024$) and project them on their first principal direction, computed by SVD.
2. Flip the sign of the score if the mean over the outer ring of 60 border patches exceeds the mean over the interior, so that the border counts as background.
3. Upsample the score bilinearly (`align_corners=False`) to $74 \times 74$ and threshold it with Otsu's method (256 bins). This gives the DINOv3 foreground.
4. Threshold the MoGe-2 mask at $0.5$.
5. Intersect the two masks.

On real images this AND rule cuts the fraction of predicted foreground outside a dilated keypoint envelope from 0.52 to 0.12 on lightbox and from 0.57 to 0.12 on sunlamp, compared with the MoGe-2 mask alone. Recall drops by only 0.020 and 0.006. The envelope is the convex hull of the 11 projected keypoints, dilated by 3 cells.

**Masked normal tokens.** The MoGe-2 featset `normals16~and` is built as follows. Let $n \in \mathbb{R}^{74\times74\times3}$ be the MoGe-2 normals and $m \in \{0, 1\}^{74\times74}$ the consensus mask. The token grid is

$$X = \left[\text{pool}_{16}(n \odot m),\ \text{pool}_{16}(m)\right] \in \mathbb{R}^{256\times4}, \qquad (13)$$

where $\text{pool}_{16}$ is adaptive average pooling to $16 \times 16$. Each token holds the three masked-normal channels and the mask fraction of its cell. The normal channels are averaged over the whole cell and are not divided by the mask fraction, so their magnitude scales with the fraction of foreground in the cell. Using the consensus mask instead of MoGe-2's own mask lowered the probe's mean error by $6.9°$ on lightbox and $13.0°$ on sunlamp in EXP-016. The reverse direction, DINOv3 tokens pooled with mask weights, did not help and is not used.

**Standardisation.** Every feature dimension (token × channel) is standardised as $(x-\mu)/\sigma$. The mean $\mu$ and standard deviation are computed in float64 on the clean training rows only, and $\sigma = \mathrm{sd} + 10^{-6}$. On SPEED+ the statistics come from the synthetic-train split. The augmented training views, the validation set and the rotated test views are standardised with the same clean-train statistics. On the other datasets the statistics are computed per object from that object's clean training rows. In EXP-122 the YCB-V synthetic composites are appended to these rows.

A floor is applied to the standard deviation in one case. Some $16 \times 16$ border cells are always empty in the clean SYMSOL-T renders, so their train sd is about $10^{-6}$. When a rolled or validation view places the object there, the standardised values reach about $1.6 \cdot 10^5$ and overflow fp16. The MoGe-2 trees for cube, cylinder and icosahedron therefore use $\sigma = \max(\sigma, 10^{-2})$; the features lie in $[-1, 1]$. No other tree uses a floor.

**Compact extractor.** Augmented views, orbit views and all non-SPEED+ datasets go through a compact extractor. It stores only the pooled $8 \times 8$ DINOv3 grid, the DINOv3 foreground, and the MoGe-2 normals and mask. On 512 SPEED+ crops this extractor did not meet the pre-registered strict tolerance against the full caches (maximum absolute difference $10^{-2}$ after standardisation). DINOv3 reached a maximum of 0.011 and MoGe-2 a maximum of 0.51, in 97 of 512 images. Predictions were unaffected: the PoE greedy leaf was identical on all 512 crops, and the single-branch leaf changed on 1 of 512 (DINOv3) and 3 of 512 (MoGe-2) crops.

## J.4. Tree network

**Input layer and positional encoding.** The model reads a flattened token grid of shape $T \times C$: $64 \times 1024$ for DINOv3 and $256 \times 4$ for MoGe-2. Each token is passed through LayerNorm$(C)$ and a linear map to $d = 256$, and a fixed 2-D sinusoidal encoding is added. For a $g \times g$ grid ($g = \sqrt{T}$), the token in row $y$ and column $x$ has $p = (y/g,\ x/g)$. With $\omega_k = 2^k \pi$ for $k = 0, \ldots, d/4 - 1 = 63$, its encoding is

$$\mathrm{PE}(y, x) = \big[\sin(p_y\omega), \cos(p_y\omega), \sin(p_x\omega), \cos(p_x\omega)\big] \in \mathbb{R}^{256}, \tag{14}$$

with each block 64-dimensional. The encoding is a fixed buffer, not a learned parameter. In exact arithmetic only the frequencies with $2^k < 2g$ separate the grid positions; for larger $k$ the argument is a multiple of $2\pi$. In float32 the products for large $k$ do not reduce exactly, so those channels hold fixed, position-dependent rounding residues rather than constants.

**Encoder and decoder.** The memory is computed by one Transformer encoder layer. The path is decoded by a causal Transformer decoder with 2 layers. All layers use pre-norm (`norm_first`), 4 attention heads, a feed-forward width of 512 with the PyTorch default ReLU activation, and dropout 0.1. There is no final LayerNorm after either stack. The decoder input sequence is $(s_0, s_1, \ldots, s_L)$:

$$s_0 = r \quad \text{(learned root token, init } \mathcal{N}(0, 0.02^2)\text{)}, \tag{15}$$

$$s_{j+1} = E_{\text{lev}}[j] + W\,\varphi(c_j) + b, \qquad j = 0, \ldots, L-1, \tag{16}$$

where $c_j$ is the centre of the cell reached after $j$ child decisions. In particular $c_0 = e_{\text{chart}}$, so the chart enters the sequence only through this first cell centre. $E_{\text{lev}}$ is a learned embedding with $L + 1 = 6$ rows. The cell encoding is

$$\varphi(q) = \big[q,\ \sin(2^k\pi q)_{k=0}^{5},\ \cos(2^k\pi q)_{k=0}^{5}\big] \in \mathbb{R}^{4+8\cdot6=52}, \tag{17}$$

and $W \in \mathbb{R}^{256\times52}$. Self-attention uses the causal mask (upper-triangular $-\infty$), and every decoder layer cross-attends to the encoded tokens. Output position 0 feeds a linear root head ($256 \to 4$ chart logits). Output position $j \geq 1$ feeds a linear child head ($256 \to 8$) that predicts the child at level $j$. Under teacher forcing the sequence is built from the ground-truth path minus its last child, so one decoder pass gives all $1 + L$ distributions. At inference the decoder is re-run on the full prefix at every level; there is no key–value cache.

**Parameter count.** The DINOv3 tree has 2,391,564 trainable parameters and the MoGe-2 tree 2,128,404. They differ only in the input layer: $\mathrm{LN}(1024) + \mathrm{Linear}(1024, 256)$ against $\mathrm{LN}(4) + \mathrm{Linear}(4, 256)$, a difference of 263,160 parameters. The positional buffer is not counted. The translation block of Appendix J.9 adds 331,011 parameters per branch.

**Geodesic soft targets.** Let $\delta(q, c) = 2\arccos|\langle q, c\rangle|$ in degrees. For a ground-truth rotation $q$ with path $(c^*, k_1, \ldots, k_L)$, the targets are

$$t_i^{(0)} = \mathrm{softmax}_i\big(-(\delta(q, e_i)/\tau_0)^2\big), \quad i = 0..3, \tag{18}$$

$$t_k^{(\ell)} = \mathrm{softmax}_k\big(-(\delta(q, c_{\ell,k})/\tau_\ell)^2\big), \quad k = 0..7, \tag{19}$$

where $e_i$ is the centre of chart $i$ and $c_{\ell,k}$ is the centre of child $k$ of the *ground-truth* prefix $(c^*, k_1, \ldots, k_{\ell-1})$. Mass therefore never leaves the ground-truth parent cell. The widths are fixed a priori. For $\ell \geq 1$, $\tau_\ell$ is the mean own-cell error at depth $\ell$ (Tab. 20); the chart level reuses the L1 value:

| level $\ell$ | 0 (chart) | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $\tau_\ell$ (°) | 39.27 | 39.27 | 19.95 | 10.01 | 5.01 | 2.51 |

The loss sums the soft cross-entropies of the six heads and averages over the batch:

$$\mathcal{L}_{\mathrm{GT}} = -\sum_{\ell=0}^{L}\sum_{k} t_k^{(\ell)} \log p_k^{(\ell)}. \tag{20}$$

**Training.** One tree is trained per backbone (and per object on the other datasets). The schedule is the same everywhere:

- 100 epochs with batch 256;
- AdamW with learning rate $10^{-3}$ and weight decay 0.05;
- a one-cycle schedule with 5% warm-up (`pct_start`=0.05) over all steps;
- bf16 autocast, with features held in fp16;
- for each sample and step, one of its three views (clean, augmented copy 1, augmented copy 2) is drawn uniformly, and the labels of that view are used.

Every 5 epochs and at the last epoch the model is evaluated with greedy decoding, and the best checkpoint is kept. On SPEED+ the criterion is the clean synthetic-validation mean error. On the other datasets it is the symmetry-aware validation mean of the greedy prediction. Without augmented views, a Transformer-tree training run (EXP-032) took about 9 minutes for DINOv3 and 19 minutes for MoGe-2.

**Beam search.** Let $\log \tilde{p}^{(\ell)}$ be the (possibly fused) conditional log-distribution at a node. The beam keeps the $k_0 = \min(B, 4)$ best charts and then, at each of the $L$ levels, expands every kept prefix into 8 children. It scores each child by the cumulative log-probability

$$S(c, k_{1:\ell}) = \log \tilde{p}^{(0)}(c) + \sum_{j=1}^{\ell} \log \tilde{p}^{(j)}(k_j \mid c, k_{1:j-1}), \tag{21}$$

and keeps the top $B$ of the $8K$ candidates. Greedy decoding is $B = 1$. Orbit voting uses $B = 8$ and keeps all 8 final leaves together with their scores. The prediction is the centre of the best leaf. EXP-034 found that beams up to 8 did not change the single-view greedy result (PoE synthetic-val mean 6.242° at $B$=1, 6.248° at $B$=8). The trained tree is decisive, and beams matter only as a source of candidates.

**Per-level product of experts.** With the two branches, the fused conditional at every node is

$$\log \tilde{p}^{(\ell)} = \log \mathrm{softmax}\Big( \textstyle\sum_b \log \mathrm{softmax}(z_b^{(\ell)}) \Big), \tag{22}$$

which is the renormalised product of experts [9]. It has no parameters and needs no training. The code also implements an equal mixture and reliability-weighted products (inverse entropy, top-2 margin), but the reported systems use only the product. The same fusion is applied during teacher-forced scoring in orbit voting and in the distillation teachers.

Table 22. **Decoder and target ablations** on SPEED+ (no augmentation, greedy, 3-seed mean). Synthetic val selects; lightbox and sunlamp are descriptive.

| exp. | decoder / targets | DINOv3 val | PoE val | PoE lightbox | PoE sunlamp |
|---|---|---|---|---|---|
| 030 | MLP / hard | 6.64 | $6.242 \pm 0.072$ | 31.1 | 43.7 |
| 031 | GRU / hard | 6.29 | $6.149 \pm 0.098$ | 31.0 | 40.6 |
| 033 | MLP / soft | 5.69 | $6.068 \pm 0.067$ | 28.2 | 42.1 |
| 035 | MLP / hard + residual | 6.58 | $6.214 \pm 0.054$ | 32.2 | 43.2 |
| 032 | Transformer / hard | 6.08 | $5.685 \pm 0.103$ | 31.6 | 41.8 |
| 036 | Transformer / soft | 5.35 | $5.540 \pm 0.063$ | 26.9 | 36.6 |

**How the tree was chosen.** Table 22 traces how the tree was arrived at. All rows use the full SPEED+ synthetic training set and no augmentation. The MLP tree reads a pooled DINOv3 $4 \times 4$ grid and the flattened `normals16~and` vector. Its trunk is LN $\to$ Linear$(\cdot, 512)$ $\to$ GELU $\to$ Linear$(512, 512)$ $\to$ GELU, with dropout 0.1. Its child head reads the trunk output together with a 128-d level embedding plus a 128-d projection of $\varphi$(parent centre). The GRU variant replaces the child head by a GRU cell over the previous decisions (state 256). The tangent residual predicts $\delta \in \mathbb{R}^3$, and the prediction becomes $q_{\text{leaf}} \otimes \mathrm{Exp}(\delta)$; the residual is trained teacher-forced on the ground-truth leaf, in units of $\tau_5$. The Transformer with soft targets (EXP-036) has the lowest PoE synthetic-validation mean, and it is the tree used throughout.

## J.5. Augmentation and the exact in-plane roll

**Augmented copies.** Two augmented copies of every training crop are extracted once, alongside the clean crop. For row $i$ of copy $j \in \{1, 2\}$, the roll angle is drawn as $\theta_i \sim \mathcal{U}(-180°, 180°)$ from a generator seeded with $j$, in row order. The photometric parameters come from a generator seeded with $[j, i]$. The synthetic validation-augmented set ("val-aug") is built in the same way, with seed 1000 for both. Each backbone receives its own rendering of the crop at its own resolution (256 px for DINOv3, 518 px for MoGe-2). The photometric generator stream is the same for both, so pixel-unit parameters (blur $\sigma$, blob radius) are relatively stronger at 256 px.

**Geometric warp.** The rotated crop is resampled directly from the full image in one warp, so no black corners appear unless the box leaves the image. With crop scale $\sigma = N/s$, the warp is

$$M = \begin{bmatrix} R_\theta & t_\theta \end{bmatrix} \circ \begin{bmatrix} \sigma & 0 & N/2 - \sigma c_x \\ 0 & \sigma & N/2 - \sigma c_y \end{bmatrix}, \tag{23}$$

where $[R_\theta \,|\, t_\theta]$ is OpenCV's `getRotationMatrix2D` about the output centre $(N/2, N/2)$ with unit scale. Positive $\theta$ rotates the content counter-clockwise on screen. Interpolation is bilinear with zero padding.

**Photometric augmentation.** The operations are applied in this order to the crop $x \in [0,1]^{H\times W(\times C)}$, after the geometric warp:

1. brightness: $x \leftarrow x \cdot a$, with $a \sim \mathcal{U}(0.5, 1.5)$;
2. contrast about the image mean $\bar{x}$: $x \leftarrow (x - \bar{x})\, b + \bar{x}$, with $b \sim \mathcal{U}(0.6, 1.4)$;
3. gamma: $x \leftarrow \mathrm{clip}(x, 0, 1)^{\gamma}$, with $\gamma \sim \mathcal{U}(0.6, 1.6)$;
4. Gaussian blur with probability 0.5, $\sigma \sim \mathcal{U}(0.01, 1.5)$ px;
5. glare with probability 0.3: 1–3 additive Gaussian blobs with uniformly random centres, radius $r \sim \mathcal{U}(0.05, 0.25)\, H$ and amplitude $\mathcal{U}(0.5, 1.5)$;
6. additive Gaussian noise with $\sigma_n \sim \mathcal{U}(0, 0.06)$, followed by clipping to $[0, 1]$;
7. random erasing with probability 0.3: one rectangle covering 5–20% of the area, aspect ratio $\mathcal{U}(0.3, 3.3)$, filled with a uniform random value;
8. JPEG re-encoding with probability 0.5, quality $\mathcal{U}\{30, \ldots, 90\}$.

Photometric operations do not change the labels. They are not applied to the rotated test views used by orbit voting.

**Label transform.** The four candidate conventions are listed below, with $d$ the unit ray through the crop centre:

| | | | |
|---|---|---|---|
| pre+ | $q' = \mathrm{Exp}(+\theta d) \otimes q$ | post+ | $q' = q \otimes \mathrm{Exp}(+\theta d)$ |
| pre− | $q' = \mathrm{Exp}(-\theta d) \otimes q$ | post− | $q' = q \otimes \mathrm{Exp}(-\theta d)$ |

We derived the convention from the camera geometry. The pose convention is $p_{\text{cam}} = R(q)\, p_{\text{body}} + t$, and the image $y$ axis points down. A camera rotation $\mathrm{Rot}(d, \phi)$ about the ray through the crop centre fixes that centre. Near the centre, and for $f_x = f_y$, it appears on screen as a *clockwise* rotation by $\phi$. Matching the counter-clockwise warp by $\theta$ therefore requires $\phi = -\theta$. This gives $R' = \mathrm{Rot}(d, -\theta)R$ and $t' = \mathrm{Rot}(d, -\theta)\, t$, *i.e.* the "pre−" convention of Eq. (2). All results use it.

On SPEED+, $d$ is computed from the *undistorted* crop centre. The pixel is undistorted with the camera's distortion coefficients, lifted to $z = 1$ and normalised. The derivation was checked without any model on 512 synthetic-validation crops at 4 angles. The 11 keypoints projected with $(q, t)$ through the real distorted camera and pushed through the rotated-crop warp agree with the projection of $(q', t')$ through the plain crop to 0.34 px mean, 1.05 px p95 and 4.75 px max in a 224-px crop. The other three conventions are off by 94–242 px. The label error, measured as the angle between $q'$ and the PnP pose that reproduces the warped points, is $0.06°$ mean, $0.20°$ p95 and $1.14°$ max. The residual comes from approximating a distorted homography by an affine warp. The round trip $q \to q' \to q$ is within $0.004°$. The pass rule (p95 label error below $0.5°$, round trip below $0.01°$, and pre− clearly best) was fixed before this measurement.

A predictor check was run afterwards. It compares the frozen EXP-036 trees' predictions on crops rotated by $90°$ and $37°$ with the ground truth under each convention. This check is descriptive only and was not used to choose the convention.

**Exact roll homography (T-LESS, YCB-V).** For objects far from the optical axis, the affine crop rotation is only an approximate camera roll. With it, the label error reached p95 $1.43°$ / $2.40°$ and max $2.5°$ / $6.7°$ on T-LESS / YCB-V, which failed the same pre-set rule. For these datasets we therefore warp with the homography of the camera roll. The images have no distortion, so $d = K^{-1}(c_x, c_y, 1)^{\top}$, normalised. The crop is

$$x_{\text{crop}} \sim C\, K\, \mathrm{Rot}(d, -\theta)\, K^{-1}\, x, \qquad C = \begin{bmatrix} \sigma & 0 & N/2 - \sigma c_x \\ 0 & \sigma & N/2 - \sigma c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{24}$$

with $\mathrm{Rot}(d, \phi) = I + \sin\phi\, [d]_{\times} + (1 - \cos\phi)[d]_{\times}^2$ (Rodrigues), bilinear interpolation and zero padding. The crop centre is a fixed point of the roll, so the labels follow exactly:

$$R'_{\text{ego}} = \mathrm{Rot}(d, -\theta) R_{\text{ego}}, \quad t' = \mathrm{Rot}(d, -\theta)\, t, \qquad R'_{\text{allo}} = R_{\text{ray}}(t')^{\top} R'_{\text{ego}}. \tag{25}$$

The allocentric target is recomputed from the rolled translation $t'$. It is not obtained by left-multiplying the old target. With this warp, the label error and the reprojection error are both 0.000 (degrees and pixels), and the mapping round trip is within $0.004°$. SYMSOL-T uses the full frame, whose centre is the principal point, so the affine rotation is already an exact roll there. It uses egocentric labels and no homography. For every dataset, right-multiplication by an object symmetry $S$ commutes with the roll (the roll multiplies on the left).

## J.6. Orbit voting

**Views.** The four views $\theta_k \in \{0°, 90°, 180°, 270°\}$ are extracted as separate crops with the warp of Appendix J.5: affine on SPEED+ and SYMSOL-T, the roll homography on T-LESS and YCB-V. Their features are standardised with the clean training statistics. For the val-aug set, view $k$ adds $\theta_k$ to the row's random angle and reuses its photometric seed. Orbit inference therefore needs $4\times$ the backbone passes of a single view.

**Candidates.** In every view the fused tree runs beam search with $B = 8$ and returns 8 leaf centres $c_{k,1..8}$ with their joint log-probabilities $S_{k,1..8}$. On SPEED+ and SYMSOL-T each leaf is mapped to view 0 by the inverse roll, $c = \mathrm{Exp}(+\theta_k d) \otimes c_k$. On the allocentric datasets

the mapping goes through the egocentric frame using the ground-truth translation of each view, $t_k = \mathrm{Rot}(d, -\theta_k)\, t$:

$$c = \mathrm{allo}_t\Big(\mathrm{Rot}(d, +\theta_k)\, \mathrm{ego}_{t_k}(c_k)\Big), \tag{26}$$

where $\mathrm{ego}_t(R) = R_{\mathrm{ray}}(t)R$ and $\mathrm{allo}_t(R) = R_{\mathrm{ray}}(t)^\top R$. This gives 32 candidates $c_{1..32}$ in the frame of view 0. All four rules below choose among these same candidates.

**Scoring rules.** Let $\mathrm{geo}(a, b)$ be the geodesic angle in degrees.

- **Orbit product** (Eq. (3)). Each candidate is mapped into every view $k$ with the forward roll and encoded to its own depth-5 path. The fused tree is scored on that path with teacher forcing, and the per-view log-probabilities are summed:

$$s(c) = \sum_k \Big[\log \tilde{p}_k^{(0)}(\text{chart}) + \textstyle\sum_{\ell=1}^{5} \log \tilde{p}_k^{(\ell)}(k_\ell \mid \text{prefix})\Big]. \tag{27}$$

  The prediction is $\arg\max_c s(c)$. In each view this costs one encoder pass and one teacher-forced decoder pass per candidate, with no beam search.
- **Max-confidence**. The candidate with the largest joint beam log-probability $S$ in its own view is chosen. These scores are not normalised across views.
- **Top-1 vote**. Only the 4 view top-1 candidates are used. Each receives the kernel support $\sum_{j=1}^{4} \exp(-\frac{1}{2}(\mathrm{geo}(c_i, c_j)/5^\circ)^2)$, including its own term. Ties are broken by adding $10^{-6}(S_i - \min_j S_j)$.
- **Weighted vote** over all 32 candidates. Within each view the beam scores are turned into weights $w = \mathrm{softmax}(S_{k,1..8})$. The support of candidate $i$ is $\sum_{j=1}^{32} w_j \exp(-\frac{1}{2}(\mathrm{geo}(c_i, c_j)/5^\circ)^2)$, also including its own term.

The kernel is a Gaussian with $\sigma = 5^\circ$, which differs from the form $\exp(-(\delta/\tau)^2)$ of the soft targets. Greedy decoding uses view 0 only. Measured in EXP-112/114, the product's re-scoring costs 8.2 ms (DINOv3-only) or 15.3 ms (PoE) per image at batch 1, and 0.4 / 1.2 ms per image batched. The weighted vote needs only a $32 \times 32$ angle matrix (below 0.1 ms).

## J.7. Symmetry-aware targets

For an object with symmetry quaternions $\{s\}$ (the matrices of Tab. 19 converted to quaternions), the three label modes of Sec. 3.3 are:

- **A0, single label.** The annotated target of the view, $q_v$, with the soft targets of Appendix J.4.
- **A1, canonical.** In every view separately, the equivalent rotation closest to the identity, $q_v \otimes s^\star$ with $s^\star = \arg\max_s |(q_v \otimes s)_w|$. Because canonicalisation is per view, the canonical labels of a clean view and its rolled copy need not be related by Eq. (2).
- **A2, multi-path** (Eq. (1)). For each sample, the set $\mathcal{S}'$ is all of $\mathcal{S}$ if $|\mathcal{S}| \le K_s = 8$. Otherwise $K_s$ members are drawn without replacement, independently per sample and step. Each $q_v \otimes s$ is encoded to its own path, gets its own soft targets, and is scored by a teacher-forced pass of the decoder over the repeated memory. The loss is the mean of $\mathcal{L}_{\mathrm{GT}}$ over the $B \cdot |\mathcal{S}'|$ rows. For $|\mathcal{S}| = 1$ this is exactly A0.

The pre-registration named the hard-path level-marginal loss for A2. We used the smoothed version above instead: the hard-path loss would neither reduce to A0 for asymmetric objects nor keep the recipe's soft targets. Before training we checked that the quaternion product $q \otimes s$ equals the matrix product $RS$ to $10^{-15}$, and that the roll commutes with right-multiplication by $S$. The trees are per object, so the existing single-label trees are a matched A0 with the same code path, recipe, seeds and checkpoint rule.

## J.8. Orbit distillation

**Teacher.** The teacher is the frozen orbit system of the same seed and variant, applied to each training image over the three views it already has: clean, augmented copy 1 and augmented copy 2, with known roll angles $\theta_v$ (where $\theta_0 = 0$). The roll angles are arbitrary, not multiples of $90^\circ$; orbit scoring only needs them to be known.

- In each view the fused tree returns its beam-8 leaves. They are mapped to the clean frame: on SPEED+ with $\mathrm{Exp}(+\theta_v d) \otimes c$; on T-LESS and SYMSOL-T exactly through `xobj`, using the ground-truth translation of view $v$. This gives 24 candidates $c_j$.
- Each candidate receives the orbit-product score $s(c_j)$, summed over the three views as in Appendix J.6, and the weight $w_j = \mathrm{softmax}_j\, s(c_j)$ (temperature 1).
- The pairs $(c_j, w_j)$ are computed once and stored. The weights are constants during student training, so no gradient flows through them.

On SPEED+ the teacher trees are the augmentation-trained trees of EXP-109. On T-LESS and SYMSOL-T they are the trees of the EXP-106 arm with the lowest validation PoE greedy mean. There are two teacher variants: a PoE teacher (DINOv3 + MoGe-2) and a DINOv3-only teacher.

**Student loss.** The student is a new tree for one branch, trained from scratch with the recipe of Appendix J.4. For the view $v$ drawn for a sample, each teacher candidate is transported into that view's frame: with $\mathrm{Exp}(-\theta_v d) \otimes c_j$ on SPEED+, and exactly through `xobj` on the other datasets. The transported candidate is encoded to its path $(c, k_1, \ldots, k_L)$. The student is teacher-forced along that path to obtain the stepwise log-probabilities $a_0 = \log p^{(0)}(c)$ and $a_\ell = \log p^{(\ell)}(k_\ell \mid \text{prefix})$. The $\log p^{(\ell)}(\mathrm{pre}_\ell(\cdot))$ of Eq. (4) is the *joint* log-probability of the

length-$\ell$ prefix, $\sum_{i\le\ell} a_i$. The distillation term is

$$\mathcal{L}_{\text{dist}} = -\sum_{j=1}^{24} w_j \frac{1}{L+1} \sum_{\ell=0}^{L} \sum_{i=0}^{\ell} a_i(c_j), \tag{28}$$

averaged over the batch. It is added to the view's ground-truth loss with $\beta = 1$, which was fixed before any run. On T-LESS and SYMSOL-T the ground-truth term uses the base arm's label mode (A0, A1 or A2). Three students are trained per seed and object:

- a DINOv3 student from the DINOv3-only teacher, for the DINOv3-only system;
- a DINOv3 student and a MoGe-2 student from the PoE teacher, which together form the PoE system.

Inference is unchanged: the students are used greedily or with orbit voting, exactly like the base trees. This is data distillation [8, 21] with the hierarchical level-marginal target.

### J.9. Translation block

**Target.** We keep the SPACE-HOP target [5] unchanged. The box is the ground-truth square crop, $(b_x, b_y, b_w, b_h) = (c_x^{\text{crop}}, c_y^{\text{crop}}, s, s)$. The image size $(W, H)$ is $1920 \times 1200$ for SPEED+, $720 \times 540$ for T-LESS, $640 \times 480$ for YCB-V and $224 \times 224$ for SYMSOL-T, and $\alpha = 1.6$. With $\kappa = \frac{1}{2}(\alpha b_w/W + b_h/H)$, which equals $\frac{1}{2}(1/s_x + 1/s_y)$ for $s_x = W/(\alpha b_w)$ and $s_y = H/b_h$, the forward and inverse maps are

$$u = \left( \frac{f_x X/Z + c_x - b_x}{b_w},\ \frac{f_y Y/Z + c_y - b_y}{b_h},\ \kappa Z \right), \tag{29}$$

$$Z = U_z/\kappa, \quad X = \frac{(U_x b_w + b_x - c_x) Z}{f_x}, \tag{30}$$

$$Y = \frac{(U_y b_h + b_y - c_y) Z}{f_y}. \tag{31}$$

The camera is pinhole, with no distortion term, as in SPACE-HOP. For a rolled view the box is unchanged and the target is recomputed from $t' = \text{Rot}(d, -\theta)\, t$. On SPEED+ the roll is built as a quaternion by the "pre−" transform of the identity and applied to $t$. Photometric augmentation leaves $t$ unchanged.

**Block.** The block reads the tree's encoded tokens $h$, the output of the encoder layer of Appendix J.4. A learned query $r_t \in \mathbb{R}^{256}$ (init $\mathcal{N}(0, 0.02^2)$) cross-attends once to $\text{LN}(h)$, using multi-head attention with 4 heads and dropout 0.1. The attended vector passes through the MLP

$$\begin{aligned} &\text{LN} \to \text{Linear}(256, 256) \to \text{GELU} \\ &\to \text{Dropout}(0.1) \to \text{Linear}(256, 3), \end{aligned} \tag{32}$$

which outputs $u$. The block has 331,011 parameters: 256 for the query, 512 for the key LayerNorm, 263,168 for the attention and 67,075 for the MLP.

**Training.**

- **Joint** (SPEED+, arm J). Tree and block are trained together from scratch with $\mathcal{L} = \mathcal{L}_{\text{GT}} + \mathcal{L}_{\text{trans}}$ and $\mathcal{L}_{\text{trans}} = \|u - u^*\|_2^2$, summed over the three components and averaged over the batch. On SPEED+, $u$ is used unstandardised. The checkpoint is chosen by the rotation criterion of Appendix J.4.
- **Frozen tree** (SPEED+ arm F, and all other datasets). The tree is loaded from its checkpoint and frozen: no gradients, and evaluation mode, so dropout is off. Only the block is trained, with the same optimiser and schedule. On T-LESS, YCB-V and SYMSOL-T, $u$ is standardised per component with the mean and standard deviation $(+10^{-8})$ of the clean training rows, because units differ (mm, m, scene units). Predictions are de-standardised before inversion. The frozen tree's own feature statistics are reused, and the code checks that they match. The checkpoint is the lowest validation mean $E_T$ of view 0.

**Inference.** For the PoE variant, the two branches' $u$ are averaged before the inverse map. For orbit inference, each view's $\hat{t}_k$ is rolled back to view 0 with $\text{Rot}(d, +\theta_k)$, and the four estimates are averaged. The rotation is predicted independently by the trees. The box-only reference uses no image: $U_x = U_y = 0$ (object centre at the box centre) and $U_z$ equal to the training mean, mapped through the same inverse.

### J.10. Flat baselines

**Model.** The flat classifier is the tree network with the path decoding removed. It keeps the same input layer, positional encoding, encoder and 2-layer decoder, and runs the decoder on the root token alone. The decoder output feeds one linear layer $256 \to K$ over $K$ anchors. The unused tree heads remain in the module, so the parameter counts are 36,078,096 (DINOv3) and 35,814,936 (MoGe-2). Of these, $256K + K = 33{,}686{,}532$ belong to the flat head for $K = 131{,}076$.

**Anchor grids.**

- **SPACE-HOP Hopf grid** [5], $N$ directions $\times$ $M$ rolls. Directions lie on a Fibonacci lattice, $\phi_i = \arccos(1 - 2(i + \frac{1}{2})/N)$ and $\vartheta_i = \pi(1 + \sqrt{5})\, i$, giving $z_i = (\cos\vartheta_i \sin\phi_i, \sin\vartheta_i \sin\phi_i, \cos\phi_i)$. Each direction gets a frame $x_i = \text{up} \times z_i$ and $y_i = z_i \times x_i$ (normalised), with $\text{up} = e_z$, or $e_x$ when $|z_{i,z}| > 0.999$. The anchors are $[x_i\, y_i\, z_i]\, R_z(2\pi j/M)$ for $j = 0..M-1$. Our NumPy reproduction matches SPACE-HOP's code to $1.6 \cdot 10^{-4}$ (float32 rounding).
- The grids used are the count-matched $10{,}923 \times 12 = 131{,}076$ (the default), the Tesseract's own 131,072 L5 centres in leaf-id order (same-grid baseline), and the balanced $1{,}771 \times 74 = 131{,}054$.

**Targets, training and decoding.** Training uses geodesic soft targets over the 32 anchors nearest to the label (by $|\langle q, g\rangle|$). The weights are softmax$(-(\delta/2.51^\circ)^2)$, with the tree's leaf-level width, and the loss is the cross-entropy against these weights. Optimiser, schedule, views, labels, standardisation and checkpoint rule are those of the tree (Appendix J.4); the checkpoint uses the argmax anchor. The single-view prediction is the argmax anchor. For orbit voting the candidates are the per-view top-8 anchors, and the orbit product sums, over views, the log-probability of the candidate's *hard anchor* in that view. For the Hopf grids the hard anchor is the nearest anchor. For the Tesseract grid it is the candidate's own L5 cell, as for the tree's teacher-forced scoring. PoE fusion adds the two branches' log-softmax outputs over all $K$ anchors and renormalises.

| Domain | Split | Images | Missing | $t_z$ range (m) |
|---|---|---|---|---|
| synthetic | train | 47,966 | 0 | 2.24–10.00 |
| synthetic | validation | 11,994 | 0 | 2.23–10.00 |
| lightbox | test | 6,740 | 0 | 2.47–9.55 |
| sunlamp | test | 2,791 | 0 | 2.48–9.53 |

Table 23. **SPEED+ splits** as read by our loader. Training uses synthetic/train only, selection uses synthetic/validation only, and lightbox and sunlamp are test-only.

# K. Datasets, protocol and training details

This appendix lists the data, protocol, training and evaluation details that Sec. 4.1 omits. Every number below comes from a file in the repository: a data manifest in `outputs/data_manifests/`, an experiment report in `outputs/reports/`, the decision log `outputs/DECISIONS.md`, a script, or the per-run records under `outputs/experiments/`. The source of each paragraph and table is given in a comment in the LaTeX source. Experiment identifiers (EXP-$nnn$) and decision identifiers (DEC-$nnn$) refer to those records. For experiments whose results were still pending when this appendix was written, we give the protocol only.

## K.1. SPEED+

**Source, splits and label checks.** We use the original SPEED+ release [19], read-only. Its four annotation files are `synthetic/train.json`, `synthetic/validation.json`, `lightbox/test.json` and `sunlamp/test.json`. The loader check (EXP-000) found no missing images, no image path shared between splits and unit-norm quaternions to within $10^{-6}$ (Tab. 23). Quaternions are scalar-first (Hamilton), and the pose convention is $p_{\text{cam}} = R(q)\,p_{\text{body}} + r$ in the OpenCV camera frame ($x$ right, $y$ down, $z$ forward). We confirmed this convention against its transpose by projecting the 11 Tango keypoints on 200 random images per domain. With $R(q)$, more of the projected keypoints land on bright spacecraft pixels (synthetic 0.482 vs. 0.261, lightbox 0.271 vs. 0.179, sunlamp 0.560 vs. 0.331). On synthetic images the keypoint box also matches an independent box table with mean IoU 0.970, against 0.519 for $R(q)^\top$. The quaternion–matrix round trip is exact to $2.96 \times 10^{-6}$ degrees.

**Camera.** Images are $1920 \times 1200$ grayscale with pixel pitch $5.86\,\mu$m. The intrinsics from `camera.json` are $f_x = 2988.580$, $f_y = 2988.340$ and $(c_x, c_y) = (960, 600)$ px, with five distortion coefficients $(-0.22383, 0.51410, -0.00066, -0.00021, -0.13124)$. Distortion is applied whenever keypoints are projected, and it is removed when the viewing ray through a crop centre is computed for the in-plane roll (Appendix K.1).

**Rotation labels and the tree target.** Each label is stored twice in the manifests: as the source quaternion (`q_src_*`) and as the sign-canonical quaternion of its tesseract chart (`q_can_*`). The manifests also store the chart, the 5-level child path, the depth-5 leaf id, the leaf centre and the quantisation error of that centre. EXP-001 checked the encoder invariants on all 69,491 labels. Over these labels, the leaf-centre quantisation error is $2.51°$ on average (median $2.49°$, maximum $5.78°$). The analytic worst case over $SO(3)$ is $6.20°$, which is the figure quoted in the main text.

**Ground-truth square crops.** The crop is built from the 11 Tango keypoints of the SPEED+ toolkit (`tangoPoints.mat`), projected with the ground-truth pose and the lens distortion. We take the tight box of the projected points, $[x_0, x_1] \times [y_0, y_1]$. The crop is the square centred on this box with side

$$s = \max(x_1 - x_0,\ y_1 - y_0,\ 32\,\text{px}) \cdot (1 + 2 \cdot 0.15),$$

*i.e.* a 15% margin per side, or $1.3\times$ the longer box edge. The square may extend beyond the image. It is resampled with an affine warp (bilinear, zero padding) to the backbone input size: $256 \times 256$ for DINOv3 ViT-L/16 (a $16 \times 16$ patch grid, ImageNet normalisation) and $518 \times 518$ for MoGe-2. The grayscale crop is replicated to three channels. The crop side has a median of 724 px on synthetic images (range 309–2,570), 738 px on lightbox (339–2,286) and 738 px on sunlamp (332–2,117). After the 2026-10-28 reboot, the keypoint file is read from a second copy of the toolkit. On 200 rows this copy reproduces every manifest crop to $< 5 \times 10^{-6}$ px.

**Uses of the keypoints.** The 11 keypoints are never a training target. We use them in four places: (i) to build the ground-truth crop boxes; (ii) to verify the pose convention (EXP-000); (iii) to verify the in-plane label transform (EXP-109); and (iv) as the box label of the EXP-124 detector (Appendix K.1).

**Manifests.** All splits, crops and labels are fixed in CSV manifests, each with a hash file (Tab. 24). The SPEED+ manifests share the columns `image_relpath`, `domain`, `split`, `q_src_{w,x,y,z}`, `q_can_{w,x,y,z}`, `root_chart`, `child_path`, `leaf_id_L5`, `leaf_centre_{w,x,y,z}`, `quant_err_L5_deg`, `t_{x,y,z}` and `source_annotation`. All of them except the label table also carry `crop_cx`, `crop_cy`, `crop_side` and `crop_margin` (= 0.15 everywhere).

| Manifest | Rows | Content | Used by |
|---|---|---|---|
| `speedplus_tesseract_L5.csv` | 69,491 | all four splits, labels and tree paths, no crop columns | EXP-001/002 |
| `subset_v1.csv` | 9,500 | seed-0 subset: 6,000 train, 1,500 val, 1,000 lightbox, 1,000 sunlamp | EXP-010–026, EXP-101 |
| `subset_full.csv` | 69,491 | 47,966 train, 11,994 val, 6,740 lightbox, 2,791 sunlamp | EXP-030 onward |
| `exp109_syn_train.csv` | 47,966 | synthetic train (source rows of the two augmented copies) | EXP-109 onward |
| `exp109_syn_val.csv` | 11,994 | synthetic validation (source of val-aug) | EXP-109 onward |
| `exp109_test3.csv` | 21,525 | synthetic val + lightbox + sunlamp (source of the 90/180/270° orbit views) | EXP-109 onward, EXP-124 |
| `exp109_gate512.csv` | 512 | synthetic-val rows of the convention gate | EXP-109, EXP-064 compact gate |
| `lightbox_val_DEC005.txt` | 1,348 | seeded 20% lightbox list of the withdrawn DEC-005 split | descriptive only |

Table 24. **SPEED+ manifests** (`outputs/data_manifests/`). The full-split hash is `c7c029785fb8458c` (margin 0.15, seed 0); the v1 hash is `bbdb3e2e3f5b70a1`.

**Subset v1 and full splits.** The early sanity runs used *subset v1*: a seed-0 draw of 6,000 synthetic-train, 1,500 synthetic-val, 1,000 lightbox and 1,000 sunlamp images, all with ground-truth crops. These are the Phase-1/2 numbers (EXP-010–026 and EXP-101). Every experiment from EXP-030 on, and every number in the paper, uses the *full* splits. The subset-v1 feature caches were later deleted to free disk space (DEC-020), so those early runs cannot be re-evaluated without re-extracting features; their records are kept.

**Evaluation set (test3).** All SPEED+ evaluations from EXP-109 on run on the 21,525 rows of `exp109_test3.csv`: the full synthetic validation set (11,994) and the full lightbox (6,740) and sunlamp (2,791) test sets. View 0 is the clean ground-truth crop. For orbit voting, its in-plane rotations by 90°, 180° and 270° are extracted once with the same extractor and cached. These test-time views have no photometric augmentation.

**Training augmentation.** The backbones are frozen, so augmentation happens at the image level and features are re-extracted. Every synthetic-train crop has two pre-extracted augmented copies. At every training step each sample uses one of three views, drawn uniformly: the clean crop, copy 1 or copy 2. Copy $j \in \{1, 2\}$ uses seed $j$ in two ways. The roll angle of row $i$ is the $i$-th draw of `default_rng(j).uniform(-180, 180)` in manifest order. The photometric stream of row $i$ is `default_rng([j,i])`. Both are fixed once and are identical for the two backbones. Augmentation is applied to the $[0, 1]$ crop in this order: (1) brightness $\times\,\mathcal{U}(0.5, 1.5)$; (2) contrast about the mean, $\times\,\mathcal{U}(0.6, 1.4)$; (3) clip, then gamma $\mathcal{U}(0.6, 1.6)$; (4) with $p = 0.5$, Gaussian blur with $\sigma \sim \mathcal{U}(0.01, 1.5)$; (5) with $p = 0.3$, glare: 1–3 additive Gaussian blobs with amplitude $\mathcal{U}(0.5, 1.5)$ and radius $\mathcal{U}(0.05, 0.25) \cdot H$; (6) always, Gaussian noise with $\sigma \sim \mathcal{U}(0, 0.06)$, then clip; (7) with $p = 0.3$, random erasing of one rectangle covering 5–20% of the area (aspect ratio $\mathcal{U}(0.3, 3.3)$, uniform random fill); (8) with $p = 0.5$, JPEG re-encoding at quality 30–90. The in-plane rotation by $\theta$ is applied in the crop warp itself. The rotated crop is sampled from the original image, so it has no black corners. The label convention $q' = \mathrm{Exp}(-\theta d) \otimes q$ was derived from the camera geometry, with $d$ the undistorted unit ray through the crop centre. It was then verified on 512 synthetic-val rows at 4 angles. The keypoints pushed through the exact crop and rotation warp match the projection of $(q', r')$ to 0.34 px mean, 1.05 px p95 and 4.75 px maximum on a 224 px crop; the other three candidate conventions are off by 94–242 px. The label error measured by PnP is $0.06°$ mean, $0.20°$ p95 and $1.14°$ maximum, and the round trip is within $0.004°$.

**Synthetic val-aug.** Real images may not be used for selection, so EXP-109 and later SPEED+ experiments add a second selection domain: *synthetic val-aug*. It is the 11,994 synthetic-validation crops with one augmented copy, generated by the same pipeline with seed 1000 for both the roll angle and the photometric stream. For orbit voting on val-aug, the views at $+90°$, $+180°$ and $+270°$ are extracted with the same seed-1000 random angle and photometric draw, plus the fixed offset.

**Detector protocol (EXP-124).** The detected-crop evaluation was pre-registered before the detector was trained and before any detected-crop result existed. *Data.* The detector label is the tight box of the 11 projected Tango keypoints, clipped to the image, with one class. This is the box the ground-truth crops are built from. Images are stored once at half resolution ($960 \times 600$, grayscale JPEG quality 95). *Training.* We use Ultralytics YOLO11s (COCO-pretrained), trained on synthetic/train only, with `imgsz` 960, 20 epochs, batch 32, seed 0 and deterministic mode. Augmentation is horizontal and vertical flips with $p = 0.5$, no rotation or shear, mosaic (switched off for the last 5 epochs), brightness jitter `hsv_v`=0.6 with no hue or saturation jitter, and the Ultralytics default albumentations. The checkpoint is the one with the best Ultralytics fitness on synthetic/validation, $0.1\,\mathrm{mAP}_{50} + 0.9\,\mathrm{mAP}_{50:95}$. *Inference.* We keep the top-1 box at confidence $\geq 0.001$, scale it to full resolution and turn it into a crop with the ground-truth rule (box centre; side $= \max(w, h, 32) \times 1.3$). When there is no de-

tection, the whole image is used as the crop, and these cases are counted. Features of the detected crops and of their $90/180/270°$ views are extracted with the same extractor and roll rule as the ground-truth test3 views. *Systems.* All systems were fixed before any result: S1 (primary) uses the EXP-117 PoE students with orbit voting for rotation and the EXP-118 joint DINOv3 translation block with the orbit mean for translation; S2 uses the EXP-118 joint PoE trees and blocks; S3 uses the EXP-117 DINOv3 student with the same translation. *Reproduction gate.* A ground-truth-crop run with the same code must reproduce EXP-117 rotation ($11.21°/11.09°$) and EXP-118 translation ($0.185/0.183$ m); otherwise the evaluator is considered wrong and nothing is reported. The ground-truth-crop run gives S1 $11.21°/11.09°$ and $0.185/0.183$ m on lightbox/sunlamp, so the gate passes. At the time of writing the detector training is still running, so no detection statistics are given here.

**Official score.** We use the SPEED+ score [19, 20] as implemented by SPNv2 [17]. Per image,

$$e_q = 2 \arccos |\langle \hat{q}, q \rangle| \text{ (rad)}, \qquad e_t = \frac{\|\hat{t} - t\|_2}{\|t\|_2},$$

and the score of a domain is the mean over its images of $e_q + e_t$. Each component is set to zero below the precision floor of the challenge: $e_q$ when the rotation error is below $0.169°$, and $e_t$ when it is below $2.173 \times 10^{-3}$ (primary rule). As a check we also compute the joint rule, which zeroes the image score only when both components are below their floors. In the ground-truth-crop reference run the two rules differ by at most $1.1 \times 10^{-4}$ in any per-seed domain score.

## K.2. T-LESS

**Data and objects.** We use the BOP release of T-LESS [10, 11], read-only, with RGB images. Intrinsics are stored per instance in the manifest: $f_x = 1075.65$ px throughout, with the principal point at $(360, 270)$ for the PBR images and varying for the real ones. The translation target uses the $720 \times 540$ image size. The 7 objects were declared before any evaluation and span the symmetry classes in the BOP metadata (Tab. 25): objects 1 and 4 have a continuous symmetry; objects 5 and 11 have one discrete $180°$ symmetry; objects 18 and 21 have no symmetry in the metadata but are visually near-symmetric; object 27 has three discrete symmetries. Symmetries come from `models_cad/models_info.json` in the model frame, and the equivalent rotations are $RS$. Discrete symmetries are used exactly. Continuous ones are discretised in 36 steps of $10°$ about their axis. The full symmetry sets have sizes 36, 36, 2, 2, 1, 1 and 4 (identity included).

| Obj. | Symmetry ($\lvert\mathcal{S}\rvert$) | PBR train | Real train | Train | Val (PBR) | Test |
|---|---|---|---|---|---|---|
| 1 | continuous (36) | 11,071 | 929 | 12,000 | 1,886 | 886 |
| 4 | continuous (36) | 11,103 | 897 | 12,000 | 1,960 | 646 |
| 5 | 180° (2) | 11,178 | 822 | 12,000 | 2,108 | 190 |
| 11 | 180° (2) | 11,210 | 790 | 12,000 | 1,589 | 183 |
| 18 | none (1) | 11,250 | 750 | 12,000 | 1,837 | 146 |
| 21 | none (1) | 11,211 | 789 | 12,000 | 1,923 | 198 |
| 27 | 3 discrete (4) | 11,236 | 764 | 12,000 | 2,727 | 96 |
| all | | 78,259 | 5,741 | 84,000 | 14,030 | 2,345 |

Table 25. **T-LESS** instances per object in our manifest. Validation is held-out PBR scenes 45–49; test is all BOP19 targets in the real Primesense test images.

**Splits.** *Train* is `train_pbr` scenes 0–44 plus the real `train_primesense` images of the 7 objects. We keep instances with visible fraction $\geq 0.3$ and cap each object at 12,000 crops (PBR and real together) with rng 0. *Validation*, which is used for selection, is the whole held-out `train_pbr` scenes 45–49, with the same visibility filter. *Test* is every BOP19 target of these objects in `test_primesense`, with no visibility filter. These are 2,345 targets, whose visible fraction ranges from 0 to 1 (median 0.92). The local `test_primesense` contains the BOP19 subset only: 20 scenes $\times$ 50 images, 4,904 targets over all 30 objects. Table 25 gives the counts per object.

**Crops and rotation target.** The crop is the ground-truth square box (margin 0.15 per side, minimum 32 px) around the 8 projected corners of the object's 3-D model bounding box. Its median side is 191 px on PBR training rows, 197 px on PBR validation rows, 212 px on real training rows and 155 px on test rows. Because the box is built from the 3-D bounding-box corners, it is looser than BOP's `bbox_obj` silhouette box: their IoU has a median of 0.61 (5th percentile 0.45). The tree predicts the *allocentric* rotation $R_{\text{ray}}(t)^\top R$. Rebuilding the egocentric rotation from it reprojects the box corners to within $3.4 \times 10^{-13}$ px.

**Augmentation and orbit views.** T-LESS objects lie far from the optical axis, so the SPEED+-style affine rotation of the crop is not an exact camera roll. With it, the label error fails the pre-set rule: p95 $1.43°$ and maximum $2.5°$ on T-LESS, and p95 $2.40°$ and maximum $6.7°$ on YCB-V. Rolled views are therefore produced by the roll homography $K\,\mathrm{Rot}(d, -\theta)\,K^{-1}$ about the crop-centre ray. Labels transform exactly:

- egocentric: $R' = \mathrm{Rot}(d, -\theta)R$ and $t' = \mathrm{Rot}(d, -\theta)t$;
- allocentric: the target is recomputed from $t'$.

The verified label error is $0.000°$, and the mapping round trip is $\leq 0.004°$. The two augmented training copies use seeds 1 and 2, as on SPEED+, and the orbit views of validation and test rows are rolled by $90/180/270°$ without pho-

tometric changes. On these allocentric datasets, mapping an orbit candidate between views needs the translation. We use the ground-truth translation, which is consistent with the ground-truth-crop protocol.

**Training-target arms (EXP-106).** All three arms are trained per object, with the same trainer, recipe, seeds and checkpoint rule. *A0* uses the raw annotated rotation; these are the EXP-064 trees. *A1* is canonical: in every view, the label is replaced by the equivalent $R\,S$ closest to the identity. *A2* is multi-path: the recipe's geodesic soft-target cross-entropy, averaged over the equivalents $R\,S$. All equivalents are used when $|\mathcal{S}| \leq 8$; otherwise 8 are drawn per row and step. The pre-registration named the hard-path level-marginal loss of EXP-044. That loss would not reduce to A0 for objects without symmetry, which the pre-registration also requires, and it drops the soft targets. The smoothed loss was therefore adopted by an amendment written before any run. Symmetry composition was unit-checked beforehand: $q\,s$ matches $R\,S$ to $10^{-15}$, and the roll commutes with right-multiplication by $S$.

**Differences from the LiePose protocol.** The T-LESS comparison in Tab. 1 is not matched. LiePose [13] differs from our setting in four ways:

1. it fine-tunes its backbone jointly on all 30 objects, whereas we train frozen-feature trees per object;
2. it multiplies the crop by the ground-truth visible mask, whereas we apply no mask;
3. its per-object results (its Tab. 8) are averaged over our 7 objects for the comparison;
4. its BOP-style symmetry handling discretises continuous symmetries more finely than our $10^\circ$ steps, so our symmetry-aware error for objects 1 and 4 can exceed the true symmetric error by up to $5^\circ$.

We also use every BOP19 target with no visibility filter. A protocol on LiePose's own instance set (every ground-truth instance with visible area $> 64$ px, BOP-style fine discretisation) is reserved as EXP-125 and has not been run.

**Symmetry handling and recall.** For a predicted target rotation $\hat{R}$, the symmetry-aware error is

$$e_{\text{sym}} = \min_{S \in \mathcal{S}} \angle\big(\hat{R},\ R\,S\big) = \arccos\left(\frac{\max_S \operatorname{tr}(\hat{R}^\top R S) - 1}{2}\right),$$

computed in the allocentric target frame (re_sym). The raw error (no minimum) is reported separately. On T-LESS it is 38–47°, which is why the symmetry-aware metric is needed. Recall at $k^\circ$ is the fraction of instances with $e_{\text{sym}} \leq k$, for $k \in \{2, 5, 10, 15, 30\}$; the main text reports $k = 2, 5, 10$. For translation (EXP-126), the BOP symmetry transforms of our 7 objects have zero translation, so te_sym equals $E_T$. Translation recall T@$c$ is the fraction with $E_T \leq c$ cm, for $c \in \{2, 5, 10\}$; T-LESS translations are in millimetres and are converted with a factor of 0.1.

## K.3. SYMSOL-T

**Data.** SYMSOL-T contains five symmetric solids (tetrahedron, cube, icosahedron, cone, cylinder) at random translations, following LiePose [13]. The local copy was rendered with LiePose's generator using seeds 12345 and 82973, so it follows the published protocol but is not guaranteed to be identical to the published image sets. We follow LiePose's protocol:

- the input is the non-colour render, the full $224 \times 224$ frame, with no crop (crop side 224, margin 0);
- the target is the egocentric quaternion of the render;
- the error is the minimum over the annotated `rotations_equivalent`.

The camera has a $60^\circ$ horizontal field of view, so the object origin is at $z = 5.25/\tan 30^\circ = 9.09$ scene units and the focal length is 193.99 px, with the principal point at the image centre. The in-plane rotation about the image centre is therefore an exact camera roll, with a verified label error of $0.000^\circ$.

**Split.** From `train-45k`, 2,000 images per shape are held out with rng 0 for validation, and the remaining 43,000 per shape are training images. A pre-registered cap uses only the first 20,000 per shape, ordered by a seeded rank, for training, to fit the GPU budget. The test set is `test-5k`, 5,000 per shape. The manifest therefore has 25,000 test, 215,000 available training and 10,000 validation rows, of which 100,000 training rows are used.

**Symmetry groups.** The groups recovered from the annotations, and checked to be constant across samples, have orders 12 (tetrahedron), 24 (cube) and 60 (icosahedron). The continuous groups are discretised by the annotations in $1^\circ$ steps: 360 elements for the cone and 720 for the cylinder, whose set also contains the flip. These sets define the symmetry-aware error. They are also the equivalents of the multi-path arm A2, of which 8 are drawn per row and step for every shape here.

**Numerical fix.** In the clean renders, some $16 \times 16$ MoGe-2 border cells are always empty, so their per-dimension training standard deviation is about $10^{-6}$. Rolled views place the object there, and the standardised values overflowed in half precision. As a result, the 9 MoGe-2 trees for the cube, cylinder and icosahedron diverged. These trees were retrained with a floor on the standard deviation, $\max(\text{sd}, 10^{-2})$ (the features lie in $[-1, 1]$), and the diverged runs are marked superseded. The same floor is used for these shapes in every later arm.

**Metrics.** We report the symmetry-aware rotation error (mean and median) and the recall at $2/5/10^\circ$ ($\leq$), per shape and averaged over shapes. For translation (EXP-126) we report the mean Euclidean distance in scene units, the unit of LiePose's Tab. 3.

## K.4. YCB-V

**Data and objects.** We use the original PoseCNN release of YCB-Video [25] (all 92 sequences, `data_syn`, models and keyframes), not the BOP version; this is a recorded deviation. Images are $640 \times 480$ RGB, with $f_x = 1066.778$, $f_y = 1067.487$ and $(c_x, c_y) = (312.99, 241.31)$. We chose 6 objects before any model result, to cover different shapes: cracker box (box), mustard bottle (bottle), banana (elongated), mug (near-symmetric; its handle is the closest analogue of the SPEED+ flips), power drill and scissors (thin). This release has no symmetry metadata, so the declared-symmetric YCB objects (bowl, wood block, clamps, foam brick) are excluded, and errors are raw geodesic errors.

**Splits.** *Training* uses every third frame of `image_sets/train.txt`, without the held-out sequences. We keep instances with $\geq$ 1,000 visible label pixels and at most 12,000 crops per object (rng 0). This gives 48,316 instances from 64 sequences, 14–20 sequences per object. *Validation*, which is used for selection, holds out one training sequence per object (rng 0) with the same stride and filter. Several objects appear in the same sequences, so the held-out set is the union of sequences 0012, 0047, 0065 and 0074, and the mug appears in two of them. Validation has 2,797 instances in 1,582 frames. *Test* is `image_sets/keyframe.txt`: the keyframes of sequences 0048–0059, all instances with no visibility filter. Eight of these sequences contain our objects (0048, 0050–0052, 0054–0056, 0059), giving 3,478 instances in 2,041 keyframes. Test sequences are never used for training, validation or synthetic backgrounds. Table 26 gives the counts per object. The crop is the ground-truth square box (margin 0.15, minimum 32 px) of the projected model points (`points.xyz`). The target is the allocentric rotation, as for T-LESS. Augmentation and orbit views use the exact roll homography, as in Appendix K.2.

**Diagnosis of the real-only failure (EXP-064).** Trained on real videos only, the trees reach about $69^\circ$ on test. We ran four CPU checks to separate a pipeline bug from a generalisation failure:

1. The trees fit their training images: for example, $3.8^\circ$ (mustard) and $3.9^\circ$ (mug) mean error on training rows.
2. Test poses are covered by training. For the cracker box, mustard and banana, the nearest training rotation is a median of about $5^\circ$ away, yet test errors are $75$–$87^\circ$.
3. All of each object's label pixels lie inside its crop.
4. The ground-truth rotations match the images: 65–98% of the projected model points land on the object mask, against 28–83% for a transposed-rotation control.

We conclude that, with 14–20 cluttered training videos per object and unmasked tokens, the per-object trees memorise video context.

| Object | Real train | Synthetic | Val | Test |
|---|---|---|---|---|
| 003 cracker box | 8,588 | 12,000 | 302 | 868 |
| 006 mustard bottle | 9,435 | 12,000 | 217 | 357 |
| 011 banana | 9,153 | 12,000 | 696 | 379 |
| 025 mug | 6,909 | 12,000 | 519 | 636 |
| 035 power drill | 6,609 | 12,000 | 367 | 1,057 |
| 037 scissors | 7,622 | 12,000 | 696 | 181 |
| all | 48,316 | 72,000 | 2,797 | 3,478 |

Table 26. **YCB-V** instances per object. "Synthetic" are the EXP-122 `data_syn` composites, which are training-only. Validation is held-out real sequences; test is the real keyframes.

**Synthetic composites (EXP-122).** `data_syn` has 80,000 renders on black backgrounds. We select instances of the 6 objects with $\geq$ 1,000 visible label pixels, at most 12,000 per object (rng 0), which is the same rule as for the real rows. This gives 72,000 instance rows from 51,253 render frames. Following DenseFusion and FFB6D, each render is pasted where `label` $= 0$ onto a real YCB-V frame. Backgrounds are drawn with rng 0, with replacement, from the EXP-064 *real training* frames only. The script asserts that no validation or test sequence is among them. In total, 24,741 distinct background frames are used, and the frame-to-background map is stored with the composites. Crops, allocentric targets and the reprojection check follow the real rows. Features are extracted for the clean composites and for two augmented copies (seeds 1 and 2, exact roll homography). The real rows reuse the EXP-064 caches. Training uses the EXP-064 per-object trainer unchanged, except that the composites are appended to the training rows. Normalisation statistics come from the combined clean training rows, and the checkpoint is selected on the held-out *real* validation sequences.

## K.5. Training details

**Features and normalisation.** Features are extracted once per view and cached.

- *DINOv3:* last-layer patch tokens of ViT-L/16, average-pooled from $16 \times 16$ to $8 \times 8$ ($64 \times 1024$, stored in fp16).
- *MoGe-2* (`normals16~and`): surface normals and mask pooled to $16 \times 16$ ($256 \times 4$). The consensus mask is the intersection of the MoGe-2 mask ($> 0.5$) and a DINOv3 foreground, obtained by Otsu thresholding the first PCA component of the patch tokens.

Each feature dimension is standardised with the mean and

standard deviation of the clean training rows, plus $10^{-6}$; the floor described in Appendix K.3 applies where noted. The statistics are per dataset on SPEED+ and per object on the other benchmarks, and the same statistics are applied to augmented views, validation and test. SPEED+ images are grayscale and replicated to three channels; T-LESS, SYMSOL-T and YCB-V are read as RGB.

**Common recipe.** All four trainers share one budget, the EXP-036 budget that was fixed at the Phase-3 gate (Tab. 27):

- 100 epochs with batch 256 and dropout 0.1;
- AdamW with learning rate $10^{-3}$ and weight decay 0.05;
- `OneCycleLR`, peak $10^{-3}$, 5% warm-up, over $100 \cdot \lceil N/256 \rceil$ steps;
- bf16 autocast;
- validation every 5 epochs and at the last epoch;
- seeds 0, 1 and 2, set with `torch.manual_seed` and `numpy.random.seed`.

In `train_probe_p3.py`, bf16 is the `--amp bf16` flag; the command line of every reported run shows it. The other three trainers always use bf16. With augmentation, every sample draws one of its three views (clean, copy 1, copy 2) uniformly at every step. The tree Transformer has a width of 256, one encoder layer and two causal decoder layers (4 heads, FFN 512), and a 6-frequency Fourier encoding of the cell centres. It has 2,391,564 parameters on DINOv3 tokens and 2,128,404 on MoGe-2 tokens. The translation block is one learned query cross-attending (4 heads) to the encoded tokens, followed by LN $\to$ 256 $\to$ GELU $\to$ 3.

**Checkpoint selection.** Every trainer keeps the epoch with the lowest validation criterion of Tab. 27, evaluated every 5 epochs on the *whole* validation split. On SPEED+, this is the clean synthetic validation set (11,994 crops) with the greedy single-branch prediction. No real image enters selection. On SPEED+ and T-LESS the selected epoch is almost always late (Tab. 28). On YCB-V it is early, which matches the overfitting diagnosed above.

## K.6. Evaluation metrics

**Rotation error.** The geodesic error is $E_R = 2 \arccos |\langle \hat{q}, q \rangle|$ in degrees. On SPEED+ and YCB-V it is the raw error. On T-LESS and SYMSOL-T it is the symmetry-aware error of Appendix K.2. On T-LESS and YCB-V, $\hat{q}$ is compared with the ground truth in the allocentric target frame. Converting both to the egocentric frame with the ground-truth translation multiplies them by the same $R_{\text{ray}}(t)$, so the geodesic error is unchanged. We report mean, median and p95. Recall acc@$k$ is the fraction of images with error below $k°$. On SPEED+, $k \in \{1, 3, 5, 10, 20\}$ with a strict $<$. On the other benchmarks, $k \in \{2, 5, 10, 15, 30\}$ with $\leq$.

**Severe errors and flips.** *Severe flips* are the fraction of images with $E_R > 150°$; we also report $E_R > 90°$. *Flip transitions* compare a system with a reference prediction on the same images. Let $n_{150}$ be the number of images with reference error $> 150°$, and $n_{20}$ the number with reference error $< 20°$. Then *corrected* $= \#\{e_0 > 150° \wedge e < 20°\}/n_{150}$ and *broken* $= \#\{e_0 < 20° \wedge e > 150°\}/n_{20}$. The denominators are always reported. In the shared evaluator (EXP-113–115, 127), the reference is the greedy prediction of the same model family and variant. In EXP-109 it is the EXP-036 PoE greedy prediction, with $n_{150} = 401$ (lightbox) and 206 (sunlamp).

**Retention.** For orbit distillation, the retention in each domain is

$$\rho = \frac{E^{\text{A}}_{\text{greedy}} - E^{\text{student}}_{\text{greedy}}}{E^{\text{A}}_{\text{greedy}} - E^{\text{A}}_{\text{orbit}}},$$

the share of the orbit-voting gain of the undistilled tree (A) that the single-view student recovers.

**Translation.** $E_T = \|\hat{t} - t\|_2$ is given in metres (SPEED+), millimetres (T-LESS), metres (YCB-V) or scene units (SYMSOL-T). We also report the normalised error $e_t = E_T / \|t\|_2$ and T@$c$cm $= \Pr[E_T \leq c\,\text{cm}]$ for $c \in \{2, 5, 10\}$ (T-LESS, YCB-V). The *box-only* baseline predicts $U_x = U_y = 0$ and $U_z$ equal to the training mean, mapped through the same inverse. It measures how much translation the ground-truth crop alone carries. With orbit voting, each view's $\hat{t}$ is rolled back to view 0 and the four estimates are averaged.

**Inference modes.** *Greedy* decodes view 0 only (beam 1). *Orbit product* collects beam-8 leaves from 4 views ($0/90/180/270°$), maps them back to view 0, and scores each of the 32 candidates by the sum over views of the teacher-forced PoE path log-probability. *Weighted vote* (orbit_vote32) scores each candidate by Gaussian-kernel support ($\sigma = 5°$) from all 32 candidates, weighted by their beam probabilities normalised within their view. On allocentric datasets, candidates are mapped between views with the ground-truth translation (Appendix K.2).

**Aggregation.** Errors are first averaged over the images of each object (or shape). Within a seed, these per-object means are averaged over objects with equal weight, which gives the object-balanced value. For a single object (SPEED+), this value is the plain mean over images. Medians and recalls are aggregated the same way. The reported value is the mean of the per-seed object-balanced values over seeds 0–2. The $\pm$ value is their sample standard deviation (ddof $= 1$) across the 3 seeds. Per-object tables average each object over the seeds.

| Trainer | Used by | Model and data | Loss | Checkpoint selection | Notes |
|---|---|---|---|---|---|
| `train_probe_p3.py` | EXP-036, 109, 115, 118 | tree per branch, SPEED+ full split; clean + 2 aug. views | soft-target CE per level (+ $\|u - u^*\|^2$ for 6-DoF) | lowest clean synthetic-val greedy mean $E_R$ | frozen-tree translation arm: lowest synthetic-val mean $E_T$ |
| `train_xobj.py` | EXP-064, 106, 119, 122, 126 | tree per branch *and object*; clean + 2 aug. views | soft-target CE; A1 canonical / A2 mean over $\leq 8$ equivalents; + $\beta$ level CE (119) | lowest symmetry-aware val greedy mean of the single-branch tree | translation: lowest val mean $E_T$ of view 0, $u$ standardised per dim. |
| `train_flat_tf.py` | EXP-113, 127 | flat classifier, same encoder input and views | CE to soft targets on the 32 nearest anchors, $\tau = 2.51°$ | lowest clean synthetic-val mean of the arg-max anchor | grids: Hopf 10923 × 12, tesseract L5 leaves, Hopf 1771 × 74 |
| `train_consistency.py` | EXP-116, 117 | tree per branch, SPEED+, clean + 2 aug. views | GT CE + $\lambda$ roll-consistency (116) or + $\beta$ orbit-teacher level CE, $\beta = 1$ (117) | lowest clean synthetic-val greedy mean | teacher: 24 candidates and softmax orbit weights per image |

All: 100 epochs, batch 256, AdamW ($10^{-3}$, wd 0.05), OneCycle (5% warm-up), dropout 0.1, bf16, validation every 5 epochs, seeds 0–2.

Table 27. **Trainers and hyper-parameters.** No hyper-parameter was tuned per dataset; the budget is the one fixed for SPEED+ at the Phase-3 gate and locked by DEC-018. "Greedy mean" is the mean rotation error of the single-view greedy prediction.

| Experiment | Dataset | Runs | Median | Min |
|---|---|---|---|---|
| EXP-109 (locked recipe) | SPEED+ | 6 | 100 | 90 |
| EXP-113 (flat) | SPEED+ | 6 | 100 | 90 |
| EXP-117 (students) | SPEED+ | 9 | 100 | 90 |
| EXP-064 | T-LESS | 42 | 90 | 55 |
| EXP-106 (A1, A2) | T-LESS | 84 | 90 | 55 |
| EXP-064 | SYMSOL-T | 30 | 95 | 90 |
| EXP-106 (A1, A2) | SYMSOL-T | 60 | 95 | 70 |
| EXP-064 (real only) | YCB-V | 36 | 35 | 5 |
| EXP-122 (real + syn.) | YCB-V | 36 | 60 | 25 |

Table 28. **Selected epoch** (out of 100) of completed, non-superseded training runs.

**Timing.** End-to-end latency (EXP-112) is measured on real lightbox crops with the image already decoded. It covers the crop warps, backbones, tree encoding and beam search, and orbit scoring. It is measured at batch 1 and batch 64 as the median of 5 repetitions after warm-up, with CUDA synchronisation. The concurrent GPU load is recorded; the GPU is shared, so all timings are upper bounds.

### K.7. Selection and pre-registration protocol

**Split policy.** We fixed the split policy at the start of the project (DEC-000) and restored it in DEC-006:

- train on SPEED+ synthetic/train only;
- select and gate on synthetic/validation only;
- treat all 6,740 lightbox and 2,791 sunlamp images as test-only and descriptive.

DEC-005 had briefly allowed a seeded 20% lightbox validation split (1,348 images). DEC-006 withdrew it on the same day, before it influenced any training run or checkpoint, and rewrote the one gate that referred to it before any result. The same policy carries over to the other benchmarks: train on the permitted training split, select on a held-out validation split built by whole scenes or sequences, and use the test split for reporting only. Synthetic validation contains almost no gross errors, so from EXP-109 on SPEED+ selection also uses the synthetic val-aug proxy (Appendix K.1).

**Recipe lock.** DEC-018 fixed the recipe before any cross-dataset run:

- frozen DINOv3 `tokens8`, plus MoGe-2 `normals16~and` in the PoE variant, on ground-truth square crops;
- one tree Transformer per branch, with the EXP-036 architecture and budget;
- training with the two augmented copies;
- equal-weight per-level PoE;
- orbit voting over 4 in-plane views.

Both variants (PoE and DINOv3-only) are reported everywhere, with and without orbit voting. The following are explicitly not part of the recipe: 16×16 tokens, the weighted PoE, discrete flow matching, flow-teacher distillation and the corner tree. DEC-023 kept this recipe after EXP-117 and recorded orbit distillation as a successful add-on, which later experiments do not use unless stated. DEC-024 added three cross-dataset fixes:

- EXP-122: synthetic YCB-V renders;
- EXP-106, amended: multi-path symmetry targets, also on SYMSOL-T;
- EXP-119: orbit distillation on T-LESS and SYMSOL-T.

DEC-025 set the matched comparisons with published results: EXP-124 (detector and official score), EXP-126 (translation on T-LESS, SYMSOL-T and YCB-V) and the reserved EXP-125. It also recorded the literature corrections to apply before submission. DEC-026 added EXP-127 (hierarchical tree vs. flat classifiers over the same 131,072 leaves and over a balanced Hopf grid). It uses the DEC-018 recipe without distillation.

**Pre-registration and the pass rule.** Each experiment has a report in `outputs/reports/`. Before any run, the report records:

- the question;
- the arms and controls;
- the selection data;
- the metrics;
- the pass criterion;
- for the main experiments, the expected outcome.

The standard pass rule compares 3-seed means on validation data: a difference passes if it exceeds the larger of the two seed standard deviations. Some criteria add a non-inferiority clause on clean synthetic validation, using the same tolerance. Real-domain and test numbers are reported for every arm, whatever their direction, and are never used for selection.

**Amendments and deviations.** A protocol change goes into the report as a dated "Amendment" section, written before the affected runs and stating what changes and why. Earlier text is not edited. Deviations found after the fact are recorded as such, with the run identifiers. Examples:

- *EXP-109.* The predictor-based convention gate was replaced, after an external review and before any augmented run, by a geometric derivation. A pixel threshold chosen before measuring was then replaced, before the degree measurement was taken, by a label-error rule; both runs are kept.
- *Compact-feature gate (EXP-064).* The gate, which compares the compact cache with the full features, was preregistered to run before use but ran after the T-LESS results. Its strict tolerance fails (MoGe-2 maximum deviation 0.51 standard deviations, p99 0.004). The pre-set functional follow-up shows no impact: on 512 crops no PoE greedy prediction changes, and 1 and 3 single-branch leaves change. Both are on record.
- *EXP-106.* The multi-path loss was changed to the smoothed version (Appendix K.2), and the arm K and best-fit stages were left unrun, by an amendment written before any run.
- *SYMSOL-T.* The NaN fix (Appendix K.3) was recorded as a bug fix, not a recipe change; only the affected trees were retrained.
- *EXP-064 zero-shot arm.* The pre-set rule was met literally on T-LESS. A post-hoc constant-prediction control, labelled as post hoc, shows that this comes from the pose prior, so no transfer is claimed.
- *EXP-127.* An amendment written before any trained-arm run defines per-system inference memory and adds a gate: the tree must reproduce EXP-113 seed 0 to within $0.02°$.

## K.8. Compute and infrastructure

**Hardware and software.** Every logged run that records its GPU (740 runs) used one NVIDIA RTX 6000 Ada Generation, in a workstation shared with other users, and Python 3.13.11. The PyTorch build differs by period. Runs of EXP-000 to EXP-042, EXP-101 and EXP-104 (and some smoke tests) used PyTorch 2.11.0 with CUDA 13.0. Runs from EXP-043 on, including every run from EXP-109 onward, used PyTorch 2.8.0 with CUDA 12.8. All runs used cuDNN 9.19. The locked-recipe runs also record timm 1.0.27 and OpenCV 5.0.0. Each run directory stores its own environment, so per-run versions can be checked.

**Memory guards.** Every heavy job runs through `scripts/guarded.sh`, which provides four protections:

- a hard kernel cap: a `systemd-run` scope with `MemoryMax` (40 GB by default, set per job), no swap, and at most 4,096 tasks;
- admission control: a job starts only when the machine has the job's cap plus 12 GB of RAM available and, by default, 8 GB of free GPU memory;
- low priority: the job runs at `nice` 10 and idle I/O priority;
- maintenance pause: new jobs wait while `outputs/experiments/QUEUE_PAUSE` exists, while running jobs are untouched.

Inside each job, `tfpose.memguard` acts as a soft guard. Every 0.25 s it polls the process's own RSS plus the USS of its child processes, and the system's available memory. It aborts the job, writes the reason into the run record and kills the job's children before the kernel cap is reached. Its limit is 85% of the cap, and it also aborts if available memory falls below 16 GB. Children were first counted by RSS, which counts pages shared with forked DataLoader workers once per worker. During EXP-124 this produced a false abort, reporting 44.3 GB against a real peak of 11 GB. The orphaned workers then stalled the queue for about 14.7 h. Children have been counted by USS since then. A watchdog checks the queues every 5 minutes. It kills orphaned DataLoader workers, marks runs whose process has died as failed, and raises alerts for dead or stuck queues, idle jobs and low memory.

**Feature caches.** Features are extracted once per view and stored on an 8 TB drive (DEC-020): the full SPEED+ caches (105 GB when moved) and the compact caches, about 170 GB for the three EXP-064 datasets. Each cache directory is named by backbone, subset tag, preprocessing hash and manifest hash, and stores its specification, git commit, timing and peak GPU memory. On the full SPEED+ split, extraction took 11.5 ms per image for DINOv3 (peak 1.6 GB GPU) and 29.2 ms per image for MoGe-2 (peak 3.0 GB). The compact extractor stores only what the recipe needs: `tokens8`, the DINOv3 foreground, and the MoGe-2 normals and mask. It was checked against the full caches as described above.

**Run records.** Every training and evaluation script opens a `tfpose.runlog.Run`. It claims a directory `outputs/experiments/EXP-nnn_<name>/RUN-<timestamp>-seed<s>` and writes the following files:

- `config.yaml`: all arguments;
- `command.txt`: the exact command;
- `environment.txt`: host, library versions, GPU, git commit and a flag for uncommitted code changes;
- `status.json`: state, PID, start and end, `wall_clock_s`, peak CPU RSS, peak GPU memory, and for trainings the selected epoch and parameter count.

Metrics are written as CSV, JSON and TSV, and checkpoints, logs and predictions go to subfolders. Runs that fail or are superseded keep their directories, with the error or the reason in `status.json`.

**Wall-clock time per tree.** Table 30 summarises `wall_clock_s` over all completed, non-superseded training runs of one tree; frozen-tree translation blocks are excluded. The times include loading and standardising

| Exp. | Pre-registered primary criterion (validation data; 3 seeds; larger-seed-std rule) | Selection data |
|---|---|---|
| EXP-109 | A or B: PoE val-aug mean below EXP-036, and clean val not worse; C: orbit lowers val-aug without worse clean val | synth. val + val-aug |
| EXP-113 | tree below flat Hopf ($10923 \times 12$), PoE + orbit (primary) and PoE greedy; also on val-aug | synth. val (+ val-aug) |
| EXP-114 | orbit product below the best of max-confidence, top-1 vote and weighted vote (PoE primary) | synth. val (+ val-aug) |
| EXP-115 | combined aug. beats photometric-only, and beats rotation-only (orbit, val-aug); orbit gain by arm descriptive | synth. val-aug |
| EXP-116 | roll-consistent tree T below A, C1 and C2 on greedy val-aug, clean val not worse; consistency above A and C1 | synth. val-aug |
| EXP-117 | student greedy val-aug below A greedy (DINOv3 primary), clean val not worse; retention descriptive | synth. val-aug |
| EXP-118 | translation block $E_T$ below box-only; joint training does not hurt rotation | synth. val |
| EXP-064 | per dataset, PoE val mean below DINOv3-only; zero-shot: pair error below chance | held-out val per dataset |
| EXP-106 | A2 (multi-path) val mean below the better of A0 and A1, PoE greedy (T-LESS, SYMSOL-T) | PBR scenes 45–49; held-out SYMSOL-T |
| EXP-119 | student greedy val mean below the base arm's greedy (DINOv3 primary); retention descriptive | held-out val per dataset |
| EXP-122 | real + synthetic PoE orbit val mean below real-only (EXP-064) | held-out real sequences |
| EXP-124 | descriptive: detected-crop official score of S1 and detection gap; GT-crop reproduction gate | – (no selection) |
| EXP-126 | DINOv3 block val $E_T$ (orbit mean) below box-only, per dataset | held-out val per dataset |
| EXP-127 | tree vs flat on the identical 131,072 leaves, PoE + orbit; flat-vs-flat and tree vs balanced Hopf ($1771 \times 74$) | synth. val (+ val-aug) |

Table 29. **Pre-registered criteria** of the experiments behind the paper. Test and real-domain numbers are reported for every arm and are never used for selection. Outcomes are reported in the main text and in the experiment reports.

| Experiment (dataset) | Branch | Runs | Median (min) | Range (min) |
|---|---|---|---|---|
| EXP-109 locked recipe (SPEED+) | DINOv3 | 3 | 13.3 | 12.9–15.8 |
| | MoGe-2 | 3 | 5.8 | 5.8–5.8 |
| EXP-115 aug. ablation (SPEED+) | DINOv3 | 6 | 15.5 | 13.8–19.3 |
| | MoGe-2 | 6 | 6.5 | 6.2–6.6 |
| EXP-113 flat Hopf (SPEED+) | DINOv3 | 3 | 17.3 | 16.8–17.5 |
| | MoGe-2 | 3 | 8.2 | 8.1–8.3 |
| EXP-117 students (SPEED+) | DINOv3 | 6 | 40.0 | 27.3–41.7 |
| | MoGe-2 | 3 | 83.6 | 51.0–93.9 |
| EXP-118 joint 6-DoF (SPEED+) | DINOv3 | 3 | 24.9 | 18.7–25.7 |
| | MoGe-2 | 3 | 12.4 | 12.3–13.8 |
| EXP-064 (T-LESS) | DINOv3 | 21 | 5.7 | 5.3–8.3 |
| | MoGe-2 | 21 | 3.4 | 3.1–6.9 |
| EXP-106 A2 (T-LESS) | DINOv3 | 21 | 11.1 | 4.6–60.5 |
| | MoGe-2 | 21 | 10.6 | 3.4–21.1 |
| EXP-064 (SYMSOL-T) | DINOv3 | 15 | 6.4 | 4.5–9.6 |
| | MoGe-2 | 15 | 4.0 | 3.1–4.8 |
| EXP-106 A2 (SYMSOL-T) | DINOv3 | 15 | 7.1 | 6.5–7.4 |
| | MoGe-2 | 15 | 9.1 | 9.0–9.3 |
| EXP-064 (YCB-V, real) | DINOv3 | 18 | 4.0 | 3.0–4.7 |
| | MoGe-2 | 18 | 2.3 | 1.9–3.3 |
| EXP-122 (YCB-V, real + syn.) | DINOv3 | 18 | 7.4 | 5.3–18.0 |
| | MoGe-2 | 18 | 4.8 | 3.6–17.0 |

Table 30. **Training wall-clock per tree** (100 epochs, one tree, shared GPU, feature loading included), from `status.json` of completed, non-superseded runs.

the cached features. They were measured on the shared GPU, so they vary with load (for example, one T-LESS A2 tree took 3,629 s against a median of 664 s). On SPEED+, a locked-recipe tree takes a median of 13.3 min for DINOv3 (range 12.9–15.8) and 5.8 min for MoGe-2. On the other benchmarks, per-object trees train on at most 12,000 crops per object (20,000 per shape on SYMSOL-T; 18,609–21,435 per object for EXP-122, real plus synthetic). Their median times are 2.3–11.1 min, and single runs on the loaded GPU took up to 60.5 min. Distilled students additionally teacher-force the 24 teacher candidates of every sample.

# L. Complete results

This appendix lists the complete numbers behind every result in the paper: all variants, inference modes, domains, per-object and per-seed breakdowns, medians, accuracies at thresholds, flip statistics, and each pre-registered criterion with its outcome. Each subsection names the evaluation run its tables are read from. Unless stated otherwise, values are 3-seed means, $\pm$ is the sample standard deviation over seeds, errors are in degrees and accuracies and failure rates are fractions. *Synth. val* is the clean synthetic validation set, *synth. val-aug* the same images with the fixed EXP-109 augmentation (one copy, seed 1000), used as a generalisation proxy for selection; lightbox and sunlamp are test-only. acc@$k$ is the fraction of images with error below $k^\circ$; $>$90 and $>$150 are the fractions above $90^\circ$ and $150^\circ$. In flip-transition columns, *fixed* is the share of the reference system's $>150^\circ$ errors that become $<20^\circ$, and *broken* the share of its $<20^\circ$ predictions that become $>150^\circ$; the reference counts are seed means and can be fractional. Two evaluation passes of the same trees can differ in the second decimal (*e.g.* the sunlamp PoE + orbit error is 12.28 in EXP-112 and 12.25 in EXP-113–117); each table uses a single run and the values are not reconciled.

## L.1. EXP-109: augmentation and orbit voting (locked recipe)

**Question.** Can the frozen-feature tree generalise better to real images when (A) training is augmented (photometric + exact in-plane roll), (B) finer $16\times16$ DINOv3 tokens are used, or (C) predictions are made consistent over the four in-plane rotations of the test crop (orbit voting)? Arm A + C is the locked recipe used throughout the paper.

**Pre-registered criteria and outcome** (selection on synthetic val and val-aug; real domains descriptive; "beats" means a 3-seed mean difference larger than the larger seed std):

- A helps generalisation (PoE val-aug below the clean EXP-036 tree, clean synth. val not worse): **PASS**. Val-aug $9.75 \pm 0.07$ vs $30.84 \pm 0.40$; clean $4.86 \pm 0.03$ vs $5.54 \pm 0.07$.
- B (tokens16) helps generalisation: **FAIL**. Val-aug $30.51 \pm 0.20$ vs $30.84 \pm 0.40$, a difference within the seed std.
- C (orbit voting) helps on top of its base: **PASS** on both bases. On A: val-aug 5.82 vs 9.75, clean 3.87 vs 4.86; on the clean EXP-036 trees: val-aug 19.84 vs 30.84.
- Real domains: descriptive only (no threshold).

**Label convention.** The roll label transform $q' = \mathrm{Exp}(-\theta d) \otimes q$ was derived from the camera geometry and verified before any augmented training: the projected Tango points match the warped crop to 0.34 px mean (1.05 px p95), the exact label error is $0.06^\circ$ mean ($0.20^\circ$ p95, $1.14^\circ$ max), and the round trip is within $0.004^\circ$; the other three candidate transforms are off by 94–242 px.

**Single branches** (training-run evaluation, greedy, synth. val / lightbox / sunlamp): DINOv3 + augmentation 4.84 / 22.5 / 22.0 against 5.34 / 37.1 / 41.5 without; MoGe-2 + augmentation 17.6 / 40.6 / 59.5 against 16.7 / 43.7 / 60.6; tokens16 DINOv3 5.37 / 37.9 / 39.7.

| System | Inference | Domain | mean | median | p95 | acc@5 | acc@10 | acc@20 | >90 | >150 |
|---|---|---|---|---|---|---|---|---|---|---|
| Clean tree (EXP-036) | greedy | Synth. val | $5.54 \pm 0.07$ | 4.62 | 10.20 | 0.565 | 0.946 | 0.990 | 0.002 | 0.001 |
| | | Synth. val-aug | $30.84 \pm 0.40$ | 10.95 | 155.50 | 0.146 | 0.455 | 0.702 | 0.107 | 0.055 |
| | | Lightbox | $26.91 \pm 0.78$ | 8.99 | 159.89 | 0.196 | 0.555 | 0.776 | 0.105 | 0.059 |
| | | Sunlamp | $36.66 \pm 0.93$ | 12.98 | 163.11 | 0.110 | 0.373 | 0.652 | 0.153 | 0.074 |
| A: augmented | greedy | Synth. val | $4.86 \pm 0.03$ | 4.36 | 8.97 | 0.618 | 0.968 | 0.998 | 0.001 | 0.000 |
| | | Synth. val-aug | $9.75 \pm 0.07$ | 5.92 | 18.37 | 0.382 | 0.825 | 0.954 | 0.018 | 0.009 |
| | | Lightbox | $17.35 \pm 0.32$ | 6.75 | 111.67 | 0.307 | 0.730 | 0.892 | 0.059 | 0.042 |
| | | Sunlamp | $20.16 \pm 0.58$ | 8.43 | 103.59 | 0.216 | 0.601 | 0.843 | 0.064 | 0.033 |
| B: tokens16 | greedy | Synth. val | $5.56 \pm 0.03$ | 4.63 | 10.15 | 0.566 | 0.947 | 0.990 | 0.002 | 0.001 |
| | | Synth. val-aug | $30.51 \pm 0.20$ | 10.90 | 153.41 | 0.146 | 0.457 | 0.705 | 0.105 | 0.053 |
| | | Lightbox | $27.01 \pm 0.38$ | 9.15 | 158.08 | 0.189 | 0.547 | 0.771 | 0.104 | 0.057 |
| | | Sunlamp | $35.97 \pm 0.96$ | 12.95 | 162.93 | 0.114 | 0.377 | 0.658 | 0.151 | 0.073 |
| C on clean tree | orbit product | Synth. val | $4.62 \pm 0.04$ | 3.91 | 9.58 | 0.688 | 0.956 | 0.995 | 0.001 | 0.000 |
| | | Synth. val-aug | $19.84 \pm 0.35$ | 7.71 | 93.26 | 0.267 | 0.632 | 0.827 | 0.054 | 0.026 |
| | | Lightbox | $20.25 \pm 0.60$ | 7.26 | 145.38 | 0.285 | 0.673 | 0.859 | 0.073 | 0.048 |
| | | Sunlamp | $24.83 \pm 0.87$ | 9.96 | 134.92 | 0.170 | 0.501 | 0.780 | 0.089 | 0.043 |
| C on A (locked) | orbit product | Synth. val | $3.87 \pm 0.03$ | 3.45 | 7.72 | 0.781 | 0.982 | 0.999 | 0.000 | 0.000 |
| | | Synth. val-aug | $5.82 \pm 0.09$ | 3.87 | 10.96 | 0.688 | 0.939 | 0.982 | 0.008 | 0.004 |
| | | Lightbox | $12.31 \pm 0.46$ | 5.26 | 27.45 | 0.465 | 0.837 | 0.939 | 0.037 | 0.030 |
| | | Sunlamp | $12.28 \pm 0.40$ | 6.32 | 31.05 | 0.358 | 0.760 | 0.931 | 0.030 | 0.017 |

Table 31. **EXP-109, all arms** (PoE, 3 seeds). p95: 95th percentile of the error.

| System | Domain | $n_{>150}$ (ref.) | fixed (n) | fixed | $n_{<20}$ (ref.) | broken (n) | broken |
|---|---|---|---|---|---|---|---|
| A: augmented | Synth. val | 12.0 | 10.7 | 0.878 | 11875.7 | 2.0 | 0.000 |
| | Synth. val-aug | 662.7 | 545.7 | 0.823 | 8415.0 | 7.3 | 0.001 |
| | Lightbox | 401.0 | 204.3 | 0.510 | 5227.0 | 49.7 | 0.010 |
| | Sunlamp | 206.0 | 121.7 | 0.590 | 1820.0 | 18.3 | 0.010 |
| B: tokens16 | Synth. val | 12.0 | 4.7 | 0.318 | 11875.7 | 6.0 | 0.001 |
| | Synth. val-aug | 662.7 | 87.7 | 0.133 | 8415.0 | 84.7 | 0.010 |
| | Lightbox | 401.0 | 57.0 | 0.142 | 5227.0 | 59.3 | 0.011 |
| | Sunlamp | 206.0 | 27.3 | 0.132 | 1820.0 | 21.7 | 0.012 |
| C on clean tree | Synth. val | 12.0 | 7.0 | 0.576 | 11875.7 | 0.7 | 0.000 |
| | Synth. val-aug | 662.7 | 313.7 | 0.473 | 8415.0 | 30.7 | 0.004 |
| | Lightbox | 401.0 | 143.0 | 0.357 | 5227.0 | 41.0 | 0.008 |
| | Sunlamp | 206.0 | 93.3 | 0.455 | 1820.0 | 15.7 | 0.009 |
| C on A (locked) | Synth. val | 12.0 | 11.0 | 0.898 | 11875.7 | 0.0 | 0.000 |
| | Synth. val-aug | 662.7 | 607.7 | 0.917 | 8415.0 | 0.7 | 0.000 |
| | Lightbox | 401.0 | 250.0 | 0.623 | 5227.0 | 19.0 | 0.004 |
| | Sunlamp | 206.0 | 164.7 | 0.800 | 1820.0 | 7.0 | 0.004 |

Table 32. **EXP-109 flip transitions** relative to the clean EXP-036 PoE greedy tree (per image, seed means of counts).

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| Clean tree (EXP-036) | greedy | 0 | 5.57 | 31.26 | 27.62 | 37.40 | 0.059 | 0.073 |
| | | 1 | 5.47 | 30.80 | 26.07 | 36.98 | 0.060 | 0.078 |
| | | 2 | 5.59 | 30.46 | 27.03 | 35.62 | 0.059 | 0.070 |
| A: augmented | greedy | 0 | 4.89 | 9.82 | 17.47 | 20.63 | 0.040 | 0.031 |
| | | 1 | 4.84 | 9.70 | 16.99 | 20.35 | 0.039 | 0.034 |
| | | 2 | 4.84 | 9.72 | 17.60 | 19.51 | 0.046 | 0.034 |
| C on clean tree | orbit product | 0 | 4.66 | 20.19 | 20.84 | 25.03 | 0.050 | 0.040 |
| | | 1 | 4.58 | 19.49 | 19.63 | 25.59 | 0.047 | 0.054 |
| | | 2 | 4.63 | 19.83 | 20.27 | 23.88 | 0.048 | 0.034 |
| C on A (locked) | orbit product | 0 | 3.88 | 5.92 | 12.13 | 12.60 | 0.028 | 0.016 |
| | | 1 | 3.90 | 5.75 | 11.96 | 12.42 | 0.029 | 0.019 |
| | | 2 | 3.84 | 5.79 | 12.82 | 11.83 | 0.032 | 0.015 |
| B: tokens16 | greedy | 0 | 5.59 | 30.68 | 27.44 | 36.12 | 0.059 | 0.068 |
| | | 1 | 5.55 | 30.28 | 26.85 | 34.94 | 0.058 | 0.071 |
| | | 2 | 5.54 | 30.57 | 26.74 | 36.85 | 0.054 | 0.078 |

Table 33. **EXP-109 per seed**: mean error per domain and severe-flip fraction on the real domains.

### L.2. EXP-112: locked-recipe evaluation and timing

**Question.** What do the four locked-recipe systems (PoE or DINOv3-only, each with one view or orbit voting) cost and achieve, all from the EXP-109 arm-A trees? This is the source of the clean-tree row of Tab. 5, of Tab. 13 and of the latency columns.

**Pre-registered criteria and outcome.** Descriptive by design (the recipe was already locked): the cost and gain of adding MoGe-2 and of orbit voting are reported as measured; there is no pass/fail rule. Timing is end-to-end on real lightbox crops (image already decoded), median of 5 after warm-up, on a shared GPU (upper bounds).

| System | Inference | Domain | mean | median | p95 | acc@5 | acc@10 | acc@20 | >90 | >150 |
|---|---|---|---|---|---|---|---|---|---|---|
| Clean PoE tree | greedy | Synth. val | $5.54 \pm 0.07$ | 4.62 | 10.20 | 0.565 | 0.946 | 0.990 | 0.002 | 0.001 |
| | | Synth. val-aug | $30.84 \pm 0.40$ | 10.95 | 155.50 | 0.146 | 0.455 | 0.702 | 0.107 | 0.055 |
| | | Lightbox | $26.91 \pm 0.78$ | 8.99 | 159.89 | 0.196 | 0.555 | 0.776 | 0.105 | 0.059 |
| | | Sunlamp | $36.66 \pm 0.93$ | 12.98 | 163.11 | 0.110 | 0.373 | 0.652 | 0.153 | 0.074 |
| DINOv3 | greedy | Synth. val | $4.84 \pm 0.05$ | 4.44 | 8.92 | 0.605 | 0.972 | 0.999 | 0.000 | 0.000 |
| | | Synth. val-aug | $9.88 \pm 0.22$ | 6.13 | 18.79 | 0.360 | 0.816 | 0.954 | 0.018 | 0.010 |
| | | Lightbox | $22.56 \pm 0.75$ | 8.37 | 162.72 | 0.211 | 0.607 | 0.842 | 0.083 | 0.060 |
| | | Sunlamp | $22.02 \pm 0.71$ | 10.17 | 110.11 | 0.148 | 0.489 | 0.801 | 0.067 | 0.032 |
| DINOv3 | orbit product | Synth. val | $4.21 \pm 0.04$ | 3.72 | 8.48 | 0.723 | 0.974 | 0.999 | 0.000 | 0.000 |
| | | Synth. val-aug | $6.25 \pm 0.11$ | 4.20 | 12.18 | 0.624 | 0.920 | 0.979 | 0.008 | 0.004 |
| | | Lightbox | $17.40 \pm 0.66$ | 6.87 | 129.30 | 0.305 | 0.708 | 0.897 | 0.058 | 0.048 |
| | | Sunlamp | $15.27 \pm 0.65$ | 8.19 | 57.04 | 0.223 | 0.627 | 0.890 | 0.036 | 0.020 |
| PoE | greedy | Synth. val | $4.86 \pm 0.03$ | 4.36 | 8.97 | 0.618 | 0.968 | 0.998 | 0.001 | 0.000 |
| | | Synth. val-aug | $9.75 \pm 0.07$ | 5.92 | 18.37 | 0.382 | 0.825 | 0.954 | 0.018 | 0.009 |
| | | Lightbox | $17.35 \pm 0.32$ | 6.75 | 111.67 | 0.307 | 0.730 | 0.892 | 0.059 | 0.042 |
| | | Sunlamp | $20.16 \pm 0.58$ | 8.43 | 103.59 | 0.216 | 0.601 | 0.843 | 0.064 | 0.033 |
| PoE | orbit product | Synth. val | $3.87 \pm 0.03$ | 3.45 | 7.72 | 0.781 | 0.982 | 0.999 | 0.000 | 0.000 |
| | | Synth. val-aug | $5.82 \pm 0.09$ | 3.87 | 10.96 | 0.688 | 0.939 | 0.982 | 0.008 | 0.004 |
| | | Lightbox | $12.31 \pm 0.46$ | 5.26 | 27.45 | 0.465 | 0.837 | 0.939 | 0.037 | 0.030 |
| | | Sunlamp | $12.28 \pm 0.40$ | 6.32 | 31.05 | 0.358 | 0.760 | 0.931 | 0.030 | 0.017 |

Table 34. **EXP-112 accuracy** of the locked-recipe systems (augmented trees except the clean reference).

| | | vs. clean PoE tree | | | | vs. own greedy | | | |
|---|---|---|---|---|---|---|---|---|---|
| System | Domain | $n_{>150}$ | fixed | $n_{<20}$ | broken | $n_{>150}$ | fixed | $n_{<20}$ | broken |
| DINOv3 greedy | Synth. val | 12.0 | 0.878 | 11875.7 | 0.000 | – | – | – | – |
| | Synth. val-aug | 662.7 | 0.819 | 8415.0 | 0.001 | – | – | – | – |
| | Lightbox | 401.0 | 0.416 | 5227.0 | 0.022 | – | – | – | – |
| | Sunlamp | 206.0 | 0.565 | 1820.0 | 0.012 | – | – | – | – |
| DINOv3 orbit product | Synth. val | 12.0 | 0.878 | 11875.7 | 0.000 | 2.7 | 0.667 | 11977.7 | 0.000 |
| | Synth. val-aug | 662.7 | 0.906 | 8415.0 | 0.000 | 116.7 | 0.585 | 11444.7 | 0.000 |
| | Lightbox | 401.0 | 0.521 | 5227.0 | 0.013 | 403.3 | 0.343 | 5676.3 | 0.007 |
| | Sunlamp | 206.0 | 0.744 | 1820.0 | 0.005 | 90.7 | 0.531 | 2236.7 | 0.005 |
| PoE greedy | Synth. val | 12.0 | 0.878 | 11875.7 | 0.000 | – | – | – | – |
| | Synth. val-aug | 662.7 | 0.823 | 8415.0 | 0.001 | – | – | – | – |
| | Lightbox | 401.0 | 0.510 | 5227.0 | 0.010 | – | – | – | – |
| | Sunlamp | 206.0 | 0.590 | 1820.0 | 0.010 | – | – | – | – |
| PoE orbit product | Synth. val | 12.0 | 0.898 | 11875.7 | 0.000 | 5.3 | 0.849 | 11964.3 | 0.000 |
| | Synth. val-aug | 662.7 | 0.917 | 8415.0 | 0.000 | 110.7 | 0.626 | 11442.7 | 0.000 |
| | Lightbox | 401.0 | 0.623 | 5227.0 | 0.004 | 280.7 | 0.436 | 6013.3 | 0.005 |
| | Sunlamp | 206.0 | 0.800 | 1820.0 | 0.004 | 91.7 | 0.656 | 2353.0 | 0.005 |

Table 35. **EXP-112 flip transitions** relative to the clean EXP-036 PoE greedy tree (left) and, for orbit systems, relative to the same variant's greedy prediction (right).

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| Clean PoE tree | greedy | 0 | 5.57 | 31.26 | 27.62 | 37.40 | 0.059 | 0.073 |
| | | 1 | 5.47 | 30.80 | 26.07 | 36.98 | 0.060 | 0.078 |
| | | 2 | 5.59 | 30.46 | 27.03 | 35.62 | 0.059 | 0.070 |
| PoE | greedy | 0 | 4.89 | 9.82 | 17.47 | 20.63 | 0.040 | 0.031 |
| | | 1 | 4.84 | 9.70 | 16.99 | 20.35 | 0.039 | 0.034 |
| | | 2 | 4.84 | 9.72 | 17.60 | 19.51 | 0.046 | 0.034 |
| PoE | orbit product | 0 | 3.88 | 5.92 | 12.13 | 12.60 | 0.028 | 0.016 |
| | | 1 | 3.90 | 5.75 | 11.96 | 12.42 | 0.029 | 0.019 |
| | | 2 | 3.84 | 5.79 | 12.82 | 11.83 | 0.032 | 0.015 |
| DINOv3 | greedy | 0 | 4.89 | 10.06 | 22.37 | 22.80 | 0.055 | 0.031 |
| | | 1 | 4.80 | 9.62 | 21.92 | 21.88 | 0.058 | 0.035 |
| | | 2 | 4.83 | 9.95 | 23.39 | 21.39 | 0.066 | 0.032 |
| DINOv3 | orbit product | 0 | 4.20 | 6.38 | 17.24 | 15.77 | 0.045 | 0.020 |
| | | 1 | 4.26 | 6.17 | 16.84 | 15.50 | 0.045 | 0.023 |
| | | 2 | 4.17 | 6.21 | 18.13 | 14.54 | 0.052 | 0.018 |

Table 36. **EXP-112 per seed**.

| System | Batch | warp | DINOv3 | MoGe-2 | MoGe post | tree(s) | orbit scoring | total | rate | peak MB |
|---|---|---|---|---|---|---|---|---|---|---|
| DINOv3, 1 view | 1 | 0.6 | 6.4 | 0.0 | 0.0 | 7.4 | 0.0 | 14.7 | 68.1 | 2463 |
| DINOv3 + orbit | 1 | 2.8 | 26.3 | 0.0 | 0.0 | 31.9 | 8.2 | 69.6 | 14.4 | 2464 |
| PoE, 1 view | 1 | 1.8 | 6.9 | 21.0 | 20.3 | 14.1 | 0.0 | 64.0 | 15.6 | 2860 |
| PoE + orbit | 1 | 7.7 | 28.1 | 84.2 | 80.8 | 61.2 | 15.3 | 278.5 | 3.6 | 2861 |
| DINOv3, 1 view | 64 | 0.3 | 2.4 | 0.0 | 0.0 | 0.1 | 0.0 | 2.9 | 346.8 | 3145 |
| DINOv3 + orbit | 64 | 1.4 | 9.6 | 0.0 | 0.0 | 0.6 | 0.4 | 12.1 | 82.4 | 3197 |
| PoE, 1 view | 64 | 1.9 | 2.4 | 26.3 | 19.1 | 0.3 | 0.0 | 50.0 | 20.0 | 28837 |
| PoE + orbit | 64 | 7.8 | 9.9 | 108.4 | 77.1 | 1.6 | 1.2 | 206.8 | 4.8 | 28906 |

Table 37. **EXP-112 end-to-end timing breakdown** (ms per image; shared RTX 6000 Ada, upper bounds). MoGe post: CPU consensus mask and post-processing. Rate: FPS at batch 1, images/s at batch 64. Peak GPU memory in MB.

### L.3. EXP-117: orbit distillation

**Question.** Can a single-view tree, distilled from the orbit system of the same seed and variant (teacher: orbit over the three training views, 24 candidates weighted by the softmax of the orbit-product score; loss $L_{\text{GT}} + \beta H$ with $\beta = 1$), recover part of the orbit gain at one-view cost? Three students per seed: a DINOv3 student from the DINOv3-only teacher, and DINOv3 and MoGe-2 students from the PoE teacher.

**Pre-registered criteria and outcome.**

- Distillation helps (student single-view greedy val-aug below the base greedy by more than the larger seed std, clean synth. val not worse; DINOv3 primary, PoE secondary): **PASS**. DINOv3 $9.33 \pm 0.18$ vs $9.88 \pm 0.22$; PoE $9.22 \pm 0.10$ vs $9.75 \pm 0.06$; clean synth. val DINOv3 4.57 vs 4.84, PoE 4.56 vs 4.85 (better).
- Retention = (base greedy − student greedy)/(base greedy − base orbit), descriptive, synth. val / val-aug / lightbox / sunlamp: DINOv3 43% / 15% / 18% / 22%; PoE 30% / 13% / 28% / 23%.
- Real domains: descriptive. Student + orbit is the best system in the paper (Tab. 5); synthetic validation ranks it first as well. By decision DEC-023 the locked recipe remains EXP-109; distillation is reported as an enhancement.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken | cons.@10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Base (EXP-109 A), DINOv3 | greedy (1 view) | Synth. val | $4.84 \pm 0.05$ | 4.43 | 0.972 | 0.000 | 0.000 | – | – | – | – | 0.701 |
| | | Synth. val-aug | $9.88 \pm 0.22$ | 6.13 | 0.816 | 0.018 | 0.010 | – | – | – | – | 0.554 |
| | | Lightbox | $22.55 \pm 0.73$ | 8.38 | 0.607 | 0.083 | 0.060 | – | – | – | – | 0.391 |
| | | Sunlamp | $22.00 \pm 0.70$ | 10.17 | 0.489 | 0.067 | 0.032 | – | – | – | – | 0.295 |
| Base (EXP-109 A), DINOv3 | orbit product | Synth. val | $4.21 \pm 0.04$ | 3.72 | 0.974 | 0.000 | 0.000 | 2.7 | 0.667 | 11977.7 | 0.000 | 0.701 |
| | | Synth. val-aug | $6.25 \pm 0.12$ | 4.20 | 0.920 | 0.007 | 0.004 | 117.0 | 0.586 | 11445.0 | 0.000 | 0.554 |
| | | Lightbox | $17.40 \pm 0.68$ | 6.88 | 0.708 | 0.058 | 0.048 | 402.7 | 0.341 | 5676.7 | 0.007 | 0.391 |
| | | Sunlamp | $15.27 \pm 0.64$ | 8.19 | 0.626 | 0.036 | 0.020 | 90.3 | 0.529 | 2236.3 | 0.005 | 0.295 |
| Base (EXP-109 A), DINOv3 | weighted vote (32) | Synth. val | $3.71 \pm 0.05$ | 3.20 | 0.983 | 0.001 | 0.000 | 2.7 | 0.583 | 11977.7 | 0.000 | 0.701 |
| | | Synth. val-aug | $6.30 \pm 0.06$ | 3.68 | 0.926 | 0.010 | 0.006 | 117.0 | 0.479 | 11445.0 | 0.001 | 0.554 |
| | | Lightbox | $18.27 \pm 0.67$ | 6.59 | 0.718 | 0.065 | 0.051 | 402.7 | 0.299 | 5676.7 | 0.009 | 0.391 |
| | | Sunlamp | $15.83 \pm 0.83$ | 7.89 | 0.636 | 0.042 | 0.023 | 90.3 | 0.467 | 2236.3 | 0.007 | 0.295 |
| Student (DINO teacher), DINOv3 | greedy (1 view) | Synth. val | $4.57 \pm 0.05$ | 4.22 | 0.983 | 0.001 | 0.000 | – | – | – | – | 0.770 |
| | | Synth. val-aug | $9.33 \pm 0.18$ | 5.82 | 0.843 | 0.017 | 0.009 | – | – | – | – | 0.619 |
| | | Lightbox | $21.61 \pm 0.82$ | 7.91 | 0.640 | 0.079 | 0.058 | – | – | – | – | 0.456 |
| | | Sunlamp | $20.55 \pm 0.94$ | 9.57 | 0.530 | 0.061 | 0.031 | – | – | – | – | 0.357 |
| Student (DINO teacher), DINOv3 | orbit product | Synth. val | $3.90 \pm 0.03$ | 3.50 | 0.984 | 0.000 | 0.000 | 3.7 | 0.417 | 11980.7 | 0.000 | 0.770 |
| | | Synth. val-aug | $5.74 \pm 0.12$ | 3.87 | 0.939 | 0.007 | 0.004 | 102.7 | 0.598 | 11523.7 | 0.000 | 0.619 |
| | | Lightbox | $16.43 \pm 0.75$ | 6.49 | 0.743 | 0.056 | 0.045 | 392.0 | 0.356 | 5763.7 | 0.007 | 0.456 |
| | | Sunlamp | $13.64 \pm 0.57$ | 7.57 | 0.671 | 0.030 | 0.016 | 86.3 | 0.535 | 2296.7 | 0.004 | 0.357 |
| Student (DINO teacher), DINOv3 | weighted vote (32) | Synth. val | $3.37 \pm 0.05$ | 2.93 | 0.989 | 0.000 | 0.000 | 3.7 | 0.417 | 11980.7 | 0.000 | 0.770 |
| | | Synth. val-aug | $5.82 \pm 0.16$ | 3.34 | 0.937 | 0.010 | 0.005 | 102.7 | 0.502 | 11523.7 | 0.001 | 0.619 |
| | | Lightbox | $16.90 \pm 0.79$ | 6.12 | 0.749 | 0.059 | 0.048 | 392.0 | 0.321 | 5763.7 | 0.007 | 0.456 |
| | | Sunlamp | $14.33 \pm 0.41$ | 7.25 | 0.674 | 0.036 | 0.019 | 86.3 | 0.533 | 2296.7 | 0.006 | 0.357 |

Table 38. **EXP-117, DINOv3-only system**: base trees and distilled students. Flip transitions (fixed/broken) are relative to the same family's greedy prediction; cons.@10: fraction of the 90/180/270° views whose back-mapped top-1 is within 10° of view 0's.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken | cons.@10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Base (EXP-109 A), PoE | greedy (1 view) | Synth. val | $4.85 \pm 0.03$ | 4.36 | 0.968 | 0.001 | 0.000 | – | – | – | – | 0.753 |
| | | Synth. val-aug | $9.75 \pm 0.06$ | 5.92 | 0.825 | 0.018 | 0.009 | – | – | – | – | 0.607 |
| | | Lightbox | $17.33 \pm 0.31$ | 6.75 | 0.730 | 0.058 | 0.042 | – | – | – | – | 0.523 |
| | | Sunlamp | $20.15 \pm 0.57$ | 8.42 | 0.601 | 0.064 | 0.033 | – | – | – | – | 0.426 |
| Base (EXP-109 A), PoE | orbit product | Synth. val | $3.87 \pm 0.03$ | 3.44 | 0.982 | 0.000 | 0.000 | 5.3 | 0.849 | 11965.0 | 0.000 | 0.753 |
| | | Synth. val-aug | $5.82 \pm 0.10$ | 3.87 | 0.939 | 0.007 | 0.004 | 110.3 | 0.631 | 11443.3 | 0.000 | 0.607 |
| | | Lightbox | $12.30 \pm 0.50$ | 5.26 | 0.836 | 0.037 | 0.030 | 280.3 | 0.437 | 6013.0 | 0.005 | 0.523 |
| | | Sunlamp | $12.25 \pm 0.37$ | 6.32 | 0.760 | 0.029 | 0.017 | 91.7 | 0.656 | 2353.3 | 0.005 | 0.426 |
| Base (EXP-109 A), PoE | weighted vote (32) | Synth. val | $3.46 \pm 0.03$ | 2.97 | 0.985 | 0.000 | 0.000 | 5.3 | 0.802 | 11965.0 | 0.000 | 0.753 |
| | | Synth. val-aug | $5.98 \pm 0.13$ | 3.43 | 0.935 | 0.011 | 0.005 | 110.3 | 0.510 | 11443.3 | 0.001 | 0.607 |
| | | Lightbox | $12.58 \pm 0.42$ | 4.98 | 0.834 | 0.039 | 0.031 | 280.3 | 0.402 | 6013.0 | 0.006 | 0.523 |
| | | Sunlamp | $12.88 \pm 0.49$ | 6.08 | 0.754 | 0.035 | 0.018 | 91.7 | 0.601 | 2353.3 | 0.005 | 0.426 |
| Student (PoE teacher), PoE | greedy (1 view) | Synth. val | $4.56 \pm 0.03$ | 4.15 | 0.981 | 0.001 | 0.000 | – | – | – | – | 0.790 |
| | | Synth. val-aug | $9.22 \pm 0.10$ | 5.66 | 0.855 | 0.017 | 0.009 | – | – | – | – | 0.650 |
| | | Lightbox | $15.90 \pm 0.53$ | 6.37 | 0.764 | 0.052 | 0.037 | – | – | – | – | 0.570 |
| | | Sunlamp | $18.33 \pm 0.66$ | 7.90 | 0.644 | 0.058 | 0.030 | – | – | – | – | 0.479 |
| Student (PoE teacher), PoE | orbit product | Synth. val | $3.69 \pm 0.02$ | 3.31 | 0.988 | 0.000 | 0.000 | 4.7 | 0.533 | 11975.3 | 0.000 | 0.790 |
| | | Synth. val-aug | $5.40 \pm 0.14$ | 3.67 | 0.950 | 0.007 | 0.003 | 105.7 | 0.596 | 11522.0 | 0.000 | 0.650 |
| | | Lightbox | $11.21 \pm 0.42$ | 5.02 | 0.858 | 0.032 | 0.026 | 247.7 | 0.453 | 6110.3 | 0.005 | 0.570 |
| | | Sunlamp | $11.09 \pm 0.49$ | 5.91 | 0.799 | 0.024 | 0.014 | 84.3 | 0.616 | 2421.0 | 0.004 | 0.479 |
| Student (PoE teacher), PoE | weighted vote (32) | Synth. val | $3.27 \pm 0.03$ | 2.83 | 0.988 | 0.000 | 0.000 | 4.7 | 0.533 | 11975.3 | 0.000 | 0.790 |
| | | Synth. val-aug | $5.59 \pm 0.07$ | 3.24 | 0.944 | 0.010 | 0.005 | 105.7 | 0.525 | 11522.0 | 0.001 | 0.650 |
| | | Lightbox | $11.35 \pm 0.36$ | 4.68 | 0.856 | 0.034 | 0.027 | 247.7 | 0.433 | 6110.3 | 0.005 | 0.570 |
| | | Sunlamp | $11.91 \pm 0.26$ | 5.74 | 0.786 | 0.032 | 0.017 | 84.3 | 0.593 | 2421.0 | 0.006 | 0.479 |

Table 39. **EXP-117, PoE system**: base trees and distilled students (columns as in Tab. 38).

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| Base (EXP-109 A), DINOv3 | greedy (1 view) | 0 | 4.89 | 10.05 | 22.37 | 22.75 | 0.055 | 0.031 |
| | | 1 | 4.80 | 9.64 | 21.92 | 21.89 | 0.058 | 0.035 |
| | | 2 | 4.83 | 9.95 | 23.36 | 21.37 | 0.066 | 0.032 |
| Base (EXP-109 A), DINOv3 | orbit product | 0 | 4.21 | 6.38 | 17.24 | 15.76 | 0.045 | 0.020 |
| | | 1 | 4.25 | 6.17 | 16.82 | 15.51 | 0.045 | 0.023 |
| | | 2 | 4.17 | 6.19 | 18.15 | 14.55 | 0.052 | 0.018 |
| Base (EXP-109 A), DINOv3 | weighted vote (32) | 0 | 3.75 | 6.34 | 18.37 | 16.47 | 0.049 | 0.023 |
| | | 1 | 3.72 | 6.23 | 17.56 | 16.13 | 0.050 | 0.027 |
| | | 2 | 3.66 | 6.35 | 18.89 | 14.88 | 0.055 | 0.020 |
| Student (DINO teacher), DINOv3 | greedy (1 view) | 0 | 4.61 | 9.53 | 21.27 | 20.99 | 0.055 | 0.031 |
| | | 1 | 4.51 | 9.19 | 21.01 | 19.47 | 0.056 | 0.028 |
| | | 2 | 4.59 | 9.26 | 22.55 | 21.19 | 0.064 | 0.034 |
| Student (DINO teacher), DINOv3 | orbit product | 0 | 3.94 | 5.87 | 16.28 | 14.23 | 0.043 | 0.019 |
| | | 1 | 3.89 | 5.62 | 15.77 | 13.09 | 0.042 | 0.014 |
| | | 2 | 3.88 | 5.73 | 17.25 | 13.59 | 0.048 | 0.015 |
| Student (DINO teacher), DINOv3 | weighted vote (32) | 0 | 3.43 | 6.00 | 16.54 | 14.56 | 0.046 | 0.019 |
| | | 1 | 3.35 | 5.70 | 16.36 | 13.86 | 0.045 | 0.018 |
| | | 2 | 3.33 | 5.76 | 17.80 | 14.57 | 0.052 | 0.021 |

Table 40. **EXP-117 per seed, DINOv3-only system**.

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| Base (EXP-109 A), PoE | greedy (1 view) | 0 | 4.89 | 9.82 | 17.44 | 20.59 | 0.040 | 0.030 |
| | | 1 | 4.84 | 9.70 | 16.98 | 20.35 | 0.039 | 0.034 |
| | | 2 | 4.84 | 9.72 | 17.58 | 19.52 | 0.046 | 0.034 |
| Base (EXP-109 A), PoE | orbit product | 0 | 3.87 | 5.93 | 12.18 | 12.50 | 0.028 | 0.016 |
| | | 1 | 3.90 | 5.75 | 11.87 | 12.43 | 0.028 | 0.019 |
| | | 2 | 3.84 | 5.78 | 12.85 | 11.83 | 0.033 | 0.015 |
| Base (EXP-109 A), PoE | weighted vote (32) | 0 | 3.48 | 6.09 | 12.42 | 13.45 | 0.029 | 0.019 |
| | | 1 | 3.48 | 5.84 | 12.26 | 12.64 | 0.030 | 0.019 |
| | | 2 | 3.43 | 6.02 | 13.06 | 12.56 | 0.034 | 0.017 |
| Student (PoE teacher), PoE | greedy (1 view) | 0 | 4.56 | 9.30 | 15.83 | 18.88 | 0.036 | 0.032 |
| | | 1 | 4.53 | 9.10 | 15.41 | 17.60 | 0.035 | 0.027 |
| | | 2 | 4.59 | 9.26 | 16.47 | 18.51 | 0.040 | 0.031 |
| Student (PoE teacher), PoE | orbit product | 0 | 3.69 | 5.56 | 11.25 | 11.57 | 0.026 | 0.017 |
| | | 1 | 3.68 | 5.30 | 10.78 | 10.59 | 0.024 | 0.013 |
| | | 2 | 3.71 | 5.34 | 11.61 | 11.11 | 0.028 | 0.013 |
| Student (PoE teacher), PoE | weighted vote (32) | 0 | 3.30 | 5.67 | 11.19 | 12.01 | 0.026 | 0.017 |
| | | 1 | 3.26 | 5.54 | 11.10 | 11.61 | 0.026 | 0.016 |
| | | 2 | 3.25 | 5.56 | 11.76 | 12.11 | 0.029 | 0.017 |

Table 41. **EXP-117 per seed, PoE system**.

### L.4. EXP-118: 6-DoF with a translation block on SPEED+

**Question.** Can the locked architecture be extended to full 6-DoF with a translation block on the tree's tokens, using the SPACE-HOP translation target [5] unchanged, and does the block add information beyond the ground-truth box without hurting rotation? Arms: J (tree and block trained jointly), F (block on the frozen EXP-109 tree), and B (box only, no image: object centre at the box centre, depth at the training mean of $U_z$). The PoE translation is the mean of the two branches' predictions; orbit translation averages the four back-rolled views.

**Pre-registered criteria and outcome.**

- The block adds information beyond the box (J and F synth. val $E_T$ below box-only by more than the larger seed std, PoE and DINOv3): **PASS**. Box only 0.561 m; J DINOv3 0.078 (1 view) / 0.059 m (orbit); J PoE 0.111 / 0.093 m.
- Joint training does not hurt rotation (J synth. val rotation within the larger seed std of EXP-109 arm A, or better): **PASS**. PoE greedy 4.78 vs 4.85, PoE orbit 3.86 vs 3.87, DINOv3 greedy 4.78 vs 4.84 (F's rotation is the EXP-109 arm A tree).
- Real domains: descriptive.

The ESA score is $e_q$ (rad) $+ E_T/\|t\|$ averaged over images, with the challenge precision floors. All rows use ground-truth crops; the box-only row shows how much translation the crop alone carries.

**External context (unmatched).** SPACE-HOP reports $E_T$ 0.096 / 0.302 / 0.282 m (synthetic / lightbox / sunlamp) with its own box source; our numbers use ground-truth crops.

| | | | | Synth. val | | | Lightbox | | | Sunlamp | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Arm | Feat. | Rotation | Translation | $E_T$ | med. | $e_t$ | $E_T$ | med. | $e_t$ | $E_T$ | med. | $e_t$ |
| B box only | – | – | box | 0.561 | 0.440 | 0.095 | 0.547 | 0.430 | 0.094 | 0.553 | 0.431 | 0.096 |
| F frozen tree | DINOv3 | greedy (1 view) | greedy (1 view) | $0.090 \pm 0.001$ | 0.069 | 0.016 | $0.233 \pm 0.003$ | 0.172 | 0.040 | $0.240 \pm 0.005$ | 0.183 | 0.040 |
| F frozen tree | DINOv3 | greedy (1 view) | orbit mean | $0.068 \pm 0.000$ | 0.052 | 0.012 | $0.205 \pm 0.003$ | 0.151 | 0.035 | $0.205 \pm 0.002$ | 0.153 | 0.035 |
| F frozen tree | DINOv3 | orbit product | greedy (1 view) | $0.090 \pm 0.001$ | 0.069 | 0.016 | $0.233 \pm 0.003$ | 0.172 | 0.040 | $0.240 \pm 0.005$ | 0.183 | 0.040 |
| F frozen tree | DINOv3 | orbit product | orbit mean | $0.068 \pm 0.000$ | 0.052 | 0.012 | $0.205 \pm 0.003$ | 0.151 | 0.035 | $0.205 \pm 0.002$ | 0.153 | 0.035 |
| F frozen tree | DINOv3 | weighted vote (32) | greedy (1 view) | $0.090 \pm 0.001$ | 0.069 | 0.016 | $0.233 \pm 0.003$ | 0.172 | 0.040 | $0.240 \pm 0.005$ | 0.183 | 0.040 |
| F frozen tree | DINOv3 | weighted vote (32) | orbit mean | $0.068 \pm 0.000$ | 0.052 | 0.012 | $0.205 \pm 0.003$ | 0.151 | 0.035 | $0.205 \pm 0.002$ | 0.153 | 0.035 |
| F frozen tree | PoE | greedy (1 view) | greedy (1 view) | $0.108 \pm 0.001$ | 0.084 | 0.020 | $0.206 \pm 0.002$ | 0.152 | 0.035 | $0.232 \pm 0.005$ | 0.178 | 0.040 |
| F frozen tree | PoE | greedy (1 view) | orbit mean | $0.089 \pm 0.001$ | 0.069 | 0.017 | $0.184 \pm 0.002$ | 0.135 | 0.032 | $0.206 \pm 0.005$ | 0.159 | 0.036 |
| F frozen tree | PoE | orbit product | greedy (1 view) | $0.108 \pm 0.001$ | 0.084 | 0.020 | $0.206 \pm 0.002$ | 0.152 | 0.035 | $0.232 \pm 0.005$ | 0.178 | 0.040 |
| F frozen tree | PoE | orbit product | orbit mean | $0.089 \pm 0.001$ | 0.069 | 0.017 | $0.184 \pm 0.002$ | 0.135 | 0.032 | $0.206 \pm 0.005$ | 0.159 | 0.036 |
| F frozen tree | PoE | weighted vote (32) | greedy (1 view) | $0.108 \pm 0.001$ | 0.084 | 0.020 | $0.206 \pm 0.002$ | 0.152 | 0.035 | $0.232 \pm 0.005$ | 0.178 | 0.040 |
| F frozen tree | PoE | weighted vote (32) | orbit mean | $0.089 \pm 0.001$ | 0.069 | 0.017 | $0.184 \pm 0.002$ | 0.135 | 0.032 | $0.206 \pm 0.005$ | 0.159 | 0.036 |
| J joint | DINOv3 | greedy (1 view) | greedy (1 view) | $0.078 \pm 0.002$ | 0.060 | 0.014 | $0.209 \pm 0.004$ | 0.152 | 0.035 | $0.212 \pm 0.003$ | 0.160 | 0.036 |
| J joint | DINOv3 | greedy (1 view) | orbit mean | $0.059 \pm 0.001$ | 0.046 | 0.011 | $0.185 \pm 0.003$ | 0.134 | 0.031 | $0.183 \pm 0.004$ | 0.137 | 0.031 |
| J joint | DINOv3 | orbit product | greedy (1 view) | $0.078 \pm 0.002$ | 0.060 | 0.014 | $0.209 \pm 0.004$ | 0.152 | 0.035 | $0.212 \pm 0.003$ | 0.160 | 0.036 |
| J joint | DINOv3 | orbit product | orbit mean | $0.059 \pm 0.001$ | 0.046 | 0.011 | $0.185 \pm 0.003$ | 0.134 | 0.031 | $0.183 \pm 0.004$ | 0.137 | 0.031 |
| J joint | DINOv3 | weighted vote (32) | greedy (1 view) | $0.078 \pm 0.002$ | 0.060 | 0.014 | $0.209 \pm 0.004$ | 0.152 | 0.035 | $0.212 \pm 0.003$ | 0.160 | 0.036 |
| J joint | DINOv3 | weighted vote (32) | orbit mean | $0.059 \pm 0.001$ | 0.046 | 0.011 | $0.185 \pm 0.003$ | 0.134 | 0.031 | $0.183 \pm 0.004$ | 0.137 | 0.031 |
| J joint | PoE | greedy (1 view) | greedy (1 view) | $0.111 \pm 0.004$ | 0.088 | 0.020 | $0.196 \pm 0.003$ | 0.148 | 0.033 | $0.221 \pm 0.001$ | 0.173 | 0.038 |
| J joint | PoE | greedy (1 view) | orbit mean | $0.093 \pm 0.004$ | 0.074 | 0.017 | $0.175 \pm 0.003$ | 0.134 | 0.030 | $0.194 \pm 0.002$ | 0.151 | 0.034 |
| J joint | PoE | orbit product | greedy (1 view) | $0.111 \pm 0.004$ | 0.088 | 0.020 | $0.196 \pm 0.003$ | 0.148 | 0.033 | $0.221 \pm 0.001$ | 0.173 | 0.038 |
| J joint | PoE | orbit product | orbit mean | $0.093 \pm 0.004$ | 0.074 | 0.017 | $0.175 \pm 0.003$ | 0.134 | 0.030 | $0.194 \pm 0.002$ | 0.151 | 0.034 |
| J joint | PoE | weighted vote (32) | greedy (1 view) | $0.111 \pm 0.004$ | 0.088 | 0.020 | $0.196 \pm 0.003$ | 0.148 | 0.033 | $0.221 \pm 0.001$ | 0.173 | 0.038 |
| J joint | PoE | weighted vote (32) | orbit mean | $0.093 \pm 0.004$ | 0.074 | 0.017 | $0.175 \pm 0.003$ | 0.134 | 0.030 | $0.194 \pm 0.002$ | 0.151 | 0.034 |

Table 42. **EXP-118 translation**, all arms and inference modes. $E_T$ in metres (mean $\pm$ seed std, median) and normalised $e_t = E_T/\|t\|$. Rotation inference does not affect the translation prediction.

| | | | | Synth. val | | | Lightbox | | | Sunlamp | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Arm | Feat. | Rotation | Translation | $E_R$ | med. | score | $E_R$ | med. | score | $E_R$ | med. | score |
| B box only | – | – | box | – | – | – | – | – | – | – | – | – |
| F frozen tree | DINOv3 | greedy (1 view) | greedy (1 view) | $4.84 \pm 0.05$ | 4.43 | 0.100 | $22.55 \pm 0.73$ | 8.38 | 0.433 | $22.00 \pm 0.70$ | 10.17 | 0.424 |
| F frozen tree | DINOv3 | greedy (1 view) | orbit mean | $4.84 \pm 0.05$ | 4.43 | 0.097 | $22.55 \pm 0.73$ | 8.38 | 0.428 | $22.00 \pm 0.70$ | 10.17 | 0.419 |
| F frozen tree | DINOv3 | orbit product | greedy (1 view) | $4.21 \pm 0.04$ | 3.72 | 0.089 | $17.40 \pm 0.68$ | 6.88 | 0.343 | $15.27 \pm 0.64$ | 8.19 | 0.307 |
| F frozen tree | DINOv3 | orbit product | orbit mean | $4.21 \pm 0.04$ | 3.72 | 0.086 | $17.40 \pm 0.68$ | 6.88 | 0.339 | $15.27 \pm 0.64$ | 8.19 | 0.301 |
| F frozen tree | DINOv3 | weighted vote (32) | greedy (1 view) | $3.71 \pm 0.05$ | 3.20 | 0.080 | $18.27 \pm 0.67$ | 6.59 | 0.358 | $15.83 \pm 0.83$ | 7.89 | 0.317 |
| F frozen tree | DINOv3 | weighted vote (32) | orbit mean | $3.71 \pm 0.05$ | 3.20 | 0.077 | $18.27 \pm 0.67$ | 6.59 | 0.354 | $15.83 \pm 0.83$ | 7.89 | 0.311 |
| F frozen tree | PoE | greedy (1 view) | greedy (1 view) | $4.85 \pm 0.03$ | 4.36 | 0.104 | $17.33 \pm 0.31$ | 6.75 | 0.338 | $20.15 \pm 0.57$ | 8.42 | 0.392 |
| F frozen tree | PoE | greedy (1 view) | orbit mean | $4.85 \pm 0.03$ | 4.36 | 0.101 | $17.33 \pm 0.31$ | 6.75 | 0.334 | $20.15 \pm 0.57$ | 8.42 | 0.388 |
| F frozen tree | PoE | orbit product | greedy (1 view) | $3.87 \pm 0.03$ | 3.44 | 0.087 | $12.30 \pm 0.50$ | 5.26 | 0.250 | $12.25 \pm 0.37$ | 6.32 | 0.254 |
| F frozen tree | PoE | orbit product | orbit mean | $3.87 \pm 0.03$ | 3.44 | 0.084 | $12.30 \pm 0.50$ | 5.26 | 0.247 | $12.25 \pm 0.37$ | 6.32 | 0.250 |
| F frozen tree | PoE | weighted vote (32) | greedy (1 view) | $3.46 \pm 0.03$ | 2.97 | 0.080 | $12.58 \pm 0.42$ | 4.98 | 0.255 | $12.88 \pm 0.49$ | 6.08 | 0.265 |
| F frozen tree | PoE | weighted vote (32) | orbit mean | $3.46 \pm 0.03$ | 2.97 | 0.077 | $12.58 \pm 0.42$ | 4.98 | 0.252 | $12.88 \pm 0.49$ | 6.08 | 0.261 |
| J joint | DINOv3 | greedy (1 view) | greedy (1 view) | $4.78 \pm 0.04$ | 4.38 | 0.097 | $21.98 \pm 0.30$ | 8.12 | 0.419 | $20.85 \pm 0.99$ | 9.83 | 0.400 |
| J joint | DINOv3 | greedy (1 view) | orbit mean | $4.78 \pm 0.04$ | 4.38 | 0.094 | $21.98 \pm 0.30$ | 8.12 | 0.415 | $20.85 \pm 0.99$ | 9.83 | 0.395 |
| J joint | DINOv3 | orbit product | greedy (1 view) | $4.19 \pm 0.03$ | 3.71 | 0.087 | $17.31 \pm 0.44$ | 6.71 | 0.337 | $14.87 \pm 0.58$ | 7.92 | 0.295 |
| J joint | DINOv3 | orbit product | orbit mean | $4.19 \pm 0.03$ | 3.71 | 0.084 | $17.31 \pm 0.44$ | 6.71 | 0.333 | $14.87 \pm 0.58$ | 7.92 | 0.290 |
| J joint | DINOv3 | weighted vote (32) | greedy (1 view) | $3.67 \pm 0.03$ | 3.19 | 0.078 | $18.08 \pm 0.49$ | 6.36 | 0.351 | $15.70 \pm 0.78$ | 7.65 | 0.310 |
| J joint | DINOv3 | weighted vote (32) | orbit mean | $3.67 \pm 0.03$ | 3.19 | 0.075 | $18.08 \pm 0.49$ | 6.36 | 0.347 | $15.70 \pm 0.78$ | 7.65 | 0.305 |
| J joint | PoE | greedy (1 view) | greedy (1 view) | $4.78 \pm 0.00$ | 4.30 | 0.103 | $17.01 \pm 0.36$ | 6.63 | 0.330 | $18.74 \pm 0.48$ | 8.22 | 0.365 |
| J joint | PoE | greedy (1 view) | orbit mean | $4.78 \pm 0.00$ | 4.30 | 0.100 | $17.01 \pm 0.36$ | 6.63 | 0.327 | $18.74 \pm 0.48$ | 8.22 | 0.361 |
| J joint | PoE | orbit product | greedy (1 view) | $3.86 \pm 0.01$ | 3.43 | 0.087 | $12.11 \pm 0.27$ | 5.14 | 0.245 | $11.73 \pm 0.28$ | 6.15 | 0.243 |
| J joint | PoE | orbit product | orbit mean | $3.86 \pm 0.01$ | 3.43 | 0.084 | $12.11 \pm 0.27$ | 5.14 | 0.241 | $11.73 \pm 0.28$ | 6.15 | 0.238 |
| J joint | PoE | weighted vote (32) | greedy (1 view) | $3.42 \pm 0.01$ | 2.94 | 0.080 | $12.17 \pm 0.44$ | 4.77 | 0.246 | $12.55 \pm 0.38$ | 5.92 | 0.257 |
| J joint | PoE | weighted vote (32) | orbit mean | $3.42 \pm 0.01$ | 2.94 | 0.077 | $12.17 \pm 0.44$ | 4.77 | 0.242 | $12.55 \pm 0.38$ | 5.92 | 0.252 |

Table 43. **EXP-118 rotation and ESA score**, all arms and inference modes ($E_R$ in degrees). For the box-only arm only translation is defined.

| Arm | Rotation | Translation | Seed | $E_T$ (m) | | | $E_R$ (°) | | | ESA score | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Syn. | LB | SL | Syn. | LB | SL | Syn. | LB | SL |
| B box only, – | – | box | 0 | 0.561 | 0.547 | 0.553 | – | – | – | – | – | – |
| F frozen tree, DINOv3 | greedy (1 view) | greedy (1 view) | 0 | 0.089 | 0.233 | 0.236 | 4.89 | 22.37 | 22.75 | 0.101 | 0.430 | 0.437 |
| | | | 1 | 0.090 | 0.231 | 0.245 | 4.80 | 21.92 | 21.89 | 0.099 | 0.422 | 0.423 |
| | | | 2 | 0.090 | 0.236 | 0.238 | 4.83 | 23.36 | 21.37 | 0.100 | 0.448 | 0.413 |
| F frozen tree, DINOv3 | greedy (1 view) | orbit mean | 0 | 0.068 | 0.206 | 0.205 | 4.89 | 22.37 | 22.75 | 0.097 | 0.425 | 0.432 |
| | | | 1 | 0.068 | 0.202 | 0.207 | 4.80 | 21.92 | 21.89 | 0.096 | 0.417 | 0.417 |
| | | | 2 | 0.068 | 0.207 | 0.203 | 4.83 | 23.36 | 21.37 | 0.096 | 0.443 | 0.407 |
| F frozen tree, DINOv3 | orbit product | greedy (1 view) | 0 | 0.089 | 0.233 | 0.236 | 4.21 | 17.24 | 15.76 | 0.089 | 0.340 | 0.315 |
| | | | 1 | 0.090 | 0.231 | 0.245 | 4.25 | 16.82 | 15.51 | 0.090 | 0.333 | 0.312 |
| | | | 2 | 0.090 | 0.236 | 0.238 | 4.17 | 18.15 | 14.55 | 0.089 | 0.357 | 0.294 |
| F frozen tree, DINOv3 | orbit product | orbit mean | 0 | 0.068 | 0.206 | 0.205 | 4.21 | 17.24 | 15.76 | 0.085 | 0.336 | 0.310 |
| | | | 1 | 0.068 | 0.202 | 0.207 | 4.25 | 16.82 | 15.51 | 0.086 | 0.328 | 0.305 |
| | | | 2 | 0.068 | 0.207 | 0.203 | 4.17 | 18.15 | 14.55 | 0.085 | 0.352 | 0.288 |
| F frozen tree, DINOv3 | weighted vote (32) | greedy (1 view) | 0 | 0.089 | 0.233 | 0.236 | 3.75 | 18.37 | 16.47 | 0.081 | 0.360 | 0.327 |
| | | | 1 | 0.090 | 0.231 | 0.245 | 3.72 | 17.56 | 16.13 | 0.081 | 0.345 | 0.323 |
| | | | 2 | 0.090 | 0.236 | 0.238 | 3.66 | 18.89 | 14.88 | 0.080 | 0.370 | 0.300 |
| F frozen tree, DINOv3 | weighted vote (32) | orbit mean | 0 | 0.068 | 0.206 | 0.205 | 3.75 | 18.37 | 16.47 | 0.077 | 0.356 | 0.322 |
| | | | 1 | 0.068 | 0.202 | 0.207 | 3.72 | 17.56 | 16.13 | 0.077 | 0.341 | 0.316 |
| | | | 2 | 0.068 | 0.207 | 0.203 | 3.66 | 18.89 | 14.88 | 0.076 | 0.365 | 0.294 |
| F frozen tree, PoE | greedy (1 view) | greedy (1 view) | 0 | 0.108 | 0.208 | 0.233 | 4.89 | 17.44 | 20.59 | 0.105 | 0.340 | 0.400 |
| | | | 1 | 0.109 | 0.204 | 0.235 | 4.84 | 16.98 | 20.35 | 0.104 | 0.332 | 0.396 |
| | | | 2 | 0.107 | 0.204 | 0.226 | 4.84 | 17.58 | 19.52 | 0.104 | 0.342 | 0.380 |
| F frozen tree, PoE | greedy (1 view) | orbit mean | 0 | 0.090 | 0.186 | 0.208 | 4.89 | 17.44 | 20.59 | 0.102 | 0.337 | 0.396 |
| | | | 1 | 0.090 | 0.183 | 0.210 | 4.84 | 16.98 | 20.35 | 0.101 | 0.328 | 0.392 |
| | | | 2 | 0.088 | 0.182 | 0.201 | 4.84 | 17.58 | 19.52 | 0.101 | 0.338 | 0.376 |
| F frozen tree, PoE | orbit product | greedy (1 view) | 0 | 0.108 | 0.208 | 0.233 | 3.87 | 12.18 | 12.50 | 0.087 | 0.248 | 0.259 |
| | | | 1 | 0.109 | 0.204 | 0.235 | 3.90 | 11.87 | 12.43 | 0.088 | 0.242 | 0.258 |
| | | | 2 | 0.107 | 0.204 | 0.226 | 3.84 | 12.85 | 11.83 | 0.086 | 0.259 | 0.246 |
| F frozen tree, PoE | orbit product | orbit mean | 0 | 0.090 | 0.186 | 0.208 | 3.87 | 12.18 | 12.50 | 0.084 | 0.245 | 0.254 |
| | | | 1 | 0.090 | 0.183 | 0.210 | 3.90 | 11.87 | 12.43 | 0.085 | 0.239 | 0.254 |
| | | | 2 | 0.088 | 0.182 | 0.201 | 3.84 | 12.85 | 11.83 | 0.083 | 0.256 | 0.242 |
| F frozen tree, PoE | weighted vote (32) | greedy (1 view) | 0 | 0.108 | 0.208 | 0.233 | 3.48 | 12.42 | 13.45 | 0.080 | 0.253 | 0.275 |
| | | | 1 | 0.109 | 0.204 | 0.235 | 3.48 | 12.26 | 12.64 | 0.081 | 0.249 | 0.261 |
| | | | 2 | 0.107 | 0.204 | 0.226 | 3.43 | 13.06 | 12.56 | 0.079 | 0.263 | 0.258 |
| F frozen tree, PoE | weighted vote (32) | orbit mean | 0 | 0.090 | 0.186 | 0.208 | 3.48 | 12.42 | 13.45 | 0.077 | 0.249 | 0.271 |
| | | | 1 | 0.090 | 0.183 | 0.210 | 3.48 | 12.26 | 12.64 | 0.078 | 0.246 | 0.257 |
| | | | 2 | 0.088 | 0.182 | 0.201 | 3.43 | 13.06 | 12.56 | 0.076 | 0.260 | 0.254 |

Table 44. **EXP-118 per seed** (part 1).

| | | | | $E_T$ (m) | | | $E_R$ (°) | | | ESA score | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Arm | Rotation | Translation | Seed | Syn. | LB | SL | Syn. | LB | SL | Syn. | LB | SL |
| J joint, DINOv3 | greedy (1 view) | greedy (1 view) | 0 | 0.077 | 0.207 | 0.213 | 4.83 | 22.33 | 19.82 | 0.098 | 0.425 | 0.382 |
| | | | 1 | 0.080 | 0.206 | 0.209 | 4.76 | 21.79 | 20.92 | 0.097 | 0.415 | 0.400 |
| | | | 2 | 0.078 | 0.214 | 0.214 | 4.76 | 21.84 | 21.79 | 0.097 | 0.417 | 0.417 |
| J joint, DINOv3 | greedy (1 view) | orbit mean | 0 | 0.058 | 0.183 | 0.183 | 4.83 | 22.33 | 19.82 | 0.095 | 0.421 | 0.377 |
| | | | 1 | 0.060 | 0.183 | 0.179 | 4.76 | 21.79 | 20.92 | 0.094 | 0.411 | 0.395 |
| | | | 2 | 0.060 | 0.189 | 0.187 | 4.76 | 21.84 | 21.79 | 0.094 | 0.413 | 0.412 |
| J joint, DINOv3 | orbit product | greedy (1 view) | 0 | 0.077 | 0.207 | 0.213 | 4.22 | 17.68 | 14.23 | 0.087 | 0.343 | 0.284 |
| | | | 1 | 0.080 | 0.206 | 0.209 | 4.16 | 17.41 | 15.01 | 0.087 | 0.339 | 0.297 |
| | | | 2 | 0.078 | 0.214 | 0.214 | 4.18 | 16.82 | 15.38 | 0.087 | 0.330 | 0.305 |
| J joint, DINOv3 | orbit product | orbit mean | 0 | 0.058 | 0.183 | 0.183 | 4.22 | 17.68 | 14.23 | 0.084 | 0.339 | 0.279 |
| | | | 1 | 0.060 | 0.183 | 0.179 | 4.16 | 17.41 | 15.01 | 0.083 | 0.335 | 0.292 |
| | | | 2 | 0.060 | 0.189 | 0.187 | 4.18 | 16.82 | 15.38 | 0.084 | 0.325 | 0.300 |
| J joint, DINOv3 | weighted vote (32) | greedy (1 view) | 0 | 0.077 | 0.207 | 0.213 | 3.70 | 18.62 | 15.02 | 0.078 | 0.360 | 0.298 |
| | | | 1 | 0.080 | 0.206 | 0.209 | 3.67 | 17.94 | 15.52 | 0.078 | 0.348 | 0.306 |
| | | | 2 | 0.078 | 0.214 | 0.214 | 3.65 | 17.68 | 16.56 | 0.077 | 0.344 | 0.325 |
| J joint, DINOv3 | weighted vote (32) | orbit mean | 0 | 0.058 | 0.183 | 0.183 | 3.70 | 18.62 | 15.02 | 0.075 | 0.356 | 0.293 |
| | | | 1 | 0.060 | 0.183 | 0.179 | 3.67 | 17.94 | 15.52 | 0.075 | 0.344 | 0.301 |
| | | | 2 | 0.060 | 0.189 | 0.187 | 3.65 | 17.68 | 16.56 | 0.074 | 0.340 | 0.320 |
| J joint, PoE | greedy (1 view) | greedy (1 view) | 0 | 0.110 | 0.194 | 0.220 | 4.78 | 17.17 | 18.20 | 0.103 | 0.333 | 0.356 |
| | | | 1 | 0.108 | 0.193 | 0.221 | 4.79 | 17.27 | 18.89 | 0.103 | 0.335 | 0.368 |
| | | | 2 | 0.116 | 0.200 | 0.222 | 4.78 | 16.60 | 19.12 | 0.104 | 0.324 | 0.372 |
| J joint, PoE | greedy (1 view) | orbit mean | 0 | 0.091 | 0.173 | 0.191 | 4.78 | 17.17 | 18.20 | 0.100 | 0.329 | 0.351 |
| | | | 1 | 0.090 | 0.173 | 0.194 | 4.79 | 17.27 | 18.89 | 0.100 | 0.331 | 0.363 |
| | | | 2 | 0.098 | 0.178 | 0.195 | 4.78 | 16.60 | 19.12 | 0.101 | 0.320 | 0.368 |
| J joint, PoE | orbit product | greedy (1 view) | 0 | 0.110 | 0.194 | 0.220 | 3.86 | 12.40 | 11.41 | 0.087 | 0.250 | 0.237 |
| | | | 1 | 0.108 | 0.193 | 0.221 | 3.85 | 12.08 | 11.93 | 0.087 | 0.244 | 0.246 |
| | | | 2 | 0.116 | 0.200 | 0.222 | 3.87 | 11.86 | 11.85 | 0.088 | 0.241 | 0.245 |
| J joint, PoE | orbit product | orbit mean | 0 | 0.091 | 0.173 | 0.191 | 3.86 | 12.40 | 11.41 | 0.084 | 0.246 | 0.232 |
| | | | 1 | 0.090 | 0.173 | 0.194 | 3.85 | 12.08 | 11.93 | 0.084 | 0.241 | 0.242 |
| | | | 2 | 0.098 | 0.178 | 0.195 | 3.87 | 11.86 | 11.85 | 0.085 | 0.237 | 0.240 |
| J joint, PoE | weighted vote (32) | greedy (1 view) | 0 | 0.110 | 0.194 | 0.220 | 3.42 | 12.49 | 12.25 | 0.079 | 0.251 | 0.252 |
| | | | 1 | 0.108 | 0.193 | 0.221 | 3.41 | 12.35 | 12.98 | 0.079 | 0.249 | 0.265 |
| | | | 2 | 0.116 | 0.200 | 0.222 | 3.42 | 11.68 | 12.41 | 0.080 | 0.238 | 0.255 |
| J joint, PoE | weighted vote (32) | orbit mean | 0 | 0.091 | 0.173 | 0.191 | 3.42 | 12.49 | 12.25 | 0.076 | 0.248 | 0.247 |
| | | | 1 | 0.090 | 0.173 | 0.194 | 3.41 | 12.35 | 12.98 | 0.076 | 0.245 | 0.260 |
| | | | 2 | 0.098 | 0.178 | 0.195 | 3.42 | 11.68 | 12.41 | 0.077 | 0.234 | 0.250 |

Table 45. **EXP-118 per seed** (part 2).

## L.5. EXP-113: tree vs. flat Hopf classifier

**Question.** Does the hierarchical Tesseract tree improve on a flat classifier over a count-matched Hopf grid (SPACE-HOP's 10,923 directions $\times$ 12 rolls $=$ 131,076 anchors), with identical features, augmentation, budget and orbit inference? The flat head uses the same trunk plus a 131k-way linear layer (36.1 M trainable parameters per branch, 33.6 M in the head, vs. 2.4 M for the tree), geodesic soft targets over the 32 nearest anchors with the tree's leaf temperature, and scores an orbit candidate by the log-probability of its nearest anchor. This comparison confounds hierarchy and grid (Appendix L.13).

**Pre-registered criteria and outcome.**

- The tree improves on the flat grid on synth. val (PoE + orbit primary, PoE greedy secondary): **PASS**. Orbit $3.87 \pm 0.03$ vs $7.53 \pm 0.01$; greedy $4.85 \pm 0.03$ vs $8.95 \pm 0.06$.
- Same rule on synth. val-aug: **PASS**. Orbit $5.82 \pm 0.10$ vs $9.80 \pm 0.09$; greedy $9.75 \pm 0.06$ vs $15.06 \pm 0.20$.
- Real domains: descriptive. The tree has the lower mean and median in every real comparison; the flat head with orbit voting has fewer $>150°$ errors on lightbox.
- The withdrawal clause (flat matches the tree) was not triggered.

Single-branch training-run numbers of the flat head (greedy, synth. val / lightbox / sunlamp): DINOv3 8.61 / 27.4 / 32.5; MoGe-2 20.1 / 44.7 / 61.6.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Flat Hopf, PoE | greedy (1 view) | Synth. val | $8.95 \pm 0.06$ | 7.74 | 0.686 | 0.004 | 0.003 | – | – | – | – |
| | | Synth. val-aug | $15.06 \pm 0.20$ | 9.22 | 0.558 | 0.034 | 0.021 | – | – | – | – |
| | | Lightbox | $20.56 \pm 0.70$ | 9.79 | 0.515 | 0.069 | 0.041 | – | – | – | – |
| | | Sunlamp | $27.16 \pm 1.04$ | 11.07 | 0.438 | 0.109 | 0.057 | – | – | – | – |
| Flat Hopf, PoE | orbit product | Synth. val | $7.53 \pm 0.01$ | 6.66 | 0.751 | 0.001 | 0.001 | 37.3 | 0.787 | 11840.7 | 0.000 |
| | | Synth. val-aug | $9.80 \pm 0.09$ | 7.34 | 0.703 | 0.012 | 0.006 | 248.3 | 0.715 | 11038.7 | 0.001 |
| | | Lightbox | $13.15 \pm 0.37$ | 8.07 | 0.639 | 0.029 | 0.017 | 276.0 | 0.654 | 5856.3 | 0.005 |
| | | Sunlamp | $16.31 \pm 0.59$ | 8.68 | 0.590 | 0.047 | 0.024 | 159.3 | 0.657 | 2235.0 | 0.007 |
| Tree, PoE | greedy (1 view) | Synth. val | $4.85 \pm 0.03$ | 4.36 | 0.968 | 0.001 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $9.75 \pm 0.06$ | 5.92 | 0.825 | 0.018 | 0.009 | – | – | – | – |
| | | Lightbox | $17.33 \pm 0.31$ | 6.75 | 0.730 | 0.058 | 0.042 | – | – | – | – |
| | | Sunlamp | $20.15 \pm 0.57$ | 8.42 | 0.601 | 0.064 | 0.033 | – | – | – | – |
| Tree, PoE | orbit product | Synth. val | $3.87 \pm 0.03$ | 3.44 | 0.982 | 0.000 | 0.000 | 5.3 | 0.849 | 11965.0 | 0.000 |
| | | Synth. val-aug | $5.82 \pm 0.10$ | 3.87 | 0.939 | 0.007 | 0.004 | 110.3 | 0.631 | 11443.3 | 0.000 |
| | | Lightbox | $12.30 \pm 0.50$ | 5.26 | 0.836 | 0.037 | 0.030 | 280.3 | 0.437 | 6013.0 | 0.005 |
| | | Sunlamp | $12.25 \pm 0.37$ | 6.32 | 0.760 | 0.029 | 0.017 | 91.7 | 0.656 | 2353.3 | 0.005 |

Table 46. **EXP-113, PoE**: tree vs. flat Hopf (10,923×12). Flip transitions relative to the same family's greedy prediction.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Flat Hopf, DINOv3 | greedy (1 view) | Synth. val | $8.61 \pm 0.09$ | 7.82 | 0.680 | 0.001 | 0.001 | – | – | – | – |
| | | Synth. val-aug | $13.97 \pm 0.17$ | 9.37 | 0.545 | 0.026 | 0.014 | – | – | – | – |
| | | Lightbox | $27.29 \pm 1.20$ | 11.35 | 0.422 | 0.105 | 0.063 | – | – | – | – |
| | | Sunlamp | $32.48 \pm 2.46$ | 12.80 | 0.357 | 0.135 | 0.063 | – | – | – | – |
| Flat Hopf, DINOv3 | orbit product | Synth. val | $7.54 \pm 0.03$ | 6.73 | 0.749 | 0.001 | 0.000 | 12.0 | 0.594 | 11883.7 | 0.000 |
| | | Synth. val-aug | $9.55 \pm 0.06$ | 7.41 | 0.697 | 0.009 | 0.005 | 165.3 | 0.638 | 11078.0 | 0.001 |
| | | Lightbox | $17.71 \pm 0.77$ | 8.99 | 0.571 | 0.055 | 0.033 | 422.0 | 0.548 | 5361.0 | 0.008 |
| | | Sunlamp | $18.61 \pm 0.47$ | 9.62 | 0.526 | 0.058 | 0.027 | 176.7 | 0.655 | 2004.0 | 0.007 |
| Tree, DINOv3 | greedy (1 view) | Synth. val | $4.84 \pm 0.05$ | 4.43 | 0.972 | 0.000 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $9.88 \pm 0.22$ | 6.13 | 0.816 | 0.018 | 0.010 | – | – | – | – |
| | | Lightbox | $22.55 \pm 0.73$ | 8.38 | 0.607 | 0.083 | 0.060 | – | – | – | – |
| | | Sunlamp | $22.00 \pm 0.70$ | 10.17 | 0.489 | 0.067 | 0.032 | – | – | – | – |
| Tree, DINOv3 | orbit product | Synth. val | $4.21 \pm 0.04$ | 3.72 | 0.974 | 0.000 | 0.000 | 2.7 | 0.667 | 11977.7 | 0.000 |
| | | Synth. val-aug | $6.25 \pm 0.12$ | 4.20 | 0.920 | 0.007 | 0.004 | 117.0 | 0.586 | 11445.0 | 0.000 |
| | | Lightbox | $17.40 \pm 0.68$ | 6.88 | 0.708 | 0.058 | 0.048 | 402.7 | 0.341 | 5676.7 | 0.007 |
| | | Sunlamp | $15.27 \pm 0.64$ | 8.19 | 0.626 | 0.036 | 0.020 | 90.3 | 0.529 | 2236.3 | 0.005 |

Table 47. **EXP-113, DINOv3-only**: tree vs. flat Hopf.

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| Tree, PoE | greedy (1 view) | 0 | 4.89 | 9.82 | 17.44 | 20.59 | 0.040 | 0.030 |
| | | 1 | 4.84 | 9.70 | 16.98 | 20.35 | 0.039 | 0.034 |
| | | 2 | 4.84 | 9.72 | 17.58 | 19.52 | 0.046 | 0.034 |
| Tree, PoE | orbit product | 0 | 3.87 | 5.93 | 12.18 | 12.50 | 0.028 | 0.016 |
| | | 1 | 3.90 | 5.75 | 11.87 | 12.43 | 0.028 | 0.019 |
| | | 2 | 3.84 | 5.78 | 12.85 | 11.83 | 0.033 | 0.015 |
| Tree, DINOv3 | greedy (1 view) | 0 | 4.89 | 10.05 | 22.37 | 22.75 | 0.055 | 0.031 |
| | | 1 | 4.80 | 9.64 | 21.92 | 21.89 | 0.058 | 0.035 |
| | | 2 | 4.83 | 9.95 | 23.36 | 21.37 | 0.066 | 0.032 |
| Tree, DINOv3 | orbit product | 0 | 4.21 | 6.38 | 17.24 | 15.76 | 0.045 | 0.020 |
| | | 1 | 4.25 | 6.17 | 16.82 | 15.51 | 0.045 | 0.023 |
| | | 2 | 4.17 | 6.19 | 18.15 | 14.55 | 0.052 | 0.018 |
| Flat Hopf, PoE | greedy (1 view) | 0 | 8.88 | 14.88 | 19.75 | 25.99 | 0.037 | 0.052 |
| | | 1 | 9.00 | 15.27 | 21.05 | 27.98 | 0.044 | 0.062 |
| | | 2 | 8.96 | 15.03 | 20.87 | 27.52 | 0.041 | 0.058 |
| Flat Hopf, PoE | orbit product | 0 | 7.51 | 9.78 | 12.87 | 15.64 | 0.016 | 0.020 |
| | | 1 | 7.53 | 9.90 | 13.01 | 16.76 | 0.016 | 0.027 |
| | | 2 | 7.54 | 9.71 | 13.57 | 16.54 | 0.019 | 0.025 |
| Flat Hopf, DINOv3 | greedy (1 view) | 0 | 8.64 | 13.78 | 26.44 | 29.77 | 0.058 | 0.051 |
| | | 1 | 8.51 | 14.10 | 26.76 | 34.58 | 0.061 | 0.074 |
| | | 2 | 8.68 | 14.04 | 28.66 | 33.08 | 0.068 | 0.065 |
| Flat Hopf, DINOv3 | orbit product | 0 | 7.58 | 9.61 | 17.42 | 18.08 | 0.031 | 0.024 |
| | | 1 | 7.53 | 9.49 | 17.13 | 18.99 | 0.031 | 0.029 |
| | | 2 | 7.52 | 9.55 | 18.59 | 18.75 | 0.036 | 0.026 |

Table 48. **EXP-113 per seed**.

## L.6. EXP-114: orbit scoring rules

**Question.** With the same four views and the same 32 back-mapped candidates, does the orbit-product score add something beyond ordinary test-time-augmentation voting? Rules: greedy (view 0 only); max-confidence (candidate with the highest beam log-probability in its own view); top-1 vote (5° Gaussian-kernel vote among the four views' top-1); weighted vote (kernel vote over all 32 candidates weighted by per-view normalised probability); orbit product (sum over views of the tree log-likelihood of the candidate mapped into each view).

**Pre-registered criteria and outcome.** The product passes if its 3-seed mean is below the best of the three simpler rules by more than the larger seed std (PoE primary, DINOv3 secondary).

- Synth. val: **FAIL** for PoE (product $3.87 \pm 0.03$ vs best other, weighted vote (32), $3.46 \pm 0.03$); **FAIL** for DINOv3 ($4.21 \pm 0.04$ vs weighted vote (32) $3.71 \pm 0.05$).
- Synth. val-aug: **PASS** for PoE, marginally ($5.82 \pm 0.10$ vs weighted vote (32) $5.98 \pm 0.13$); **FAIL** for DINOv3 ($6.25 \pm 0.12$ vs weighted vote (32) $6.30 \pm 0.06$).
- Real domains: descriptive; the product has the lowest mean on both real domains for both variants.
- Consequence (pre-registered clause): no product-specific claim; the claim is multi-view orbit consistency over hierarchical candidates.

Cost: the weighted vote needs the per-view beam scores and a $32 \times 32$ angle matrix (<0.1 ms per image); the product re-scores 32 candidates in 4 views (8.2 ms DINOv3-only, 15.3 ms PoE at batch 1, Tab. 37).

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken |
|---|---|---|---|---|---|---|---|---|---|---|---|
| PoE | greedy (1 view) | Synth. val | $4.85 \pm 0.03$ | 4.36 | 0.968 | 0.001 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $9.75 \pm 0.06$ | 5.92 | 0.825 | 0.018 | 0.009 | – | – | – | – |
| | | Lightbox | $17.33 \pm 0.31$ | 6.75 | 0.730 | 0.058 | 0.042 | – | – | – | – |
| | | Sunlamp | $20.15 \pm 0.57$ | 8.42 | 0.601 | 0.064 | 0.033 | – | – | – | – |
| PoE | max-confidence | Synth. val | $5.51 \pm 0.03$ | 4.84 | 0.922 | 0.000 | 0.000 | 5.3 | 0.595 | 11965.0 | 0.000 |
| | | Synth. val-aug | $8.43 \pm 0.03$ | 5.67 | 0.845 | 0.012 | 0.006 | 110.3 | 0.474 | 11443.3 | 0.002 |
| | | Lightbox | $14.63 \pm 0.42$ | 6.80 | 0.736 | 0.042 | 0.032 | 280.3 | 0.417 | 6013.0 | 0.008 |
| | | Sunlamp | $15.92 \pm 0.15$ | 8.07 | 0.634 | 0.042 | 0.023 | 91.7 | 0.566 | 2353.3 | 0.009 |
| PoE | top-1 vote | Synth. val | $4.37 \pm 0.03$ | 3.84 | 0.971 | 0.000 | 0.000 | 5.3 | 0.849 | 11965.0 | 0.000 |
| | | Synth. val-aug | $6.85 \pm 0.11$ | 4.50 | 0.912 | 0.010 | 0.005 | 110.3 | 0.535 | 11443.3 | 0.001 |
| | | Lightbox | $13.24 \pm 0.42$ | 5.86 | 0.800 | 0.038 | 0.031 | 280.3 | 0.391 | 6013.0 | 0.005 |
| | | Sunlamp | $13.38 \pm 0.57$ | 6.93 | 0.718 | 0.032 | 0.017 | 91.7 | 0.613 | 2353.3 | 0.005 |
| PoE | weighted vote (32) | Synth. val | $3.46 \pm 0.03$ | 2.97 | 0.985 | 0.000 | 0.000 | 5.3 | 0.802 | 11965.0 | 0.000 |
| | | Synth. val-aug | $5.98 \pm 0.13$ | 3.43 | 0.935 | 0.011 | 0.005 | 110.3 | 0.510 | 11443.3 | 0.001 |
| | | Lightbox | $12.58 \pm 0.42$ | 4.98 | 0.834 | 0.039 | 0.031 | 280.3 | 0.402 | 6013.0 | 0.006 |
| | | Sunlamp | $12.88 \pm 0.49$ | 6.08 | 0.754 | 0.035 | 0.018 | 91.7 | 0.601 | 2353.3 | 0.005 |
| PoE | orbit product | Synth. val | $3.87 \pm 0.03$ | 3.44 | 0.982 | 0.000 | 0.000 | 5.3 | 0.849 | 11965.0 | 0.000 |
| | | Synth. val-aug | $5.82 \pm 0.10$ | 3.87 | 0.939 | 0.007 | 0.004 | 110.3 | 0.631 | 11443.3 | 0.000 |
| | | Lightbox | $12.30 \pm 0.50$ | 5.26 | 0.836 | 0.037 | 0.030 | 280.3 | 0.437 | 6013.0 | 0.005 |
| | | Sunlamp | $12.25 \pm 0.37$ | 6.32 | 0.760 | 0.029 | 0.017 | 91.7 | 0.656 | 2353.3 | 0.005 |

Table 49. **EXP-114, PoE**: all selection rules over the same 32 candidates. Flip transitions relative to greedy.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DINOv3 | greedy (1 view) | Synth. val | $4.84 \pm 0.05$ | 4.43 | 0.972 | 0.000 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $9.88 \pm 0.22$ | 6.13 | 0.816 | 0.018 | 0.010 | – | – | – | – |
| | | Lightbox | $22.55 \pm 0.73$ | 8.38 | 0.607 | 0.083 | 0.060 | – | – | – | – |
| | | Sunlamp | $22.00 \pm 0.70$ | 10.17 | 0.489 | 0.067 | 0.032 | – | – | – | – |
| DINOv3 | max-confidence | Synth. val | $5.92 \pm 0.02$ | 5.21 | 0.896 | 0.000 | 0.000 | 2.7 | 0.750 | 11977.7 | 0.000 |
| | | Synth. val-aug | $8.71 \pm 0.13$ | 6.08 | 0.817 | 0.011 | 0.006 | 117.0 | 0.496 | 11445.0 | 0.001 |
| | | Lightbox | $20.02 \pm 0.61$ | 8.69 | 0.594 | 0.065 | 0.051 | 402.7 | 0.318 | 5676.7 | 0.010 |
| | | Sunlamp | $17.75 \pm 0.48$ | 9.80 | 0.510 | 0.044 | 0.025 | 90.3 | 0.473 | 2236.3 | 0.009 |
| DINOv3 | top-1 vote | Synth. val | $4.65 \pm 0.03$ | 4.11 | 0.961 | 0.000 | 0.000 | 2.7 | 0.750 | 11977.7 | 0.000 |
| | | Synth. val-aug | $7.27 \pm 0.13$ | 4.83 | 0.896 | 0.010 | 0.006 | 117.0 | 0.508 | 11445.0 | 0.001 |
| | | Lightbox | $18.68 \pm 0.73$ | 7.59 | 0.662 | 0.062 | 0.050 | 402.7 | 0.304 | 5676.7 | 0.008 |
| | | Sunlamp | $16.50 \pm 0.52$ | 8.84 | 0.580 | 0.041 | 0.023 | 90.3 | 0.460 | 2236.3 | 0.007 |
| DINOv3 | weighted vote (32) | Synth. val | $3.71 \pm 0.05$ | 3.20 | 0.983 | 0.001 | 0.000 | 2.7 | 0.583 | 11977.7 | 0.000 |
| | | Synth. val-aug | $6.30 \pm 0.06$ | 3.68 | 0.926 | 0.010 | 0.006 | 117.0 | 0.479 | 11445.0 | 0.001 |
| | | Lightbox | $18.27 \pm 0.67$ | 6.59 | 0.718 | 0.065 | 0.051 | 402.7 | 0.299 | 5676.7 | 0.009 |
| | | Sunlamp | $15.83 \pm 0.83$ | 7.89 | 0.636 | 0.042 | 0.023 | 90.3 | 0.467 | 2236.3 | 0.007 |
| DINOv3 | orbit product | Synth. val | $4.21 \pm 0.04$ | 3.72 | 0.974 | 0.000 | 0.000 | 2.7 | 0.667 | 11977.7 | 0.000 |
| | | Synth. val-aug | $6.25 \pm 0.12$ | 4.20 | 0.920 | 0.007 | 0.004 | 117.0 | 0.586 | 11445.0 | 0.000 |
| | | Lightbox | $17.40 \pm 0.68$ | 6.88 | 0.708 | 0.058 | 0.048 | 402.7 | 0.341 | 5676.7 | 0.007 |
| | | Sunlamp | $15.27 \pm 0.64$ | 8.19 | 0.626 | 0.036 | 0.020 | 90.3 | 0.529 | 2236.3 | 0.005 |

Table 50. **EXP-114, DINOv3**: all selection rules over the same 32 candidates. Flip transitions relative to greedy.

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| DINOv3 | greedy (1 view) | 0 | 4.89 | 10.05 | 22.37 | 22.75 | 0.055 | 0.031 |
| | | 1 | 4.80 | 9.64 | 21.92 | 21.89 | 0.058 | 0.035 |
| | | 2 | 4.83 | 9.95 | 23.36 | 21.37 | 0.066 | 0.032 |
| DINOv3 | max-confidence | 0 | 5.93 | 8.86 | 19.89 | 18.18 | 0.050 | 0.024 |
| | | 1 | 5.89 | 8.66 | 19.49 | 17.83 | 0.050 | 0.026 |
| | | 2 | 5.94 | 8.60 | 20.69 | 17.24 | 0.055 | 0.024 |
| DINOv3 | top-1 vote | 0 | 4.67 | 7.41 | 18.40 | 16.82 | 0.047 | 0.022 |
| | | 1 | 4.67 | 7.15 | 18.13 | 16.78 | 0.048 | 0.025 |
| | | 2 | 4.62 | 7.25 | 19.51 | 15.91 | 0.055 | 0.021 |
| DINOv3 | weighted vote (32) | 0 | 3.75 | 6.34 | 18.37 | 16.47 | 0.049 | 0.023 |
| | | 1 | 3.72 | 6.23 | 17.56 | 16.13 | 0.050 | 0.027 |
| | | 2 | 3.66 | 6.35 | 18.89 | 14.88 | 0.055 | 0.020 |
| DINOv3 | orbit product | 0 | 4.21 | 6.38 | 17.24 | 15.76 | 0.045 | 0.020 |
| | | 1 | 4.25 | 6.17 | 16.82 | 15.51 | 0.045 | 0.023 |
| | | 2 | 4.17 | 6.19 | 18.15 | 14.55 | 0.052 | 0.018 |
| PoE | greedy (1 view) | 0 | 4.89 | 9.82 | 17.44 | 20.59 | 0.040 | 0.030 |
| | | 1 | 4.84 | 9.70 | 16.98 | 20.35 | 0.039 | 0.034 |
| | | 2 | 4.84 | 9.72 | 17.58 | 19.52 | 0.046 | 0.034 |
| PoE | max-confidence | 0 | 5.53 | 8.46 | 14.24 | 16.02 | 0.030 | 0.022 |
| | | 1 | 5.53 | 8.41 | 14.59 | 16.01 | 0.032 | 0.025 |
| | | 2 | 5.48 | 8.40 | 15.07 | 15.75 | 0.035 | 0.023 |
| PoE | top-1 vote | 0 | 4.39 | 6.92 | 12.93 | 13.97 | 0.029 | 0.018 |
| | | 1 | 4.38 | 6.72 | 13.08 | 13.34 | 0.031 | 0.020 |
| | | 2 | 4.34 | 6.90 | 13.71 | 12.84 | 0.034 | 0.015 |
| PoE | weighted vote (32) | 0 | 3.48 | 6.09 | 12.42 | 13.45 | 0.029 | 0.019 |
| | | 1 | 3.48 | 5.84 | 12.26 | 12.64 | 0.030 | 0.019 |
| | | 2 | 3.43 | 6.02 | 13.06 | 12.56 | 0.034 | 0.017 |
| PoE | orbit product | 0 | 3.87 | 5.93 | 12.18 | 12.50 | 0.028 | 0.016 |
| | | 1 | 3.90 | 5.75 | 11.87 | 12.43 | 0.028 | 0.019 |
| | | 2 | 3.84 | 5.78 | 12.85 | 11.83 | 0.033 | 0.015 |

Table 51. **EXP-114 per seed**.

### L.7. EXP-115: augmentation ablation

**Question.** Which part of the augmentation causes the real-domain flip recovery: photometric only, in-plane roll only, or both, each with and without orbit inference? The clean EXP-036 trees are the no-augmentation reference.

**Pre-registered criteria and outcome** (orbit product, synth. val-aug):

- Roll augmentation matters (both beats photometric only): **PASS**. PoE $5.82 \pm 0.10$ vs $11.16 \pm 0.08$; DINOv3 $6.25 \pm 0.12$ vs $8.68 \pm 0.19$. Caveat: val-aug itself contains random rolls and favours roll-trained arms by construction.
- Photometric augmentation matters (both beats roll only): **PASS**. PoE $10.49 \pm 0.17$ for roll only; DINOv3 $12.58 \pm 0.03$.
- Orbit gain by arm (greedy − orbit, PoE, val-aug; descriptive): none (clean) 11.00, photometric only 5.91, in-plane roll only 7.06, both (EXP-109 A) 3.93.
- Real domains (descriptive): photometric only matches the combined recipe with orbit voting; roll only stays close to the clean trees. The real-domain gain is therefore attributed to photometric augmentation plus orbit voting (Tab. 9).

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken |
|---|---|---|---|---|---|---|---|---|---|---|---|
| none (clean) | greedy (1 view) | Synth. val | $5.54 \pm 0.07$ | 4.62 | 0.947 | 0.002 | 0.001 | – | – | – | – |
| | | Synth. val-aug | $30.84 \pm 0.44$ | 10.95 | 0.455 | 0.107 | 0.055 | – | – | – | – |
| | | Lightbox | $26.90 \pm 0.80$ | 8.99 | 0.555 | 0.105 | 0.059 | – | – | – | – |
| | | Sunlamp | $36.66 \pm 0.94$ | 12.98 | 0.373 | 0.153 | 0.074 | – | – | – | – |
| none (clean) | orbit product | Synth. val | $4.62 \pm 0.04$ | 3.91 | 0.956 | 0.001 | 0.000 | 12.0 | 0.595 | 11875.7 | 0.000 |
| | | Synth. val-aug | $19.84 \pm 0.34$ | 7.71 | 0.631 | 0.054 | 0.026 | 662.0 | 0.471 | 8414.7 | 0.004 |
| | | Lightbox | $20.25 \pm 0.56$ | 7.28 | 0.673 | 0.073 | 0.048 | 400.0 | 0.356 | 5226.0 | 0.008 |
| | | Sunlamp | $24.83 \pm 0.96$ | 9.98 | 0.501 | 0.089 | 0.043 | 206.3 | 0.458 | 1822.3 | 0.009 |
| none (clean) | weighted vote (32) | Synth. val | $4.27 \pm 0.06$ | 3.37 | 0.960 | 0.001 | 0.001 | 12.0 | 0.497 | 11875.7 | 0.000 |
| | | Synth. val-aug | $22.35 \pm 0.41$ | 7.57 | 0.625 | 0.073 | 0.038 | 662.0 | 0.404 | 8414.7 | 0.006 |
| | | Lightbox | $20.82 \pm 0.84$ | 7.02 | 0.671 | 0.077 | 0.049 | 400.0 | 0.337 | 5226.0 | 0.009 |
| | | Sunlamp | $27.62 \pm 0.93$ | 9.93 | 0.504 | 0.109 | 0.054 | 206.3 | 0.398 | 1822.3 | 0.012 |
| photometric only | greedy (1 view) | Synth. val | $5.00 \pm 0.03$ | 4.46 | 0.966 | 0.001 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $17.07 \pm 0.08$ | 7.99 | 0.640 | 0.038 | 0.020 | – | – | – | – |
| | | Lightbox | $17.38 \pm 0.41$ | 6.76 | 0.724 | 0.059 | 0.039 | – | – | – | – |
| | | Sunlamp | $19.68 \pm 1.34$ | 8.22 | 0.615 | 0.063 | 0.032 | – | – | – | – |
| photometric only | orbit product | Synth. val | $4.25 \pm 0.06$ | 3.76 | 0.972 | 0.000 | 0.000 | 5.3 | 0.667 | 11959.3 | 0.000 |
| | | Synth. val-aug | $11.16 \pm 0.08$ | 5.94 | 0.796 | 0.018 | 0.008 | 245.3 | 0.598 | 10414.3 | 0.001 |
| | | Lightbox | $12.29 \pm 0.38$ | 5.47 | 0.828 | 0.036 | 0.027 | 262.3 | 0.438 | 6002.0 | 0.005 |
| | | Sunlamp | $12.37 \pm 0.79$ | 6.47 | 0.752 | 0.029 | 0.015 | 89.7 | 0.639 | 2362.7 | 0.004 |
| photometric only | weighted vote (32) | Synth. val | $3.76 \pm 0.01$ | 3.23 | 0.977 | 0.000 | 0.000 | 5.3 | 0.750 | 11959.3 | 0.000 |
| | | Synth. val-aug | $11.84 \pm 0.01$ | 5.66 | 0.797 | 0.023 | 0.012 | 245.3 | 0.531 | 10414.3 | 0.002 |
| | | Lightbox | $12.56 \pm 0.40$ | 5.11 | 0.834 | 0.039 | 0.029 | 262.3 | 0.414 | 6002.0 | 0.006 |
| | | Sunlamp | $12.91 \pm 1.04$ | 6.17 | 0.756 | 0.034 | 0.017 | 89.7 | 0.584 | 2362.7 | 0.005 |
| in-plane roll only | greedy (1 view) | Synth. val | $4.91 \pm 0.00$ | 4.31 | 0.967 | 0.001 | 0.001 | – | – | – | – |
| | | Synth. val-aug | $17.55 \pm 0.15$ | 7.01 | 0.695 | 0.057 | 0.028 | – | – | – | – |
| | | Lightbox | $24.30 \pm 0.67$ | 7.93 | 0.620 | 0.093 | 0.060 | – | – | – | – |
| | | Sunlamp | $31.52 \pm 0.59$ | 11.35 | 0.438 | 0.129 | 0.062 | – | – | – | – |
| in-plane roll only | orbit product | Synth. val | $3.78 \pm 0.03$ | 3.33 | 0.983 | 0.000 | 0.000 | 7.7 | 0.597 | 11946.7 | 0.000 |
| | | Synth. val-aug | $10.49 \pm 0.17$ | 4.65 | 0.841 | 0.028 | 0.012 | 332.0 | 0.512 | 10405.3 | 0.001 |
| | | Lightbox | $18.69 \pm 0.77$ | 6.31 | 0.738 | 0.069 | 0.052 | 406.7 | 0.349 | 5488.0 | 0.007 |
| | | Sunlamp | $20.70 \pm 0.26$ | 8.39 | 0.589 | 0.070 | 0.038 | 174.0 | 0.454 | 1985.7 | 0.007 |
| in-plane roll only | weighted vote (32) | Synth. val | $3.36 \pm 0.07$ | 2.84 | 0.984 | 0.000 | 0.000 | 7.7 | 0.486 | 11946.7 | 0.000 |
| | | Synth. val-aug | $11.48 \pm 0.07$ | 4.25 | 0.834 | 0.036 | 0.016 | 332.0 | 0.443 | 10405.3 | 0.002 |
| | | Lightbox | $19.02 \pm 0.71$ | 6.05 | 0.732 | 0.071 | 0.052 | 406.7 | 0.336 | 5488.0 | 0.009 |
| | | Sunlamp | $22.63 \pm 1.14$ | 8.35 | 0.585 | 0.085 | 0.045 | 174.0 | 0.399 | 1985.7 | 0.011 |
| both (EXP-109 A) | greedy (1 view) | Synth. val | $4.85 \pm 0.03$ | 4.36 | 0.968 | 0.001 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $9.75 \pm 0.06$ | 5.92 | 0.825 | 0.018 | 0.009 | – | – | – | – |
| | | Lightbox | $17.33 \pm 0.31$ | 6.75 | 0.730 | 0.058 | 0.042 | – | – | – | – |
| | | Sunlamp | $20.15 \pm 0.57$ | 8.42 | 0.601 | 0.064 | 0.033 | – | – | – | – |
| both (EXP-109 A) | orbit product | Synth. val | $3.87 \pm 0.03$ | 3.44 | 0.982 | 0.000 | 0.000 | 5.3 | 0.849 | 11965.0 | 0.000 |
| | | Synth. val-aug | $5.82 \pm 0.10$ | 3.87 | 0.939 | 0.007 | 0.004 | 110.3 | 0.631 | 11443.3 | 0.000 |
| | | Lightbox | $12.30 \pm 0.50$ | 5.26 | 0.836 | 0.037 | 0.030 | 280.3 | 0.437 | 6013.0 | 0.005 |
| | | Sunlamp | $12.25 \pm 0.37$ | 6.32 | 0.760 | 0.029 | 0.017 | 91.7 | 0.656 | 2353.3 | 0.005 |
| both (EXP-109 A) | weighted vote (32) | Synth. val | $3.46 \pm 0.03$ | 2.97 | 0.985 | 0.000 | 0.000 | 5.3 | 0.802 | 11965.0 | 0.000 |
| | | Synth. val-aug | $5.98 \pm 0.13$ | 3.43 | 0.935 | 0.011 | 0.005 | 110.3 | 0.510 | 11443.3 | 0.001 |
| | | Lightbox | $12.58 \pm 0.42$ | 4.98 | 0.834 | 0.039 | 0.031 | 280.3 | 0.402 | 6013.0 | 0.006 |
| | | Sunlamp | $12.88 \pm 0.49$ | 6.08 | 0.754 | 0.035 | 0.018 | 91.7 | 0.601 | 2353.3 | 0.005 |

Table 52. **EXP-115, PoE**: training augmentation × inference. Flip transitions relative to the same arm's greedy prediction.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken |
|---|---|---|---|---|---|---|---|---|---|---|---|
| none (clean) | greedy (1 view) | Synth. val | $5.35 \pm 0.11$ | 4.80 | 0.955 | 0.001 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $24.73 \pm 0.84$ | 10.79 | 0.455 | 0.080 | 0.032 | – | – | – | – |
| | | Lightbox | $37.07 \pm 2.05$ | 13.48 | 0.358 | 0.153 | 0.093 | – | – | – | – |
| | | Sunlamp | $41.46 \pm 0.99$ | 20.46 | 0.203 | 0.159 | 0.062 | – | – | – | – |
| none (clean) | orbit product | Synth. val | $4.98 \pm 0.08$ | 4.27 | 0.942 | 0.000 | 0.000 | 5.3 | 0.511 | 11951.0 | 0.000 |
| | | Synth. val-aug | $17.22 \pm 0.52$ | 8.50 | 0.591 | 0.042 | 0.016 | 378.7 | 0.326 | 9037.0 | 0.001 |
| | | Lightbox | $30.98 \pm 1.33$ | 11.22 | 0.437 | 0.121 | 0.086 | 623.7 | 0.172 | 4380.7 | 0.008 |
| | | Sunlamp | $32.12 \pm 1.65$ | 15.71 | 0.280 | 0.107 | 0.049 | 172.3 | 0.180 | 1369.3 | 0.005 |
| none (clean) | weighted vote (32) | Synth. val | $4.36 \pm 0.11$ | 3.59 | 0.960 | 0.001 | 0.000 | 5.3 | 0.633 | 11951.0 | 0.000 |
| | | Synth. val-aug | $19.53 \pm 0.69$ | 7.99 | 0.611 | 0.062 | 0.026 | 378.7 | 0.264 | 9037.0 | 0.003 |
| | | Lightbox | $32.23 \pm 1.81$ | 11.07 | 0.451 | 0.131 | 0.087 | 623.7 | 0.149 | 4380.7 | 0.009 |
| | | Sunlamp | $35.53 \pm 1.72$ | 16.35 | 0.281 | 0.131 | 0.054 | 172.3 | 0.148 | 1369.3 | 0.007 |
| photometric only | greedy (1 view) | Synth. val | $4.91 \pm 0.04$ | 4.52 | 0.973 | 0.001 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $12.33 \pm 0.26$ | 7.67 | 0.682 | 0.021 | 0.010 | – | – | – | – |
| | | Lightbox | $22.80 \pm 1.32$ | 8.33 | 0.609 | 0.085 | 0.060 | – | – | – | – |
| | | Sunlamp | $22.31 \pm 1.31$ | 9.96 | 0.505 | 0.071 | 0.037 | – | – | – | – |
| photometric only | orbit product | Synth. val | $4.50 \pm 0.04$ | 3.97 | 0.963 | 0.000 | 0.000 | 3.0 | 0.750 | 11980.3 | 0.000 |
| | | Synth. val-aug | $8.68 \pm 0.19$ | 6.06 | 0.808 | 0.010 | 0.004 | 122.3 | 0.575 | 11127.3 | 0.000 |
| | | Lightbox | $17.59 \pm 0.73$ | 7.09 | 0.697 | 0.059 | 0.045 | 404.7 | 0.358 | 5660.3 | 0.008 |
| | | Sunlamp | $15.42 \pm 1.31$ | 8.31 | 0.618 | 0.038 | 0.021 | 103.0 | 0.503 | 2234.0 | 0.005 |
| photometric only | weighted vote (32) | Synth. val | $3.92 \pm 0.06$ | 3.37 | 0.974 | 0.000 | 0.000 | 3.0 | 0.750 | 11980.3 | 0.000 |
| | | Synth. val-aug | $8.72 \pm 0.27$ | 5.56 | 0.829 | 0.013 | 0.006 | 122.3 | 0.515 | 11127.3 | 0.001 |
| | | Lightbox | $18.05 \pm 0.77$ | 6.68 | 0.714 | 0.064 | 0.048 | 404.7 | 0.321 | 5660.3 | 0.008 |
| | | Sunlamp | $15.82 \pm 1.39$ | 7.85 | 0.637 | 0.042 | 0.022 | 103.0 | 0.461 | 2234.0 | 0.006 |
| in-plane roll only | greedy (1 view) | Synth. val | $4.89 \pm 0.05$ | 4.43 | 0.971 | 0.001 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $19.17 \pm 0.34$ | 7.91 | 0.629 | 0.062 | 0.028 | – | – | – | – |
| | | Lightbox | $35.07 \pm 1.20$ | 11.23 | 0.445 | 0.148 | 0.099 | – | – | – | – |
| | | Sunlamp | $37.64 \pm 0.31$ | 16.83 | 0.265 | 0.145 | 0.060 | – | – | – | – |
| in-plane roll only | orbit product | Synth. val | $4.07 \pm 0.05$ | 3.57 | 0.978 | 0.000 | 0.000 | 3.7 | 0.617 | 11966.7 | 0.000 |
| | | Synth. val-aug | $12.58 \pm 0.03$ | 5.67 | 0.757 | 0.032 | 0.013 | 333.0 | 0.384 | 10047.7 | 0.001 |
| | | Lightbox | $28.86 \pm 1.07$ | 9.34 | 0.535 | 0.117 | 0.086 | 664.0 | 0.213 | 4730.0 | 0.008 |
| | | Sunlamp | $28.41 \pm 1.40$ | 13.08 | 0.363 | 0.096 | 0.048 | 168.0 | 0.238 | 1583.7 | 0.005 |
| in-plane roll only | weighted vote (32) | Synth. val | $3.53 \pm 0.05$ | 3.03 | 0.984 | 0.000 | 0.000 | 3.7 | 0.467 | 11966.7 | 0.000 |
| | | Synth. val-aug | $13.95 \pm 0.13$ | 5.13 | 0.763 | 0.043 | 0.019 | 333.0 | 0.324 | 10047.7 | 0.002 |
| | | Lightbox | $30.35 \pm 1.13$ | 9.13 | 0.545 | 0.127 | 0.091 | 664.0 | 0.167 | 4730.0 | 0.010 |
| | | Sunlamp | $30.28 \pm 1.18$ | 13.33 | 0.363 | 0.105 | 0.051 | 168.0 | 0.191 | 1583.7 | 0.007 |
| both (EXP-109 A) | greedy (1 view) | Synth. val | $4.84 \pm 0.05$ | 4.43 | 0.972 | 0.000 | 0.000 | – | – | – | – |
| | | Synth. val-aug | $9.88 \pm 0.22$ | 6.13 | 0.816 | 0.018 | 0.010 | – | – | – | – |
| | | Lightbox | $22.55 \pm 0.73$ | 8.38 | 0.607 | 0.083 | 0.060 | – | – | – | – |
| | | Sunlamp | $22.00 \pm 0.70$ | 10.17 | 0.489 | 0.067 | 0.032 | – | – | – | – |
| both (EXP-109 A) | orbit product | Synth. val | $4.21 \pm 0.04$ | 3.72 | 0.974 | 0.000 | 0.000 | 2.7 | 0.667 | 11977.7 | 0.000 |
| | | Synth. val-aug | $6.25 \pm 0.12$ | 4.20 | 0.920 | 0.007 | 0.004 | 117.0 | 0.586 | 11445.0 | 0.000 |
| | | Lightbox | $17.40 \pm 0.68$ | 6.88 | 0.708 | 0.058 | 0.048 | 402.7 | 0.341 | 5676.7 | 0.007 |
| | | Sunlamp | $15.27 \pm 0.64$ | 8.19 | 0.626 | 0.036 | 0.020 | 90.3 | 0.529 | 2236.3 | 0.005 |
| both (EXP-109 A) | weighted vote (32) | Synth. val | $3.71 \pm 0.05$ | 3.20 | 0.983 | 0.001 | 0.000 | 2.7 | 0.583 | 11977.7 | 0.000 |
| | | Synth. val-aug | $6.30 \pm 0.06$ | 3.68 | 0.926 | 0.010 | 0.006 | 117.0 | 0.479 | 11445.0 | 0.001 |
| | | Lightbox | $18.27 \pm 0.67$ | 6.59 | 0.718 | 0.065 | 0.051 | 402.7 | 0.299 | 5676.7 | 0.009 |
| | | Sunlamp | $15.83 \pm 0.83$ | 7.89 | 0.636 | 0.042 | 0.023 | 90.3 | 0.467 | 2236.3 | 0.007 |

Table 53. **EXP-115, DINOv3**: training augmentation × inference. Flip transitions relative to the same arm's greedy prediction.

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| none (clean) | greedy (1 view) | 0 | 5.57 | 31.29 | 27.61 | 37.43 | 0.059 | 0.073 |
| | | 1 | 5.47 | 30.81 | 26.04 | 36.93 | 0.059 | 0.078 |
| | | 2 | 5.59 | 30.42 | 27.05 | 35.61 | 0.059 | 0.070 |
| none (clean) | orbit product | 0 | 4.66 | 20.18 | 20.80 | 25.10 | 0.049 | 0.040 |
| | | 1 | 4.58 | 19.51 | 19.68 | 25.63 | 0.047 | 0.054 |
| | | 2 | 4.62 | 19.84 | 20.26 | 23.76 | 0.048 | 0.034 |
| none (clean) | weighted vote (32) | 0 | 4.26 | 22.83 | 21.75 | 28.21 | 0.051 | 0.053 |
| | | 1 | 4.22 | 22.11 | 20.10 | 28.11 | 0.047 | 0.062 |
| | | 2 | 4.33 | 22.12 | 20.60 | 26.55 | 0.048 | 0.046 |
| photometric only | greedy (1 view) | 0 | 5.01 | 16.99 | 17.42 | 21.09 | 0.038 | 0.037 |
| | | 1 | 5.01 | 17.09 | 16.95 | 18.42 | 0.038 | 0.029 |
| | | 2 | 4.96 | 17.13 | 17.76 | 19.55 | 0.041 | 0.030 |
| photometric only | orbit product | 0 | 4.30 | 11.20 | 12.48 | 13.07 | 0.027 | 0.018 |
| | | 1 | 4.19 | 11.08 | 11.85 | 11.52 | 0.025 | 0.013 |
| | | 2 | 4.27 | 11.22 | 12.55 | 12.52 | 0.028 | 0.014 |
| photometric only | weighted vote (32) | 0 | 3.78 | 11.83 | 12.64 | 13.69 | 0.029 | 0.020 |
| | | 1 | 3.75 | 11.83 | 12.13 | 11.72 | 0.028 | 0.013 |
| | | 2 | 3.76 | 11.85 | 12.92 | 13.31 | 0.030 | 0.019 |
| in-plane roll only | greedy (1 view) | 0 | 4.91 | 17.59 | 24.75 | 32.18 | 0.059 | 0.065 |
| | | 1 | 4.91 | 17.38 | 23.53 | 31.05 | 0.059 | 0.063 |
| | | 2 | 4.90 | 17.67 | 24.63 | 31.33 | 0.063 | 0.059 |
| in-plane roll only | orbit product | 0 | 3.79 | 10.31 | 18.73 | 20.99 | 0.051 | 0.039 |
| | | 1 | 3.75 | 10.51 | 17.91 | 20.64 | 0.048 | 0.039 |
| | | 2 | 3.80 | 10.65 | 19.45 | 20.49 | 0.056 | 0.037 |
| in-plane roll only | weighted vote (32) | 0 | 3.43 | 11.41 | 19.43 | 23.51 | 0.053 | 0.049 |
| | | 1 | 3.30 | 11.47 | 18.20 | 21.34 | 0.049 | 0.039 |
| | | 2 | 3.34 | 11.54 | 19.42 | 23.04 | 0.053 | 0.046 |
| both (EXP-109 A) | greedy (1 view) | 0 | 4.89 | 9.82 | 17.44 | 20.59 | 0.040 | 0.030 |
| | | 1 | 4.84 | 9.70 | 16.98 | 20.35 | 0.039 | 0.034 |
| | | 2 | 4.84 | 9.72 | 17.58 | 19.52 | 0.046 | 0.034 |
| both (EXP-109 A) | orbit product | 0 | 3.87 | 5.93 | 12.18 | 12.50 | 0.028 | 0.016 |
| | | 1 | 3.90 | 5.75 | 11.87 | 12.43 | 0.028 | 0.019 |
| | | 2 | 3.84 | 5.78 | 12.85 | 11.83 | 0.033 | 0.015 |
| both (EXP-109 A) | weighted vote (32) | 0 | 3.48 | 6.09 | 12.42 | 13.45 | 0.029 | 0.019 |
| | | 1 | 3.48 | 5.84 | 12.26 | 12.64 | 0.030 | 0.019 |
| | | 2 | 3.43 | 6.02 | 13.06 | 12.56 | 0.034 | 0.017 |

Table 54. **EXP-115 per seed, PoE**.

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| none (clean) | greedy (1 view) | 0 | 5.46 | 25.54 | 38.85 | 42.59 | 0.093 | 0.058 |
| | | 1 | 5.25 | 23.86 | 34.82 | 41.01 | 0.089 | 0.066 |
| | | 2 | 5.34 | 24.78 | 37.53 | 40.77 | 0.096 | 0.061 |
| none (clean) | orbit product | 0 | 5.07 | 17.80 | 32.32 | 32.36 | 0.087 | 0.040 |
| | | 1 | 4.91 | 16.79 | 29.66 | 33.63 | 0.085 | 0.061 |
| | | 2 | 4.97 | 17.08 | 30.95 | 30.37 | 0.086 | 0.047 |
| none (clean) | weighted vote (32) | 0 | 4.46 | 20.25 | 33.71 | 37.33 | 0.091 | 0.054 |
| | | 1 | 4.24 | 18.86 | 30.22 | 35.33 | 0.081 | 0.059 |
| | | 2 | 4.38 | 19.49 | 32.77 | 33.92 | 0.089 | 0.048 |
| photometric only | greedy (1 view) | 0 | 4.95 | 12.55 | 23.14 | 23.52 | 0.059 | 0.039 |
| | | 1 | 4.86 | 12.05 | 21.34 | 20.91 | 0.056 | 0.036 |
| | | 2 | 4.91 | 12.40 | 23.92 | 22.49 | 0.065 | 0.036 |
| photometric only | orbit product | 0 | 4.54 | 8.88 | 17.43 | 16.61 | 0.043 | 0.023 |
| | | 1 | 4.46 | 8.50 | 16.95 | 14.01 | 0.044 | 0.019 |
| | | 2 | 4.50 | 8.66 | 18.38 | 15.64 | 0.047 | 0.021 |
| photometric only | weighted vote (32) | 0 | 3.99 | 8.96 | 18.33 | 16.96 | 0.049 | 0.024 |
| | | 1 | 3.88 | 8.42 | 17.18 | 14.27 | 0.046 | 0.020 |
| | | 2 | 3.91 | 8.77 | 18.64 | 16.22 | 0.050 | 0.023 |
| in-plane roll only | greedy (1 view) | 0 | 4.94 | 19.23 | 36.13 | 37.96 | 0.101 | 0.062 |
| | | 1 | 4.84 | 18.80 | 33.76 | 37.34 | 0.095 | 0.063 |
| | | 2 | 4.88 | 19.48 | 35.33 | 37.62 | 0.100 | 0.056 |
| in-plane roll only | orbit product | 0 | 4.12 | 12.59 | 29.51 | 29.62 | 0.088 | 0.050 |
| | | 1 | 4.03 | 12.55 | 27.63 | 26.87 | 0.082 | 0.043 |
| | | 2 | 4.07 | 12.60 | 29.45 | 28.74 | 0.089 | 0.051 |
| in-plane roll only | weighted vote (32) | 0 | 3.60 | 13.80 | 31.50 | 30.88 | 0.095 | 0.054 |
| | | 1 | 3.50 | 14.00 | 29.24 | 28.92 | 0.088 | 0.044 |
| | | 2 | 3.51 | 14.04 | 30.30 | 31.05 | 0.090 | 0.054 |
| both (EXP-109 A) | greedy (1 view) | 0 | 4.89 | 10.05 | 22.37 | 22.75 | 0.055 | 0.031 |
| | | 1 | 4.80 | 9.64 | 21.92 | 21.89 | 0.058 | 0.035 |
| | | 2 | 4.83 | 9.95 | 23.36 | 21.37 | 0.066 | 0.032 |
| both (EXP-109 A) | orbit product | 0 | 4.21 | 6.38 | 17.24 | 15.76 | 0.045 | 0.020 |
| | | 1 | 4.25 | 6.17 | 16.82 | 15.51 | 0.045 | 0.023 |
| | | 2 | 4.17 | 6.19 | 18.15 | 14.55 | 0.052 | 0.018 |
| both (EXP-109 A) | weighted vote (32) | 0 | 3.75 | 6.34 | 18.37 | 16.47 | 0.049 | 0.023 |
| | | 1 | 3.72 | 6.23 | 17.56 | 16.13 | 0.050 | 0.027 |
| | | 2 | 3.66 | 6.35 | 18.89 | 14.88 | 0.055 | 0.020 |

Table 55. **EXP-115 per seed, DINOv3**.

### L.8. EXP-116: roll-consistent tree

**Question.** Can training learn the cross-view consistency that orbit voting enforces at test time, at single-view cost? Arms (per branch, 3 seeds): C1 paired views of the same image, GT loss only (data control); C2 a transported consistency loss on the leaf distribution (a standard equivariance regulariser); T the hierarchical transported loss at every tree level ($\lambda = 1$, beam 8); A the EXP-109 base recipe. Transport maps each view's beam-8 leaves into the other view's frame with the exact roll.

**Pre-registered criteria and outcome.**

- T beats A, C1 and C2 on single-view greedy val-aug (DINOv3 primary, PoE secondary) without hurting clean synth. val: **FAIL**. DINOv3: T $9.57 \pm 0.06$ vs A $9.88 \pm 0.22$, C1 $10.04 \pm 0.21$, C2 $9.43 \pm 0.14$; PoE: T $9.17 \pm 0.12$ vs C2 $9.08 \pm 0.05$. T beats A and C1 but not C2.
- Consistency improves over A and C1 (DINOv3 val-aug cons.@10): **PASS**. T 0.603 vs A 0.554, C1 0.541; C2 is higher still (0.621).
- Real domains: descriptive; differences between A, C2 and T are within the seed std. By the pre-registered rule no hierarchical-transport claim is made.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken | cons.@10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A (base) | greedy (1 view) | Synth. val | $4.85 \pm 0.03$ | 4.36 | 0.968 | 0.001 | 0.000 | – | – | – | – | 0.753 |
| | | Synth. val-aug | $9.75 \pm 0.06$ | 5.92 | 0.825 | 0.018 | 0.009 | – | – | – | – | 0.607 |
| | | Lightbox | $17.33 \pm 0.31$ | 6.75 | 0.730 | 0.058 | 0.042 | – | – | – | – | 0.523 |
| | | Sunlamp | $20.15 \pm 0.57$ | 8.42 | 0.601 | 0.064 | 0.033 | – | – | – | – | 0.426 |
| A (base) | orbit product | Synth. val | $3.87 \pm 0.03$ | 3.44 | 0.982 | 0.000 | 0.000 | 5.3 | 0.849 | 11965.0 | 0.000 | 0.753 |
| | | Synth. val-aug | $5.82 \pm 0.10$ | 3.87 | 0.939 | 0.007 | 0.004 | 110.3 | 0.631 | 11443.3 | 0.000 | 0.607 |
| | | Lightbox | $12.30 \pm 0.50$ | 5.26 | 0.836 | 0.037 | 0.030 | 280.3 | 0.437 | 6013.0 | 0.005 | 0.523 |
| | | Sunlamp | $12.25 \pm 0.37$ | 6.32 | 0.760 | 0.029 | 0.017 | 91.7 | 0.656 | 2353.3 | 0.005 | 0.426 |
| A (base) | weighted vote (32) | Synth. val | $3.46 \pm 0.03$ | 2.97 | 0.985 | 0.000 | 0.000 | 5.3 | 0.802 | 11965.0 | 0.000 | 0.753 |
| | | Synth. val-aug | $5.98 \pm 0.13$ | 3.43 | 0.935 | 0.011 | 0.005 | 110.3 | 0.510 | 11443.3 | 0.001 | 0.607 |
| | | Lightbox | $12.58 \pm 0.42$ | 4.98 | 0.834 | 0.039 | 0.031 | 280.3 | 0.402 | 6013.0 | 0.006 | 0.523 |
| | | Sunlamp | $12.88 \pm 0.49$ | 6.08 | 0.754 | 0.035 | 0.018 | 91.7 | 0.601 | 2353.3 | 0.005 | 0.426 |
| C1 paired views | greedy (1 view) | Synth. val | $4.89 \pm 0.04$ | 4.37 | 0.966 | 0.001 | 0.001 | – | – | – | – | 0.749 |
| | | Synth. val-aug | $10.00 \pm 0.11$ | 5.99 | 0.819 | 0.019 | 0.009 | – | – | – | – | 0.597 |
| | | Lightbox | $17.88 \pm 0.63$ | 6.88 | 0.719 | 0.060 | 0.043 | – | – | – | – | 0.509 |
| | | Sunlamp | $20.34 \pm 0.43$ | 8.55 | 0.594 | 0.066 | 0.033 | – | – | – | – | 0.409 |
| C1 paired views | orbit product | Synth. val | $3.94 \pm 0.02$ | 3.47 | 0.982 | 0.000 | 0.000 | 6.3 | 0.681 | 11959.3 | 0.000 | 0.749 |
| | | Synth. val-aug | $6.04 \pm 0.08$ | 3.91 | 0.932 | 0.008 | 0.004 | 113.7 | 0.572 | 11407.0 | 0.001 | 0.597 |
| | | Lightbox | $12.38 \pm 0.51$ | 5.32 | 0.830 | 0.037 | 0.029 | 290.3 | 0.436 | 5973.7 | 0.005 | 0.509 |
| | | Sunlamp | $12.45 \pm 0.26$ | 6.44 | 0.753 | 0.029 | 0.016 | 92.3 | 0.599 | 2343.0 | 0.004 | 0.409 |
| C1 paired views | weighted vote (32) | Synth. val | $3.48 \pm 0.01$ | 2.98 | 0.984 | 0.000 | 0.000 | 6.3 | 0.681 | 11959.3 | 0.000 | 0.749 |
| | | Synth. val-aug | $6.19 \pm 0.04$ | 3.45 | 0.931 | 0.011 | 0.006 | 113.7 | 0.503 | 11407.0 | 0.001 | 0.597 |
| | | Lightbox | $12.72 \pm 0.61$ | 5.00 | 0.832 | 0.041 | 0.031 | 290.3 | 0.420 | 5973.7 | 0.006 | 0.509 |
| | | Sunlamp | $13.33 \pm 0.47$ | 6.18 | 0.753 | 0.036 | 0.020 | 92.3 | 0.566 | 2343.0 | 0.006 | 0.409 |
| C2 leaf consistency | greedy (1 view) | Synth. val | $4.52 \pm 0.04$ | 4.11 | 0.981 | 0.001 | 0.000 | – | – | – | – | 0.794 |
| | | Synth. val-aug | $9.08 \pm 0.05$ | 5.58 | 0.856 | 0.017 | 0.009 | – | – | – | – | 0.659 |
| | | Lightbox | $17.28 \pm 1.12$ | 6.45 | 0.750 | 0.061 | 0.043 | – | – | – | – | 0.564 |
| | | Sunlamp | $18.85 \pm 0.60$ | 8.02 | 0.626 | 0.061 | 0.031 | – | – | – | – | 0.471 |
| C2 leaf consistency | orbit product | Synth. val | $3.71 \pm 0.00$ | 3.32 | 0.987 | 0.000 | 0.000 | 3.0 | 0.333 | 11974.0 | 0.000 | 0.794 |
| | | Synth. val-aug | $5.39 \pm 0.06$ | 3.63 | 0.949 | 0.007 | 0.003 | 106.0 | 0.581 | 11529.7 | 0.000 | 0.659 |
| | | Lightbox | $12.41 \pm 0.93$ | 5.07 | 0.850 | 0.040 | 0.032 | 293.0 | 0.410 | 6037.7 | 0.005 | 0.564 |
| | | Sunlamp | $11.74 \pm 0.18$ | 6.04 | 0.783 | 0.028 | 0.016 | 85.3 | 0.624 | 2403.0 | 0.004 | 0.471 |
| C2 leaf consistency | weighted vote (32) | Synth. val | $3.29 \pm 0.02$ | 2.82 | 0.988 | 0.000 | 0.000 | 3.0 | 0.417 | 11974.0 | 0.000 | 0.794 |
| | | Synth. val-aug | $5.53 \pm 0.18$ | 3.18 | 0.942 | 0.010 | 0.005 | 106.0 | 0.507 | 11529.7 | 0.001 | 0.659 |
| | | Lightbox | $12.63 \pm 1.19$ | 4.78 | 0.847 | 0.042 | 0.033 | 293.0 | 0.395 | 6037.7 | 0.006 | 0.564 |
| | | Sunlamp | $12.27 \pm 0.38$ | 5.89 | 0.769 | 0.033 | 0.016 | 85.3 | 0.588 | 2403.0 | 0.004 | 0.471 |
| T hierarchical | greedy (1 view) | Synth. val | $4.56 \pm 0.02$ | 4.13 | 0.981 | 0.001 | 0.000 | – | – | – | – | 0.788 |
| | | Synth. val-aug | $9.17 \pm 0.12$ | 5.66 | 0.851 | 0.017 | 0.008 | – | – | – | – | 0.646 |
| | | Lightbox | $17.39 \pm 0.70$ | 6.58 | 0.744 | 0.061 | 0.044 | – | – | – | – | 0.553 |
| | | Sunlamp | $19.01 \pm 0.64$ | 8.26 | 0.616 | 0.061 | 0.029 | – | – | – | – | 0.455 |
| T hierarchical | orbit product | Synth. val | $3.74 \pm 0.03$ | 3.34 | 0.985 | 0.000 | 0.000 | 4.3 | 0.528 | 11974.0 | 0.000 | 0.788 |
| | | Synth. val-aug | $5.59 \pm 0.03$ | 3.71 | 0.945 | 0.007 | 0.004 | 101.7 | 0.626 | 11521.3 | 0.000 | 0.646 |
| | | Lightbox | $12.49 \pm 0.59$ | 5.19 | 0.842 | 0.039 | 0.032 | 297.3 | 0.421 | 6038.7 | 0.005 | 0.553 |
| | | Sunlamp | $11.73 \pm 0.80$ | 6.20 | 0.772 | 0.026 | 0.014 | 81.0 | 0.554 | 2393.3 | 0.003 | 0.455 |
| T hierarchical | weighted vote (32) | Synth. val | $3.29 \pm 0.01$ | 2.83 | 0.987 | 0.000 | 0.000 | 4.3 | 0.583 | 11974.0 | 0.000 | 0.788 |
| | | Synth. val-aug | $5.65 \pm 0.08$ | 3.23 | 0.939 | 0.010 | 0.005 | 101.7 | 0.563 | 11521.3 | 0.001 | 0.646 |
| | | Lightbox | $12.55 \pm 0.88$ | 4.86 | 0.841 | 0.040 | 0.032 | 297.3 | 0.406 | 6038.7 | 0.006 | 0.553 |
| | | Sunlamp | $12.67 \pm 0.72$ | 6.09 | 0.755 | 0.033 | 0.017 | 81.0 | 0.499 | 2393.3 | 0.004 | 0.455 |

Table 56. **EXP-116, PoE**: training-time consistency arms. Flip transitions relative to the same arm's greedy prediction; cons.@10 as in Tab. 38.

| System | Inference | Domain | mean | median | acc@10 | >90 | >150 | $n_{>150}$ | fixed | $n_{<20}$ | broken | cons.@10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A (base) | greedy (1 view) | Synth. val | $4.84 \pm 0.05$ | 4.43 | 0.972 | 0.000 | 0.000 | – | – | – | – | 0.701 |
| | | Synth. val-aug | $9.88 \pm 0.22$ | 6.13 | 0.816 | 0.018 | 0.010 | – | – | – | – | 0.554 |
| | | Lightbox | $22.55 \pm 0.73$ | 8.38 | 0.607 | 0.083 | 0.060 | – | – | – | – | 0.391 |
| | | Sunlamp | $22.00 \pm 0.70$ | 10.17 | 0.489 | 0.067 | 0.032 | – | – | – | – | 0.295 |
| A (base) | orbit product | Synth. val | $4.21 \pm 0.04$ | 3.72 | 0.974 | 0.000 | 0.000 | 2.7 | 0.667 | 11977.7 | 0.000 | 0.701 |
| | | Synth. val-aug | $6.25 \pm 0.12$ | 4.20 | 0.920 | 0.007 | 0.004 | 117.0 | 0.586 | 11445.0 | 0.000 | 0.554 |
| | | Lightbox | $17.40 \pm 0.68$ | 6.88 | 0.708 | 0.058 | 0.048 | 402.7 | 0.341 | 5676.7 | 0.007 | 0.391 |
| | | Sunlamp | $15.27 \pm 0.64$ | 8.19 | 0.626 | 0.036 | 0.020 | 90.3 | 0.529 | 2236.3 | 0.005 | 0.295 |
| A (base) | weighted vote (32) | Synth. val | $3.71 \pm 0.05$ | 3.20 | 0.983 | 0.001 | 0.000 | 2.7 | 0.583 | 11977.7 | 0.000 | 0.701 |
| | | Synth. val-aug | $6.30 \pm 0.06$ | 3.68 | 0.926 | 0.010 | 0.006 | 117.0 | 0.479 | 11445.0 | 0.001 | 0.554 |
| | | Lightbox | $18.27 \pm 0.67$ | 6.59 | 0.718 | 0.065 | 0.051 | 402.7 | 0.299 | 5676.7 | 0.009 | 0.391 |
| | | Sunlamp | $15.83 \pm 0.83$ | 7.89 | 0.636 | 0.042 | 0.023 | 90.3 | 0.467 | 2236.3 | 0.007 | 0.295 |
| C1 paired views | greedy (1 view) | Synth. val | $4.96 \pm 0.06$ | 4.52 | 0.970 | 0.001 | 0.000 | – | – | – | – | 0.692 |
| | | Synth. val-aug | $10.04 \pm 0.21$ | 6.24 | 0.804 | 0.018 | 0.009 | – | – | – | – | 0.541 |
| | | Lightbox | $23.75 \pm 1.61$ | 8.62 | 0.590 | 0.088 | 0.064 | – | – | – | – | 0.370 |
| | | Sunlamp | $22.72 \pm 0.97$ | 10.65 | 0.464 | 0.069 | 0.034 | – | – | – | – | 0.272 |
| C1 paired views | orbit product | Synth. val | $4.34 \pm 0.00$ | 3.81 | 0.970 | 0.000 | 0.000 | 4.3 | 0.528 | 11973.0 | 0.000 | 0.692 |
| | | Synth. val-aug | $6.52 \pm 0.19$ | 4.35 | 0.913 | 0.008 | 0.004 | 109.3 | 0.531 | 11424.0 | 0.000 | 0.541 |
| | | Lightbox | $18.47 \pm 0.88$ | 7.14 | 0.687 | 0.064 | 0.051 | 430.3 | 0.320 | 5603.0 | 0.007 | 0.370 |
| | | Sunlamp | $15.71 \pm 0.11$ | 8.49 | 0.602 | 0.036 | 0.022 | 94.0 | 0.448 | 2189.3 | 0.005 | 0.272 |
| C1 paired views | weighted vote (32) | Synth. val | $3.75 \pm 0.02$ | 3.24 | 0.981 | 0.000 | 0.000 | 4.3 | 0.639 | 11973.0 | 0.000 | 0.692 |
| | | Synth. val-aug | $6.53 \pm 0.20$ | 3.73 | 0.921 | 0.012 | 0.006 | 109.3 | 0.448 | 11424.0 | 0.001 | 0.541 |
| | | Lightbox | $19.16 \pm 1.06$ | 6.67 | 0.709 | 0.070 | 0.056 | 430.3 | 0.273 | 5603.0 | 0.009 | 0.370 |
| | | Sunlamp | $16.46 \pm 0.71$ | 8.17 | 0.612 | 0.044 | 0.023 | 94.0 | 0.392 | 2189.3 | 0.006 | 0.272 |
| C2 leaf consistency | greedy (1 view) | Synth. val | $4.59 \pm 0.04$ | 4.22 | 0.982 | 0.001 | 0.000 | – | – | – | – | 0.771 |
| | | Synth. val-aug | $9.43 \pm 0.14$ | 5.84 | 0.836 | 0.017 | 0.009 | – | – | – | – | 0.621 |
| | | Lightbox | $22.88 \pm 1.42$ | 8.13 | 0.626 | 0.087 | 0.062 | – | – | – | – | 0.442 |
| | | Sunlamp | $21.64 \pm 0.69$ | 9.95 | 0.504 | 0.068 | 0.032 | – | – | – | – | 0.349 |
| C2 leaf consistency | orbit product | Synth. val | $3.92 \pm 0.01$ | 3.48 | 0.983 | 0.000 | 0.000 | 3.0 | 0.533 | 11974.7 | 0.000 | 0.771 |
| | | Synth. val-aug | $5.66 \pm 0.05$ | 3.81 | 0.940 | 0.007 | 0.003 | 107.0 | 0.613 | 11498.7 | 0.000 | 0.621 |
| | | Lightbox | $17.40 \pm 1.36$ | 6.57 | 0.739 | 0.062 | 0.049 | 419.3 | 0.343 | 5687.0 | 0.007 | 0.442 |
| | | Sunlamp | $14.47 \pm 0.47$ | 7.75 | 0.659 | 0.035 | 0.019 | 88.3 | 0.540 | 2255.3 | 0.005 | 0.349 |
| C2 leaf consistency | weighted vote (32) | Synth. val | $3.40 \pm 0.02$ | 2.93 | 0.988 | 0.000 | 0.000 | 3.0 | 0.700 | 11974.7 | 0.000 | 0.771 |
| | | Synth. val-aug | $5.77 \pm 0.11$ | 3.31 | 0.934 | 0.010 | 0.005 | 107.0 | 0.509 | 11498.7 | 0.001 | 0.621 |
| | | Lightbox | $17.64 \pm 1.56$ | 6.32 | 0.737 | 0.064 | 0.050 | 419.3 | 0.317 | 5687.0 | 0.008 | 0.442 |
| | | Sunlamp | $15.29 \pm 0.14$ | 7.65 | 0.650 | 0.040 | 0.022 | 88.3 | 0.517 | 2255.3 | 0.006 | 0.349 |
| T hierarchical | greedy (1 view) | Synth. val | $4.65 \pm 0.03$ | 4.27 | 0.979 | 0.000 | 0.000 | – | – | – | – | 0.761 |
| | | Synth. val-aug | $9.57 \pm 0.06$ | 5.94 | 0.829 | 0.017 | 0.009 | – | – | – | – | 0.603 |
| | | Lightbox | $22.84 \pm 1.31$ | 8.32 | 0.611 | 0.085 | 0.061 | – | – | – | – | 0.427 |
| | | Sunlamp | $22.20 \pm 0.58$ | 10.27 | 0.485 | 0.068 | 0.033 | – | – | – | – | 0.329 |
| T hierarchical | orbit product | Synth. val | $4.02 \pm 0.04$ | 3.57 | 0.981 | 0.000 | 0.000 | 2.7 | 0.417 | 11977.0 | 0.000 | 0.761 |
| | | Synth. val-aug | $5.91 \pm 0.02$ | 4.00 | 0.932 | 0.007 | 0.003 | 104.0 | 0.618 | 11471.7 | 0.000 | 0.603 |
| | | Lightbox | $17.55 \pm 0.66$ | 6.81 | 0.717 | 0.061 | 0.049 | 413.0 | 0.329 | 5681.7 | 0.007 | 0.427 |
| | | Sunlamp | $14.66 \pm 0.68$ | 8.03 | 0.634 | 0.033 | 0.017 | 93.3 | 0.490 | 2218.7 | 0.005 | 0.329 |
| T hierarchical | weighted vote (32) | Synth. val | $3.45 \pm 0.04$ | 2.98 | 0.987 | 0.000 | 0.000 | 2.7 | 0.250 | 11977.0 | 0.000 | 0.761 |
| | | Synth. val-aug | $5.95 \pm 0.05$ | 3.42 | 0.930 | 0.010 | 0.005 | 104.0 | 0.545 | 11471.7 | 0.001 | 0.603 |
| | | Lightbox | $17.84 \pm 0.80$ | 6.52 | 0.726 | 0.063 | 0.050 | 413.0 | 0.310 | 5681.7 | 0.008 | 0.427 |
| | | Sunlamp | $15.72 \pm 1.03$ | 7.86 | 0.633 | 0.041 | 0.021 | 93.3 | 0.443 | 2218.7 | 0.008 | 0.329 |

Table 57. **EXP-116, DINOv3**: training-time consistency arms. Flip transitions relative to the same arm's greedy prediction; cons.@10 as in Tab. 38.

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| A (base) | greedy (1 view) | 0 | 4.89 | 9.82 | 17.44 | 20.59 | 0.040 | 0.030 |
| | | 1 | 4.84 | 9.70 | 16.98 | 20.35 | 0.039 | 0.034 |
| | | 2 | 4.84 | 9.72 | 17.58 | 19.52 | 0.046 | 0.034 |
| A (base) | orbit product | 0 | 3.87 | 5.93 | 12.18 | 12.50 | 0.028 | 0.016 |
| | | 1 | 3.90 | 5.75 | 11.87 | 12.43 | 0.028 | 0.019 |
| | | 2 | 3.84 | 5.78 | 12.85 | 11.83 | 0.033 | 0.015 |
| A (base) | weighted vote (32) | 0 | 3.48 | 6.09 | 12.42 | 13.45 | 0.029 | 0.019 |
| | | 1 | 3.48 | 5.84 | 12.26 | 12.64 | 0.030 | 0.019 |
| | | 2 | 3.43 | 6.02 | 13.06 | 12.56 | 0.034 | 0.017 |
| C1 paired views | greedy (1 view) | 0 | 4.93 | 10.10 | 17.91 | 20.69 | 0.041 | 0.031 |
| | | 1 | 4.89 | 10.00 | 17.23 | 19.86 | 0.040 | 0.034 |
| | | 2 | 4.85 | 9.89 | 18.49 | 20.45 | 0.048 | 0.034 |
| C1 paired views | orbit product | 0 | 3.92 | 6.14 | 12.56 | 12.16 | 0.029 | 0.014 |
| | | 1 | 3.95 | 5.98 | 11.81 | 12.56 | 0.027 | 0.016 |
| | | 2 | 3.96 | 6.02 | 12.78 | 12.64 | 0.031 | 0.018 |
| C1 paired views | weighted vote (32) | 0 | 3.48 | 6.23 | 12.80 | 13.04 | 0.031 | 0.019 |
| | | 1 | 3.47 | 6.16 | 12.08 | 13.07 | 0.028 | 0.020 |
| | | 2 | 3.48 | 6.18 | 13.29 | 13.88 | 0.034 | 0.022 |
| C2 leaf consistency | greedy (1 view) | 0 | 4.55 | 9.12 | 17.81 | 18.62 | 0.044 | 0.028 |
| | | 1 | 4.48 | 9.02 | 15.99 | 18.40 | 0.039 | 0.031 |
| | | 2 | 4.51 | 9.12 | 18.05 | 19.53 | 0.048 | 0.033 |
| C2 leaf consistency | orbit product | 0 | 3.71 | 5.46 | 12.76 | 11.86 | 0.033 | 0.015 |
| | | 1 | 3.71 | 5.36 | 11.35 | 11.53 | 0.026 | 0.015 |
| | | 2 | 3.71 | 5.36 | 13.11 | 11.83 | 0.036 | 0.016 |
| C2 leaf consistency | weighted vote (32) | 0 | 3.29 | 5.71 | 13.21 | 12.31 | 0.035 | 0.016 |
| | | 1 | 3.26 | 5.51 | 11.26 | 11.87 | 0.027 | 0.015 |
| | | 2 | 3.30 | 5.35 | 13.40 | 12.63 | 0.037 | 0.018 |
| T hierarchical | greedy (1 view) | 0 | 4.58 | 9.25 | 18.12 | 19.24 | 0.047 | 0.030 |
| | | 1 | 4.55 | 9.04 | 16.72 | 18.29 | 0.040 | 0.027 |
| | | 2 | 4.54 | 9.24 | 17.32 | 19.50 | 0.045 | 0.030 |
| T hierarchical | orbit product | 0 | 3.78 | 5.59 | 13.16 | 12.47 | 0.035 | 0.018 |
| | | 1 | 3.74 | 5.56 | 12.12 | 10.88 | 0.029 | 0.010 |
| | | 2 | 3.71 | 5.62 | 12.18 | 11.84 | 0.031 | 0.014 |
| T hierarchical | weighted vote (32) | 0 | 3.30 | 5.73 | 13.52 | 13.37 | 0.037 | 0.020 |
| | | 1 | 3.30 | 5.57 | 11.78 | 11.93 | 0.028 | 0.015 |
| | | 2 | 3.28 | 5.63 | 12.36 | 12.70 | 0.032 | 0.015 |

Table 58. **EXP-116 per seed, PoE**.

| System | Inference | Seed | Synth. val | Synth. val-aug | Lightbox | Sunlamp | >150 LB | >150 SL |
|---|---|---|---|---|---|---|---|---|
| A (base) | greedy (1 view) | 0 | 4.89 | 10.05 | 22.37 | 22.75 | 0.055 | 0.031 |
| | | 1 | 4.80 | 9.64 | 21.92 | 21.89 | 0.058 | 0.035 |
| | | 2 | 4.83 | 9.95 | 23.36 | 21.37 | 0.066 | 0.032 |
| A (base) | orbit product | 0 | 4.21 | 6.38 | 17.24 | 15.76 | 0.045 | 0.020 |
| | | 1 | 4.25 | 6.17 | 16.82 | 15.51 | 0.045 | 0.023 |
| | | 2 | 4.17 | 6.19 | 18.15 | 14.55 | 0.052 | 0.018 |
| A (base) | weighted vote (32) | 0 | 3.75 | 6.34 | 18.37 | 16.47 | 0.049 | 0.023 |
| | | 1 | 3.72 | 6.23 | 17.56 | 16.13 | 0.050 | 0.027 |
| | | 2 | 3.66 | 6.35 | 18.89 | 14.88 | 0.055 | 0.020 |
| C1 paired views | greedy (1 view) | 0 | 5.02 | 10.27 | 24.54 | 23.52 | 0.064 | 0.036 |
| | | 1 | 4.90 | 9.88 | 21.89 | 21.64 | 0.056 | 0.030 |
| | | 2 | 4.94 | 9.98 | 24.81 | 23.01 | 0.071 | 0.035 |
| C1 paired views | orbit product | 0 | 4.34 | 6.73 | 18.44 | 15.58 | 0.049 | 0.020 |
| | | 1 | 4.34 | 6.36 | 17.61 | 15.80 | 0.047 | 0.024 |
| | | 2 | 4.34 | 6.47 | 19.37 | 15.76 | 0.057 | 0.023 |
| C1 paired views | weighted vote (32) | 0 | 3.73 | 6.76 | 19.20 | 16.45 | 0.054 | 0.021 |
| | | 1 | 3.77 | 6.40 | 18.07 | 15.76 | 0.051 | 0.021 |
| | | 2 | 3.75 | 6.42 | 20.19 | 17.18 | 0.063 | 0.027 |
| C2 leaf consistency | greedy (1 view) | 0 | 4.63 | 9.35 | 23.59 | 21.73 | 0.065 | 0.033 |
| | | 1 | 4.60 | 9.34 | 21.25 | 20.91 | 0.055 | 0.029 |
| | | 2 | 4.55 | 9.59 | 23.79 | 22.27 | 0.066 | 0.032 |
| C2 leaf consistency | orbit product | 0 | 3.92 | 5.67 | 17.90 | 14.60 | 0.051 | 0.021 |
| | | 1 | 3.91 | 5.61 | 15.86 | 13.95 | 0.042 | 0.017 |
| | | 2 | 3.92 | 5.71 | 18.44 | 14.86 | 0.054 | 0.020 |
| C2 leaf consistency | weighted vote (32) | 0 | 3.42 | 5.89 | 18.37 | 15.45 | 0.052 | 0.023 |
| | | 1 | 3.41 | 5.68 | 15.86 | 15.18 | 0.041 | 0.021 |
| | | 2 | 3.39 | 5.73 | 18.70 | 15.25 | 0.055 | 0.022 |
| T hierarchical | greedy (1 view) | 0 | 4.68 | 9.61 | 24.04 | 22.77 | 0.066 | 0.035 |
| | | 1 | 4.62 | 9.50 | 21.44 | 21.60 | 0.055 | 0.032 |
| | | 2 | 4.65 | 9.60 | 23.05 | 22.24 | 0.063 | 0.033 |
| T hierarchical | orbit product | 0 | 4.06 | 5.93 | 18.24 | 15.38 | 0.051 | 0.020 |
| | | 1 | 4.01 | 5.92 | 16.93 | 14.04 | 0.046 | 0.015 |
| | | 2 | 3.98 | 5.89 | 17.48 | 14.54 | 0.049 | 0.016 |
| T hierarchical | weighted vote (32) | 0 | 3.49 | 5.93 | 18.65 | 16.86 | 0.054 | 0.028 |
| | | 1 | 3.44 | 6.01 | 17.04 | 14.85 | 0.044 | 0.019 |
| | | 2 | 3.42 | 5.90 | 17.84 | 15.46 | 0.051 | 0.018 |

Table 59. **EXP-116 per seed, DINOv3**.

## L.9. EXP-104: flip audit

**Question.** Do the $\sim$180° real-domain errors of the clean trees come from a bug (encoder/decoder, metric, rotation convention, augmentation), from chart seams, from hierarchical routing, or from visual near-symmetry of the spacecraft? Analysis only (no training, no selection), on the frozen EXP-036 seed-0 greedy predictions of each branch.

**Pre-registered criteria and outcome.** This audit was a user-requested side analysis with no pass/fail criterion; each candidate cause has a test and a verdict. Encoder/decoder, metric, convention and augmentation bugs are ruled out; chart seams and hierarchical routing are not the main cause (flat Hopf heads on the same features flip as often or more: $P(>150° \mid >90°) = 0.42$–$0.58$ vs. 0.35–0.40 for the tree on the same subset); visual near-symmetry is the main cause. For uniformly random wrong rotations about 20% of errors above 90° fall in 165–180°; the observed share is 36–41%. Among the tree's top-8 beam paths, one lies within 20° of the ground truth for 86% of lightbox images.

**Decoder round trip.** Over all ground-truth rotations, decoding the encoded path gives a maximum error of 5.78° (NumPy decoder) and 5.78° (GPU decoder used at inference), mean 2.51° (the depth-5 cell-centre quantisation); manifest paths match the encoder for a fraction 1.000 of rows, and $q$ and $-q$ encode identically (yes).

| | DINOv3 | | | MoGe-2 | | |
|---|---|---|---|---|---|---|
| Quantity | Synth. | LB | SL | Synth. | LB | SL |
| images | 11994 | 6740 | 2791 | 11994 | 6740 | 2791 |
| fraction $>150°$ | 0.001 | 0.093 | 0.058 | 0.031 | 0.114 | 0.187 |
| mean error (°) | 5.45 | 38.88 | 42.59 | 16.81 | 43.41 | 60.28 |
| mean error, best 180° body flip allowed | 5.36 | 23.51 | 30.71 | 11.62 | 25.24 | 31.27 |
| $>150$ failures within 20° after 180° about body $x$ | 0.000 | 0.005 | 0.000 | 0.024 | 0.032 | 0.042 |
| ... about body $y$ | 0.000 | 0.169 | 0.025 | 0.111 | 0.116 | 0.141 |
| ... about body $z$ | 0.500 | 0.118 | 0.172 | 0.300 | 0.226 | 0.218 |
| ... about any body axis | 0.500 | 0.292 | 0.196 | 0.435 | 0.374 | 0.402 |
| $>150$ failures fixed by transposing the prediction | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| max \|quaternion − matrix geodesic\| (°) | 3.5e-05 | 8.7e-06 | 3.9e-06 | 5.2e-05 | 1.3e-05 | 8.5e-06 |
| max \|error$(-\hat{q})$ − error$(\hat{q})$\| (°) | 0.0e+00 | 0.0e+00 | 0.0e+00 | 0.0e+00 | 0.0e+00 | 0.0e+00 |
| failures first diverging from GT path at the root | 0.833 | 0.986 | 0.982 | 0.936 | 0.943 | 0.935 |
| ... at level 1 | 0.167 | 0.014 | 0.018 | 0.064 | 0.057 | 0.065 |
| median GT chart margin, failures | 0.115 | 0.312 | 0.235 | 0.179 | 0.199 | 0.222 |
| median GT chart margin, all images | 0.257 | 0.258 | 0.269 | 0.257 | 0.258 | 0.269 |
| chart margin $<0.05$, failures | 0.333 | 0.101 | 0.166 | 0.188 | 0.148 | 0.159 |
| chart margin $<0.05$, all images | 0.115 | 0.113 | 0.113 | 0.115 | 0.113 | 0.113 |

Table 60. **EXP-104 flip audit**, clean EXP-036 trees, seed 0, greedy, per branch. Failures: errors $>150°$ (very few on synth. val for DINOv3, so its failure fractions there are noisy). Chart margin: difference between the largest and second-largest absolute quaternion components of the GT rotation.

## L.10. EXP-064: T-LESS, SYMSOL-T and YCB-V (incl. zero-shot)

**Question.** Do the locked recipe and its components (orbit voting, MoGe-2 fusion) carry over to other objects when per-object trees are retrained with the unchanged recipe? Datasets: T-LESS [10] (7 objects, PBR validation scenes 45–49, real BOP19 test targets), SYMSOL-T (5 shapes, LiePose split [13], 20k training images per shape, 2k held-out validation, test-5k) and YCB-V [25] (6 objects, real training videos only, held-out sequences for validation, keyframes for test). T-LESS and SYMSOL-T use the symmetry-aware error (continuous symmetries discretised in 10° steps); YCB-V uses the raw error (no symmetry metadata in this release). These are the single-label (A0) trees. A zero-shot arm evaluates the frozen SPEED+ trees without retraining.

**Pre-registered criteria and outcome.** Per dataset, fusion helps if the object-balanced PoE validation mean is below DINOv3-only by more than the larger seed std; test is reported for every arm. Zero-shot transfer is claimed only if the frame-free pair error is below its uniform-chance level by more than the seed std.

- T-LESS: **PASS**. PoE vs DINOv3 validation, greedy $12.30 \pm 0.13$ vs $12.67 \pm 0.06$; orbit $9.25 \pm 0.07$ vs $9.94 \pm 0.06$.
- SYMSOL-T: **FAIL**. PoE vs DINOv3 validation, greedy $3.87 \pm 0.03$ vs $3.78 \pm 0.03$; orbit $2.65 \pm 0.01$ vs $2.61 \pm 0.00$.
- YCB-V: **FAIL**. PoE vs DINOv3 validation, greedy $73.52 \pm 6.81$ vs $64.32 \pm 11.63$; orbit $74.06 \pm 7.43$ vs $62.99 \pm 2.40$.
- Zero-shot: the pre-set rule is literally met on T-LESS and SYMSOL-T, but a post-hoc constant-prediction control (one rotation for every image, same pairs) is better than the frozen trees on T-LESS (88.7°) and YCB-V (57.8°); on SYMSOL-T the control equals chance (53.8°) and the signal comes from the cube only. No general zero-shot transfer is claimed (Tab. 18).

**Runs.** The first SYMSOL-T evaluation (RUN-20261019-165420) is superseded: nine MoGe-2 trees (cube, cylinder, icosahedron) had trained to NaN and were retrained with a standard-deviation floor; only the corrected run is reported. The YCB-V run is the re-run after the manifest path fix.

**T-LESS.**

| Split | Feat. | Inference | mean | median | acc@2 | acc@5 | acc@10 | acc@15 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|---|
| validation | DINOv3 | greedy (1 view) | $12.67 \pm 0.06$ | 8.07 | 0.022 | 0.242 | 0.660 | 0.822 | 0.939 | 0.016 | 43.01 |
| | DINOv3 | orbit product | $9.94 \pm 0.06$ | 6.45 | 0.039 | 0.361 | 0.766 | 0.884 | 0.963 | 0.011 | 37.91 |
| | DINOv3 | weighted vote (32) | $10.10 \pm 0.12$ | 6.15 | 0.046 | 0.389 | 0.769 | 0.878 | 0.958 | 0.012 | 38.70 |
| | PoE | greedy (1 view) | $12.30 \pm 0.13$ | 7.37 | 0.029 | 0.297 | 0.702 | 0.839 | 0.938 | 0.018 | 43.22 |
| | PoE | orbit product | $9.25 \pm 0.07$ | 5.80 | 0.051 | 0.442 | 0.808 | 0.900 | 0.963 | 0.011 | 37.84 |
| | PoE | weighted vote (32) | $9.65 \pm 0.11$ | 5.61 | 0.059 | 0.457 | 0.805 | 0.890 | 0.957 | 0.013 | 38.60 |
| test | DINOv3 | greedy (1 view) | $18.59 \pm 0.18$ | 8.43 | 0.016 | 0.206 | 0.620 | 0.781 | 0.884 | 0.052 | 47.11 |
| | DINOv3 | orbit product | $17.88 \pm 0.35$ | 7.42 | 0.026 | 0.278 | 0.674 | 0.794 | 0.887 | 0.053 | 43.97 |
| | DINOv3 | weighted vote (32) | $17.90 \pm 0.47$ | 7.10 | 0.033 | 0.321 | 0.687 | 0.797 | 0.883 | 0.054 | 44.59 |
| | PoE | greedy (1 view) | $17.92 \pm 0.16$ | 7.38 | 0.023 | 0.270 | 0.681 | 0.805 | 0.887 | 0.055 | 47.24 |
| | PoE | orbit product | $16.57 \pm 0.25$ | 6.33 | 0.040 | 0.360 | 0.735 | 0.826 | 0.893 | 0.051 | 43.65 |
| | PoE | weighted vote (32) | $16.64 \pm 0.34$ | 6.08 | 0.053 | 0.399 | 0.736 | 0.821 | 0.889 | 0.053 | 44.38 |

Table 61. **EXP-064 T-LESS (A0), object-balanced** (RUN-20261019-063238; symmetry-aware error). acc@2/5/15/30: per-object seed means averaged over objects (as acc@10). raw: error without symmetry handling.

| | DINOv3 | | | PoE | | |
|---|---|---|---|---|---|---|
| Object | greedy | orbit prod. | vote (32) | greedy | orbit prod. | vote (32) |
| *validation* | | | | | | |
| obj. 1 | $6.92 \pm 0.13$ | $5.88 \pm 0.14$ | $6.54 \pm 0.13$ | $6.41 \pm 0.16$ | $5.36 \pm 0.05$ | $6.07 \pm 0.16$ |
| obj. 4 | $7.19 \pm 0.14$ | $6.04 \pm 0.05$ | $6.17 \pm 0.04$ | $6.70 \pm 0.13$ | $5.42 \pm 0.03$ | $5.80 \pm 0.05$ |
| obj. 5 | $9.91 \pm 0.13$ | $7.74 \pm 0.26$ | $7.40 \pm 0.28$ | $9.63 \pm 0.07$ | $7.04 \pm 0.12$ | $7.07 \pm 0.06$ |
| obj. 11 | $12.70 \pm 0.21$ | $10.35 \pm 0.21$ | $9.94 \pm 0.36$ | $12.36 \pm 0.04$ | $9.49 \pm 0.07$ | $9.48 \pm 0.12$ |
| obj. 18 | $22.25 \pm 0.57$ | $17.04 \pm 0.35$ | $17.53 \pm 0.14$ | $22.61 \pm 0.23$ | $16.81 \pm 0.39$ | $17.48 \pm 0.11$ |
| obj. 21 | $17.64 \pm 0.32$ | $13.60 \pm 0.55$ | $13.53 \pm 0.25$ | $18.15 \pm 0.82$ | $13.33 \pm 0.49$ | $13.86 \pm 0.75$ |
| obj. 27 | $12.09 \pm 0.40$ | $8.92 \pm 0.19$ | $9.60 \pm 0.32$ | $10.27 \pm 0.16$ | $7.31 \pm 0.06$ | $7.78 \pm 0.06$ |
| *test* | | | | | | |
| obj. 1 | $24.82 \pm 0.14$ | $25.00 \pm 0.49$ | $25.60 \pm 0.28$ | $24.19 \pm 0.16$ | $23.82 \pm 0.37$ | $24.22 \pm 0.41$ |
| obj. 4 | $17.27 \pm 0.87$ | $15.69 \pm 0.73$ | $15.70 \pm 0.50$ | $15.77 \pm 0.58$ | $14.50 \pm 0.53$ | $14.54 \pm 0.14$ |
| obj. 5 | $8.29 \pm 0.61$ | $7.07 \pm 0.23$ | $6.77 \pm 0.20$ | $8.16 \pm 0.48$ | $6.22 \pm 0.26$ | $5.92 \pm 0.52$ |
| obj. 11 | $14.26 \pm 1.00$ | $12.92 \pm 0.67$ | $12.80 \pm 0.67$ | $13.26 \pm 0.80$ | $11.90 \pm 0.57$ | $11.73 \pm 0.64$ |
| obj. 18 | $16.31 \pm 0.46$ | $14.78 \pm 1.32$ | $15.35 \pm 2.55$ | $15.88 \pm 1.43$ | $14.38 \pm 1.02$ | $16.10 \pm 1.67$ |
| obj. 21 | $36.14 \pm 2.65$ | $37.24 \pm 3.79$ | $35.24 \pm 2.19$ | $35.46 \pm 2.65$ | $34.78 \pm 3.00$ | $33.06 \pm 1.41$ |
| obj. 27 | $13.06 \pm 1.27$ | $12.43 \pm 0.85$ | $13.85 \pm 1.45$ | $12.74 \pm 2.39$ | $10.36 \pm 0.57$ | $10.94 \pm 1.36$ |

Table 62. **EXP-064 T-LESS (A0), per object**: mean symmetry-aware error $\pm$ seed std for every variant and inference mode.

| System | Object | $n$ | mean | median | acc@2 | acc@5 | acc@10 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|
| DINOv3, greedy (1 view) | obj. 1 | 886 | 24.82 | 9.62 | 0.013 | 0.162 | 0.515 | 0.790 | 0.083 | 83.79 |
| | obj. 4 | 646 | 17.27 | 8.74 | 0.010 | 0.167 | 0.593 | 0.883 | 0.038 | 59.59 |
| | obj. 5 | 190 | 8.29 | 6.23 | 0.028 | 0.339 | 0.826 | 0.979 | 0.004 | 19.80 |
| | obj. 11 | 183 | 14.26 | 7.56 | 0.024 | 0.262 | 0.663 | 0.907 | 0.020 | 22.65 |
| | obj. 18 | 146 | 16.31 | 8.26 | 0.009 | 0.221 | 0.619 | 0.897 | 0.030 | 16.31 |
| | obj. 21 | 198 | 36.14 | 9.88 | 0.013 | 0.133 | 0.503 | 0.781 | 0.167 | 36.14 |
| | obj. 27 | 96 | 13.06 | 8.72 | 0.017 | 0.156 | 0.618 | 0.948 | 0.021 | 91.52 |
| DINOv3, orbit product | obj. 1 | 886 | 25.00 | 8.49 | 0.016 | 0.208 | 0.572 | 0.783 | 0.091 | 82.65 |
| | obj. 4 | 646 | 15.69 | 7.76 | 0.012 | 0.204 | 0.656 | 0.898 | 0.037 | 52.99 |
| | obj. 5 | 190 | 7.07 | 5.52 | 0.056 | 0.433 | 0.867 | 0.982 | 0.000 | 16.34 |
| | obj. 11 | 183 | 12.92 | 6.97 | 0.031 | 0.319 | 0.698 | 0.925 | 0.020 | 20.99 |
| | obj. 18 | 146 | 14.78 | 7.98 | 0.023 | 0.269 | 0.632 | 0.906 | 0.021 | 14.78 |
| | obj. 21 | 198 | 37.24 | 8.41 | 0.022 | 0.251 | 0.567 | 0.766 | 0.175 | 37.24 |
| | obj. 27 | 96 | 12.43 | 6.83 | 0.024 | 0.260 | 0.729 | 0.944 | 0.031 | 82.82 |
| PoE, greedy (1 view) | obj. 1 | 886 | 24.19 | 7.99 | 0.020 | 0.225 | 0.594 | 0.790 | 0.089 | 83.76 |
| | obj. 4 | 646 | 15.77 | 6.92 | 0.025 | 0.270 | 0.705 | 0.890 | 0.038 | 59.88 |
| | obj. 5 | 190 | 8.16 | 6.08 | 0.037 | 0.374 | 0.865 | 0.972 | 0.004 | 20.61 |
| | obj. 11 | 183 | 13.26 | 6.50 | 0.036 | 0.324 | 0.729 | 0.923 | 0.029 | 23.04 |
| | obj. 18 | 146 | 15.88 | 7.55 | 0.007 | 0.276 | 0.655 | 0.916 | 0.037 | 15.88 |
| | obj. 21 | 198 | 35.46 | 8.76 | 0.015 | 0.209 | 0.557 | 0.769 | 0.163 | 35.46 |
| | obj. 27 | 96 | 12.74 | 7.87 | 0.024 | 0.215 | 0.660 | 0.948 | 0.024 | 92.07 |
| PoE, orbit product | obj. 1 | 886 | 23.82 | 7.21 | 0.026 | 0.278 | 0.640 | 0.793 | 0.091 | 82.83 |
| | obj. 4 | 646 | 14.50 | 6.14 | 0.028 | 0.321 | 0.762 | 0.896 | 0.036 | 53.85 |
| | obj. 5 | 190 | 6.22 | 4.73 | 0.084 | 0.539 | 0.905 | 0.982 | 0.000 | 15.14 |
| | obj. 11 | 183 | 11.90 | 6.17 | 0.040 | 0.370 | 0.776 | 0.931 | 0.022 | 21.54 |
| | obj. 18 | 146 | 14.38 | 6.85 | 0.039 | 0.324 | 0.669 | 0.918 | 0.027 | 14.38 |
| | obj. 21 | 198 | 34.78 | 7.30 | 0.025 | 0.305 | 0.623 | 0.774 | 0.160 | 34.78 |
| | obj. 27 | 96 | 10.36 | 5.94 | 0.038 | 0.382 | 0.767 | 0.958 | 0.024 | 82.99 |

Table 63. **EXP-064 T-LESS (A0), per-object test detail** (seed means of the per-seed statistics; $n$: test images).

| | | DINOv3 | | PoE | |
|---|---|---|---|---|---|
| Inference | Seed | val | test | val | test |
| greedy (1 view) | 0 | 12.67 | 18.64 | 12.33 | 17.81 |
| | 1 | 12.73 | 18.74 | 12.42 | 17.85 |
| | 2 | 12.61 | 18.40 | 12.16 | 18.11 |
| orbit product | 0 | 9.93 | 18.15 | 9.33 | 16.83 |
| | 1 | 10.00 | 17.48 | 9.21 | 16.54 |
| | 2 | 9.88 | 18.00 | 9.22 | 16.33 |
| weighted vote (32) | 0 | 10.22 | 17.93 | 9.66 | 16.66 |
| | 1 | 10.10 | 17.42 | 9.75 | 16.29 |
| | 2 | 9.99 | 18.35 | 9.54 | 16.98 |

Table 64. **EXP-064 T-LESS (A0), per seed**: object-balanced mean symmetry-aware error (mean over objects of each seed; these average to the object-balanced table).

**SYMSOL-T.**

| Split | Feat. | Inference | mean | median | acc@2 | acc@5 | acc@10 | acc@15 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|---|
| validation | DINOv3 | greedy (1 view) | $3.78 \pm 0.03$ | 3.47 | 0.256 | 0.759 | 0.971 | 0.996 | 1.000 | 0.000 | 121.45 |
| | DINOv3 | orbit product | $2.61 \pm 0.00$ | 2.47 | 0.395 | 0.931 | 1.000 | 1.000 | 1.000 | 0.000 | 121.67 |
| | DINOv3 | weighted vote (32) | $2.85 \pm 0.02$ | 2.67 | 0.346 | 0.901 | 0.998 | 1.000 | 1.000 | 0.000 | 121.70 |
| | PoE | greedy (1 view) | $3.87 \pm 0.03$ | 3.52 | 0.248 | 0.747 | 0.967 | 0.995 | 1.000 | 0.000 | 121.19 |
| | PoE | orbit product | $2.65 \pm 0.01$ | 2.50 | 0.390 | 0.923 | 0.999 | 1.000 | 1.000 | 0.000 | 121.39 |
| | PoE | weighted vote (32) | $2.91 \pm 0.00$ | 2.70 | 0.349 | 0.883 | 0.996 | 1.000 | 1.000 | 0.000 | 120.89 |
| test | DINOv3 | greedy (1 view) | $3.82 \pm 0.03$ | 3.51 | 0.253 | 0.753 | 0.970 | 0.996 | 1.000 | 0.000 | 121.33 |
| | DINOv3 | orbit product | $2.62 \pm 0.01$ | 2.49 | 0.388 | 0.931 | 1.000 | 1.000 | 1.000 | 0.000 | 121.39 |
| | DINOv3 | weighted vote (32) | $2.86 \pm 0.01$ | 2.68 | 0.341 | 0.899 | 0.997 | 1.000 | 1.000 | 0.000 | 121.08 |
| | PoE | greedy (1 view) | $3.88 \pm 0.01$ | 3.57 | 0.248 | 0.742 | 0.967 | 0.996 | 1.000 | 0.000 | 121.12 |
| | PoE | orbit product | $2.67 \pm 0.00$ | 2.53 | 0.386 | 0.920 | 0.999 | 1.000 | 1.000 | 0.000 | 121.38 |
| | PoE | weighted vote (32) | $2.92 \pm 0.01$ | 2.71 | 0.346 | 0.881 | 0.996 | 1.000 | 1.000 | 0.000 | 121.10 |

Table 65. **EXP-064 SYMSOL-T (A0), object-balanced** (RUN-20261019-233710; symmetry-aware error). acc@2/5/15/30: per-object seed means averaged over objects (as acc@10). raw: error without symmetry handling.

| | DINOv3 | | | PoE | | |
|---|---|---|---|---|---|---|
| Object | greedy | orbit prod. | vote (32) | greedy | orbit prod. | vote (32) |
| *validation* | | | | | | |
| cone | $2.21 \pm 0.02$ | $1.75 \pm 0.01$ | $2.01 \pm 0.04$ | $2.34 \pm 0.02$ | $1.82 \pm 0.01$ | $1.97 \pm 0.01$ |
| cube | $4.11 \pm 0.08$ | $2.92 \pm 0.02$ | $3.07 \pm 0.04$ | $3.91 \pm 0.06$ | $2.73 \pm 0.02$ | $3.00 \pm 0.01$ |
| cylinder | $2.11 \pm 0.05$ | $1.64 \pm 0.01$ | $1.84 \pm 0.04$ | $2.07 \pm 0.02$ | $1.66 \pm 0.02$ | $1.80 \pm 0.04$ |
| icosahedron | $6.11 \pm 0.12$ | $3.71 \pm 0.01$ | $4.26 \pm 0.03$ | $6.18 \pm 0.13$ | $3.76 \pm 0.08$ | $4.29 \pm 0.02$ |
| tetrahedron | $4.36 \pm 0.02$ | $3.04 \pm 0.01$ | $3.05 \pm 0.02$ | $4.86 \pm 0.09$ | $3.28 \pm 0.03$ | $3.49 \pm 0.04$ |
| *test* | | | | | | |
| cone | $2.18 \pm 0.01$ | $1.79 \pm 0.02$ | $2.04 \pm 0.01$ | $2.28 \pm 0.01$ | $1.83 \pm 0.01$ | $1.95 \pm 0.02$ |
| cube | $4.17 \pm 0.05$ | $2.90 \pm 0.01$ | $3.06 \pm 0.01$ | $3.97 \pm 0.02$ | $2.72 \pm 0.03$ | $3.04 \pm 0.02$ |
| cylinder | $2.12 \pm 0.02$ | $1.65 \pm 0.02$ | $1.87 \pm 0.02$ | $2.06 \pm 0.02$ | $1.67 \pm 0.03$ | $1.80 \pm 0.01$ |
| icosahedron | $6.22 \pm 0.15$ | $3.71 \pm 0.06$ | $4.25 \pm 0.04$ | $6.22 \pm 0.13$ | $3.81 \pm 0.02$ | $4.31 \pm 0.03$ |
| tetrahedron | $4.39 \pm 0.03$ | $3.06 \pm 0.03$ | $3.09 \pm 0.01$ | $4.88 \pm 0.05$ | $3.35 \pm 0.03$ | $3.50 \pm 0.03$ |

Table 66. **EXP-064 SYMSOL-T (A0), per object**: mean symmetry-aware error $\pm$ seed std for every variant and inference mode.

| System | Object | $n$ | mean | median | acc@2 | acc@5 | acc@10 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|
| DINOv3, greedy (1 view) | cone | 5000 | 2.18 | 2.03 | 0.490 | 0.978 | 1.000 | 1.000 | 0.000 | 90.96 |
| | cube | 5000 | 4.17 | 3.83 | 0.117 | 0.725 | 0.983 | 1.000 | 0.000 | 129.44 |
| | cylinder | 5000 | 2.12 | 1.99 | 0.504 | 0.983 | 1.000 | 1.000 | 0.000 | 134.30 |
| | icosahedron | 5000 | 6.22 | 5.61 | 0.051 | 0.411 | 0.884 | 0.999 | 0.000 | 127.56 |
| | tetrahedron | 5000 | 4.39 | 4.08 | 0.102 | 0.669 | 0.983 | 1.000 | 0.000 | 124.40 |
| DINOv3, orbit product | cone | 5000 | 1.79 | 1.62 | 0.650 | 0.990 | 1.000 | 1.000 | 0.000 | 91.10 |
| | cube | 5000 | 2.90 | 2.80 | 0.239 | 0.946 | 1.000 | 1.000 | 0.000 | 129.37 |
| | cylinder | 5000 | 1.65 | 1.54 | 0.701 | 0.996 | 1.000 | 1.000 | 0.000 | 133.63 |
| | icosahedron | 5000 | 3.71 | 3.56 | 0.134 | 0.802 | 0.999 | 1.000 | 0.000 | 128.27 |
| | tetrahedron | 5000 | 3.06 | 2.93 | 0.217 | 0.919 | 1.000 | 1.000 | 0.000 | 124.60 |
| PoE, greedy (1 view) | cone | 5000 | 2.28 | 2.12 | 0.460 | 0.970 | 1.000 | 1.000 | 0.000 | 90.15 |
| | cube | 5000 | 3.97 | 3.65 | 0.127 | 0.760 | 0.988 | 1.000 | 0.000 | 129.01 |
| | cylinder | 5000 | 2.06 | 1.94 | 0.522 | 0.987 | 1.000 | 1.000 | 0.000 | 134.51 |
| | icosahedron | 5000 | 6.22 | 5.69 | 0.044 | 0.399 | 0.889 | 1.000 | 0.000 | 127.62 |
| | tetrahedron | 5000 | 4.88 | 4.42 | 0.085 | 0.595 | 0.957 | 1.000 | 0.000 | 124.33 |
| PoE, orbit product | cone | 5000 | 1.83 | 1.68 | 0.629 | 0.992 | 1.000 | 1.000 | 0.000 | 90.17 |
| | cube | 5000 | 2.72 | 2.63 | 0.282 | 0.967 | 1.000 | 1.000 | 0.000 | 129.40 |
| | cylinder | 5000 | 1.67 | 1.56 | 0.693 | 0.996 | 1.000 | 1.000 | 0.000 | 134.59 |
| | icosahedron | 5000 | 3.81 | 3.67 | 0.127 | 0.780 | 1.000 | 1.000 | 0.000 | 127.87 |
| | tetrahedron | 5000 | 3.35 | 3.13 | 0.197 | 0.864 | 0.997 | 1.000 | 0.000 | 124.89 |

Table 67. **EXP-064 SYMSOL-T (A0), per-object test detail** (seed means of the per-seed statistics; $n$: test images).

| | | DINOv3 | | PoE | |
|---|---|---|---|---|---|
| Inference | Seed | val | test | val | test |
| greedy (1 view) | 0 | 3.75 | 3.78 | 3.85 | 3.87 |
| | 1 | 3.80 | 3.85 | 3.91 | 3.89 |
| | 2 | 3.78 | 3.82 | 3.86 | 3.89 |
| orbit product | 0 | 2.61 | 2.61 | 2.64 | 2.67 |
| | 1 | 2.62 | 2.63 | 2.65 | 2.68 |
| | 2 | 2.61 | 2.62 | 2.65 | 2.67 |
| weighted vote (32) | 0 | 2.85 | 2.86 | 2.91 | 2.93 |
| | 1 | 2.86 | 2.85 | 2.91 | 2.92 |
| | 2 | 2.82 | 2.88 | 2.91 | 2.90 |

Table 68. **EXP-064 SYMSOL-T (A0), per seed**: object-balanced mean symmetry-aware error (mean over objects of each seed; these average to the object-balanced table).

**YCB-V (real training videos only).** The YCB-V diagnosis (training fit, pose coverage, crop and label checks) is in the supplementary YCB-V section.

| Split | Feat. | Inference | mean | median | acc@2 | acc@5 | acc@10 | acc@15 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|---|
| validation | DINOv3 | greedy (1 view) | $64.32 \pm 11.63$ | 64.98 | 0.001 | 0.014 | 0.086 | 0.221 | 0.502 | 0.317 | 64.32 |
| | DINOv3 | orbit product | $62.99 \pm 2.40$ | 58.32 | 0.001 | 0.017 | 0.109 | 0.225 | 0.479 | 0.297 | 62.99 |
| | DINOv3 | weighted vote (32) | $65.36 \pm 3.41$ | 61.12 | 0.000 | 0.012 | 0.098 | 0.218 | 0.474 | 0.325 | 65.36 |
| | PoE | greedy (1 view) | $73.52 \pm 6.81$ | 75.14 | 0.004 | 0.049 | 0.176 | 0.313 | 0.484 | 0.398 | 73.52 |
| | PoE | orbit product | $74.06 \pm 7.43$ | 71.30 | 0.003 | 0.031 | 0.168 | 0.291 | 0.483 | 0.401 | 74.06 |
| | PoE | weighted vote (32) | $73.85 \pm 6.57$ | 70.95 | 0.003 | 0.039 | 0.168 | 0.299 | 0.483 | 0.396 | 73.85 |
| test | DINOv3 | greedy (1 view) | $71.19 \pm 6.97$ | 63.08 | 0.003 | 0.019 | 0.080 | 0.121 | 0.313 | 0.332 | 71.19 |
| | DINOv3 | orbit product | $69.45 \pm 5.22$ | 63.30 | 0.003 | 0.021 | 0.087 | 0.156 | 0.356 | 0.305 | 69.45 |
| | DINOv3 | weighted vote (32) | $72.87 \pm 3.07$ | 65.82 | 0.002 | 0.019 | 0.079 | 0.134 | 0.312 | 0.332 | 72.87 |
| | PoE | greedy (1 view) | $81.15 \pm 7.27$ | 78.53 | 0.003 | 0.048 | 0.159 | 0.234 | 0.402 | 0.422 | 81.15 |
| | PoE | orbit product | $79.92 \pm 3.20$ | 72.77 | 0.003 | 0.043 | 0.180 | 0.265 | 0.435 | 0.418 | 79.92 |
| | PoE | weighted vote (32) | $84.20 \pm 4.55$ | 77.85 | 0.003 | 0.051 | 0.167 | 0.235 | 0.373 | 0.444 | 84.20 |

Table 69. **EXP-064 YCB-V, real only, object-balanced** (RUN-20261019-224920; raw error). acc@2/5/15/30: per-object seed means averaged over objects (as acc@10). raw: error without symmetry handling.

| | DINOv3 | | | PoE | | |
|---|---|---|---|---|---|---|
| Object | greedy | orbit prod. | vote (32) | greedy | orbit prod. | vote (32) |
| *validation* | | | | | | |
| cracker box | $36.17 \pm 3.43$ | $36.39 \pm 3.28$ | $38.85 \pm 3.18$ | $56.94 \pm 6.42$ | $47.91 \pm 1.72$ | $51.82 \pm 3.68$ |
| mustard bottle | $147.51 \pm 5.19$ | $153.89 \pm 7.09$ | $145.57 \pm 3.38$ | $159.06 \pm 6.79$ | $166.35 \pm 7.53$ | $160.84 \pm 9.49$ |
| banana | $103.20 \pm 17.17$ | $96.89 \pm 7.02$ | $104.12 \pm 5.90$ | $107.72 \pm 18.11$ | $104.55 \pm 9.74$ | $101.40 \pm 4.98$ |
| mug | $21.49 \pm 2.10$ | $20.99 \pm 0.31$ | $21.44 \pm 1.57$ | $18.26 \pm 1.23$ | $18.40 \pm 1.28$ | $17.82 \pm 1.11$ |
| power drill | $14.76 \pm 2.22$ | $15.33 \pm 2.41$ | $15.51 \pm 1.64$ | $10.70 \pm 0.79$ | $10.76 \pm 0.22$ | $11.11 \pm 0.22$ |
| scissors | $62.77 \pm 47.01$ | $54.43 \pm 9.69$ | $66.69 \pm 18.68$ | $88.43 \pm 19.24$ | $96.39 \pm 32.31$ | $100.09 \pm 29.25$ |
| *test* | | | | | | |
| cracker box | $77.73 \pm 24.77$ | $74.61 \pm 5.91$ | $78.23 \pm 4.73$ | $78.72 \pm 11.92$ | $85.27 \pm 7.21$ | $93.92 \pm 10.30$ |
| mustard bottle | $101.53 \pm 18.57$ | $87.29 \pm 10.97$ | $84.53 \pm 10.37$ | $102.43 \pm 9.10$ | $97.96 \pm 16.46$ | $112.26 \pm 14.58$ |
| banana | $79.86 \pm 17.78$ | $83.23 \pm 9.93$ | $86.61 \pm 1.72$ | $77.59 \pm 30.18$ | $61.63 \pm 3.69$ | $74.66 \pm 19.16$ |
| mug | $24.56 \pm 8.92$ | $22.76 \pm 0.57$ | $28.24 \pm 1.15$ | $28.45 \pm 11.14$ | $24.25 \pm 1.53$ | $29.92 \pm 4.49$ |
| power drill | $30.27 \pm 2.25$ | $25.07 \pm 3.94$ | $29.07 \pm 3.12$ | $38.18 \pm 3.14$ | $41.34 \pm 1.04$ | $42.15 \pm 0.96$ |
| scissors | $113.16 \pm 5.79$ | $123.72 \pm 51.70$ | $130.54 \pm 28.17$ | $161.52 \pm 6.36$ | $169.07 \pm 6.19$ | $152.28 \pm 17.34$ |

Table 70. **EXP-064 YCB-V, real only, per object**: mean raw error $\pm$ seed std for every variant and inference mode.

| System | Object | $n$ | mean | median | acc@2 | acc@5 | acc@10 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|
| DINOv3, greedy (1 view) | cracker box | 868 | 77.73 | 62.58 | 0.000 | 0.001 | 0.003 | 0.159 | 0.364 | 77.73 |
| | mustard bottle | 357 | 101.53 | 102.38 | 0.000 | 0.006 | 0.012 | 0.088 | 0.641 | 101.53 |
| | banana | 379 | 79.86 | 61.58 | 0.000 | 0.000 | 0.001 | 0.193 | 0.318 | 79.86 |
| | mug | 636 | 24.56 | 22.60 | 0.010 | 0.073 | 0.293 | 0.784 | 0.031 | 24.56 |
| | power drill | 1057 | 30.27 | 22.71 | 0.006 | 0.036 | 0.174 | 0.653 | 0.021 | 30.27 |
| | scissors | 181 | 113.16 | 106.61 | 0.000 | 0.000 | 0.000 | 0.000 | 0.617 | 113.16 |
| DINOv3, orbit product | cracker box | 868 | 74.61 | 52.29 | 0.000 | 0.002 | 0.028 | 0.287 | 0.388 | 74.61 |
| | mustard bottle | 357 | 87.29 | 75.20 | 0.000 | 0.003 | 0.014 | 0.147 | 0.431 | 87.29 |
| | banana | 379 | 83.23 | 77.37 | 0.000 | 0.004 | 0.030 | 0.245 | 0.415 | 83.23 |
| | mug | 636 | 22.76 | 22.40 | 0.013 | 0.082 | 0.244 | 0.742 | 0.002 | 22.76 |
| | power drill | 1057 | 25.07 | 18.65 | 0.003 | 0.037 | 0.208 | 0.715 | 0.004 | 25.07 |
| | scissors | 181 | 123.72 | 133.92 | 0.000 | 0.000 | 0.000 | 0.000 | 0.591 | 123.72 |
| PoE, greedy (1 view) | cracker box | 868 | 78.72 | 67.18 | 0.000 | 0.001 | 0.006 | 0.174 | 0.387 | 78.72 |
| | mustard bottle | 357 | 102.43 | 126.84 | 0.000 | 0.002 | 0.049 | 0.229 | 0.556 | 102.43 |
| | banana | 379 | 77.59 | 74.97 | 0.000 | 0.004 | 0.092 | 0.500 | 0.366 | 77.59 |
| | mug | 636 | 28.45 | 21.33 | 0.012 | 0.159 | 0.368 | 0.743 | 0.078 | 28.45 |
| | power drill | 1057 | 38.18 | 11.06 | 0.009 | 0.122 | 0.437 | 0.768 | 0.176 | 38.18 |
| | scissors | 181 | 161.52 | 169.77 | 0.000 | 0.000 | 0.000 | 0.000 | 0.967 | 161.52 |
| PoE, orbit product | cracker box | 868 | 85.27 | 77.85 | 0.000 | 0.002 | 0.019 | 0.246 | 0.482 | 85.27 |
| | mustard bottle | 357 | 97.96 | 127.39 | 0.000 | 0.007 | 0.075 | 0.318 | 0.534 | 97.96 |
| | banana | 379 | 61.63 | 28.09 | 0.003 | 0.027 | 0.185 | 0.545 | 0.294 | 61.63 |
| | mug | 636 | 24.25 | 22.06 | 0.009 | 0.105 | 0.327 | 0.724 | 0.013 | 24.25 |
| | power drill | 1057 | 41.34 | 10.44 | 0.007 | 0.120 | 0.476 | 0.773 | 0.186 | 41.34 |
| | scissors | 181 | 169.07 | 170.76 | 0.000 | 0.000 | 0.000 | 0.004 | 0.996 | 169.07 |

Table 71. **EXP-064 YCB-V, real only, per-object test detail** (seed means of the per-seed statistics; $n$: test images).

| | | DINOv3 | | PoE | |
|---|---|---|---|---|---|
| Inference | Seed | val | test | val | test |
| greedy (1 view) | 0 | 77.35 | 78.82 | 79.46 | 82.98 |
| | 1 | 55.00 | 65.17 | 66.08 | 73.13 |
| | 2 | 60.59 | 69.56 | 75.02 | 87.32 |
| orbit product | 0 | 65.48 | 63.42 | 82.53 | 81.77 |
| | 1 | 60.69 | 72.19 | 68.66 | 76.23 |
| | 2 | 62.78 | 72.73 | 71.00 | 81.77 |
| weighted vote (32) | 0 | 69.29 | 69.39 | 81.38 | 87.43 |
| | 1 | 63.14 | 74.00 | 69.31 | 79.00 |
| | 2 | 63.66 | 75.21 | 70.86 | 86.17 |

Table 72. **EXP-064 YCB-V, real only, per seed**: object-balanced mean raw error (mean over objects of each seed; these average to the object-balanced table).

**Zero-shot arm.** Frozen SPEED+ EXP-109 arm-A trees, no retraining. (a) raw symmetry-aware error and (b) its uniform-chance level; (c) frame-free pair error $\min_S \angle(\hat{R}_i \hat{R}_j^\top, R_i S R_j^\top)$ over 2,000 random same-object test pairs and its uniform-chance level; (d) one constant offset per object chosen on validation, reported on test and labelled calibrated (not zero-shot). Standardisation uses the SPEED+ statistics (primary) or the target dataset's own training statistics (label-free).

**Post-hoc constant-prediction control.** On T-LESS a constant prediction gives an object-balanced pair error of 88.7° (per object: 1: 80.6, 4: 81.6, 5: 71.7, 11: 100.8, 18: 126.6, 21: 124.3, 27: 35.5), below every frozen-tree system; on SYMSOL-T it is 53.8°, equal to chance because the rotations are uniform. On YCB-V the control was computed in-script (last column of Tab. 77).

| Stats, split | Feat. | Inference | (a) raw | median | acc@10 | (b) chance | (c) pair | chance | (d) calib. |
|---|---|---|---|---|---|---|---|---|---|
| SPEED+, validation | DINOv3 | greedy (1 view) | $104.5 \pm 1.1$ | 104.4 | 0.004 | 104.2 | – | – | – |
| | DINOv3 | orbit product | $103.7 \pm 1.5$ | 102.7 | 0.006 | 104.2 | – | – | – |
| | DINOv3 | weighted vote (32) | $104.0 \pm 1.5$ | 103.8 | 0.004 | 104.2 | – | – | – |
| | PoE | greedy (1 view) | $105.2 \pm 0.5$ | 105.4 | 0.008 | 104.2 | – | – | – |
| | PoE | orbit product | $104.2 \pm 0.5$ | 104.3 | 0.015 | 104.2 | – | – | – |
| | PoE | weighted vote (32) | $104.5 \pm 0.6$ | 104.4 | 0.012 | 104.2 | – | – | – |
| SPEED+, test | DINOv3 | greedy (1 view) | $106.0 \pm 4.5$ | 108.7 | 0.002 | 103.6 | $96.4 \pm 1.5$ | 104.5 | 104.9 |
| | DINOv3 | orbit product | $110.3 \pm 7.0$ | 114.4 | 0.003 | 103.6 | $96.6 \pm 2.7$ | 104.5 | 104.7 |
| | DINOv3 | weighted vote (32) | $108.0 \pm 8.8$ | 110.9 | 0.003 | 103.6 | $98.3 \pm 2.1$ | 104.5 | 102.4 |
| | PoE | greedy (1 view) | $113.8 \pm 1.7$ | 118.4 | 0.000 | 103.6 | $95.6 \pm 3.5$ | 104.5 | 106.5 |
| | PoE | orbit product | $117.5 \pm 3.7$ | 123.0 | 0.006 | 103.6 | $95.4 \pm 3.0$ | 104.5 | 107.2 |
| | PoE | weighted vote (32) | $115.5 \pm 4.0$ | 121.3 | 0.002 | 103.6 | $96.6 \pm 1.9$ | 104.5 | 106.4 |
| own, validation | DINOv3 | greedy (1 view) | $103.6 \pm 1.2$ | 103.8 | 0.004 | 104.2 | – | – | – |
| | PoE | greedy (1 view) | $104.6 \pm 0.8$ | 105.9 | 0.007 | 104.2 | – | – | – |
| own, test | DINOv3 | greedy (1 view) | $107.7 \pm 3.8$ | 111.0 | 0.003 | 103.6 | $97.8 \pm 0.4$ | 104.5 | 102.8 |
| | PoE | greedy (1 view) | $116.0 \pm 2.3$ | 123.2 | 0.002 | 103.6 | $97.0 \pm 0.9$ | 104.5 | 108.1 |

Table 73. **EXP-064 zero-shot, T-LESS**, object-balanced. Missing cells: not computed for that split or statistics. The constant-prediction pair control was computed post hoc (see text).

| | DINOv3 greedy | | PoE greedy | | PoE orbit prod. | | chance | |
|---|---|---|---|---|---|---|---|---|
| Object | raw | pair | raw | pair | raw | pair | raw | pair |
| obj. 1 | 95.2 | 88.8 | 101.0 | 87.3 | 103.4 | 86.4 | 88.0 | 90.2 |
| obj. 4 | 101.2 | 85.6 | 102.7 | 86.7 | 106.9 | 85.5 | 85.4 | 90.3 |
| obj. 5 | 102.3 | 78.3 | 104.2 | 78.4 | 113.5 | 80.3 | 102.8 | 102.3 |
| obj. 11 | 88.9 | 99.2 | 97.8 | 100.9 | 100.9 | 100.6 | 105.0 | 101.8 |
| obj. 18 | 122.9 | 121.1 | 129.1 | 123.5 | 132.0 | 123.2 | 127.4 | 126.7 |
| obj. 21 | 125.5 | 125.5 | 135.7 | 124.3 | 133.7 | 125.8 | 126.0 | 126.4 |
| obj. 27 | 105.9 | 76.4 | 126.1 | 68.0 | 132.2 | 66.2 | 90.9 | 94.2 |

Table 74. **EXP-064 zero-shot, T-LESS, per object** (test, SPEED+ statistics, seed means).

| Stats, split | Feat. | Inference | (a) raw | median | acc@10 | (b) chance | (c) pair | chance | (d) calib. |
|---|---|---|---|---|---|---|---|---|---|
| SPEED+, validation | DINOv3 | greedy (1 view) | $52.2 \pm 1.1$ | 53.2 | 0.014 | 53.9 | – | – | – |
| | DINOv3 | orbit product | $51.3 \pm 1.1$ | 52.0 | 0.023 | 53.9 | – | – | – |
| | DINOv3 | weighted vote (32) | $52.7 \pm 1.5$ | 53.4 | 0.015 | 53.9 | – | – | – |
| | PoE | greedy (1 view) | $51.5 \pm 1.0$ | 52.7 | 0.038 | 53.9 | – | – | – |
| | PoE | orbit product | $49.5 \pm 2.0$ | 52.0 | 0.062 | 53.9 | – | – | – |
| | PoE | weighted vote (32) | $51.5 \pm 1.1$ | 52.9 | 0.050 | 53.9 | – | – | – |
| SPEED+, test | DINOv3 | greedy (1 view) | $51.9 \pm 1.1$ | 52.8 | 0.016 | 53.7 | $52.5 \pm 0.4$ | 53.7 | 46.7 |
| | DINOv3 | orbit product | $50.9 \pm 1.6$ | 51.5 | 0.022 | 53.7 | $50.5 \pm 0.4$ | 53.7 | 43.2 |
| | DINOv3 | weighted vote (32) | $52.4 \pm 1.3$ | 53.1 | 0.016 | 53.7 | $51.6 \pm 0.2$ | 53.7 | 45.6 |
| | PoE | greedy (1 view) | $51.4 \pm 1.1$ | 52.8 | 0.037 | 53.7 | $49.9 \pm 0.4$ | 53.7 | 44.1 |
| | PoE | orbit product | $49.4 \pm 2.1$ | 52.0 | 0.060 | 53.7 | $47.8 \pm 0.5$ | 53.7 | 40.9 |
| | PoE | weighted vote (32) | $51.4 \pm 1.2$ | 52.8 | 0.049 | 53.7 | $48.4 \pm 1.0$ | 53.7 | 42.1 |
| own, validation | DINOv3 | greedy (1 view) | $48.1 \pm 0.7$ | 49.0 | 0.023 | 53.9 | – | – | – |
| | PoE | greedy (1 view) | $49.8 \pm 0.5$ | 51.3 | 0.043 | 53.9 | – | – | – |
| own, test | DINOv3 | greedy (1 view) | $48.0 \pm 0.4$ | 48.6 | 0.025 | 53.7 | $51.6 \pm 0.1$ | 53.7 | 46.3 |
| | PoE | greedy (1 view) | $49.8 \pm 0.4$ | 51.1 | 0.043 | 53.7 | $50.2 \pm 0.2$ | 53.7 | 44.9 |

Table 75. **EXP-064 zero-shot, SYMSOL-T**, object-balanced. Missing cells: not computed for that split or statistics. The constant-prediction pair control was computed post hoc (see text).

| | DINOv3 greedy | | PoE greedy | | PoE orbit prod. | | chance | |
|---|---|---|---|---|---|---|---|---|
| Object | raw | pair | raw | pair | raw | pair | raw | pair |
| cone | 80.7 | 87.9 | 82.8 | 87.8 | 77.9 | 84.4 | 90.4 | 90.0 |
| cube | 31.5 | 38.3 | 19.9 | 27.8 | 15.2 | 21.4 | 40.5 | 40.5 |
| cylinder | 64.3 | 55.7 | 70.1 | 54.5 | 69.8 | 54.8 | 56.8 | 57.2 |
| icosahedron | 29.5 | 29.6 | 29.7 | 29.6 | 29.6 | 29.2 | 29.5 | 29.5 |
| tetrahedron | 53.3 | 50.9 | 54.6 | 50.0 | 54.5 | 48.9 | 51.4 | 51.3 |

Table 76. **EXP-064 zero-shot, SYMSOL-T, per object** (test, SPEED+ statistics, seed means).

| Stats, split | Feat. | Inference | (a) raw | median | acc@10 | (b) chance | (c) pair | chance | (d) calib. | constant |
|---|---|---|---|---|---|---|---|---|---|---|
| SPEED+, validation | DINOv3 | greedy (1 view) | $115.7 \pm 5.5$ | 115.2 | 0.004 | 126.2 | – | – | – | – |
| | DINOv3 | orbit product | $116.9 \pm 2.8$ | 119.3 | 0.001 | 126.2 | – | – | – | – |
| | DINOv3 | weighted vote (32) | $116.4 \pm 1.4$ | 118.9 | 0.001 | 126.2 | – | – | – | – |
| | PoE | greedy (1 view) | $120.4 \pm 4.0$ | 119.4 | 0.001 | 126.2 | – | – | – | – |
| | PoE | orbit product | $124.3 \pm 4.5$ | 127.9 | 0.000 | 126.2 | – | – | – | – |
| | PoE | weighted vote (32) | $124.2 \pm 1.1$ | 128.4 | 0.000 | 126.2 | – | – | – | – |
| SPEED+, test | DINOv3 | greedy (1 view) | $119.5 \pm 1.7$ | 119.5 | 0.000 | 125.8 | $95.9 \pm 4.0$ | 125.8 | 126.7 | 57.8 |
| | DINOv3 | orbit product | $118.8 \pm 1.4$ | 118.2 | 0.000 | 125.8 | $92.6 \pm 6.0$ | 125.8 | 127.9 | 57.8 |
| | DINOv3 | weighted vote (32) | $121.4 \pm 4.1$ | 121.6 | 0.000 | 125.8 | $102.1 \pm 2.8$ | 125.8 | 123.9 | 57.8 |
| | PoE | greedy (1 view) | $119.9 \pm 4.8$ | 120.5 | 0.002 | 125.8 | $104.1 \pm 4.7$ | 125.8 | 126.5 | 57.8 |
| | PoE | orbit product | $115.6 \pm 3.8$ | 118.5 | 0.003 | 125.8 | $101.6 \pm 2.8$ | 125.8 | 127.6 | 57.8 |
| | PoE | weighted vote (32) | $114.5 \pm 2.7$ | 117.6 | 0.001 | 125.8 | $107.0 \pm 2.3$ | 125.8 | 121.8 | 57.8 |
| own, validation | DINOv3 | greedy (1 view) | $110.4 \pm 7.0$ | 109.4 | 0.000 | 126.2 | – | – | – | – |
| | PoE | greedy (1 view) | $118.7 \pm 6.3$ | 118.2 | 0.001 | 126.2 | – | – | – | – |
| own, test | DINOv3 | greedy (1 view) | $121.1 \pm 4.3$ | 124.5 | 0.000 | 125.8 | $103.0 \pm 1.1$ | 125.8 | 126.8 | 57.8 |
| | PoE | greedy (1 view) | $117.2 \pm 3.2$ | 119.0 | 0.002 | 125.8 | $109.2 \pm 1.9$ | 125.8 | 125.0 | 57.8 |

Table 77. **EXP-064 zero-shot, YCB-V**, object-balanced. Missing cells: not computed for that split or statistics.

| | DINOv3 greedy | | PoE greedy | | PoE orbit prod. | | chance | | |
|---|---|---|---|---|---|---|---|---|---|
| Object | raw | pair | raw | pair | raw | pair | raw | pair | constant pair |
| cracker box | 110.3 | 107.9 | 116.2 | 120.5 | 112.1 | 112.9 | 126.2 | 126.3 | 68.1 |
| mustard bottle | 115.5 | 96.6 | 113.9 | 104.6 | 104.5 | 93.8 | 121.1 | 125.5 | 90.2 |
| banana | 133.8 | 103.5 | 120.4 | 104.9 | 112.4 | 98.7 | 129.7 | 127.3 | 49.1 |
| mug | 110.7 | 96.2 | 133.3 | 104.9 | 134.6 | 99.5 | 128.0 | 124.9 | 27.5 |
| power drill | 118.8 | 112.5 | 121.4 | 112.5 | 121.9 | 110.7 | 124.5 | 126.8 | 95.3 |
| scissors | 128.2 | 58.7 | 114.3 | 77.6 | 107.9 | 94.1 | 125.4 | 124.1 | 16.6 |

Table 78. **EXP-064 zero-shot, YCB-V, per object** (test, SPEED+ statistics, seed means).

| | | (a) raw | | | (c) pair | | |
|---|---|---|---|---|---|---|---|
| Dataset | System | s0 | s1 | s2 | s0 | s1 | s2 |
| T-LESS | DINOv3, greedy (1 view) | 102.1 | 105.0 | 110.9 | 96.7 | 97.8 | 94.8 |
| | DINOv3, orbit product | 102.7 | 111.8 | 116.4 | 98.8 | 97.3 | 93.6 |
| | PoE, greedy (1 view) | 112.1 | 113.9 | 115.5 | 95.3 | 99.2 | 92.3 |
| | PoE, orbit product | 113.4 | 118.6 | 120.6 | 98.4 | 95.6 | 92.3 |
| SYMSOL-T | DINOv3, greedy (1 view) | 52.3 | 52.6 | 50.7 | 52.8 | 52.0 | 52.7 |
| | DINOv3, orbit product | 51.5 | 52.1 | 49.1 | 50.8 | 50.6 | 50.0 |
| | PoE, greedy (1 view) | 51.8 | 52.3 | 50.2 | 50.1 | 49.5 | 50.2 |
| | PoE, orbit product | 49.9 | 51.2 | 47.2 | 48.2 | 47.2 | 47.9 |
| YCB-V | DINOv3, greedy (1 view) | 119.4 | 121.3 | 118.0 | 100.2 | 92.3 | 95.1 |
| | DINOv3, orbit product | 118.5 | 120.4 | 117.5 | 97.5 | 86.0 | 94.4 |
| | PoE, greedy (1 view) | 125.4 | 116.4 | 117.9 | 104.3 | 99.4 | 108.8 |
| | PoE, orbit product | 113.2 | 113.6 | 120.0 | 102.6 | 98.5 | 103.7 |

Table 79. **EXP-064 zero-shot per seed** (test, SPEED+ statistics): object-balanced raw symmetry-aware error and pair error for each seed's frozen trees.

## L.11. EXP-106: symmetry-aware training targets

**Question.** For symmetric objects, does a multi-path target, in which every equivalent rotation gets its own tree path and the probability mass is divided between them, predict better than the single annotated label (A0) or a canonical label (A1, the equivalent rotation closest to identity)? A2 averages the recipe's geodesic soft-target loss over the equivalent rotations (all of them when $|S| \leq 8$, otherwise 8 drawn per row and step); it reduces exactly to A0 for objects without symmetry. Per-object trees, 2 branches, 3 seeds, the locked recipe otherwise. A0 is the EXP-064 tree set re-evaluated under this experiment (identical numbers). Arm K (Kurz corner formulation) and the best-fit stage were not run.

**Pre-registered criteria and outcome.** Criterion 1 (as amended before any run): A2's object-balanced validation PoE greedy mean is below the better of A0 and A1 by more than the larger seed std. Evaluated here from the CSVs against that rule.

- T-LESS: **PASS**. A2 $10.82 \pm 0.11$ vs A1 $11.46 \pm 0.04$ (A0 $12.30 \pm 0.13$). Descriptive, PoE orbit product: A0 $9.25 \pm 0.07$, A1 $10.71 \pm 0.03$, A2 $8.28 \pm 0.05$.
- SYMSOL-T: **PASS**. A2 $2.72 \pm 0.01$ vs A1 $3.40 \pm 0.01$ (A0 $3.87 \pm 0.03$). Descriptive, PoE orbit product: A0 $2.65 \pm 0.01$, A1 $9.73 \pm 0.05$, A2 $2.10 \pm 0.01$.
- Criterion 2 (best-fit re-ranking): not run. Criterion 3: test reported for every arm below.

| | | | validation | | | test | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Target | Feat. | Inference | mean | median | acc@10 | mean | median | acc@10 | >90 | raw |
| A0 | DINOv3 | greedy (1 view) | $12.67 \pm 0.06$ | 8.07 | 0.660 | $18.59 \pm 0.18$ | 8.43 | 0.620 | 0.052 | 47.11 |
| | DINOv3 | orbit product | $9.94 \pm 0.06$ | 6.45 | 0.766 | $17.88 \pm 0.35$ | 7.42 | 0.674 | 0.053 | 43.97 |
| | DINOv3 | weighted vote (32) | $10.10 \pm 0.12$ | 6.15 | 0.769 | $17.90 \pm 0.47$ | 7.10 | 0.687 | 0.054 | 44.59 |
| | PoE | greedy (1 view) | $12.30 \pm 0.13$ | 7.37 | 0.702 | $17.92 \pm 0.16$ | 7.38 | 0.681 | 0.055 | 47.24 |
| | PoE | orbit product | $9.25 \pm 0.07$ | 5.80 | 0.808 | $16.57 \pm 0.25$ | 6.33 | 0.735 | 0.051 | 43.65 |
| | PoE | weighted vote (32) | $9.65 \pm 0.11$ | 5.61 | 0.805 | $16.64 \pm 0.34$ | 6.08 | 0.736 | 0.053 | 44.38 |
| A1 | DINOv3 | greedy (1 view) | $11.39 \pm 0.02$ | 7.11 | 0.733 | $16.84 \pm 0.27$ | 7.24 | 0.707 | 0.050 | 73.31 |
| | DINOv3 | orbit product | $11.25 \pm 0.07$ | 7.69 | 0.667 | $18.72 \pm 0.44$ | 8.64 | 0.596 | 0.053 | 74.87 |
| | DINOv3 | weighted vote (32) | $9.39 \pm 0.04$ | 5.78 | 0.811 | $16.61 \pm 0.13$ | 6.56 | 0.731 | 0.050 | 74.93 |
| | PoE | greedy (1 view) | $11.46 \pm 0.04$ | 6.69 | 0.755 | $16.71 \pm 0.11$ | 6.59 | 0.734 | 0.053 | 73.07 |
| | PoE | orbit product | $10.71 \pm 0.03$ | 6.94 | 0.721 | $18.24 \pm 0.66$ | 7.50 | 0.659 | 0.055 | 74.33 |
| | PoE | weighted vote (32) | $9.24 \pm 0.07$ | 5.39 | 0.827 | $16.21 \pm 0.28$ | 5.86 | 0.762 | 0.053 | 73.34 |
| A2 | DINOv3 | greedy (1 view) | $10.89 \pm 0.08$ | 6.75 | 0.754 | $16.11 \pm 0.30$ | 6.74 | 0.729 | 0.047 | 74.76 |
| | DINOv3 | orbit product | $8.61 \pm 0.01$ | 5.52 | 0.828 | $16.05 \pm 0.42$ | 6.04 | 0.754 | 0.050 | 74.51 |
| | DINOv3 | weighted vote (32) | $8.77 \pm 0.04$ | 5.24 | 0.837 | $15.78 \pm 0.14$ | 5.69 | 0.770 | 0.050 | 73.17 |
| | PoE | greedy (1 view) | $10.82 \pm 0.11$ | 6.35 | 0.777 | $15.71 \pm 0.16$ | 6.07 | 0.765 | 0.049 | 73.58 |
| | PoE | orbit product | $8.28 \pm 0.05$ | 5.13 | 0.849 | $15.19 \pm 0.27$ | 5.39 | 0.790 | 0.049 | 72.67 |
| | PoE | weighted vote (32) | $8.59 \pm 0.04$ | 4.93 | 0.848 | $15.41 \pm 0.25$ | 5.17 | 0.790 | 0.052 | 73.10 |

Table 80. **EXP-106, T-LESS**, object-balanced symmetry-aware error for the three training targets.

| Target | Feat. | Inference | R@2 | R@5 | R@10 | R@15 | R@30 |
|---|---|---|---|---|---|---|---|
| A0 | DINOv3 | greedy (1 view) | 1.6 | 20.6 | 62.0 | 78.1 | 88.4 |
| | DINOv3 | orbit product | 2.6 | 27.8 | 67.4 | 79.4 | 88.7 |
| | DINOv3 | weighted vote (32) | 3.3 | 32.1 | 68.7 | 79.7 | 88.3 |
| | PoE | greedy (1 view) | 2.3 | 27.0 | 68.1 | 80.5 | 88.7 |
| | PoE | orbit product | 4.0 | 36.0 | 73.5 | 82.6 | 89.3 |
| | PoE | weighted vote (32) | 5.3 | 39.9 | 73.6 | 82.1 | 88.9 |
| A1 | DINOv3 | greedy (1 view) | 2.6 | 27.8 | 70.7 | 82.1 | 89.5 |
| | DINOv3 | orbit product | 1.9 | 20.7 | 59.6 | 77.8 | 89.0 |
| | DINOv3 | weighted vote (32) | 3.7 | 34.4 | 73.1 | 83.1 | 89.7 |
| | PoE | greedy (1 view) | 3.6 | 33.8 | 73.4 | 82.4 | 89.3 |
| | PoE | orbit product | 2.7 | 26.7 | 65.9 | 80.7 | 88.9 |
| | PoE | weighted vote (32) | 6.1 | 41.8 | 76.2 | 83.4 | 89.6 |
| A2 | DINOv3 | greedy (1 view) | 3.2 | 33.4 | 72.9 | 82.7 | 89.9 |
| | DINOv3 | orbit product | 4.6 | 41.4 | 75.4 | 83.5 | 89.6 |
| | DINOv3 | weighted vote (32) | 6.3 | 44.7 | 77.0 | 83.8 | 89.7 |
| | PoE | greedy (1 view) | 4.5 | 40.6 | 76.5 | 83.6 | 89.9 |
| | PoE | orbit product | 5.7 | 48.1 | 79.0 | 85.0 | 90.4 |
| | PoE | weighted vote (32) | 7.6 | 50.6 | 79.0 | 84.3 | 89.7 |

Table 81. **EXP-106, T-LESS test recall** R@$k$ (%, symmetry-aware; mean over seeds per object, then over objects, as in Tab. 1).

| | greedy (1 view) | | | orbit product | | |
|---|---|---|---|---|---|---|
| Object | A0 | A1 | A2 | A0 | A1 | A2 |
| *PoE* | | | | | | |
| obj. 1 | $6.41 \pm 0.16$ | $5.82 \pm 0.07$ | $4.69 \pm 0.13$ | $5.36 \pm 0.05$ | $7.09 \pm 0.09$ | $4.26 \pm 0.05$ |
| obj. 4 | $6.70 \pm 0.13$ | $5.85 \pm 0.05$ | $4.84 \pm 0.08$ | $5.42 \pm 0.03$ | $7.17 \pm 0.15$ | $4.26 \pm 0.09$ |
| obj. 5 | $9.63 \pm 0.07$ | $8.41 \pm 0.14$ | $7.96 \pm 0.08$ | $7.04 \pm 0.12$ | $9.08 \pm 0.06$ | $6.07 \pm 0.08$ |
| obj. 11 | $12.36 \pm 0.04$ | $11.26 \pm 0.31$ | $10.67 \pm 0.18$ | $9.49 \pm 0.07$ | $10.87 \pm 0.29$ | $8.09 \pm 0.10$ |
| obj. 18 | $22.61 \pm 0.23$ | $22.61 \pm 0.23$ | $22.61 \pm 0.23$ | $16.81 \pm 0.39$ | $16.81 \pm 0.39$ | $16.81 \pm 0.39$ |
| obj. 21 | $18.15 \pm 0.82$ | $18.15 \pm 0.82$ | $18.15 \pm 0.82$ | $13.33 \pm 0.49$ | $13.33 \pm 0.49$ | $13.33 \pm 0.49$ |
| obj. 27 | $10.27 \pm 0.16$ | $8.11 \pm 0.30$ | $6.80 \pm 0.11$ | $7.31 \pm 0.06$ | $10.58 \pm 0.44$ | $5.15 \pm 0.05$ |
| *DINOv3* | | | | | | |
| obj. 1 | $6.92 \pm 0.13$ | $5.55 \pm 0.02$ | $4.78 \pm 0.07$ | $5.88 \pm 0.14$ | $7.54 \pm 0.04$ | $4.37 \pm 0.04$ |
| obj. 4 | $7.19 \pm 0.14$ | $5.79 \pm 0.17$ | $4.89 \pm 0.04$ | $6.04 \pm 0.05$ | $7.54 \pm 0.12$ | $4.41 \pm 0.03$ |
| obj. 5 | $9.91 \pm 0.13$ | $8.60 \pm 0.13$ | $8.30 \pm 0.05$ | $7.74 \pm 0.26$ | $9.91 \pm 0.37$ | $6.43 \pm 0.09$ |
| obj. 11 | $12.70 \pm 0.21$ | $11.37 \pm 0.29$ | $10.91 \pm 0.18$ | $10.35 \pm 0.21$ | $11.68 \pm 0.08$ | $8.65 \pm 0.12$ |
| obj. 18 | $22.25 \pm 0.57$ | $22.25 \pm 0.57$ | $22.25 \pm 0.57$ | $17.04 \pm 0.35$ | $17.04 \pm 0.35$ | $17.04 \pm 0.35$ |
| obj. 21 | $17.64 \pm 0.32$ | $17.64 \pm 0.32$ | $17.64 \pm 0.32$ | $13.60 \pm 0.55$ | $13.60 \pm 0.55$ | $13.60 \pm 0.55$ |
| obj. 27 | $12.09 \pm 0.40$ | $8.57 \pm 0.16$ | $7.49 \pm 0.13$ | $8.92 \pm 0.19$ | $11.48 \pm 0.11$ | $5.76 \pm 0.07$ |

Table 82. **EXP-106, T-LESS, per object, validation**: mean symmetry-aware error $\pm$ seed std.

| | greedy (1 view) | | | orbit product | | |
|---|---|---|---|---|---|---|
| Object | A0 | A1 | A2 | A0 | A1 | A2 |
| *PoE* | | | | | | |
| obj. 1 | $24.19 \pm 0.16$ | $23.62 \pm 0.29$ | $20.83 \pm 0.10$ | $23.82 \pm 0.37$ | $25.93 \pm 0.34$ | $20.82 \pm 0.41$ |
| obj. 4 | $15.77 \pm 0.58$ | $14.90 \pm 0.54$ | $13.08 \pm 1.08$ | $14.50 \pm 0.53$ | $15.97 \pm 0.52$ | $12.90 \pm 0.20$ |
| obj. 5 | $8.16 \pm 0.48$ | $6.44 \pm 0.21$ | $6.88 \pm 0.30$ | $6.22 \pm 0.26$ | $8.86 \pm 0.15$ | $5.69 \pm 0.13$ |
| obj. 11 | $13.26 \pm 0.80$ | $10.98 \pm 0.48$ | $11.31 \pm 0.36$ | $11.90 \pm 0.57$ | $12.63 \pm 0.73$ | $10.55 \pm 0.20$ |
| obj. 18 | $15.88 \pm 1.43$ | $15.88 \pm 1.43$ | $15.88 \pm 1.43$ | $14.38 \pm 1.02$ | $14.38 \pm 1.02$ | $14.38 \pm 1.02$ |
| obj. 21 | $35.46 \pm 2.65$ | $35.46 \pm 2.65$ | $35.46 \pm 2.65$ | $34.78 \pm 3.00$ | $34.78 \pm 3.00$ | $34.78 \pm 3.00$ |
| obj. 27 | $12.74 \pm 2.39$ | $9.71 \pm 1.71$ | $6.49 \pm 0.50$ | $10.36 \pm 0.57$ | $15.15 \pm 3.89$ | $7.18 \pm 0.71$ |
| *DINOv3* | | | | | | |
| obj. 1 | $24.82 \pm 0.14$ | $22.82 \pm 0.65$ | $21.11 \pm 0.36$ | $25.00 \pm 0.49$ | $25.96 \pm 0.22$ | $21.26 \pm 0.30$ |
| obj. 4 | $17.27 \pm 0.87$ | $15.33 \pm 0.69$ | $13.69 \pm 1.06$ | $15.69 \pm 0.73$ | $16.76 \pm 0.46$ | $13.41 \pm 0.39$ |
| obj. 5 | $8.29 \pm 0.61$ | $6.77 \pm 0.32$ | $6.93 \pm 0.40$ | $7.07 \pm 0.23$ | $8.79 \pm 0.42$ | $6.05 \pm 0.09$ |
| obj. 11 | $14.26 \pm 1.00$ | $12.18 \pm 0.29$ | $11.55 \pm 0.34$ | $12.92 \pm 0.67$ | $13.56 \pm 0.80$ | $11.38 \pm 0.20$ |
| obj. 18 | $16.31 \pm 0.46$ | $16.31 \pm 0.46$ | $16.31 \pm 0.46$ | $14.78 \pm 1.32$ | $14.78 \pm 1.32$ | $14.78 \pm 1.32$ |
| obj. 21 | $36.14 \pm 2.65$ | $36.14 \pm 2.65$ | $36.14 \pm 2.65$ | $37.24 \pm 3.79$ | $37.24 \pm 3.79$ | $37.24 \pm 3.79$ |
| obj. 27 | $13.06 \pm 1.27$ | $8.35 \pm 0.68$ | $7.05 \pm 1.11$ | $12.43 \pm 0.85$ | $13.94 \pm 0.78$ | $8.26 \pm 0.75$ |

Table 83. **EXP-106, T-LESS, per object, test**: mean symmetry-aware error $\pm$ seed std.

| | | | DINOv3 | | PoE | |
|---|---|---|---|---|---|---|
| Target | Inference | Seed | val | test | val | test |
| A0 | greedy (1 view) | 0 | 12.67 | 18.64 | 12.33 | 17.81 |
| | | 1 | 12.73 | 18.74 | 12.42 | 17.85 |
| | | 2 | 12.61 | 18.40 | 12.16 | 18.11 |
| | orbit product | 0 | 9.93 | 18.15 | 9.33 | 16.83 |
| | | 1 | 10.00 | 17.48 | 9.21 | 16.54 |
| | | 2 | 9.88 | 18.00 | 9.22 | 16.33 |
| | weighted vote (32) | 0 | 10.22 | 17.93 | 9.66 | 16.66 |
| | | 1 | 10.10 | 17.42 | 9.75 | 16.29 |
| | | 2 | 9.99 | 18.35 | 9.54 | 16.98 |
| A1 | greedy (1 view) | 0 | 11.37 | 16.81 | 11.48 | 16.69 |
| | | 1 | 11.39 | 17.13 | 11.49 | 16.61 |
| | | 2 | 11.42 | 16.59 | 11.41 | 16.83 |
| | orbit product | 0 | 11.22 | 19.20 | 10.67 | 18.62 |
| | | 1 | 11.34 | 18.35 | 10.74 | 18.63 |
| | | 2 | 11.20 | 18.60 | 10.70 | 17.48 |
| | weighted vote (32) | 0 | 9.44 | 16.76 | 9.20 | 16.50 |
| | | 1 | 9.38 | 16.52 | 9.31 | 15.95 |
| | | 2 | 9.36 | 16.56 | 9.20 | 16.17 |
| A2 | greedy (1 view) | 0 | 10.81 | 16.33 | 10.81 | 15.55 |
| | | 1 | 10.96 | 16.23 | 10.93 | 15.87 |
| | | 2 | 10.91 | 15.77 | 10.71 | 15.70 |
| | orbit product | 0 | 8.60 | 16.49 | 8.33 | 15.50 |
| | | 1 | 8.63 | 15.65 | 8.26 | 15.08 |
| | | 2 | 8.60 | 16.02 | 8.25 | 14.99 |
| | weighted vote (32) | 0 | 8.79 | 15.63 | 8.60 | 15.42 |
| | | 1 | 8.80 | 15.79 | 8.63 | 15.15 |
| | | 2 | 8.72 | 15.92 | 8.55 | 15.65 |

Table 84. **EXP-106, T-LESS, per seed**: object-balanced mean symmetry-aware error.

| | | | validation | | | test | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Target | Feat. | Inference | mean | median | acc@10 | mean | median | acc@10 | >90 | raw |
| A0 | DINOv3 | greedy (1 view) | $3.78 \pm 0.03$ | 3.47 | 0.971 | $3.82 \pm 0.03$ | 3.51 | 0.970 | 0.000 | 121.33 |
| | DINOv3 | orbit product | $2.61 \pm 0.00$ | 2.47 | 1.000 | $2.62 \pm 0.01$ | 2.49 | 1.000 | 0.000 | 121.39 |
| | DINOv3 | weighted vote (32) | $2.85 \pm 0.02$ | 2.67 | 0.998 | $2.86 \pm 0.01$ | 2.68 | 0.997 | 0.000 | 121.08 |
| | PoE | greedy (1 view) | $3.87 \pm 0.03$ | 3.52 | 0.967 | $3.88 \pm 0.01$ | 3.57 | 0.967 | 0.000 | 121.12 |
| | PoE | orbit product | $2.65 \pm 0.01$ | 2.50 | 0.999 | $2.67 \pm 0.00$ | 2.53 | 0.999 | 0.000 | 121.38 |
| | PoE | weighted vote (32) | $2.91 \pm 0.00$ | 2.70 | 0.996 | $2.92 \pm 0.01$ | 2.71 | 0.996 | 0.000 | 121.10 |
| A1 | DINOv3 | greedy (1 view) | $3.31 \pm 0.02$ | 3.20 | 0.998 | $3.30 \pm 0.01$ | 3.20 | 0.997 | 0.000 | 121.03 |
| | DINOv3 | orbit product | $10.32 \pm 0.07$ | 7.17 | 0.660 | $10.26 \pm 0.11$ | 7.16 | 0.661 | 0.000 | 120.38 |
| | DINOv3 | weighted vote (32) | $3.00 \pm 0.01$ | 2.90 | 0.999 | $3.02 \pm 0.00$ | 2.91 | 0.999 | 0.000 | 120.90 |
| | PoE | greedy (1 view) | $3.40 \pm 0.01$ | 3.26 | 0.995 | $3.39 \pm 0.01$ | 3.25 | 0.996 | 0.000 | 121.04 |
| | PoE | orbit product | $9.73 \pm 0.05$ | 6.93 | 0.683 | $9.69 \pm 0.05$ | 6.94 | 0.682 | 0.000 | 120.15 |
| | PoE | weighted vote (32) | $3.04 \pm 0.01$ | 2.93 | 0.998 | $3.06 \pm 0.00$ | 2.95 | 0.999 | 0.000 | 120.71 |
| A2 | DINOv3 | greedy (1 view) | $2.54 \pm 0.01$ | 2.38 | 0.998 | $2.56 \pm 0.01$ | 2.39 | 0.998 | 0.000 | 121.14 |
| | DINOv3 | orbit product | $2.07 \pm 0.00$ | 1.98 | 1.000 | $2.07 \pm 0.00$ | 1.97 | 1.000 | 0.000 | 121.45 |
| | DINOv3 | weighted vote (32) | $2.18 \pm 0.01$ | 2.09 | 1.000 | $2.20 \pm 0.00$ | 2.10 | 1.000 | 0.000 | 121.38 |
| | PoE | greedy (1 view) | $2.72 \pm 0.01$ | 2.48 | 0.994 | $2.71 \pm 0.01$ | 2.49 | 0.995 | 0.000 | 121.29 |
| | PoE | orbit product | $2.10 \pm 0.01$ | 2.00 | 1.000 | $2.11 \pm 0.01$ | 2.00 | 1.000 | 0.000 | 121.31 |
| | PoE | weighted vote (32) | $2.25 \pm 0.01$ | 2.12 | 0.999 | $2.27 \pm 0.00$ | 2.12 | 0.999 | 0.000 | 121.26 |

Table 85. **EXP-106, SYMSOL-T**, object-balanced symmetry-aware error for the three training targets.

| Target | Feat. | Inference | R@2 | R@5 | R@10 | R@15 | R@30 |
|---|---|---|---|---|---|---|---|
| A0 | DINOv3 | greedy (1 view) | 25.3 | 75.3 | 97.0 | 99.6 | 100.0 |
| | DINOv3 | orbit product | 38.8 | 93.1 | 100.0 | 100.0 | 100.0 |
| | DINOv3 | weighted vote (32) | 34.1 | 89.9 | 99.7 | 100.0 | 100.0 |
| | PoE | greedy (1 view) | 24.8 | 74.2 | 96.7 | 99.6 | 100.0 |
| | PoE | orbit product | 38.6 | 92.0 | 99.9 | 100.0 | 100.0 |
| | PoE | weighted vote (32) | 34.6 | 88.1 | 99.6 | 100.0 | 100.0 |
| A1 | DINOv3 | greedy (1 view) | 18.0 | 90.0 | 99.7 | 100.0 | 100.0 |
| | DINOv3 | orbit product | 5.6 | 34.3 | 66.1 | 80.5 | 95.4 |
| | DINOv3 | weighted vote (32) | 22.3 | 94.0 | 99.9 | 100.0 | 100.0 |
| | PoE | greedy (1 view) | 17.6 | 88.4 | 99.6 | 100.0 | 100.0 |
| | PoE | orbit product | 5.6 | 35.2 | 68.2 | 82.9 | 96.7 |
| | PoE | weighted vote (32) | 21.6 | 93.4 | 99.9 | 100.0 | 100.0 |
| A2 | DINOv3 | greedy (1 view) | 39.4 | 94.6 | 99.8 | 100.0 | 100.0 |
| | DINOv3 | orbit product | 51.5 | 99.3 | 100.0 | 100.0 | 100.0 |
| | DINOv3 | weighted vote (32) | 47.4 | 98.3 | 100.0 | 100.0 | 100.0 |
| | PoE | greedy (1 view) | 37.5 | 92.2 | 99.5 | 100.0 | 100.0 |
| | PoE | orbit product | 50.7 | 98.7 | 100.0 | 100.0 | 100.0 |
| | PoE | weighted vote (32) | 46.4 | 97.0 | 99.9 | 100.0 | 100.0 |

Table 86. **EXP-106, SYMSOL-T test recall** R@$k$ (%, symmetry-aware; mean over seeds per object, then over objects, as in Tab. 1).

| | greedy (1 view) | | | orbit product | | |
|---|---|---|---|---|---|---|
| Object | A0 | A1 | A2 | A0 | A1 | A2 |
| *PoE* | | | | | | |
| cone | $2.34 \pm 0.02$ | $3.22 \pm 0.02$ | $1.86 \pm 0.01$ | $1.82 \pm 0.01$ | $5.61 \pm 0.12$ | $1.60 \pm 0.00$ |
| cube | $3.91 \pm 0.06$ | $3.26 \pm 0.01$ | $3.10 \pm 0.02$ | $2.73 \pm 0.02$ | $14.10 \pm 0.12$ | $2.33 \pm 0.00$ |
| cylinder | $2.07 \pm 0.02$ | $2.95 \pm 0.02$ | $1.69 \pm 0.01$ | $1.66 \pm 0.02$ | $9.55 \pm 0.11$ | $1.53 \pm 0.01$ |
| icosahedron | $6.18 \pm 0.13$ | $3.44 \pm 0.02$ | $3.04 \pm 0.01$ | $3.76 \pm 0.08$ | $10.87 \pm 0.08$ | $2.45 \pm 0.02$ |
| tetrahedron | $4.86 \pm 0.09$ | $4.11 \pm 0.03$ | $3.89 \pm 0.04$ | $3.28 \pm 0.03$ | $8.51 \pm 0.14$ | $2.59 \pm 0.02$ |
| *DINOv3* | | | | | | |
| cone | $2.21 \pm 0.02$ | $3.10 \pm 0.05$ | $1.79 \pm 0.01$ | $1.75 \pm 0.01$ | $6.09 \pm 0.16$ | $1.58 \pm 0.01$ |
| cube | $4.11 \pm 0.08$ | $3.27 \pm 0.00$ | $2.95 \pm 0.04$ | $2.92 \pm 0.02$ | $15.30 \pm 0.57$ | $2.31 \pm 0.01$ |
| cylinder | $2.11 \pm 0.05$ | $2.89 \pm 0.03$ | $1.67 \pm 0.01$ | $1.64 \pm 0.01$ | $9.03 \pm 0.19$ | $1.46 \pm 0.02$ |
| icosahedron | $6.11 \pm 0.12$ | $3.45 \pm 0.02$ | $3.01 \pm 0.01$ | $3.71 \pm 0.01$ | $11.69 \pm 0.17$ | $2.54 \pm 0.01$ |
| tetrahedron | $4.36 \pm 0.02$ | $3.83 \pm 0.02$ | $3.28 \pm 0.02$ | $3.04 \pm 0.01$ | $9.50 \pm 0.10$ | $2.47 \pm 0.01$ |

Table 87. **EXP-106, SYMSOL-T, per object, validation**: mean symmetry-aware error $\pm$ seed std.

| | greedy (1 view) | | | orbit product | | |
|---|---|---|---|---|---|---|
| Object | A0 | A1 | A2 | A0 | A1 | A2 |
| *PoE* | | | | | | |
| cone | $2.28 \pm 0.01$ | $3.17 \pm 0.03$ | $1.84 \pm 0.01$ | $1.83 \pm 0.01$ | $5.65 \pm 0.05$ | $1.60 \pm 0.00$ |
| cube | $3.97 \pm 0.02$ | $3.27 \pm 0.00$ | $3.15 \pm 0.03$ | $2.72 \pm 0.03$ | $13.77 \pm 0.14$ | $2.31 \pm 0.01$ |
| cylinder | $2.06 \pm 0.02$ | $2.99 \pm 0.02$ | $1.71 \pm 0.01$ | $1.67 \pm 0.03$ | $9.29 \pm 0.26$ | $1.56 \pm 0.00$ |
| icosahedron | $6.22 \pm 0.13$ | $3.42 \pm 0.02$ | $3.06 \pm 0.01$ | $3.81 \pm 0.02$ | $11.02 \pm 0.05$ | $2.45 \pm 0.02$ |
| tetrahedron | $4.88 \pm 0.05$ | $4.08 \pm 0.04$ | $3.82 \pm 0.02$ | $3.35 \pm 0.03$ | $8.70 \pm 0.06$ | $2.62 \pm 0.02$ |
| *DINOv3* | | | | | | |
| cone | $2.18 \pm 0.01$ | $3.05 \pm 0.02$ | $1.78 \pm 0.01$ | $1.79 \pm 0.02$ | $6.06 \pm 0.01$ | $1.57 \pm 0.01$ |
| cube | $4.17 \pm 0.05$ | $3.28 \pm 0.00$ | $3.01 \pm 0.01$ | $2.90 \pm 0.01$ | $15.01 \pm 0.33$ | $2.30 \pm 0.01$ |
| cylinder | $2.12 \pm 0.02$ | $2.95 \pm 0.03$ | $1.69 \pm 0.02$ | $1.65 \pm 0.02$ | $9.01 \pm 0.17$ | $1.48 \pm 0.01$ |
| icosahedron | $6.22 \pm 0.15$ | $3.42 \pm 0.02$ | $3.02 \pm 0.01$ | $3.71 \pm 0.06$ | $11.61 \pm 0.13$ | $2.52 \pm 0.01$ |
| tetrahedron | $4.39 \pm 0.03$ | $3.82 \pm 0.02$ | $3.33 \pm 0.02$ | $3.06 \pm 0.03$ | $9.63 \pm 0.13$ | $2.46 \pm 0.01$ |

Table 88. **EXP-106, SYMSOL-T, per object, test**: mean symmetry-aware error $\pm$ seed std.

| | | | DINOv3 | | PoE | |
|---|---|---|---|---|---|---|
| Target | Inference | Seed | val | test | val | test |
| A0 | greedy (1 view) | 0 | 3.75 | 3.78 | 3.85 | 3.87 |
| | | 1 | 3.80 | 3.85 | 3.91 | 3.89 |
| | | 2 | 3.78 | 3.82 | 3.86 | 3.89 |
| | orbit product | 0 | 2.61 | 2.61 | 2.64 | 2.67 |
| | | 1 | 2.62 | 2.63 | 2.65 | 2.68 |
| | | 2 | 2.61 | 2.62 | 2.65 | 2.67 |
| | weighted vote (32) | 0 | 2.85 | 2.86 | 2.91 | 2.93 |
| | | 1 | 2.86 | 2.85 | 2.91 | 2.92 |
| | | 2 | 2.82 | 2.88 | 2.91 | 2.90 |
| A1 | greedy (1 view) | 0 | 3.31 | 3.31 | 3.40 | 3.38 |
| | | 1 | 3.32 | 3.31 | 3.41 | 3.39 |
| | | 2 | 3.29 | 3.29 | 3.39 | 3.38 |
| | orbit product | 0 | 10.32 | 10.30 | 9.70 | 9.65 |
| | | 1 | 10.39 | 10.35 | 9.78 | 9.74 |
| | | 2 | 10.25 | 10.14 | 9.70 | 9.67 |
| | weighted vote (32) | 0 | 3.00 | 3.02 | 3.03 | 3.06 |
| | | 1 | 3.01 | 3.02 | 3.05 | 3.06 |
| | | 2 | 2.99 | 3.02 | 3.04 | 3.05 |
| A2 | greedy (1 view) | 0 | 2.54 | 2.57 | 2.73 | 2.72 |
| | | 1 | 2.53 | 2.56 | 2.70 | 2.71 |
| | | 2 | 2.54 | 2.57 | 2.72 | 2.70 |
| | orbit product | 0 | 2.07 | 2.07 | 2.09 | 2.10 |
| | | 1 | 2.07 | 2.07 | 2.10 | 2.11 |
| | | 2 | 2.07 | 2.06 | 2.11 | 2.11 |
| | weighted vote (32) | 0 | 2.18 | 2.21 | 2.25 | 2.27 |
| | | 1 | 2.17 | 2.20 | 2.25 | 2.27 |
| | | 2 | 2.19 | 2.21 | 2.26 | 2.27 |

Table 89. **EXP-106, SYMSOL-T, per seed**: object-balanced mean symmetry-aware error.

**Leaf quantisation on SYMSOL-T.** Post hoc, the symmetry-aware error of the ground-truth rotation's own depth-5 leaf centre on the SYMSOL-T test set is 2.31° on average (cone 2.02, cube 2.50, cylinder 2.02, icosahedron 2.51, tetrahedron 2.51); A2's orbit error can fall below it because the tree may select the leaf closest to any equivalent rotation.

### L.12. EXP-122: YCB-V with synthetic training renders

**Question.** The EXP-064 YCB-V trees, trained on real videos only, fit their training frames but did not generalise. Does adding the standard `data_syn` renders (composited onto real frames drawn only from the training sequences; at most 12,000 per object, same visibility rule) make the frozen-feature tree work? One arm (B: real + synthetic), the EXP-064 real-only trees as control; validation and test rows unchanged.

**Pre-registered criteria and outcome.**

- Synthetic data helps (B's validation PoE orbit-product mean below real-only by more than the larger seed std), evaluated here from the CSVs against that rule: **PASS**. $12.74 \pm 1.03$ vs $74.06 \pm 7.43$.
- Test numbers for every variant and mode, and per-object test errors: below.

| | | | validation (held-out real sequences) | | | | test (real keyframes) | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Training data | Feat. | Inference | mean | median | acc@10 | >90 | mean | median | acc@10 | >90 |
| real only (EXP-064) | DINOv3 | greedy (1 view) | $64.32 \pm 11.63$ | 64.98 | 0.086 | 0.317 | $71.19 \pm 6.97$ | 63.08 | 0.080 | 0.332 |
| | DINOv3 | orbit product | $62.99 \pm 2.40$ | 58.32 | 0.109 | 0.297 | $69.45 \pm 5.22$ | 63.30 | 0.087 | 0.305 |
| | DINOv3 | weighted vote (32) | $65.36 \pm 3.41$ | 61.12 | 0.098 | 0.325 | $72.87 \pm 3.07$ | 65.82 | 0.079 | 0.332 |
| | PoE | greedy (1 view) | $73.52 \pm 6.81$ | 75.14 | 0.176 | 0.398 | $81.15 \pm 7.27$ | 78.53 | 0.159 | 0.422 |
| | PoE | orbit product | $74.06 \pm 7.43$ | 71.30 | 0.168 | 0.401 | $79.92 \pm 3.20$ | 72.77 | 0.180 | 0.418 |
| | PoE | weighted vote (32) | $73.85 \pm 6.57$ | 70.95 | 0.168 | 0.396 | $84.20 \pm 4.55$ | 77.85 | 0.167 | 0.444 |
| + synthetic (EXP-122) | DINOv3 | greedy (1 view) | $8.97 \pm 1.05$ | 6.96 | 0.778 | 0.006 | $12.92 \pm 6.90$ | 13.83 | 0.851 | 0.044 |
| | DINOv3 | orbit product | $11.53 \pm 1.79$ | 8.57 | 0.680 | 0.014 | $13.43 \pm 4.73$ | 5.59 | 0.834 | 0.043 |
| | DINOv3 | weighted vote (32) | $10.68 \pm 1.17$ | 6.60 | 0.778 | 0.016 | $12.11 \pm 3.71$ | 4.81 | 0.878 | 0.039 |
| | PoE | greedy (1 view) | $10.52 \pm 0.92$ | 6.39 | 0.790 | 0.021 | $10.90 \pm 4.12$ | 4.56 | 0.871 | 0.032 |
| | PoE | orbit product | $12.74 \pm 1.03$ | 8.40 | 0.705 | 0.026 | $12.16 \pm 5.53$ | 5.40 | 0.840 | 0.037 |
| | PoE | weighted vote (32) | $12.02 \pm 1.39$ | 6.55 | 0.766 | 0.028 | $12.04 \pm 5.18$ | 4.47 | 0.875 | 0.040 |

Table 90. **EXP-122 vs. EXP-064 on YCB-V**, object-balanced raw rotation error.

| Split | Feat. | Inference | mean | median | acc@2 | acc@5 | acc@10 | acc@15 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|---|
| validation | DINOv3 | greedy (1 view) | $8.97 \pm 1.05$ | 6.96 | 0.052 | 0.414 | 0.778 | 0.887 | 0.956 | 0.006 | 8.97 |
| | DINOv3 | orbit product | $11.53 \pm 1.79$ | 8.57 | 0.031 | 0.306 | 0.680 | 0.832 | 0.935 | 0.014 | 11.53 |
| | DINOv3 | weighted vote (32) | $10.68 \pm 1.17$ | 6.60 | 0.042 | 0.424 | 0.778 | 0.879 | 0.928 | 0.016 | 10.68 |
| | PoE | greedy (1 view) | $10.52 \pm 0.92$ | 6.39 | 0.064 | 0.448 | 0.790 | 0.892 | 0.939 | 0.021 | 10.52 |
| | PoE | orbit product | $12.74 \pm 1.03$ | 8.40 | 0.029 | 0.318 | 0.705 | 0.832 | 0.915 | 0.026 | 12.74 |
| | PoE | weighted vote (32) | $12.02 \pm 1.39$ | 6.55 | 0.046 | 0.416 | 0.766 | 0.870 | 0.919 | 0.028 | 12.02 |
| test | DINOv3 | greedy (1 view) | $12.92 \pm 6.90$ | 13.83 | 0.095 | 0.569 | 0.851 | 0.912 | 0.956 | 0.044 | 12.92 |
| | DINOv3 | orbit product | $13.43 \pm 4.73$ | 5.59 | 0.060 | 0.452 | 0.834 | 0.913 | 0.957 | 0.043 | 13.43 |
| | DINOv3 | weighted vote (32) | $12.11 \pm 3.71$ | 4.81 | 0.087 | 0.552 | 0.878 | 0.931 | 0.960 | 0.039 | 12.11 |
| | PoE | greedy (1 view) | $10.90 \pm 4.12$ | 4.56 | 0.113 | 0.610 | 0.871 | 0.920 | 0.963 | 0.032 | 10.90 |
| | PoE | orbit product | $12.16 \pm 5.53$ | 5.40 | 0.077 | 0.518 | 0.840 | 0.922 | 0.963 | 0.037 | 12.16 |
| | PoE | weighted vote (32) | $12.04 \pm 5.18$ | 4.47 | 0.109 | 0.618 | 0.875 | 0.926 | 0.959 | 0.040 | 12.04 |

Table 91. **EXP-122 YCB-V, real + synthetic, object-balanced** (RUN-20261027-200852; raw error). acc@2/5/15/30: per-object seed means averaged over objects (as acc@10). raw: error without symmetry handling.

| | DINOv3 | | | PoE | | |
|---|---|---|---|---|---|---|
| Object | greedy | orbit prod. | vote (32) | greedy | orbit prod. | vote (32) |
| *validation* | | | | | | |
| cracker box | $7.02 \pm 0.55$ | $8.22 \pm 0.41$ | $7.13 \pm 0.68$ | $7.44 \pm 1.60$ | $7.63 \pm 0.45$ | $6.52 \pm 0.96$ |
| mustard bottle | $6.97 \pm 0.69$ | $10.17 \pm 1.74$ | $7.51 \pm 1.31$ | $6.85 \pm 0.25$ | $9.55 \pm 1.60$ | $7.69 \pm 1.38$ |
| banana | $17.56 \pm 6.43$ | $25.52 \pm 11.42$ | $27.12 \pm 7.85$ | $26.73 \pm 5.30$ | $33.29 \pm 7.29$ | $33.59 \pm 7.59$ |
| mug | $12.02 \pm 0.77$ | $12.63 \pm 0.44$ | $12.20 \pm 0.48$ | $12.21 \pm 0.52$ | $12.85 \pm 0.39$ | $13.15 \pm 0.57$ |
| power drill | $5.04 \pm 0.09$ | $6.29 \pm 0.47$ | $5.28 \pm 0.42$ | $4.97 \pm 0.10$ | $6.11 \pm 0.23$ | $5.28 \pm 0.28$ |
| scissors | $5.20 \pm 0.19$ | $6.38 \pm 0.41$ | $4.83 \pm 0.39$ | $4.93 \pm 0.24$ | $7.02 \pm 0.16$ | $5.89 \pm 0.51$ |
| *test* | | | | | | |
| cracker box | $5.08 \pm 0.18$ | $5.56 \pm 0.31$ | $4.76 \pm 0.24$ | $5.67 \pm 0.41$ | $5.98 \pm 0.09$ | $5.49 \pm 0.23$ |
| mustard bottle | $6.13 \pm 0.97$ | $11.08 \pm 3.86$ | $10.24 \pm 3.25$ | $7.56 \pm 3.84$ | $11.22 \pm 3.36$ | $10.28 \pm 3.40$ |
| banana | $6.10 \pm 1.09$ | $7.20 \pm 0.42$ | $6.13 \pm 0.24$ | $5.33 \pm 0.56$ | $5.74 \pm 0.28$ | $5.51 \pm 0.44$ |
| mug | $6.45 \pm 0.57$ | $7.55 \pm 1.11$ | $7.24 \pm 0.95$ | $5.56 \pm 0.42$ | $7.10 \pm 0.14$ | $6.05 \pm 0.49$ |
| power drill | $4.34 \pm 0.39$ | $4.91 \pm 0.43$ | $4.25 \pm 0.54$ | $4.39 \pm 0.27$ | $4.64 \pm 0.34$ | $4.01 \pm 0.44$ |
| scissors | $49.40 \pm 40.91$ | $44.27 \pm 28.92$ | $40.02 \pm 21.87$ | $36.92 \pm 26.33$ | $38.27 \pm 34.39$ | $40.93 \pm 31.99$ |

Table 92. **EXP-122 YCB-V, real + synthetic, per object**: mean raw error $\pm$ seed std for every variant and inference mode.

| System | Object | $n$ | mean | median | acc@2 | acc@5 | acc@10 | acc@30 | >90 | raw |
|---|---|---|---|---|---|---|---|---|---|---|
| DINOv3, greedy (1 view) | cracker box | 868 | 5.08 | 4.19 | 0.105 | 0.660 | 0.929 | 0.998 | 0.002 | 5.08 |
| | mustard bottle | 357 | 6.13 | 4.31 | 0.099 | 0.645 | 0.950 | 0.993 | 0.007 | 6.13 |
| | banana | 379 | 6.10 | 4.97 | 0.109 | 0.512 | 0.858 | 0.999 | 0.001 | 6.10 |
| | mug | 636 | 6.45 | 4.81 | 0.091 | 0.521 | 0.830 | 1.000 | 0.000 | 6.45 |
| | power drill | 1057 | 4.34 | 3.80 | 0.102 | 0.699 | 0.973 | 1.000 | 0.000 | 4.34 |
| | scissors | 181 | 49.40 | 60.88 | 0.066 | 0.379 | 0.567 | 0.748 | 0.252 | 49.40 |
| DINOv3, orbit product | cracker box | 868 | 5.56 | 5.02 | 0.073 | 0.498 | 0.914 | 1.000 | 0.000 | 5.56 |
| | mustard bottle | 357 | 11.08 | 4.76 | 0.058 | 0.540 | 0.923 | 0.965 | 0.035 | 11.08 |
| | banana | 379 | 7.20 | 6.27 | 0.047 | 0.337 | 0.817 | 1.000 | 0.000 | 7.20 |
| | mug | 636 | 7.55 | 5.57 | 0.068 | 0.432 | 0.749 | 1.000 | 0.000 | 7.55 |
| | power drill | 1057 | 4.91 | 4.27 | 0.084 | 0.611 | 0.948 | 1.000 | 0.000 | 4.91 |
| | scissors | 181 | 44.27 | 7.63 | 0.033 | 0.293 | 0.654 | 0.779 | 0.221 | 44.27 |
| PoE, greedy (1 view) | cracker box | 868 | 5.67 | 4.20 | 0.082 | 0.642 | 0.886 | 0.997 | 0.002 | 5.67 |
| | mustard bottle | 357 | 7.56 | 3.99 | 0.102 | 0.664 | 0.951 | 0.982 | 0.018 | 7.56 |
| | banana | 379 | 5.33 | 4.20 | 0.169 | 0.592 | 0.914 | 0.998 | 0.002 | 5.33 |
| | mug | 636 | 5.56 | 4.29 | 0.131 | 0.605 | 0.885 | 1.000 | 0.000 | 5.56 |
| | power drill | 1057 | 4.39 | 3.62 | 0.109 | 0.726 | 0.968 | 0.997 | 0.000 | 4.39 |
| | scissors | 181 | 36.92 | 7.04 | 0.088 | 0.429 | 0.624 | 0.803 | 0.171 | 36.92 |
| PoE, orbit product | cracker box | 868 | 5.98 | 5.10 | 0.084 | 0.497 | 0.850 | 1.000 | 0.000 | 5.98 |
| | mustard bottle | 357 | 11.22 | 4.33 | 0.070 | 0.626 | 0.941 | 0.962 | 0.038 | 11.22 |
| | banana | 379 | 5.74 | 4.69 | 0.097 | 0.536 | 0.901 | 1.000 | 0.000 | 5.74 |
| | mug | 636 | 7.10 | 5.44 | 0.082 | 0.442 | 0.794 | 0.999 | 0.000 | 7.10 |
| | power drill | 1057 | 4.64 | 4.01 | 0.085 | 0.657 | 0.956 | 1.000 | 0.000 | 4.64 |
| | scissors | 181 | 38.27 | 8.80 | 0.044 | 0.352 | 0.599 | 0.816 | 0.184 | 38.27 |

Table 93. **EXP-122 YCB-V, real + synthetic, per-object test detail** (seed means of the per-seed statistics; $n$: test images).

| | | DINOv3 | | PoE | |
|---|---|---|---|---|---|
| Inference | Seed | val | test | val | test |
| greedy (1 view) | 0 | 10.08 | 8.42 | 11.36 | 6.64 |
| | 1 | 8.85 | 20.87 | 10.66 | 14.85 |
| | 2 | 7.98 | 9.47 | 9.54 | 11.22 |
| orbit product | 0 | 13.60 | 11.96 | 13.91 | 9.84 |
| | 1 | 10.66 | 18.72 | 12.38 | 18.47 |
| | 2 | 10.34 | 9.60 | 11.95 | 8.17 |
| weighted vote (32) | 0 | 11.95 | 12.53 | 12.45 | 9.44 |
| | 1 | 10.45 | 15.59 | 13.14 | 18.01 |
| | 2 | 9.63 | 8.20 | 10.47 | 8.67 |

Table 94. **EXP-122 YCB-V, real + synthetic, per seed**: object-balanced mean raw error (mean over objects of each seed; these average to the object-balanced table).

### L.13. EXP-127: hierarchy vs. flat on the same grid

**Question.** EXP-113 compared the tree with a flat classifier over a *different* grid, so hierarchy and discretisation were confounded. EXP-127 trains two further flat heads with the identical recipe and orbit inference: FL over the tree's own 131,072 leaf centres (scoring a candidate by the probability of its Tesseract cell, as the tree does), and FH over a balanced Hopf grid (1,771 Fibonacci directions $\times$ 74 rolls $=$ 131,054 anchors). The tree arm reuses the EXP-109 arm-A checkpoints; orbit distillation is excluded from all arms.

**Pre-registered criteria.** A difference passes if the 3-seed synthetic-val means differ by more than the larger seed std. (1) Hierarchy, T vs. FL, PoE + orbit primary: if FL ties or wins, the claim that the tree's advantage comes from hierarchical prediction is withdrawn. (2) Discretisation, FL vs. FH, descriptive. (3) System, T vs. FH, PoE + orbit. (4) Real domains and cost descriptive. A tree-reproduction gate against EXP-113 seed 0 (within 0.02°) precedes the trainings.

**Outcome.** *Pending*: the trainings and the evaluation are queued and have not run; no criterion has been evaluated. Only the quantisation floor, which needs no learning, is available.

| | Synth. val | | | | Synth. val-aug | | | |
|---|---|---|---|---|---|---|---|---|
| Grid | mean | median | p95 | max | mean | median | p95 | max |
| Tesseract L5 encode→decode (tree leaf; FL hard anchor) | 2.507 | 2.486 | 3.950 | 5.776 | 2.504 | 2.480 | 3.963 | 5.855 |
| Tesseract L5 nearest centre | 2.476 | 2.449 | 3.914 | 5.776 | 2.474 | 2.449 | 3.907 | 5.820 |
| Balanced Hopf 1,771×74, nearest | 2.291 | 2.374 | 3.214 | 3.840 | 2.305 | 2.384 | 3.235 | 3.811 |
| SPACE-HOP Hopf 10,923×12, nearest (EXP-113 grid) | 6.057 | 5.436 | 12.947 | 14.973 | 6.032 | 5.394 | 12.899 | 15.004 |

Table 95. **EXP-127 quantisation floor**: error (°) of the nearest anchor (tree: its leaf) to each ground-truth rotation, on the identical 11,994 synthetic-val and val-aug rotations (CPU, no learning).

## L.14. Pending experiments: EXP-119, EXP-124, EXP-126

The following pre-registered experiments are running or queued at the time of writing. No result is reported for them except the completed ground-truth-crop reference evaluation of EXP-124.

- **EXP-119, orbit distillation on T-LESS and SYMSOL-T.** Base arm: per dataset, the EXP-106 target with the lowest validation PoE greedy mean; DINOv3-only and PoE students as in EXP-117 ($\beta = 1$). Criterion: student single-view greedy validation mean below the base arm's by more than the larger seed std (DINOv3 primary). Status: queued; no run directory exists. The distillation rows of Tab. 6 and Tab. 1 remain blank.
- **EXP-124, SPEED+ with a synthetic-only detector and the official score.** A YOLO11s detector trained on synthetic images only provides the crops; systems S1 (EXP-117 PoE students + orbit, EXP-118 joint DINOv3 translation, orbit mean), S2 (EXP-118 joint PoE) and S3 (EXP-117 DINOv3 student + EXP-118 joint DINOv3 translation). Criteria are descriptive: the detected-crop score per domain, and the detection gap (detected − GT-crop score). Status: the detector training is running (the first attempt was aborted by the memory guard and the queue retried with identical settings); no detected-crop result exists, and the last row of Tab. 4 remains blank. The ground-truth-crop reference evaluation of the same systems with the same code is complete and is reported below.
- **EXP-126, translation on T-LESS, SYMSOL-T and YCB-V.** Frozen trees plus a SPACE-HOP translation block per object; criterion: DINOv3 validation $E_T$ (orbit mean) below the box-only baseline by more than the larger seed std. Status: queued; no run directory exists.

**EXP-124 ground-truth-crop reference.** Its pre-registered gate requires the GT-crop S1 rotation to reproduce EXP-117 (11.21/11.09° on lightbox/sunlamp) and the S1 translation to reproduce EXP-118 joint DINOv3 (0.185/0.183 m). The run gives 11.21/11.09° and 0.185/0.183 m, so the gate passes. The score is the official SPEED+ score [19]: the mean over images of $e_q$ (rad) + $e_t = \|\hat{t} - t\|/\|t\|$, with the precision floors applied per component; *joint rule* applies the floor to the sum (a check; the sources disagree on the rule).

S2 with orbit rotation and orbit-mean translation reproduces the EXP-118 joint PoE ESA score (0.241/0.238 here vs. 0.241/0.238 in Tab. 43), the GT-crop score quoted in the main text; S1, which combines the distilled PoE rotation with the DINOv3 translation block, scores 0.227/0.224.

**Systems S1–S3.** S1 uses the EXP-117 PoE students for rotation and the EXP-118 joint DINOv3 block for translation; S2 is the EXP-118 joint PoE model (rotation and translation of one family); S3 uses the EXP-117 DINOv3 student (DINOv3-only teacher) with the EXP-118 joint DINOv3 translation block.

| | | | Synth. val | | | Lightbox | | | Sunlamp | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Rotation | Translation | $E_R$ (°) | $E_T$ (m) | score | $E_R$ (°) | $E_T$ (m) | score | $E_R$ (°) | $E_T$ (m) | score |
| S1 | greedy (1 view) | greedy (1 view) | $4.56 \pm 0.03$ | $0.078 \pm 0.002$ | $0.093 \pm 0.000$ | $15.90 \pm 0.53$ | $0.209 \pm 0.004$ | $0.313 \pm 0.010$ | $18.33 \pm 0.66$ | $0.212 \pm 0.003$ | $0.356 \pm 0.012$ |
| S1 | greedy (1 view) | orbit mean | $4.56 \pm 0.03$ | $0.059 \pm 0.001$ | $0.090 \pm 0.000$ | $15.90 \pm 0.53$ | $0.185 \pm 0.003$ | $0.309 \pm 0.010$ | $18.33 \pm 0.66$ | $0.183 \pm 0.004$ | $0.351 \pm 0.012$ |
| S1 | orbit product | greedy (1 view) | $3.69 \pm 0.02$ | $0.078 \pm 0.002$ | $0.078 \pm 0.000$ | $11.21 \pm 0.42$ | $0.209 \pm 0.004$ | $0.231 \pm 0.008$ | $11.09 \pm 0.49$ | $0.212 \pm 0.003$ | $0.229 \pm 0.009$ |
| S1 | orbit product | orbit mean | $3.69 \pm 0.02$ | $0.059 \pm 0.001$ | $0.075 \pm 0.000$ | $11.21 \pm 0.42$ | $0.185 \pm 0.003$ | $0.227 \pm 0.008$ | $11.09 \pm 0.49$ | $0.183 \pm 0.004$ | $0.224 \pm 0.009$ |
| S1 | weighted vote (32) | greedy (1 view) | $3.27 \pm 0.03$ | $0.078 \pm 0.002$ | $0.071 \pm 0.000$ | $11.35 \pm 0.36$ | $0.209 \pm 0.004$ | $0.233 \pm 0.007$ | $11.91 \pm 0.26$ | $0.212 \pm 0.003$ | $0.244 \pm 0.005$ |
| S1 | weighted vote (32) | orbit mean | $3.27 \pm 0.03$ | $0.059 \pm 0.001$ | $0.068 \pm 0.000$ | $11.35 \pm 0.36$ | $0.185 \pm 0.003$ | $0.229 \pm 0.007$ | $11.91 \pm 0.26$ | $0.183 \pm 0.004$ | $0.239 \pm 0.005$ |
| S2 | greedy (1 view) | greedy (1 view) | $4.78 \pm 0.00$ | $0.111 \pm 0.004$ | $0.103 \pm 0.001$ | $17.01 \pm 0.36$ | $0.196 \pm 0.003$ | $0.330 \pm 0.006$ | $18.74 \pm 0.48$ | $0.221 \pm 0.001$ | $0.365 \pm 0.009$ |
| S2 | greedy (1 view) | orbit mean | $4.78 \pm 0.00$ | $0.093 \pm 0.004$ | $0.100 \pm 0.001$ | $17.01 \pm 0.36$ | $0.175 \pm 0.003$ | $0.327 \pm 0.006$ | $18.74 \pm 0.48$ | $0.194 \pm 0.002$ | $0.361 \pm 0.009$ |
| S2 | orbit product | greedy (1 view) | $3.86 \pm 0.01$ | $0.111 \pm 0.004$ | $0.087 \pm 0.001$ | $12.11 \pm 0.27$ | $0.196 \pm 0.003$ | $0.245 \pm 0.004$ | $11.73 \pm 0.28$ | $0.221 \pm 0.001$ | $0.243 \pm 0.005$ |
| S2 | orbit product | orbit mean | $3.86 \pm 0.01$ | $0.093 \pm 0.004$ | $0.084 \pm 0.001$ | $12.11 \pm 0.27$ | $0.175 \pm 0.003$ | $0.241 \pm 0.004$ | $11.73 \pm 0.28$ | $0.194 \pm 0.002$ | $0.238 \pm 0.005$ |
| S2 | weighted vote (32) | greedy (1 view) | $3.42 \pm 0.01$ | $0.111 \pm 0.004$ | $0.080 \pm 0.001$ | $12.17 \pm 0.44$ | $0.196 \pm 0.003$ | $0.246 \pm 0.007$ | $12.55 \pm 0.38$ | $0.221 \pm 0.001$ | $0.257 \pm 0.007$ |
| S2 | weighted vote (32) | orbit mean | $3.42 \pm 0.01$ | $0.093 \pm 0.004$ | $0.077 \pm 0.001$ | $12.17 \pm 0.44$ | $0.175 \pm 0.003$ | $0.242 \pm 0.007$ | $12.55 \pm 0.38$ | $0.194 \pm 0.002$ | $0.252 \pm 0.007$ |
| S3 | greedy (1 view) | greedy (1 view) | $4.57 \pm 0.05$ | $0.078 \pm 0.002$ | $0.094 \pm 0.001$ | $21.61 \pm 0.82$ | $0.209 \pm 0.004$ | $0.412 \pm 0.015$ | $20.55 \pm 0.94$ | $0.212 \pm 0.003$ | $0.394 \pm 0.017$ |
| S3 | greedy (1 view) | orbit mean | $4.57 \pm 0.05$ | $0.059 \pm 0.001$ | $0.090 \pm 0.001$ | $21.61 \pm 0.82$ | $0.185 \pm 0.003$ | $0.408 \pm 0.015$ | $20.55 \pm 0.94$ | $0.183 \pm 0.004$ | $0.389 \pm 0.017$ |
| S3 | orbit product | greedy (1 view) | $3.90 \pm 0.03$ | $0.078 \pm 0.002$ | $0.082 \pm 0.000$ | $16.43 \pm 0.75$ | $0.209 \pm 0.004$ | $0.322 \pm 0.014$ | $13.64 \pm 0.57$ | $0.212 \pm 0.003$ | $0.274 \pm 0.010$ |
| S3 | orbit product | orbit mean | $3.90 \pm 0.03$ | $0.059 \pm 0.001$ | $0.079 \pm 0.000$ | $16.43 \pm 0.75$ | $0.185 \pm 0.003$ | $0.318 \pm 0.014$ | $13.64 \pm 0.57$ | $0.183 \pm 0.004$ | $0.269 \pm 0.010$ |
| S3 | weighted vote (32) | greedy (1 view) | $3.37 \pm 0.05$ | $0.078 \pm 0.002$ | $0.073 \pm 0.001$ | $16.90 \pm 0.79$ | $0.209 \pm 0.004$ | $0.330 \pm 0.014$ | $14.33 \pm 0.41$ | $0.212 \pm 0.003$ | $0.286 \pm 0.008$ |
| S3 | weighted vote (32) | orbit mean | $3.37 \pm 0.05$ | $0.059 \pm 0.001$ | $0.069 \pm 0.001$ | $16.90 \pm 0.79$ | $0.185 \pm 0.003$ | $0.326 \pm 0.014$ | $14.33 \pm 0.41$ | $0.183 \pm 0.004$ | $0.281 \pm 0.008$ |

Table 96. **EXP-124, ground-truth-crop reference** (3 seeds). Detected-crop results are pending.

| | | | Synth. val | | | | | | Lightbox | | | | | | Sunlamp | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Rot. | Trans. | $E_R$ med. | $E_T$ med. | $s_q$ | $s_t$ | joint | >150 | $E_R$ med. | $E_T$ med. | $s_q$ | $s_t$ | joint | >150 | $E_R$ med. | $E_T$ med. | $s_q$ | $s_t$ | joint | >150 |
| S1 | greedy (1 view) | greedy (1 view) | 4.15 | 0.060 | 0.080 | 0.014 | 0.093 | 0.000 | 6.37 | 0.152 | 0.278 | 0.035 | 0.313 | 0.037 | 7.90 | 0.160 | 0.320 | 0.036 | 0.356 | 0.030 |
| S1 | greedy (1 view) | orbit mean | 4.15 | 0.046 | 0.080 | 0.010 | 0.090 | 0.000 | 6.37 | 0.134 | 0.278 | 0.031 | 0.309 | 0.037 | 7.90 | 0.137 | 0.320 | 0.031 | 0.351 | 0.030 |
| S1 | orbit product | greedy (1 view) | 3.31 | 0.060 | 0.064 | 0.014 | 0.078 | 0.000 | 5.02 | 0.152 | 0.196 | 0.035 | 0.231 | 0.026 | 5.91 | 0.160 | 0.194 | 0.036 | 0.229 | 0.014 |
| S1 | orbit product | orbit mean | 3.31 | 0.046 | 0.064 | 0.010 | 0.075 | 0.000 | 5.02 | 0.134 | 0.196 | 0.031 | 0.227 | 0.026 | 5.91 | 0.137 | 0.194 | 0.031 | 0.224 | 0.014 |
| S1 | weighted vote (32) | greedy (1 view) | 2.83 | 0.060 | 0.057 | 0.014 | 0.071 | 0.000 | 4.68 | 0.152 | 0.198 | 0.035 | 0.233 | 0.027 | 5.74 | 0.160 | 0.208 | 0.036 | 0.244 | 0.017 |
| S1 | weighted vote (32) | orbit mean | 2.83 | 0.046 | 0.057 | 0.010 | 0.068 | 0.000 | 4.68 | 0.134 | 0.198 | 0.031 | 0.229 | 0.027 | 5.74 | 0.137 | 0.208 | 0.031 | 0.239 | 0.017 |
| S2 | greedy (1 view) | greedy (1 view) | 4.30 | 0.088 | 0.083 | 0.020 | 0.103 | 0.000 | 6.63 | 0.148 | 0.297 | 0.033 | 0.330 | 0.041 | 8.22 | 0.173 | 0.327 | 0.038 | 0.365 | 0.030 |
| S2 | greedy (1 view) | orbit mean | 4.30 | 0.074 | 0.083 | 0.017 | 0.100 | 0.000 | 6.63 | 0.134 | 0.297 | 0.030 | 0.327 | 0.041 | 8.22 | 0.151 | 0.327 | 0.034 | 0.361 | 0.030 |
| S2 | orbit product | greedy (1 view) | 3.43 | 0.088 | 0.067 | 0.020 | 0.087 | 0.000 | 5.14 | 0.148 | 0.211 | 0.033 | 0.245 | 0.030 | 6.15 | 0.173 | 0.205 | 0.038 | 0.243 | 0.015 |
| S2 | orbit product | orbit mean | 3.43 | 0.074 | 0.067 | 0.017 | 0.084 | 0.000 | 5.14 | 0.134 | 0.211 | 0.030 | 0.241 | 0.030 | 6.15 | 0.151 | 0.205 | 0.034 | 0.238 | 0.015 |
| S2 | weighted vote (32) | greedy (1 view) | 2.94 | 0.088 | 0.060 | 0.020 | 0.080 | 0.000 | 4.77 | 0.148 | 0.212 | 0.033 | 0.246 | 0.030 | 5.92 | 0.173 | 0.219 | 0.038 | 0.257 | 0.019 |
| S2 | weighted vote (32) | orbit mean | 2.94 | 0.074 | 0.060 | 0.017 | 0.077 | 0.000 | 4.77 | 0.134 | 0.212 | 0.030 | 0.243 | 0.030 | 5.92 | 0.151 | 0.219 | 0.034 | 0.252 | 0.019 |
| S3 | greedy (1 view) | greedy (1 view) | 4.22 | 0.060 | 0.080 | 0.014 | 0.094 | 0.000 | 7.91 | 0.152 | 0.377 | 0.035 | 0.412 | 0.058 | 9.57 | 0.160 | 0.359 | 0.036 | 0.394 | 0.031 |
| S3 | greedy (1 view) | orbit mean | 4.22 | 0.046 | 0.080 | 0.010 | 0.090 | 0.000 | 7.91 | 0.134 | 0.377 | 0.031 | 0.408 | 0.058 | 9.57 | 0.137 | 0.359 | 0.031 | 0.389 | 0.031 |
| S3 | orbit product | greedy (1 view) | 3.50 | 0.060 | 0.068 | 0.014 | 0.082 | 0.000 | 6.49 | 0.152 | 0.287 | 0.035 | 0.322 | 0.045 | 7.57 | 0.160 | 0.238 | 0.036 | 0.274 | 0.016 |
| S3 | orbit product | orbit mean | 3.50 | 0.046 | 0.068 | 0.010 | 0.079 | 0.000 | 6.49 | 0.134 | 0.287 | 0.031 | 0.318 | 0.045 | 7.57 | 0.137 | 0.238 | 0.031 | 0.269 | 0.016 |
| S3 | weighted vote (32) | greedy (1 view) | 2.93 | 0.060 | 0.059 | 0.014 | 0.073 | 0.000 | 6.12 | 0.152 | 0.295 | 0.035 | 0.330 | 0.048 | 7.25 | 0.160 | 0.250 | 0.036 | 0.286 | 0.019 |
| S3 | weighted vote (32) | orbit mean | 2.93 | 0.046 | 0.059 | 0.010 | 0.069 | 0.000 | 6.12 | 0.134 | 0.295 | 0.031 | 0.326 | 0.048 | 7.25 | 0.137 | 0.250 | 0.031 | 0.281 | 0.019 |

Table 97. **EXP-124 GT-crop reference, components**: median errors, the rotation ($s_q$) and translation ($s_t$) parts of the score, the score under the joint floor rule, and the fraction of $E_R > 150°$.

| | | | $E_R$ (°) | | | $E_T$ (m) | | | score | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Rotation | Seed | Syn. | LB | SL | Syn. | LB | SL | Syn. | LB | SL |
| S1 | greedy (1 view) | 0 | 4.56 | 15.83 | 18.88 | 0.058 | 0.183 | 0.183 | 0.090 | 0.307 | 0.360 |
| | | 1 | 4.53 | 15.41 | 17.60 | 0.060 | 0.183 | 0.179 | 0.090 | 0.300 | 0.337 |
| | | 2 | 4.59 | 16.47 | 18.51 | 0.060 | 0.189 | 0.187 | 0.091 | 0.319 | 0.354 |
| S1 | orbit product | 0 | 3.69 | 11.25 | 11.57 | 0.058 | 0.183 | 0.183 | 0.075 | 0.227 | 0.233 |
| | | 1 | 3.68 | 10.78 | 10.59 | 0.060 | 0.183 | 0.179 | 0.075 | 0.219 | 0.215 |
| | | 2 | 3.71 | 11.61 | 11.11 | 0.060 | 0.189 | 0.187 | 0.075 | 0.234 | 0.225 |
| S1 | weighted vote (32) | 0 | 3.30 | 11.19 | 12.01 | 0.058 | 0.183 | 0.183 | 0.068 | 0.226 | 0.240 |
| | | 1 | 3.26 | 11.10 | 11.61 | 0.060 | 0.183 | 0.179 | 0.067 | 0.225 | 0.233 |
| | | 2 | 3.25 | 11.76 | 12.11 | 0.060 | 0.189 | 0.187 | 0.067 | 0.237 | 0.243 |
| S2 | greedy (1 view) | 0 | 4.78 | 17.17 | 18.20 | 0.091 | 0.173 | 0.191 | 0.100 | 0.329 | 0.351 |
| | | 1 | 4.79 | 17.27 | 18.89 | 0.090 | 0.173 | 0.194 | 0.100 | 0.331 | 0.363 |
| | | 2 | 4.78 | 16.60 | 19.12 | 0.098 | 0.178 | 0.195 | 0.101 | 0.320 | 0.368 |
| S2 | orbit product | 0 | 3.86 | 12.40 | 11.41 | 0.091 | 0.173 | 0.191 | 0.084 | 0.246 | 0.232 |
| | | 1 | 3.85 | 12.08 | 11.93 | 0.090 | 0.173 | 0.194 | 0.084 | 0.241 | 0.242 |
| | | 2 | 3.87 | 11.86 | 11.85 | 0.098 | 0.178 | 0.195 | 0.085 | 0.237 | 0.240 |
| S2 | weighted vote (32) | 0 | 3.42 | 12.49 | 12.25 | 0.091 | 0.173 | 0.191 | 0.076 | 0.248 | 0.247 |
| | | 1 | 3.41 | 12.35 | 12.98 | 0.090 | 0.173 | 0.194 | 0.076 | 0.245 | 0.260 |
| | | 2 | 3.42 | 11.68 | 12.41 | 0.098 | 0.178 | 0.195 | 0.077 | 0.234 | 0.250 |
| S3 | greedy (1 view) | 0 | 4.61 | 21.27 | 20.99 | 0.058 | 0.183 | 0.183 | 0.091 | 0.402 | 0.397 |
| | | 1 | 4.51 | 21.01 | 19.47 | 0.060 | 0.183 | 0.179 | 0.089 | 0.398 | 0.370 |
| | | 2 | 4.59 | 22.55 | 21.19 | 0.060 | 0.189 | 0.187 | 0.091 | 0.425 | 0.401 |
| S3 | orbit product | 0 | 3.94 | 16.28 | 14.23 | 0.058 | 0.183 | 0.183 | 0.079 | 0.315 | 0.279 |
| | | 1 | 3.89 | 15.77 | 13.09 | 0.060 | 0.183 | 0.179 | 0.078 | 0.306 | 0.259 |
| | | 2 | 3.88 | 17.25 | 13.59 | 0.060 | 0.189 | 0.187 | 0.078 | 0.333 | 0.269 |
| S3 | weighted vote (32) | 0 | 3.43 | 16.54 | 14.56 | 0.058 | 0.183 | 0.183 | 0.070 | 0.319 | 0.285 |
| | | 1 | 3.35 | 16.36 | 13.86 | 0.060 | 0.183 | 0.179 | 0.069 | 0.316 | 0.272 |
| | | 2 | 3.33 | 17.80 | 14.57 | 0.060 | 0.189 | 0.187 | 0.069 | 0.342 | 0.286 |

Table 98. **EXP-124 GT-crop reference per seed** (orbit-mean translation; the single-view-translation rows are in the run's per-seed file).

# M. Complete experiment log

This section is the complete research log of the project: every experiment identifier that was allocated, what it asked, how it ended, and which decision it triggered. It extends the short summaries in Appendix B (Tab. 11) and Appendix C (Tab. 12) with the abandoned and negative directions that do not appear in the main paper. Most experiments were pre-registered in a per-experiment report before their first result, with a written pass rule, and amendments are dated and were written before the runs they affect. The descriptive analyses (EXP-020, 104, 107, 112) had no pass rule, and EXP-045 collated results that were already known when it was written.

**Conventions.** Unless stated otherwise, errors are mean geodesic rotation errors in degrees, mean over 3 seeds. "Synth" is SPEED+ [19] synthetic validation, "LB" lightbox and "SL" sunlamp; "val-aug" is the augmented synthetic-validation proxy introduced with EXP-109. Training always uses synthetic/train only, selection uses synthetic validation only, and lightbox and sunlamp are test-only and never used for selection or tuning (DEC-000, restored by DEC-006). Phase-1/2 numbers (EXP-010–026 and EXP-101) come from a sanity subset (6,000 synthetic-train, 1,500 synthetic-val, 1,000 lightbox and 1,000 sunlamp images); EXP-030 onward uses the full splits. The *keep score* is the project's 0–5 rubric: 5 strong support (pre-registered goal met with a large effect, adopt); 4 support (goal met, carry forward); 3 partial (mixed or one-sided evidence); 2 weak (goal not met, informative side result); 1 negative (fails, reject); 0 inconclusive or not run.

**Where the sources disagree.** We list the known inconsistencies between the master index, the per-experiment reports and the paper tables, and which source we follow.

- **EXP-106 and EXP-122.** The index lists both as *queued* and both reports end with "Runs / Results / Decision: Not started". However, completed runs exist (EXP-106: T-LESS and SYMSOL-T arms A0–A2; EXP-122: evaluation `RUN-20261027-200852`), and their numbers are in the paper tables (`tables/sym.tex`, `tables/tless_sota.tex`, `tables/symsol_sota.tex`, `tables/ycbv.tex`). We report those numbers, mark the status as "runs completed; report and index not yet updated", and record no keep score or verdict that the reports do not contain. `paper/SOURCES.md` still describes the T-LESS A2 rows as blank, but `tables/sym.tex` has them filled; we follow the table.
- **EXP-064 and EXP-117.** The status header of each report is stale ("Running" and "Pre-registered … before any code or run"), but the report bodies contain the full results and the decisions. We follow the bodies and the index.
- **Sunlamp 12.28 vs 12.25.** The same locked-recipe trees (PoE + orbit) give 12.28 in the EXP-109/112 evaluation and 12.25 in the EXP-113/115/117 evaluations; two evaluation passes of the same trees differ in the second decimal. Each row below quotes the number of the experiment it describes.
- **Keep scores.** For EXP-116, 117 and 118 the index puts a PASS/FAIL verdict in the keep-score column instead of a 0–5 number; for EXP-064, 106, 119, 122, 124, 126 and 127 it has none. We copy these entries literally and do not assign scores.

## M.1. Phases and decisions

**Plan.** The project followed a phase-wise execution plan in which each phase answers one question and ends with a written gate: Phase 0 (ground truth and rotation ge-

ometry), 1 (single-branch representations), 2 (fusion), 3 (fixed-depth Tesseract predictor), 4 (continuous rotation and $SO(3)$ flow), 5 (adaptive depth), 6 (rotation milestone, including broader objects), 7 (translation and 6-DoF), 8 (joint $SE(3)$ flow) and 9 (optional extensions). The plan originally used VGGT as the geometric branch; MoGe-2 [24] replaced it after Phase 1 (DEC-001). Phases 0–4 were run as planned; Phase 5 was reduced to an oracle check; Phases 6–8 were largely replaced by the real-domain and cross-dataset experiments of the EXP-1xx series, which the user requested as results came in.

**Phase 0: geometry and ground truth (EXP-000–003).** EXP-000 fixed the SPEED+ pose convention ($p_{\text{cam}} = R(q)\,p + r$, scalar-first quaternions): projected keypoint boxes match a cross-check with IoU 0.97 for $R$ against 0.52 for $R^\top$. EXP-001 verified that every one of the 69,491 SPEED+ labels encodes to a valid, antipodally invariant, invertible Tesseract path. EXP-002 measured the fixed-depth precision of the recursive tesseract grid [15]: the error halves per level and depth 5 (131,072 cells) has a 2.51° mean, 3.97° p95 and 6.20° exact worst-case quantisation error, with cell volumes varying by up to 14.4×. EXP-003 compared it with Hopf [27] and Super-Fibonacci grids: per hypothesis the Tesseract is *less* precise (Super-Fibonacci needs only 0.61–0.88× as many points for equal p95 error), and its defensible property is the cheap hierarchy (44 scored nodes for a greedy descent to depth 5). The Phase-0 gate set $L_{\max} = 5$ and kept a Hopf grid as the mandatory matched baseline.

**Phase 1: branch studies on the subset (EXP-010–020).** A hierarchical MLP probe on frozen DINOv3 [22] features (EXP-010) selected ViT-L with $4 \times 4$ spatial pooling (`grid4`): synthetic-validation mean 27.7° (median 17.2°), with real-domain medians of 62.2° (LB) and 70.3° (SL), so the synthetic-to-real gap was about 4× from the start. VGGT tokens and geometry maps (EXP-011) were weaker everywhere (best synthetic mean 48.7°) at about 17× the extraction time; the VGGT complementarity audit (EXP-012) found an oracle-of-two gain of +7.7 acc@10 points on synthetic data but almost none on real images. MoGe-2 was then tested as a single-image geometry branch (EXP-013, 39.6° synthetic mean) and adopted in place of VGGT (DEC-001), although its complementarity to DINOv3 (EXP-014) was no larger than VGGT's and a non-geometric DINOv3-B control was more complementary still. Four user-requested side studies followed: a training-free two-way foreground consensus (EXP-015, kept), its effect on pose (EXP-016, one-way: it fixes MoGe-2 normals but not DINOv3 pooling), and three CASS-inspired spectral options (EXP-017–020), all rejected.

**Phase 2: fusion (EXP-025, 026, 029).** A training-free per-level product of experts [9] (PoE) of the DINOv3 `grid4` probe and the consensus-masked MoGe-2 normals probe (`normals16~and`) beat the strongest single branch (EXP-025: synthetic 25.9° vs 27.7°; descriptively LB 62.5° vs 76.1°). No training-free reliability weighting beat equal weights (EXP-026), so the combination and learned-gate experiments of the plan were skipped and the fusion was frozen (DEC-003). The plan's learned-fusion slot was later reopened at the user's request as EXP-029 (DEC-007), run at full data after Phase 3 with the Transformer tree and compared with a PoE of two single-branch trees on the same $8\times8$ grid: it passed its literal rule on synthetic data but equalled DINOv3 alone and was much worse on real images, and PoE was kept (DEC-008).

**Phase 3: the tree predictor at full data (EXP-030–036).** Training on all 47,966 synthetic-train images cut the error about 4× (EXP-030: PoE 6.24°, DINOv3 alone 6.64°) and kept the fusion gain. Five decoder and target variants were then tested against this control: a GRU path decoder (EXP-031, failed its rule by 0.005°), a cross-attentive Transformer decoder on $8 \times 8$ tokens (EXP-032, 5.685°), geodesic soft targets (EXP-033, 6.068°), beam search (EXP-034, no gain) and a tangent residual trained at the ground-truth leaf (EXP-035, no gain). The Transformer with soft targets (EXP-036) was best (PoE 5.540°; LB 26.9°, SL 36.6°) and was frozen (DEC-004 update). From EXP-033 on, DINOv3 alone was better than PoE on synthetic validation (5.35° vs 5.54° for EXP-036) while PoE was far better on real images; this selection-policy conflict led to DEC-005 and its withdrawal in DEC-006.

**Phase 4: continuous refinement, flows and distillation (EXP-040–045), with the flip audit.** Phase 4 asked whether a continuous $SO(3)$ model could refine the tree and fix its real-domain failures. A tangent residual at the *predicted* leaf (EXP-040) gained 0.13° on synthetic data only; a separate cell-conditioned Riemannian flow-matching refiner [2] (EXP-041) gained 0.30° but under-covered; an image-conditioned $SO(3)$ flow posterior from a uniform base (EXP-042) gave the best samples (best-of-32 9.7° on LB) but a worse mode estimate on real images. Flow-induced Tesseract masses (EXP-043) were exactly consistent but a poor decoder, and distilling them into the tree (EXP-044) did not pass its criterion. In parallel, a user-requested audit of the ∼180° failures (EXP-104) ruled out encoder, metric, convention and seam bugs and attributed them to visual near-symmetry: the right mode is among the tree's hypotheses but is not selected. A candidate-expansion study (EXP-105) confirmed that no scorer on the clean features selects the right flip on real images, and also tested the original corner formulation of Kurz et al. (arm K). The

matched collation EXP-045 kept the flow posterior only as a re-noise verifier over tree candidates. Phase 5 was reduced to the oracle check EXP-050 (DEC-012), whose small synthetic headroom (0.51°) and real gains that only hedge towards coarser cells made adaptive depth a low priority.

**The real-domain phase (EXP-107–118).** After EXP-104/105 showed that the failures are a mode-selection problem, four real-domain levers were proposed (DEC-013): photometric and in-plane rotation augmentation (A), $16\times16$ tokens (B), in-plane orbit voting at test time (C) and a flip verifier on augmented features (D). A, B and C were run as EXP-109 (DEC-014). A discrete flow-matching revision of the tree path (EXP-108) was run first and failed. EXP-109 A + C reached 3.87° / 12.30° / 12.28° (synthetic / LB / SL) against 5.54° / 26.9° / 36.7° for EXP-036; B failed. A weighted PoE (EXP-110) and a flow head on the augmented DINOv3 tree (EXP-111) were not adopted, the recipe was locked on the EXP-109 methodology in a PoE and a DINOv3-only variant (DEC-018), and EXP-112 timed it. The user then asked for three strengthening studies (DEC-019): the tree against a count-matched flat Hopf classifier (EXP-113, tree far better), orbit scoring rules (EXP-114, the product does not beat weighted voting on synthetic data) and an augmentation ablation (EXP-115, photometric augmentation plus orbit voting carries the real-domain gain, not the rotation augmentation). Two training-time alternatives to orbit voting followed (DEC-021): a roll-consistent hierarchical loss (EXP-116, no gain over a leaf-only equivariance loss) and orbit distillation into a single-view student (EXP-117, the best system: 3.69° / 11.21° / 11.09°). EXP-118 added the SPACE-HOP [5] translation block for full 6-DoF. EXP-107 produced the descriptive figures, log-likelihoods and timings for the Phase-4 system.

**The cross-dataset phase (EXP-064, 106, 119, 122, 124, 126, 127).** EXP-064 retrained per-object trees with the locked recipe on T-LESS, SYMSOL-T and YCB-Video (DEC-009, DEC-018) and added a zero-shot arm with the frozen SPEED+ trees (DEC-022). Orbit voting transferred to T-LESS and SYMSOL-T, MoGe-2 fusion helped only on T-LESS, YCB-V trained on real videos alone failed to generalise, and there was no zero-shot transfer. Three fixes were then approved together (DEC-024): synthetic YCB-V renders (EXP-122), symmetry-aware multi-path training targets on T-LESS and SYMSOL-T (EXP-106, amended) and cross-dataset orbit distillation (EXP-119). For matched comparisons the user chose a "core matched set" (DEC-025): SPEED+ with a synthetic-only detector and the official score (EXP-124) and translation on the other datasets (EXP-126); the LiePose instance-set protocol (EXP-125) is reserved but not yet pre-registered. Finally, EXP-127 (DEC-026) was queued to separate hierarchical prediction from grid geometry, because EXP-113's flat baseline used a grid with a much higher quantisation floor. At the time of writing EXP-106 and EXP-122 have completed runs, EXP-127 has its quantisation-floor stage, and EXP-119, 124, 126 and the EXP-127 trainings are queued.

**Decision log.** Every architecture, scope and split decision was recorded with the evidence behind it. The entries below give date, decision and reason; "user" marks decisions taken by the user, the others were proposed by us and accepted.

- **DEC-000 (2026-10-05, user). Split policy.** Train on SPEED+ synthetic/train only; select on synthetic/validation; lightbox and sunlamp are test-only and never used for selection or tuning.
- **DEC-001 (2026-10-08, user). MoGe-2 replaces VGGT.** MoGe-2 was the better single geometric branch on synthetic validation (39.6° vs 48.7° mean) and about equally complementary to DINOv3 (oracle acc@10 0.289 vs 0.278), with a slightly stronger agreement cue (3.2× vs 2.6×). Recorded caveats: a non-geometric DINOv3-B branch was more complementary (0.308), and MoGe-2's own foreground mask fails on real images.
- **DEC-002 (2026-10-09). Numbering.** Side experiments EXP-015–020 had collided with the plan's Phase-2 IDs, so the plan's Phase-2 experiments became EXP-025–029 and new user-requested side experiments use the EXP-1xx series; IDs are never renamed.
- **DEC-003 (2026-10-09). Phase-2 gate.** The representation is the training-free equal-weight PoE of DINOv3-L `grid4` and MoGe-2 `normals16~and` (EXP-025: 25.9° vs 27.7°). EXP-027 was skipped because no reliability signal helped alone; EXP-028/029 (learned gates, cross-attention) were conditional on training-free fusion failing, which it did not.
- **DEC-004 (2026-10-09). Phase-3 gate.** First froze the MLP tree with geodesic soft targets and greedy decoding (EXP-033, PoE 6.068° vs 6.242°). It flagged the selection-policy problem: DINOv3 alone was lower on synthetic validation (5.69° vs 6.07°) while PoE was far better on lightbox and sunlamp, and three options were put to the user.
- **DEC-004 update (2026-10-09). Phase-3 gate re-decided.** After EXP-032 and EXP-036, the Transformer decoder with geodesic soft targets (PoE 5.540°) replaced EXP-033. The selection-policy question remained open (DINOv3 alone 5.35° vs PoE 5.54° on synthetic data; PoE 26.9° vs 37.1° on LB).
- **DEC-005 (2026-10-10, user). Real-domain validation split.** A seeded 20% of lightbox (1,348 images) was designated for selection because synthetic validation has almost no errors above 90° (0.1% for EXP-036 DINOv3 seed 0, vs 16% on real images). *Withdrawn the same day by DEC-006*; it never influenced a training run or checkpoint.

- **DEC-006 (2026-10-10, user). Strict split policy restored.** The user clarified that lightbox and sunlamp must stay test-only; DEC-000 applies unchanged, and the pre-registered Phase-4 gate was rewritten to synthetic validation before any result. Recorded limitation: gains in multimodality or flip recovery can only appear in the descriptive real-domain numbers.
- **DEC-007 (2026-10-10, user). Two plan additions.** EXP-029 (PanSt3R-inspired learned feature fusion) reopens the plan's skipped learned-fusion slot; EXP-064 (broader objects) is added to Phase 6. The cancelled LoRA experiment (EXP-102) stays cancelled.
- **DEC-008 (2026-10-13, user). Learned fusion not adopted.** EXP-029's concatenation + MLP passed its literal rule against the $8 \times 8$ PoE (5.341° vs 5.582°) but equalled DINOv3 alone (5.345°) and was much worse on real images (LB 40.8° vs 27.9°); adopting it on synthetic data alone would contradict the ground rule against selecting only by the best synthetic result.
- **DEC-009 (2026-10-13, user). Datasets for EXP-064.** T-LESS, SYMSOL-T and YCB-Video, located on a second drive. Recorded deviations: YCB-V is the original PoseCNN release rather than the BOP version, the local T-LESS test set is the BOP19 subset, and SYMSOL-T was rendered locally.
- **DEC-010 (2026-10-13, user). EXP-105 and EXP-106.** Candidate expansion and best-fit selection on SPEED+ (absorbing the proposed EXP-103 "rotate and match" as a template arm), and a T-LESS multi-path tree. The EXP-042 flow serves as the diffusion model because on $SO(3)$ the diffusion score equals the flow-matching target. Addendum: arm K, the original corner formulation of Kurz et al. [15], added to both, with the stated expectation that redundancy between $q$ and $-q$ cannot by itself fix 180° flips.
- **DEC-011 (2026-10-13, user). EXP-106 on hold.** SPEED+ phases first; cross-dataset work (EXP-064, 106) last, after the recipe is locked, retraining only the heads.
- **DEC-012 (2026-10-13, user). Run order and scope.** Cross-dataset block at the end of Phase 6; a recipe-lock decision before any cross-dataset run; early CPU-only data pipelines; Phase 5 reduced to the oracle check EXP-050 because errors are dominated by 180° flips, not depth; LiePose-style visualisations as a separate analysis experiment (EXP-107). LoRA stays cancelled ("might take too much time and only give us a trivial positive result").
- **DEC-013 (2026-10-15, user). EXP-108 added; real-domain levers proposed.** A direct path flow that revises root and child decisions; real-domain options A (augmentation), B ($16 \times 16$ tokens), C (orbit voting), D (flip verifier) proposed. Unlabelled real-image adaptation was not proposed because DEC-006 keeps real images test-only.
- **DEC-014 (2026-10-15, user). EXP-109.** A + B + C as one pre-registered experiment, order EXP-050 → EXP-109 → EXP-108 → EXP-107; selection on synthetic validation and the val-aug proxy only.
- **DEC-015 (2026-10-15, user). EXP-108 becomes discrete flow matching; baselines dropped.** The iterative path denoiser and its no-geometry matched control were not run (code kept, disabled). Consequence recorded: whether the geometry token adds value cannot be tested.
- **DEC-016 (2026-10-15, user). EXP-110.** Weighted per-level PoE of the frozen EXP-036 trees, selected on the val-aug proxy because clean synthetic validation is known to favour DINOv3 alone.
- **DEC-017 (2026-10-16, user). EXP-111.** A DINOv3-only augmented tree with the EXP-042 flow head and DINOv3-only timing, later moved to the end of the queue.
- **DEC-018 (2026-10-16, user). Recipe lock.** Frozen DINOv3-L `tokens8` (+ MoGe-2 `normals16~and` in the PoE variant), GT square crops, one Tesseract Transformer per branch trained with EXP-109 arm A (2 augmented copies, photometric + in-plane rotation with transformed labels), equal per-level PoE, and EXP-109 C orbit voting (4 views, beam-8 candidates mapped back, product of tree likelihoods). Two variants are reported everywhere: PoE and DINOv3-only, each with and without orbit voting. Excluded: $16 \times 16$ tokens, weighted PoE, discrete flow matching, flow distillation, arm K; the flow is optional and judged in EXP-111.
- **DEC-019 (2026-10-16, user). Strengthening experiments.** EXP-113 (tree vs matched flat Hopf), EXP-114 (orbit scoring rules) and EXP-115 (augmentation ablation), all with the locked recipe and a shared evaluator, then EXP-064. Optional work (EXP-051–055, EXP-107 re-run, EXP-106) deferred.
- **DEC-020 (2026-10-16, user). Caches moved; obsolete caches deleted.** 105 GB of caches moved to an 8 TB drive; 55 GB deleted, including the $16 \times 16$ token memmap of the rejected EXP-109 arm B and all subset-v1 caches of Phases 1–2, so those can no longer be re-evaluated without re-extraction. A relative-path bug introduced by the move broke several queued runs (EXP-111 seed 2, EXP-112, EXP-114, the EXP-113 trainings); they were fixed and re-queued, and the failed runs kept as records.
- **DEC-021 (2026-10-17, user). EXP-116 and EXP-117 pre-registered; EXP-064 paused for disk space.**
- **DEC-022 (2026-10-19, user). EXP-064 zero-shot arm and EXP-118.** EXP-064 keeps per-object trees with the unchanged recipe and adds a CPU zero-shot arm with the frozen SPEED+ trees; EXP-118 adds the SPACE-HOP translation formulation unchanged; queue order EXP-064 → EXP-118 → EXP-117.
- **DEC-023 (2026-10-21, user). Recipe unchanged; EXP-117 recorded as an enhancement.** Orbit distillation (PoE

+ orbit 3.69° / 11.21° / 11.09° vs 3.87° / 12.30° / 12.25°) is reported as a successful add-on; later experiments use it only if the user says so.

- **DEC-024 (2026-10-27, user). Cross-dataset fixes.** EXP-122 (YCB-V with synthetic renders), EXP-106 (amended: per-object A0/A1/A2 on T-LESS and SYMSOL-T) and EXP-119 (cross-dataset orbit distillation); IDs EXP-120 (3-D geometry into DINO), EXP-121 (smaller backbones / Jetson) and EXP-123 (flip-confusion probe) reserved for on-hold ideas.
- **DEC-025 (2026-10-28, user). Core matched set.** EXP-124 (detector + official score on SPEED+), EXP-125 (LiePose T-LESS instance set; to be pre-registered later) and EXP-126 (translation on the other datasets, pre-registered by amendment). A literature check found paper corrections to apply before submission (swapped SPNv2 numbers, SPACE-HOP's swapped light-box/sunlamp baseline columns, two missing recent crop-based results, and LiePose's use of GT visible masks).
- **DEC-026 (2026-10-29, user). EXP-127.** Hierarchical tree vs a flat classifier over the same 131,072 leaves vs a flat balanced-Hopf ($1771 \times 74$) classifier, with the locked recipe and without orbit distillation; 12 new flat trainings, queued after the other queues.

## M.2. Experiment table

Tables 99 to 102 list every experiment in the index, in ID order, with the index wording shortened. Table 103 lists the identifiers that were allocated by the plan or a decision but never run. Abbreviations: tr.-free = training-free; pred. = trainable predictor; LB/SL = lightbox/sunlamp; "a/b/c" triples are synthetic/LB/SL unless stated.

| ID | Question | Track | | Status | Key result | Decision | Keep |
|---|---|---|---|---|---|---|---|
| 000 | SPEED+ loader and pose convention | tr.-free | | done | Convention $R$ ($p_{\text{cam}} = R(q)p + r$); bbox IoU 0.97 vs 0.52 for $R^\top$ | Keep | 5 |
| 001 | GT quaternion → Tesseract path | tr.-free | | done | All invariants hold on 69,491 labels | Keep | 5 |
| 002 | Precision by depth | tr.-free | | done | L5: 2.51 mean, 3.97 p95, 6.20 exact max; cell-volume ratio 14.4× | Keep; $L_{\max} = 5$ | 4 |
| 003 | Tesseract vs Hopf vs Super-Fibonacci | tr.-free | | done | Less precise per hypothesis (1.2–1.6× the points of Super-Fibonacci for equal p95); cheapest hierarchy (44 nodes at L5) | Keep Tesseract; Hopf as matched baseline | 3 |
| 010 | DINOv3 representation + MLP tree probe (subset) | pred. | | done | ViT-L `grid4`: synth median 17.2, LB 62.2, SL 70.3 | Primary branch | 4 |
| 011 | VGGT token / geometry controls (subset) | pred. | | done | Best tokens synth median 25–27, LB 84–105; weaker than DINOv3 everywhere, 17× slower | Kept conditionally for 012; retired by DEC-001 | 1 |
| 012 | DINOv3–VGGT complementarity, frequency audit | analysis | | done (one attempt crashed the host) | Synth oracle-of-two acc@10 +7.7 pts; agreement makes DINOv3 correct 2.6× more often; real gains small | VGGT kept conditionally (synth only) | 2 |
| 013 | MoGe-2 single-image geometry controls | pred. | | done | `grid4` synth 39.6 mean / 21.2 median (VGGT 48.7 / 25.4); `normals16` best geometric acc@10 (0.179); own mask fails on real | MoGe-2 replaces VGGT (DEC-001) | 3 |
| 014 | DINOv3–MoGe-2 complementarity | analysis | | done | Oracle acc@10 0.289±0.020 vs VGGT 0.278±0.022; non-geometric DINOv3-B control 0.308 | MoGe-2 adopted; geometric value re-tested later | 2 |
| 015 | Two-way DINOv3/MoGe-2 foreground consensus | tr.-free | | done | AND mask cuts MoGe-2 background leakage 0.52→0.12 (LB), 0.57→0.12 (SL) | Two-way, asymmetric; AND kept | 4 |
| 016 | Does the consensus mask help pose? | tr.-free probe | + | done | Consensus normals mean 84.0→77.1 (LB), 100.1→87.1 (SL); masked DINOv3 pooling no gain | One-way (DINOv3 → MoGe-2) | 4 |
| 017 | CASS 1: spectral foreground from fused graph | tr.-free | | done | Worse than AND (leakage +0.10/+0.16, recall −0.10); splits faces, not figure/ground | Reject | 1 |
| 018 | CASS 2: spectral attention injection | tr.-free | | done | No mask gain; pose worse both ways (MoGe-2 SL 83.9→91.5; DINOv3 synth 27.7→34.2) | Reject | 1 |
| 019 | CASS 3: spectral part pooling | tr.-free probe | + | done | `parts4` vs matched `grid2`: synth 53.0 vs 28.5, LB 96.9 vs 79.9 | Reject | 1 |
| 020 | CASS 4: spectral head-matching complementarity | analysis | | done | Weak inverted signal on real ($\rho \approx -0.25$), none on synth (AUROC 0.48); disagreement AUROC 0.68–0.75 | Not adopted | 2 |

Table 99. **Experiment log, part 1**: Phase 0 and Phase 1 (subset v1 for EXP-010–020). Mean errors in degrees.

| ID | Question | Track | Status | Key result | Decision | Keep |
|---|---|---|---|---|---|---|
| 025 | Single branches vs training-free posterior fusion (plan 020) | tr.-free fusion | done | PoE synth 25.9 vs 27.7 (DINOv3); LB 62.5 vs 76.1 (median 41.3 vs 62.2); SL 71.4 vs 79.1 | Keep fusion (gate passed) | 5 |
| 026 | Reliability signals one at a time (plan 021) | tr.-free fusion | done | None beats equal PoE (margin 25.8 vs 25.9, entropy 26.1, agreement gate 27.4) | Equal PoE; 027–028 skipped | 2 |
| 029 | PanSt3R-inspired learned fusion vs PoE, $8 \times 8$ grid | trainable fusion | done | Concat+MLP synth $5.341 \pm 0.031$ vs PoE8 5.582 (rule PASS) but = DINOv3 alone 5.345; LB 40.8 vs 27.9 | Not adopted (DEC-008) | 2 |
| 030 | MLP tree baseline on full data | pred. + fusion | done | PoE synth 6.24 vs DINOv3 6.64; LB 31.1 vs 41.8; SL 43.7 vs 53.4; full data cuts error $\sim 4\times$ | Phase-3 control | 5 |
| 031 | GRU path decoder | pred. | done | PoE 6.149 vs 6.242 ($\Delta$ 0.093 $<$ std 0.098, narrow FAIL); DINOv3 6.29 vs 6.64; SL 40.6 vs 43.7 | Not frozen | 3 |
| 032 | Cross-attentive Tesseract Transformer, hard targets | pred. | done | PoE 5.685 vs 6.242; DINOv3 6.08; LB 31.6, SL 41.8 | Keep | 4 |
| 033 | Hard vs geodesic soft targets | pred. | done | PoE 6.068 vs 6.242 (PASS); DINOv3 alone 5.69; LB 28.2, SL 42.1 | Frozen (DEC-004); selection issue flagged | 4 |
| 034 | Greedy vs beam decoding | inference | done | No gain at any width (6.242 greedy vs 6.248 beam 8; up to $6.6\times$ nodes) | Greedy | 1 |
| 035 | Tangent residual (GT-leaf teacher forcing) | pred. | done | Refines 0.05 (6.27→6.21); wrong-cell errors dominate | Reject | 1 |
| 036 | Transformer + geodesic soft targets | pred. | done | PoE 5.540; LB 26.9 (median 9.0), SL 36.6 (median 13.0); DINOv3 alone 5.35 | Frozen (DEC-004 update) | 5 |
| 040 | Tangent residual at the predicted leaf | pred. | done | Fused $5.414 \pm 0.043$ vs 5.544 (PASS, all from DINOv3); LB 26.88 vs 26.91, SL 36.63 vs 36.66 | Sub-ms fast fallback | 3 |
| 041 | Cell-conditioned $SO(3)$ flow refiner (base $\sigma = 3°$) | RFM | done | Fused $5.244 \pm 0.077$ (PASS vs leaf and 040); real unchanged; 90% regions cover 80%; 19 ms/img | Dropped in 045 | 3 |
| 042 | Image-conditioned $SO(3)$ flow posterior (uniform base) | RFM | done | Fused $5.219 \pm 0.149$ (PASS; DINOv3 head 4.57); LB best-of-32 9.7 vs greedy 26.9 but KDE mode 29.8; coverage 0.98 at 0.9 | Kept as verifier; samples for 043/044/107 | 4 |
| 043 | Flow-induced Tesseract masses | analysis | done | Parent–child error 0 (PASS); synth flow top-1 beats tree at L2–5 (L2 0.832 vs 0.787); flow-mass greedy 6.21 vs 5.54 (FAIL); real: tree better at all levels | Masses as 044 target; counting decoder rejected | 3 |
| 044 | Tree distilled from flow masses, $\beta \in \{0, 0.1, 0.3, 1\}$ | pred. | done (seed 0; stopped by rule) | $\beta = 0$ best (5.589; $\beta = 1$ 5.622, all within 0.1) → FAIL; fit 4.38→2.86; $\beta = 1$ LB 27.6→26.2 but $>90°$ 0.21%→0.49% | EXP-036 stays frozen | 2 |

Table 100. **Experiment log, part 2**: Phase 2 (EXP-025/026 on the subset; EXP-029 at full data, run after Phase 3), Phase 3 and Phase 4 (full data). Mean errors in degrees on synthetic validation unless a domain is named.

| ID | Question | Track | Status | Key result | Decision | Keep |
|---|---|---|---|---|---|---|
| 045 | Phase-4 matched comparison | collation | done | Verifier (tree candidates + re-noise $\tau$ 0.8) synth 4.78, SL 32.7; KDE 5.22; RFM 5.24 with coverage 0.80 vs 0.98; residual 5.41 at 0.33 ms | Posterior kept as verifier; RFM dropped | 4 |
| 050 | Oracle adaptive-depth targets | analysis | done | Synth stop headroom 0.51±0.02 (PASS; oracle bound); LB 4.21, SL 5.83 by hedging, flips unchanged; BRANCH oracle 3.26 synth | 051–055 eligible, wait for user | 3 |
| 064 | Broader objects: T-LESS, SYMSOL-T, YCB-V (heads retrained) | full system | done[a] | Test: T-LESS PoE+orbit 16.57 (val 9.25); SYMSOL-T DINOv3+orbit 2.62 (PoE 2.67); YCB-V DINOv3+orbit 69.4. Zero-shot: none | Fusion PASS on T-LESS only; claim narrowed | – |
| 101 | Tesseract vs SPACE-HOP Hopf: grid, prediction, timing (subset) | grid + head | done | Tree head faster from 131k hypotheses (A400: 1.2 vs 3.3 ms; CPU 1.7 vs 12.2 ms; 9 M vs 76 M params); flat faster at 3,072; DINOv3 L3 29.2 vs 50.7; consensus feature ties | Tree for L5–6; flat Hopf-3k as baseline | 3 |
| 102 | LoRA adapters for DINOv3 / MoGe-2 | backbone adapt. | cancelled by user before any run | – | Code kept; not run | 0 |
| 104 | Audit of $\sim$180° failures | analysis | done | Decoders exact (max 5.78); metric, convention, augmentation ruled out; flat heads flip too; 180° failures at 2× chance with clustered body axes; 180° allowance LB 38.9→23.5 | Cause: visual near-symmetry → mode selection | 4 |
| 105 | Candidate expansion + best fit; absorbs 103; arm K (Kurz corners) | verifier | done | Re-noise $\tau$ 0.8 on C3 4.78±0.11 (PASS); SL 36.6→32.1–32.7, LB 26.6→25.3–26.5; flip oracle LB 9.8 but no scorer repairs real flips; templates fail (flip acc. 0.63/0.78); arm K 5.20±0.09 (PASS) but LB 28.8, SL 39.1 | Re-noise as verifier; templates rejected; K not in recipe | 3 |
| 106 | Symmetry-aware multi-path targets (A0 single, A1 canonical, A2 all equivalents) on T-LESS, SYMSOL-T | pred. | runs done[b] | PoE val/test, orbit: T-LESS A0 9.3/16.6, A1 10.7/18.2, A2 8.3/15.2; SYMSOL-T A0 2.65/2.67, A1 9.73/9.69, A2 2.10/2.11; arm K and best fit not run | A2 best; A1 breaks orbit voting | –[b] |
| 107 | Paper visuals, metrics, log-likelihood, NFE/M sweep, timing | analysis | done (descriptive) | NFE 10–20 plateau; $M = 64$ best (synth 5.05, SL 35.1); GT log-lik. flow 3.85 vs tree 0.98 nats (LB); DINOv3 6 ms, MoGe-2 20.5 ms, trees 12.8 ms at batch 1 | Templates; re-run for final recipe | 3 |
| 108 | Discrete flow matching on the Tesseract path | pred. (flow) | done | NFE 4 / $M$ 8: 7.22±0.12 vs 5.54 (FAIL); LB 29.8, SL 39.8; repairs $\sim$5% of >150° failures | Not adopted | 1 |
| 109 | Augmentation (A), 16 × 16 tokens (B), orbit voting (C) | pred. + inference | done | A+C synth 3.87, val-aug 5.82, LB 12.30, SL 12.28 (EXP-036: 5.54/30.84/26.91/36.66); A alone 4.86/9.75/17.35/20.16; B FAIL (30.51 vs 30.84); C on 036 4.62/19.84/20.25/24.83; A+C fixes 62%/80% of >150°, breaks 0.4% | Recipe A + C (DEC-018); B rejected | 5 |
| 110 | Weighted per-level PoE, $w \in [0, 1]$ | tr.-free fusion | done | $w^* = 0.8$ on val-aug (23.65 vs 30.84, PASS; clean 5.14) but LB 32.0 vs 26.9 (real best $w \approx$0.5–0.7) | Not adopted | 2 |
| 111 | Augmented DINOv3 tree + $SO(3)$ flow (KDE, verifier) | RFM | done | KDE synth 4.10 vs 4.84 (PASS) but real worse (25.2/24.4 vs 22.6/22.1); verifier 4.42, real $\approx$ tree; 21.6 vs 13.1 ms | Not adopted; orbit voting better everywhere | 2 |
| 112 | Locked recipe, PoE vs DINOv3-only, timing | evaluation | done | PoE+orbit 3.87/12.30/12.28 at 279 ms; DINOv3+orbit 4.21/17.40/15.27 at 70 ms; DINOv3 greedy 4.84/22.56/22.02 at 15 ms; MoGe-2 $\approx$41 ms/view incl. 20 ms CPU mask | Report both variants | 4 |

Table 101. **Experiment log, part 3**: end of Phase 4, Phase 5 and the real-domain phase. [a]The EXP-064 report's status header still says "Running"; its body and the index record completion on 2026-10-20. [b]Index: "Queued"; report: "Not started". Completed runs exist and their numbers are taken from the paper tables; no keep score has been assigned.

| ID | Question | Track | Status | Key result | Decision | Keep |
|---|---|---|---|---|---|---|
| 113 | Tree vs count-matched flat Hopf classifier (131,076 anchors), same recipe | grid + pred. | done | PoE+orbit synth 3.87 vs 7.53, val-aug 5.82 vs 9.80; greedy 4.85 vs 8.95; real: tree better in all 8 cases (LB 12.30 vs 13.15, SL 12.25 vs 16.31); 2.4 M vs 36.1 M params | Supported (precision); grid confound open (127) | 5 |
| 114 | Orbit product vs max-confidence vs cluster voting | inference | done | Synth: weighted vote 3.46 $<$ product 3.87 (FAIL); val-aug 5.82 vs 5.98 (marginal PASS, PoE); real: product best 12.30/12.25 vs 12.58/12.88 | Claim limited to orbit consistency | 3 |
| 115 | Photometric-only vs rotation-only vs combined augmentation | ablation | done | PoE+orbit: combined 3.87/12.30/12.25, photometric 4.25/12.29/12.37, rotation 3.78/18.69/20.70, none 4.62/20.25/24.83 | Real gain = photometric + orbit; keep combined | 4 |
| 116 | Roll-consistent hierarchical consistency loss (T) vs paired views (C1), leaf-only equivariance (C2) | pred. | done | Greedy val-aug (DINOv3): C2 9.43, T 9.57, A 9.88, C1 10.04; PoE+orbit synth C2 3.71, T 3.74, A 3.87; real $\approx$ A | No hierarchical claim; C2 not adopted | FAIL[c] |
| 117 | Orbit distillation into a single-view student | pred. (distill.) | done[a] | Student greedy val-aug DINOv3 9.33 vs 9.88, PoE 9.22 vs 9.75; student PoE+orbit 3.69/11.21/11.09 (vs 3.87/12.30/12.25) | Enhancement; recipe unchanged (DEC-023) | PASS[c] |
| 118 | Full 6-DoF: SPACE-HOP translation block on the tree tokens | pred. (6-DoF) | done | $E_T$ (joint, orbit) DINOv3 0.059/0.185/0.183 m, PoE 0.093/0.175/0.194 m; $E_R$ PoE 3.86/12.11/11.73; box only 0.56 m | Block $\gg$ box only; rotation not hurt | PASS[c] |
| 119 | Orbit distillation on T-LESS and SYMSOL-T | pred. (distill.) | queued | – | Pending | – |
| 122 | YCB-V with `data_syn` renders on real train backgrounds | training data | run done[b] | Val PoE+orbit 12.7$\pm$1.0 vs 74.1$\pm$7.4 (real only); test PoE greedy 10.9, PoE+orbit 12.2; orbit worse than greedy on val (PoE 10.5$\rightarrow$12.7) | No verdict recorded yet | –[b] |
| 124 | SPEED+ with a synthetic-only detector, official score | full system | queued | First detector run aborted by the memory guard (reported 44.3 GB vs 11 GB real); retried with identical settings | Pending | – |
| 126 | Translation on T-LESS, SYMSOL-T, YCB-V vs box only | full system | queued | – | Pending | – |
| 127 | Hierarchy vs flat at the same 131,072 leaves; flat balanced Hopf (1771 $\times$ 74) | ablation | queued (floors done) | Quantisation floor on synth-val rotations: Tesseract L5 2.51 mean / 5.78 max; balanced Hopf 2.29 / 3.84; SPACE-HOP 10923 $\times$ 12 6.06 / 14.97 | Pending | – |

Table 102. **Experiment log, part 4**: strengthening experiments, 6-DoF and the cross-dataset phase. [a,b]As in Tab. 101. [c]The index puts the pre-registered verdict in the keep-score column for these experiments and assigns no 0–5 score.

| ID(s) | Planned content | Fate and source |
|---|---|---|
| 027 | Best training-free combination of reliability signals (plan 022) | Skipped: no signal helped individually (DEC-003, EXP-026) |
| 028 | Learned gate (plan 023) | Skipped: conditional on training-free fusion failing, which it did not (DEC-003) |
| 029 (slot) | Shallow channel projection (plan 024) | Initially skipped (DEC-003), reopened as the PanSt3R-inspired EXP-029 (DEC-007) |
| 051–055 | Adaptive depth: feature validation, rule-based and MLP controllers, fixed vs adaptive | Eligible after EXP-050 passed; not started, waiting for the user (DEC-014); judged low priority |
| 060–063 | Phase-6 SPEED+ milestone: domain evaluation, robustness, confidence, efficiency | Proposed / on hold; efficiency covered in part by EXP-107 and EXP-112 |
| 070–085 | Phases 7–8: translation variants, octree, joint and $SE(3)$ flows | Not run; 6-DoF was obtained with the SPACE-HOP block of EXP-118 instead |
| 103 | "Rotate and match" templates | Merged into EXP-105 as its synthetic-template arm (DEC-010) |
| 120, 121, 123 | 3-D geometry into DINOv3; smaller backbones / Jetson; DINOv3/MoGe-2 flip-confusion probe | Reserved for on-hold ideas (DEC-024) |
| 125 | T-LESS on LiePose's instance set and metric | Reserved; to be pre-registered after EXP-124's code (DEC-025, DEC-026) |

Table 103. **Identifiers allocated but never run.**

## M.3. Negative and abandoned directions

Table 12 in Appendix C summarises the refiners and verifiers that failed to fix real flips. Here we record every dropped idea, in roughly chronological order, with what was tried, the numbers and the reason it was dropped. We group them by the question they addressed.

### M.3.1. Backbones and features

**VGGT as the geometric branch (EXP-011, 012).** The plan's original second branch was VGGT. With the predictor, budget, subset and seeds of EXP-010, its best tokens reached a synthetic-validation mean of 48.7° (median 25.4°) against 27.7° for DINOv3 `grid4`, its lightbox medians were 84–105°, and extraction took about 17× longer. Its late layers were worse than its middle layers (medians 44–57° for layers 17 and 23), the opposite of DINOv3. The complementarity audit showed a real but synthetic-only benefit: errors were weakly correlated (Spearman 0.31), the oracle-of-two acc@10 rose from 0.201 to 0.278, and agreement made DINOv3 correct 2.6× more often, but 95–98% of real images were solved by neither branch. VGGT is no more domain-invariant than DINOv3 (feature drift 0.54/0.96 vs 0.48/0.93 on LB/SL). It was replaced by MoGe-2 (DEC-001). One attempt of this audit computed a band-energy FFT over all 9,500 DINOv3-L grids at once (an estimated peak of about 100 GB on a shared 123 GB host) and took the machine down; this led to the memory watchdog and cgroup cap used for every later job.

**Geometry-specific complementarity was never shown (EXP-014).** MoGe-2 was adopted as the better *single* geometric branch (39.6° vs 48.7°), but on the subset it was not more complementary to DINOv3-L than VGGT (oracle acc@10 0.289±0.020 vs 0.278±0.022), and an appearance-only DINOv3-B control was more complementary than both (0.308±0.022, agreement ratio 3.67× vs 3.18×). Complementarity tracked the second model's accuracy, not whether it carried geometry. The full-data fusion results (EXP-030, 036) and later the cross-dataset results (EXP-064: PoE helps on T-LESS but not on SYMSOL-T or YCB-V) are the only evidence for the geometric branch, and they are dataset-dependent.

**Mask-weighted DINOv3 pooling (EXP-016).** The consensus mask that fixes MoGe-2 normals did not help DINOv3: mask-weighted `grid4` pooling made synthetic validation worse (27.7°→31.1°, failing the guard) and changed the real domains by less than a degree (LB 76.1→77.0°, SL 79.1→78.2°). DINOv3's patch tokens use silhouette and near-boundary context that masking removes. The pose-level consensus is one-way.

**CASS-inspired spectral fusion (EXP-017–020).** Four user-requested options were inspired by CASS (spectral object-structure graphs injected across models); all were rejected. (1) A spectral foreground from a fused DINOv3 + MoGe-2 patch graph was worse than the simple AND consensus (leakage +0.098 on LB, +0.160 on SL; recall −0.10) and failed the synthetic recall guard (0.784 vs 0.989): the geometric affinity separates spacecraft faces, not figure from ground. A spectral DINOv3-only mask was slightly better than the PCA mask, but was never registered as a separate variant. (2) Injecting one model's spectral attention graph into the other's last encoder block did not reduce MoGe-2's mask leakage (0.523→0.524 on LB) and made pose worse on every domain for all three injected feature sets, e.g. MoGe-2 `grid4` SL 83.9→91.5° and DINOv3 `grid4` synthetic 27.7→34.2°. Unlike CASS, whose edited block feeds a similarity directly, our edited block feeds downstream heads and a probe that expect the original attention statistics. (3) Pooling DINOv3 tokens into spectral parts (`parts4`, 4,116 dims) was far worse than a size-matched 2 × 2 grid (synthetic 53.0° vs 28.5°, LB 96.9° vs 79.9°, SL 104.2° vs 85.8°): Laplacian eigenvectors are not ordered consistently across images, so "part 1" is not the same physical part from view to view. (4) The spectral head-matching distance between the two backbones carried no signal on synthetic validation (AUROC 0.481 for DINOv3 errors >20°) and a weak inverted one on real images (Spearman about −0.25), while plain prediction disagreement reached AUROC 0.68–0.75.

**Finer image tokens (EXP-109 B).** Replacing the 8 × 8 DINOv3 tokens by the unpooled 16×16 grid (4× the tokens, about 2 h of training per seed) gave no gain: val-aug 30.51° vs 30.84° (seed std 0.40), clean synthetic 5.56° vs 5.54°, LB 27.01°, SL 35.97°. The 16 × 16 cache was deleted under DEC-020.

**Backbone adaptation with LoRA (EXP-102, cancelled).** LoRA adapters for DINOv3, MoGe-2 and their fusion were pre-registered as a separately named backbone-adaptation experiment, so that its results would not be mixed into the frozen-backbone comparison. The user cancelled it on 2026-10-09 before any run, and kept it cancelled in DEC-007 and DEC-012 ("might take too much time and only give us a trivial positive result"). The code, tests and an initialisation-equivalence check are kept. All results in this paper therefore use frozen backbones.

### M.3.2. Fusion

**Reliability weighting (EXP-026).** Weighting each branch by a fixed, training-free reliability signal did not beat equal-weight PoE by the pre-registered margin (0.46°): margin weighting 25.8° (−0.08), entropy weighting 26.1° (+0.18), and an agreement gate (PoE if the branches agree within 20°,

else DINOv3) 27.4° (+1.48). Equal-weight PoE already lets the peaked branch dominate; the gate hurts because it discards PoE exactly on the images where the branches disagree, which are most real images.

**PanSt3R-inspired learned feature fusion (EXP-029).** Concatenating DINOv3 and MoGe-2 tokens (linear or MLP projection) into one Transformer tree passed the literal rule against an $8 \times 8$ PoE on synthetic validation (5.341±0.031° vs 5.582±0.103°), but equalled DINOv3 alone (5.345°; per-seed differences −0.15/+0.10/+0.04°) and was much worse on real images (LB 40.8° vs 27.9°, SL 46.2° vs 37.9°; on LB it fixed 2.6% of >90° failures but newly broke 9.0%). The "gain" over PoE is the known DINOv3-vs-PoE synthetic gap. It was not adopted (DEC-008). A source-rebalancing follow-up (per-source normalisation or a gate) was suggested but never run.

**Weighted PoE (EXP-110).** A per-level weight on the DINOv3 tree, selected on the val-aug proxy, chose $w^* = 0.8$ (val-aug 23.65° vs 30.84° at $w = 0.5$; clean synthetic 5.14° vs 5.54°), but lightbox became 5.1° worse (32.0° vs 26.9°) and sunlamp did not change (36.7°). The best real weights were about 0.5 (LB) and 0.7 (SL). Following DEC-006 we did not use real data to choose another weight. The experiment was run on the clean EXP-036 trees; whether the balance shifts for the augmentation-trained trees was not tested.

**MoGe-2 fusion does not transfer to every dataset (EXP-064).** On T-LESS, PoE beat DINOv3 alone (validation, orbit 9.25° vs 9.94°; test 16.57° vs 17.88°). On SYMSOL-T it did not (orbit validation 2.65° vs 2.61°): the textureless renders give MoGe-2 normals nothing DINOv3 lacks. On YCB-V trained on real videos it was worse (validation 73.5° vs 64.3°). The fusion claim is therefore restricted to SPEED+ and T-LESS.

### M.3.3. Decoder, targets and continuous refinement

**GRU path decoder (EXP-031).** Conditioning each level on the decoded history with a GRU improved DINOv3 alone (6.64°→6.29°) and the real domains (SL 43.7°→40.6°), but the fused gain (6.242°→6.149°) missed the rule by 0.005° (gain 0.093 vs seed std 0.098). The planned GRU + soft-target combination was never run, because the cross-attentive Transformer (EXP-032, 036) superseded both.

**Beam search (EXP-034).** At full data, beams of width 2, 4 and 8 changed the fused synthetic error by at most +0.006° (6.242° greedy vs 6.248° beam 8) at up to 6.6× the scored nodes (292 vs 44). The trained tree is decisive: wrong branches are confidently wrong. On the subset (EXP-010) beam search had also changed the error by at most 0.4°.

**Tangent residual (EXP-035, 040).** A residual trained teacher-forced on the ground-truth leaf moved the fused error by only 0.055° (6.269° discrete to 6.214° refined), because the predicted leaf is correct only about 18% of the time. Retrained on the *predicted* leaf (EXP-040), it passed on synthetic data (5.414° vs 5.544°), but the gain came entirely from the DINOv3 head (the MoGe-2 residual learned essentially nothing, so averaging halved the correction) and the real domains moved by at most 0.03°. It survives only as the sub-millisecond fast fallback.

**Cell-conditioned flow refiner (EXP-041).** A separate Riemannian flow [2] started from a 3° Gaussian around the predicted cell gave the best within-mode refinement (fused 5.244°, best-of-32 2.64°), but a wrong-mode greedy choice cannot be escaped from that base, so lightbox (27.12°) and sunlamp (36.65°) did not improve. Its nominal 90% regions covered 80% on synthetic validation and 34%/22% on LB/SL, and it cost 19.4 ms per image against 0.33 ms for the residual. It was dropped in EXP-045 as dominated by the coupled posterior on coverage.

**Image-conditioned $SO(3)$ posterior as a point estimator (EXP-042, 107, 111).** A global flow posterior from a uniform base, conditioned on the tree's encoder memory, was the most accurate single head on synthetic data (DINOv3 head 4.57°; fused KDE mode 5.219°) and contained the right mode on real images (best-of-32 9.7° LB, 10.6° SL), but its samples spread over several modes (fused spread about 41° on LB) and the KDE mode chose the wrong one more often than the tree (LB 29.85° vs 26.88°). Its coverage was over-dispersed (0.98 at the 0.9 level), so it is not calibrated, and the cross-branch velocity average is a heuristic rather than a product of experts. More samples helped ($M = 64$: 5.05° / 29.6° / 35.1°), more steps did not beyond NFE 10–20 (EXP-107). Repeated on the augmentation-trained DINOv3 tree (EXP-111), the KDE arm improved synthetic error (4.10° vs 4.84°) but was worse on both real domains (25.16° vs 22.55°, 24.39° vs 22.07°), and the verifier arm was about equal to the tree on real images; orbit voting on the same tree (4.21° / 17.40° / 15.27°) beat both. The flow is kept only for its density: its ground-truth log-likelihood is much higher than the tree's on real images (3.85 vs 0.98 nats on LB, 3.12 vs −0.76 on SL).

**Flow-induced masses and distillation into the tree (EXP-043, 044).** Tesseract masses counted from the flow's 32 samples are exactly parent–child consistent, but this holds by construction and is not a finding. On synthetic data the flow's most massive cell is the ground-truth cell more often than the tree's greedy choice at levels 2–5 (level 2: 0.832 vs 0.787), but greedy descent over counted masses is worse than the tree (6.210° vs 5.541°) because leaf counts from

32 samples are noisy (TV distance 0.26 between 16 and 32 samples), and on real images the tree is better at every level. Distilling the masses into the tree with a KL term took (fit 4.38→2.86) and improved each branch alone (DINOv3 5.45→5.07°, MoGe-2 16.81→14.48° at $\beta = 1$), yet the fused synthetic error did not improve (5.589° at $\beta = 0$ vs 5.622° at $\beta = 1$) and the experiment stopped at seed 0 by rule. Descriptively, $\beta = 1$ improved lightbox (27.63→26.17°) and top-8 leaf recall (0.69→0.77) but more than doubled the synthetic >90° rate (0.21%→0.49%). The $\beta = 1$ checkpoints were kept but never used.

**Adaptive depth (EXP-050, 051–055).** An oracle that stops decoding at the best depth per image would gain 0.51±0.02° on synthetic validation (5.54°→5.03°), which passes the 0.25° eligibility rule. On real images the oracle gains 4.21° (LB) and 5.83° (SL), but by hedging towards coarser cells, which leaves the flips unchanged. The controllers EXP-051–055 were never started; augmentation and orbit voting were pursued instead.

**Discrete flow matching on the tree path (EXP-108).** The user asked for a "direct path flow" that iteratively revises root and child decisions. It was first designed as an iterative path denoiser with a geometry-aware corruption kernel and a no-geometry matched control; the user then replaced it with discrete flow matching [6] (DEC-015), so the control was never run and the value of the geometry token cannot be identified. The final model (source = samples of the frozen PoE tree, factorised mixture path $\kappa(t) = t$, time-conditioned posterior, CTMC Euler sampler) reached 7.22±0.12° at NFE 4 / $M$ 8, against 5.54° for the tree, and was worse on both real domains (29.80°, 39.84°). It changed 66–80% of paths, almost always at fine levels (root changes 3–7%), and corrected only 4–5% of the >150° failures. Two reasons are recorded: the source samples are noisier than the greedy path (training-set leaf mass on the ground truth about 0.35), and token-wise mixing is not geometry-preserving (at $t = 0.3$–$0.6$, 45–53% of corrupted paths are more than 30° from both endpoints).

### M.3.4. Selecting the right flip

**Candidate expansion and best-fit scorers (EXP-105).** After EXP-104 showed that the large real errors are 180° body-axis flips caused by visual near-symmetry (they occur in the 165–180° band at about twice the Haar rate, 0.36–0.41 vs 0.20, and a 180° allowance about one body axis cuts the DINOv3 lightbox mean from 38.9° to 23.5°), each prediction was expanded into candidates (tree beams, 180° body-axis flips, leaf-cell corners) and re-scored. The correct flip is usually present (flip oracle: LB 26.6→9.8°, where 26.6° is EXP-105's own evaluation of the EXP-036 baseline), but no scorer selects it on real images: the flow prefers the ground truth over its flips 91–92% of the time on real images, against 87–91% for the tree, and the re-ranked flip fractions stay at about 10%. The flow log-density as a re-ranker was worse than the tree on synthetic data (5.86°). Synthetic-template matching, which stood in for render-and-compare because SPEED+ has no 3-D model (absorbing the proposed EXP-103), failed even on synthetic data (flip-discrimination accuracy 0.63 for DINOv3 tokens and 0.78 for normals, vs 0.997 for the tree), because the nearest-rotation renders differ in lighting and background. Re-noise/return re-ranking over the expanded set was the one useful outcome (synthetic 4.78°, SL 32.1–32.7°), at about 100 ms per image; it improved precision rather than repairing flips, and was superseded by orbit voting.

**Kurz corner tree (EXP-105 arm K).** At the user's request we implemented the original tesseract formulation of Kurz et al. [15]: all 8 facets (so $q$ and $-q$ are both represented), corners of the subdivided cubes as representatives, and a multi-path target spread over every leaf cube sharing the corner, decoded by corner voting over the top-8 leaves. It passed on synthetic data (5.20±0.09° vs 5.54±0.07°), almost entirely from the 8-facet multi-path training rather than from corner decoding (top-1 centre 5.216°), but it was worse on both real domains (LB 28.8°, SL 39.1°; the DINOv3-alone K tree 43.1° vs 37.1° on LB). The arm changed both the targets and the loss (level-marginal instead of per-level soft cross-entropy), so the two cannot be separated. It was excluded from the locked recipe (DEC-018), and the T-LESS arm K of EXP-106 was descoped (DEC-024 amendment).

**The rotation augmentation was not what fixed the flips (EXP-115).** Our initial hypothesis, recorded in the contribution inventory after EXP-109, was that the geometry-correct in-plane rotation augmentation causes the real-domain flip repair. The ablation refuted it: with orbit voting, photometric-only training matches the combined recipe on real images (LB 12.29° vs 12.30°, SL 12.37° vs 12.25°), while rotation-only training stays near the clean trees (18.69° / 20.70° vs 20.25° / 24.83°). Rotation augmentation gives the best synthetic precision (3.78°) and is favoured by the val-aug proxy, which itself contains random in-plane rotations and so over-rewards rotation training by construction. The roll transform remains necessary for orbit voting at inference.

**Learning orbit consistency with a hierarchical transport loss (EXP-116).** Transporting the tree's level-wise distribution under a known camera roll ($p(T_\theta I) \approx K_\theta\, p(I)$) beat the base recipe and a paired-view data control on single-view val-aug (DINOv3 greedy 9.57° vs 9.88° and 10.04°), but not a plain leaf-level transported equivariance loss (9.43°); the pre-registered criterion therefore fails and we make no

hierarchical-transport claim. The leaf-level loss is a known equivariance regulariser; it improved synthetic precision (PoE + orbit 3.71° vs 3.87°) at zero inference cost, left lightbox unchanged and severe flips unchanged, and was not adopted without a user decision.

**The orbit-product scoring rule (EXP-114).** The pre-registered claim that the product of tree likelihoods over the orbit adds something beyond simpler voting *failed* on synthetic validation: a kernel-weighted vote over the same 32 candidates was more precise (PoE 3.46±0.03° vs 3.87±0.03°; DINOv3-only 3.71° vs 4.21°). The product passed only marginally on val-aug for PoE (5.82° vs 5.98°) and not for DINOv3-only (6.25° vs 6.30°), and was best on the real domains descriptively. Max-confidence and top-1 voting were clearly worse than both. The claim is therefore "multi-view orbit consistency over hierarchical candidates"; the product remains in the recipe only because DEC-018 fixed it before this experiment.

### M.3.5. Beyond SPEED+

**Zero-shot transfer of the SPEED+ trees (EXP-064 zero-shot arm).** The frozen SPEED+ trees were applied to the other datasets without retraining, with metrics fixed in advance: raw error, uniform chance, a frame-free pair error that cancels the unknown offset between object frames, and a 3-DOF calibrated error. On T-LESS the pre-set rule was literally met (PoE greedy pair error 95.6° vs 104.5° uniform chance), but a post-hoc control showed that predicting one constant rotation does better (88.7°), because T-LESS objects are viewed from above a table; the same holds on YCB-V (pair error 95.9° for DINOv3 greedy and 101.6° for PoE + orbit, vs 57.8° for a constant). On SYMSOL-T, where rotations are uniform, PoE + orbit beat chance (47.8° vs 53.7°) only because of the cube (PoE greedy pair 27.8° vs 41.2° constant); every other shape was at chance. There is no general zero-shot transfer, and the trees must be retrained per object. The constant-prediction control was added after the T-LESS result and is labelled post hoc.

**YCB-V trained on real videos only (EXP-064).** With per-object trees trained on the real YCB-V training videos (16–20 sequences per object), the test error was 69.4° (DINOv3 + orbit) against about 126° for random rotations, and PoE was worse than DINOv3 alone. Four checks ruled out a pipeline error: the trees fit their training frames (3.8° and 3.9° for mustard bottle and mug), test poses are covered by training (median nearest training rotation about 5° for three objects whose test errors are 75–87°), every crop contains all object label pixels, and 65–98% of projected model points fall on the object mask (28–83% for a transposed-rotation control). The trees memorise the cluttered training videos. Adding the synthetic `data_syn` renders composited on real training backgrounds (EXP-122) reduced the validation PoE + orbit error from 74.1° to 12.7° and the test error from 79.9° to 12.2°; scissors remain the outlier (49.4° / 36.9° for DINOv3 / PoE).

**Orbit voting does not always help (EXP-064, EXP-122).** Orbit voting is the most consistent gain on SPEED+, T-LESS and SYMSOL-T, but there are two exceptions. On T-LESS real test images it removes few gross errors (>90° fraction 0.055→0.051), unlike on SPEED+. On YCB-V with synthetic training data it makes validation *worse* for both variants (PoE greedy 10.52° vs orbit 12.74°; DINOv3 8.97° vs 11.53°) and does not help the test mean (PoE 10.90° vs 12.16°), with large seed spread on test (DINOv3 greedy ±6.9°). "Orbit voting transfers wherever the base model generalises" is therefore too strong as stated in the EXP-064 decision; the EXP-122 report has not yet recorded a verdict.

**Canonical labels break orbit voting (EXP-106 A1).** For symmetric objects, replacing each label by its equivalent rotation closest to identity (A1) removes contradictory labels and helps a single view (T-LESS validation 12.3°→11.5°; SYMSOL-T 3.87°→3.40°), but the canonical label is not related across in-plane rotations by the roll transform, so the orbit views disagree: SYMSOL-T orbit error rises to 9.73° (A0: 2.65°) and T-LESS test orbit error to 18.2° (A0: 16.6°; T-LESS R@10 73.4%→65.9%). Training on all equivalent rotations (A2) keeps the relation and is best with or without orbit voting (SYMSOL-T orbit 2.10°; T-LESS validation orbit 8.3°, test 15.2°). This negative result motivated keeping A2. Arm K and the render-and-compare and flow best-fit stages of EXP-106 were descoped (DEC-024 amendment). We also record a protocol deviation: the pre-registration named the hard-path level-marginal loss for A2, but the amendment used the recipe's geodesic soft targets averaged over equivalents, because the former would not reduce to A0 for objects without symmetry.

**Remaining gap to fine-tuned methods.** On SYMSOL-T our best orbit result (A2 + orbit, 2.07° averaged over the five shapes) is about 4× LiePose's [13] 0.55°; on T-LESS R@2 is 5.7% against 34.9%. The depth-5 quantisation floor (2.31° mean on SYMSOL-T, post hoc), the frozen backbone and LiePose's GT visible masks are confounded in this gap.

### M.3.6. Process failures and policy reversals

**Real-domain validation split, adopted and withdrawn (DEC-005, 006).** Because synthetic validation contains almost no errors above 90°, a seeded 20% of lightbox was briefly designated for selection (DEC-005). The user withdrew it the same day; it never influenced a training run or checkpoint, but the per-run files of that period still list

`lightbox_val`/`lightbox_test` rows, which are descriptive only. The price of the strict policy is that flip repair can never be selected for directly; the val-aug proxy of EXP-109 was introduced for this reason, and EXP-110 and EXP-115 show it does not always agree with real data.

**Compact-feature gate run late (EXP-064).** The augmented and orbit views used since EXP-109 come from a compact feature extractor, and the pre-registered equivalence gate against the full caches was only run after the T-LESS results. It failed its strict tolerance (MoGe-2 max standardised difference 0.51, p99 0.004; DINOv3 0.0108), and its pre-set functional follow-up found no PoE prediction change on 512 crops (single-branch greedy leaves changed for 1 and 3 crops). The failure is kept on record.

**Infrastructure incidents.** Besides the EXP-012 host crash, a SYMSOL-T bug produced NaN losses in 9 MoGe-2 trees (always-empty border cells with standard deviation near $10^{-6}$ overflowing float16); they were retrained with a standard-deviation floor and the invalid first evaluation was not reported. The cache move of DEC-020 broke several queued runs through a relative path. For EXP-124, the memory guard aborted the detector training at a reported 44.3 GB because it summed the shared pages of six DataLoader workers (real peak 11 GB); the orphaned workers stalled the queue for about 14.7 h. The guard now counts child processes by unique memory and kills them on abort, and the run was retried with identical settings.

### M.4. Claim boundaries

The project keeps a living claim document. Its contribution inventory and shortlist were written on 2026-10-16 and were later narrowed by dated updates; we give the boundaries as they stand after those updates, and say which earlier entries were superseded.

**What is claimed.**

- **A lightweight hierarchical rotation predictor on frozen features.** One Tesseract-path Transformer per frozen backbone, fused per level by a training-free PoE, about 4.5 M trainable parameters. Under the locked recipe it is far more accurate than a count-matched flat classifier over SPACE-HOP's Hopf grid (EXP-113: 3.87° vs 7.53° synthetic with orbit voting; better in all 8 real comparisons; 2.4 M vs 36.1 M parameters per branch).
- **Real-domain error on SPEED+ is a mode-selection problem.** The large errors are 180° body-axis flips whose correct mode is already among the hypotheses (EXP-104, 105), and scorers that read the clean-trained features cannot select it (EXP-042, 045, 105, 108).
- **Lighting-robust training plus multi-hypothesis orbit voting resolves most flips.** PoE + orbit reaches 3.87° / 12.30° / 12.28° against 5.54° / 26.9° / 36.7°, fixing 62%/80% of the >150° failures and breaking 0.4% of good predictions (EXP-109); the real-domain share comes from photometric augmentation and orbit voting (EXP-115). The claim is multi-view orbit consistency over hierarchical candidates, not a specific scoring rule (EXP-114).
- **Orbit distillation** improves both the single-view tree and the orbit system on SPEED+ (EXP-117: 3.69° / 11.21° / 11.09°), reported as an enhancement; self-distillation from test-time-augmentation ensembles is known, so the novelty is limited to its use with hierarchical rotation targets.
- **6-DoF** with the SPACE-HOP translation formulation on the tree tokens, trained jointly without hurting rotation (EXP-118).
- **Cross-dataset**: with per-object retraining, the tree and orbit recipe transfers to T-LESS and SYMSOL-T, and symmetry-aware multi-path targets (EXP-106 A2) improve both and keep orbit voting effective (Sec. 5).

**What is not claimed.**

- **No "first" claims.** Hierarchical $SO(3)$ grids, the recursive tesseract and its corner formulation [15], flow matching for pose [2, 13], discrete flow matching [6], grid-based $SO(3)$ densities [16], augmentation against the SPEED+ domain gap, test-time voting and PoE [9] all pre-exist. Our tree is a cell-centre, unique-path variant of the tesseract.
- **Not a better discretisation.** Per hypothesis the Tesseract is less precise than Hopf and Super-Fibonacci grids (EXP-003, 101). The tree's head is faster than a flat head only from about $10^5$ hypotheses and depends on hardware: on a large shared GPU at batch 1 the flat 131k-way head was about 3× faster (0.40 vs 1.23 ms), and at SPACE-HOP's 3,072 anchors the flat head is faster on every device (EXP-101).
- **Hierarchy vs grid is open.** EXP-113's flat baseline used a grid whose quantisation floor is 6.06° against 2.51° for the tree's leaves, so the gain is confounded with grid geometry. Until EXP-127's same-grid arm runs (Sec. 5), we do not attribute the gain to hierarchical prediction; if the same-grid flat head ties the tree, the claim is withdrawn and the gain is attributed to the grid.
- **The flow is not calibrated.** The coupled posterior overcovers (0.98 at the 0.9 level) and the cell-conditioned refiner under-covers (0.80); re-noise/return consistency is not a normalised likelihood (only the CNF log-density is); parent–child consistency of counted masses holds by construction.
- **Flip candidates do not repair real flips**; only augmentation plus orbit voting does, and only for 62–80% of them.
- **No geometry claim for the path flow (EXP-108)**: token interpolation is not geometry-preserving, the no-geometry control was not run, and the method did not help.

- **No hierarchical-transport claim** (EXP-116) and no claim that the orbit product beats weighted voting (EXP-114).
- **Not training-free and not zero-shot.** The backbones are frozen but the trees are trained; cross-object heads are retrained per object, and the SPEED+ trees do not transfer zero-shot.
- **MoGe-2 fusion is not general**: it helps on SPEED+ and T-LESS, not on SYMSOL-T or YCB-V.
- **External comparisons are unmatched**: SPEED+ numbers use ground-truth crops, the translation relies on the ground-truth box (which alone gives 0.56 m), and published methods use detected boxes and the official score (EXP-124 is queued for this); our T-LESS crops are not masked, unlike LiePose's.
- **Do not combine experiments never evaluated together**, e.g. the locked recipe with the flow verifier or with $w = 0.8$.

**Superseded entries of the claim document.** Several entries of the 2026-10-16 inventory no longer hold. "A tree-vs-flat comparison under the final recipe is missing" was answered by EXP-113 (and is now qualified by the EXP-127 confound). The claim that the geometry-correct rotation augmentation drives flip repair, and the "missing photometric-only control", were answered by EXP-115, which moved the real-domain credit to photometric augmentation. "Cross-object validation: none yet" was answered by EXP-064 and EXP-106. The DINOv3-only flow variant listed as running (EXP-111) was completed and not adopted, and the unmeasured $4\times$ inference cost was measured in EXP-112 (279 ms per image at batch 1 for PoE + orbit, 70 ms for DINOv3-only + orbit).